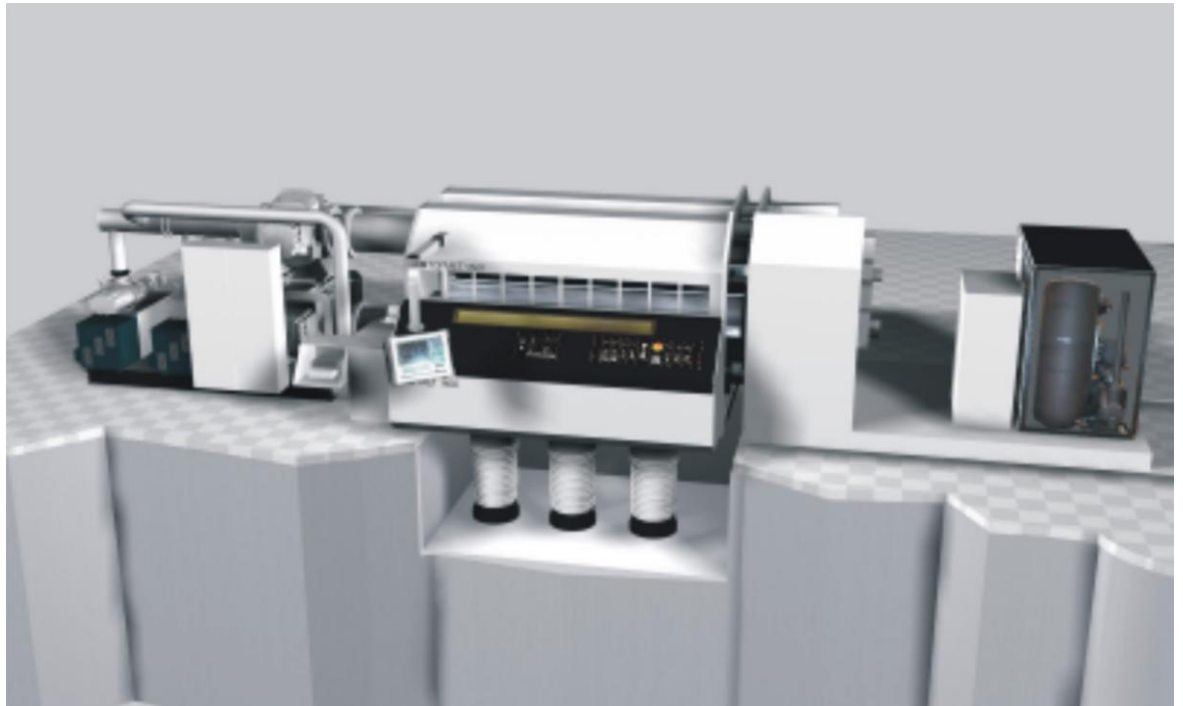




TOPMET 2450

Operating Instructions

Edition 02/2012 (P004834)

Applied Materials GmbH & Co. KG
Siemensstrasse 100
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1

Use and Technical Data

The chapter "Use and Technical Data" defines the proper use of the TOPMET web coating system, with all related specifications.

1.1 Copyrights

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We reserve the right to make changes as technical progress may warrant.

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1.2 Proper use

The TOPMET web coating system serves exclusively to coat film-shaped, flexible materials (in the following referred to as substrates) on one side with aluminum.

The safety precautions stated in Chapter 2, Safety of these instructions must be complied with for all processes performed using the TOPMET web coating system.

If coating processes developed by APPLIED MATERIALS GmbH & Co. KG (APPLIED MATERIALS) are used, then additionally the safety precautions in the corresponding process instructions must be adhered to.

Any and every use beyond these specifications is considered improper and can result in serious injuries or equipment damage. APPLIED MATERIALS will assume absolutely no liability for any such injuries or damage.

	Note!
	The process materials listed below were specified for operation of the TOPMET web coating system. The vacuum system is designed only for the specified gases. The vacuum system is not suited for toxic or explosive gases. No other target materials may be evaporated nor may other process gases may be used without the consent of APPLIED MATERIALS. Use of other materials or gases is considered to be improper. No liability will be assumed for any resulting damage.

Specified process materials:

Process materials	Designation	Symbol/formula
Gases:	- Argon - Oxygen	Ar O ₂
Evaporating materials:	Aluminum (wire)	Al

1.3 Hazards

	⚠ DANGER Operating and maintenance personnel must have the correct training before adjusting and maintaining high-voltage electrical equipment. This equipment must be adjusted according to applicable requirements.
---	--

The TOPMET web coating system is equipped with protective equipment, and its safety was tested and accepted. Despite this, in case of misuse, there are hazards:

- for the life and limb of the user.
- for the TOPMET web coating system and other equipment of the owner.
- for efficient work with the TOPMET web coating system.

All persons who deal with setting up, starting up, operating, maintaining or servicing the TOPMET web coating system must carefully read and adhere to these instructions.

It is a matter of your safety!

1.4 Safety Responsibilities

For the safety of the TOPMET web coating system the information provided in Chapter 2 of these operating instructions apply. The responsibilities for the safety of the TOPMET web coating system are as follows:

- APPLIED MATERIALS is responsible for the safety of the TOPMET web coating system!
- The owner (customer) is responsible for safety around the TOPMET web coating system!

1.4.1 APPLIED MATERIALS assumes responsibility for:

TOPMET web coating system	Scope of responsibilities
Exhaust gases of the pump system	Up to the interface between the high vacuum pumping station of the TOPMET web coating system and the fore vacuum pumping station.
Substrate handling	Entire substrate transport within the TOPMET web coating system.
Grounding	Starting at the interface between the grounding connection lead and the TOPMET web coating system.
Supply of electrical power	Starting at the point where electrical power is fed to the TOPMET web coating system.
Water	Starting at the interface between the water line and the TOPMET web coating system.
Supply of compressed air	Starting at the interface between the supply line for compressed air and the TOPMET web coating system.
Supply of process gas	Starting at the interface between the supply line for process gas and the TOPMET web coating system.



1.4.2 The customer assumes responsibility for:

Around the TOPMET web coating system	Scope of responsibility
Exhaust gases of the pump system	Starting at the interface between the high vacuum pumping station of the TOPMET web coating system and the fore vacuum pumping station, and extending to the point of gas disposal. Providing the exhaust system, in accordance with the specifications of chapter 1.6.8 of these operating instructions.
Substrate handling	Loading and unloading the winding system with substrates. Feeding and removing substrates from the TOPMET web coating system.
Grounding	Grounding connection. Grounding lead to the grounding bar of the TOPMET web coating system.
Supply of electrical power	Providing the electrical power, in accordance with the specifications of chapter 1.6.5 of these operating instructions. Connection leads to the terminals of the TOPMET web coating system.
Water warm and cold	Providing the water, in accordance with the specifications of chapter 1.6.2 of these operating instructions. Water line to the TOPMET web coating system.
Supply of compressed air	Providing the compressed air, in accordance with the specifications of chapter 1.6.3 of these operating instructions. Compressed air supply line to the maintenance unit of the TOPMET web coating system.
Process gases	Providing the process gas, in accordance with the specifications of chapter 1.6.7 of these operating instructions. Process gas supply lines to the TOPMET web coating system.



1.5 System Identification

When contacting APPLIED MATERIALS, please have the information on the system label (or the label of the individual components).

The system label is located on the front side of the feed cabinet + E1 (see Fig. 1- 1). It states the specifications of the system.

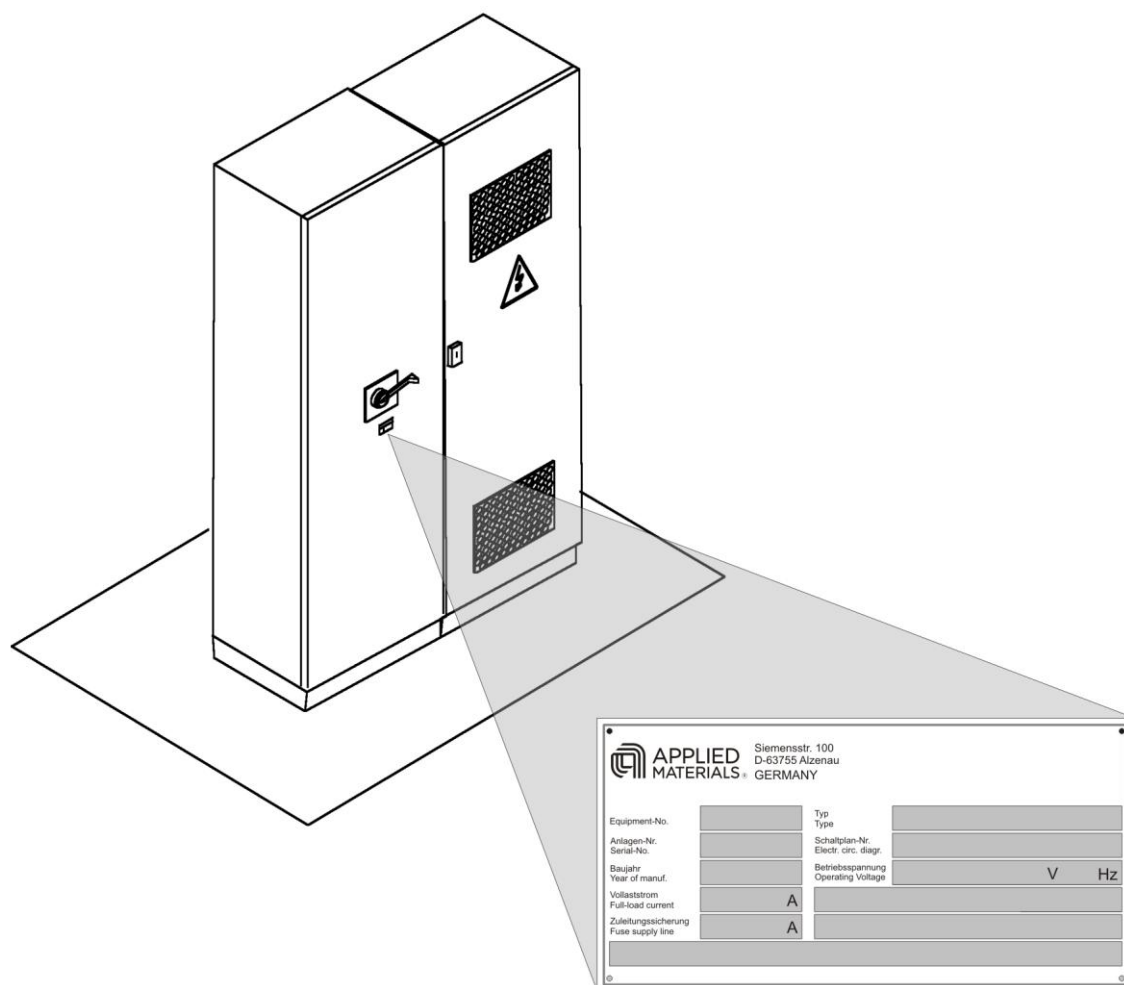


Fig. 1- 1 System label

Equipment-No.:	Is the predefined system ID number of the plant.
Serial-No.:	Is the material number (top parts list number) of the overall system (not the material number for “electrical equipment”).
Full-load current:	Nominal current of the calculated connection power.
Electric circuit diagram	Is the wiring diagram number.

1.6 Technical data

1.6.1 Dimensions and Weights

- Weight approx. 44 t
- Chamber cross-section approx. 2.4 m x 2.6 m

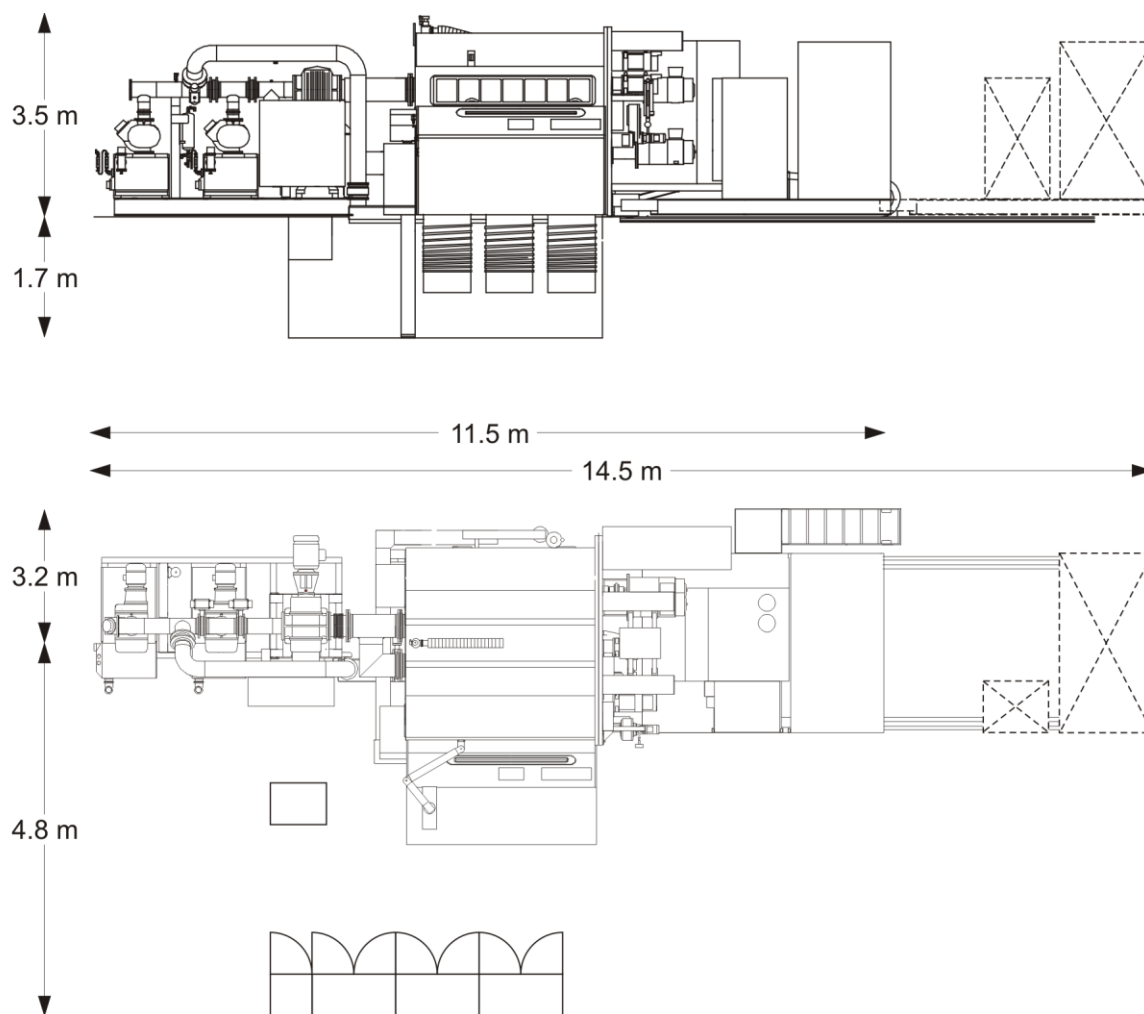


Fig. 1-2 Standard layout of the TOPMET web coating system

For the ground load and the loading points when setting up the TOPMET web coating system, please refer to the supplied ground load and foundation plan.

1.6.2 Cooling water specification

- Pressure 4 - 6 bar
- Pressure drop min. 3 bar
- Maximum flow approx. 40 m³/h
- Mean flow approx. 26 m³/h
- Temperature (inlet) min. 20° C / max. 25° C
- Temperature rise approx. 6° C
- Carbonate hardness < 4° dh
- ph value 8.5 – 10
- Particle concentration max. 10 particles/cm³
- Particle size max. 100 µm
- Cl content max. 20 mg/l
- CO₂ content max. 15 mg/l
- O₂ content 8 - 9 mg/l
- Fe content < 0.2 mg/l
- Mn content < 0.1 mg/l
- M-NO₃ content (nitrate) < 10 mg/l
- M-SO₄ content (sulfate) < 50 mg/l
- Recommended water additive Nalco 00GE056 or similar equivalent

1.6.3 Compressed air specification

- Pressure 6 - 7 bar
- Mean consumption approx. 1.0 m³/h
- Particle size max. 5 µm
- Particle concentration max. 5 mg/m³
- Oil content max. 5 mg/m³
- Dew point 2° C (at working pressure)



1.6.4 Ambient conditions

- Temperature min. 15°C / max. 35°C
- Humidity max. 90% at 20°C / max. 50% at 35°C

1.6.5 Electrical connection values

- Voltage 3 phases, PEN, 400 V, + 6 / - 10 %
- Frequency 50 Hz
- Connection power Approx. 750 kVA
- Mean consumption Approx. 270 kVA
- Input fuse per phase 1000 A
- Short-circuit current max. 50 kA
- Internal control voltage 24 VDC

1.6.6 Requirements for substrates and drums**Nature of the film**

The system is specified to coat the following materials:

Material:	Film thickness (min. – max.) :
• BOPP	12 – 50 µm
• CPP	20 – 60 µm
• HDPE	25 – 60 µm
• LDPE	25 – 60 µm
• PA (Nylon)	12 – 25 µm
• PET	8 – 50 µm
• PVC, unplasticized	20 – 60 µm

Quality of the material

All specified data only applies for films and film brands whose quality for film coating is generally recognized, and who fulfill the standards of internationally known manufacturers.

Each and every film or film brand used for acceptance tests must fulfill the above standards. On demand, films not up to the standards will be replaced by such films which do fulfill the standards. No partner will refuse to agree to replace the films without stating his reasons.



Roll dimensions

The system is specified for coating rolls with the following dimensions:

- Roll width min. 1,270 mm / max. 2,470 mm
- Roll diameter max. 1,000 mm
- Tube length max. 2,480 mm
- Tube inner diameter 6" / 152 mm
- Tube material Plastic or impregnated paper

1.6.7 Process data**Vacuum**

- Working pressure approx. 4×10^{-4} mbar
- Pumping time (dry, clean and empty chamber) < 6 minutes for film

Winding

- Winding speed 1 – 14 m/s
- Acceleration/braking 45 sec
- Winding tension 60 – 600 N for unwinder
60 – 600 N for rewinder
40 – 400 N for tension roll

Coating

- Number of evaporator boats 26
- Nature of coating Single-sided
- Coating area Full web width, with 10 mm edge masking
- Coating width min. 1,250 mm / max. 2,450 mm
- Coating material Aluminum, purity better than 99.7%
(wire diameter d: $1.5 \text{ mm} < d < 1.8 \text{ mm}$)
- Coating uniformity $\pm 5\%$ for film, $\pm 10\%$ for paper
- Coating thickness 0.5 - 3 OD (thickness measurement system LMS-OD)
0.8 - 10 Ohm/area (thickness measurement system LMS-R)



Process gases

- Argon Purity 99.998%
- Oxygen Purity 99.999%

1.6.8 Exhaust air system

The specifications for the exhaust air system depend on the local regulations wherever the system is installed.

1.6.9 Noise level

	Note !
	After setting up the system noise measurements must be taken. Depending on the noise level the customer must take all required steps to protect operating personnel in accordance with noise protection regulations and standards.

Information concerning noise level can be found in this documentation as follows:

- drawing showing measuring points for noise level measurements on the spare parts catalog CD-ROM or in the mechanical drawings folder.
- measuring report (Vacuum measuring device verification) in the OEM documentation under supplier APPLIED MATERIALS.

1.7 Vacuum system

1.7.1 Pumping system

The pumping system of the TOPMET web coating system consists of the following main components:

- Fore vacuum pumps:
 - rotary vane pump SOGEVAC SV 750 B(F)
 - roots pump RUVAC RA 3001 / 7001 / 9001
 - roots pump WAU 1001/2001 / RA 3001
- High-vacuum pumps:
 - oil diffusion pump DIP 20000
- Polycold coldtrap



1.7.2 Vacuum diagram

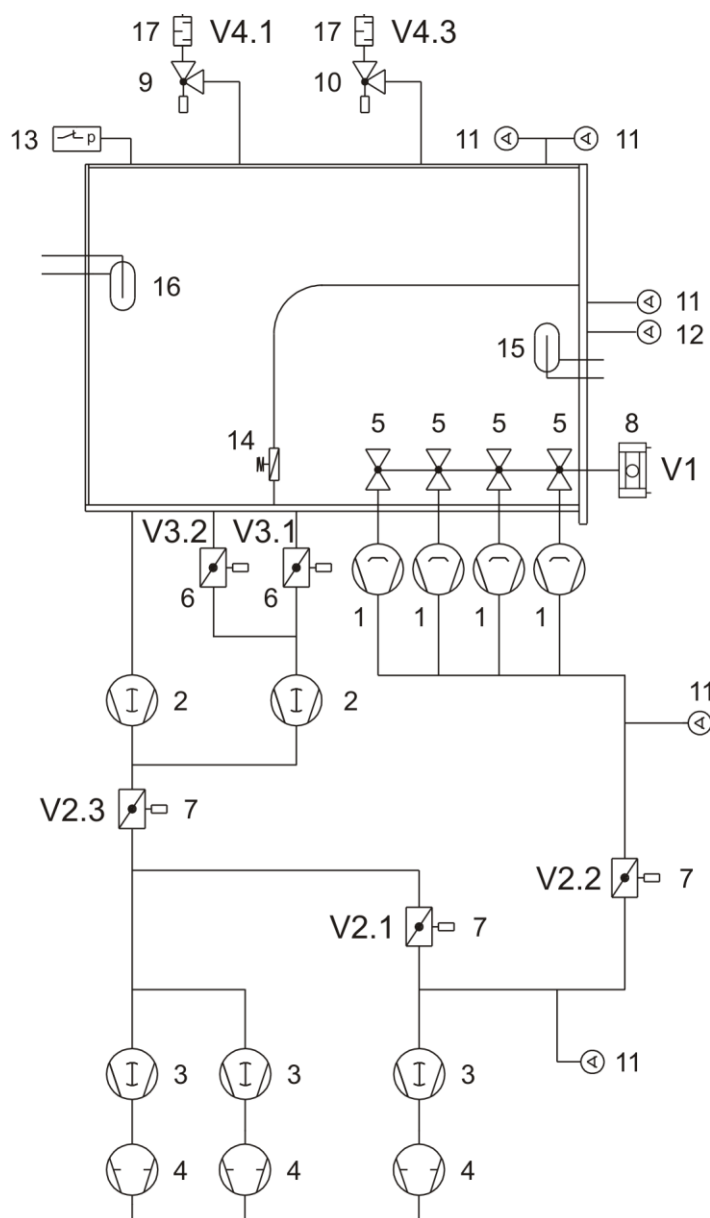


Fig. 1- 3 Vacuum diagram of the TOPMET plant



Oil diffusion pump



Roots pump



Rotary vane pump



Cooling trap

For a detailed explanation of the vacuum diagram, refer to chapter 3.

2

Safety and Access Rights

The chapter "Safety and Access Rights" illustrates the safety philosophy of the TOPMET web coating system. It provides information on the residual hazards, and explains all safety measures and warnings.

2.1 Be sure to read!

	Note!
	It is imperative that all persons involved in commissioning, operating, maintaining and servicing the TOPMET web coating system read Chapter 2, Safety and access rights!

Areas of responsibility of the system owner.

The system owner (customer) is responsible for ensuring,

- that each person who deals with starting up, operating, maintaining and servicing the TOPMET web coating system has read and understood those parts of these operating instructions of relevance to him or her, in particular Chapter 2.
- that all persons who deal with the TOPMET web coating system are technically qualified, appropriately trained and instructed, and have been informed of the hazards and risks.
- that unauthorized persons are not able to access the TOPMET web coating system.
- that the user always complies with the general safety regulations in section 2.2 and all notes related to safety and warnings contained in the following sections.
- that persons who load and unload the TOPMET web coating system with substrates are adequately trained and sufficiently informed about potential hazards.

2.2 General safety regulations

In addition to the safety regulations and the notes related to safety and warnings in the individual chapters, the following safety regulations absolutely must be complied with:

- **Proper use**

see chapter 1.2.

- **Operational regulations of APPLIED MATERIALS**

Proper use includes compliance with the installation, commissioning, operational, maintenance and service regulations prescribed by APPLIED MATERIALS.

- **Training the personnel**

The TOPMET web coating system may only be operated, maintained and serviced by authorized, instructed persons. These persons must be specifically informed about potential hazards.

- **Improper behavior**

Any operation which impairs safety of the user and of the TOPMET web coating system is prohibited.

- **Unauthorized modifications**

All modifications by end users which affect the safety of the TOPMET web coating system are forbidden. Contact APPLIED MATERIALS first.

- **Obligation to report modifications**

The end user is obliged to promptly report any changes to the TOPMET web coating system to APPLIED MATERIALS.

- **Maintenance and servicing obligation**

The system owner is obliged to operate the TOPMET web coating system only in proper condition.

- **Workplace and surroundings**

By means of instructions and inspections, the system owner must ensure that the workplace and surroundings of the TOPMET web coating system are uncluttered and clean.

- **Shipping and commissioning**

The TOPMET web coating system may only be shipped in the original packing. Only technically qualified and authorized personnel of APPLIED MATERIALS is permitted to commission the system.

- **Shutdown and disconnecting**

For all maintenance and service work, the TOPMET web coating system must be switched to the required operational status as specified in Chapter 7 to ensure that the effected parts of the system are disconnected from electrical power or are secured.

- **Removing protective equipment**

Before protective equipment may be removed, the TOPMET web coating system must first be shut down completely and then disconnected. Before connecting again, make sure all of the protective equipment is reinstalled.

- **Inspections after service work**

After service work, make sure all protective equipment is in place and works flawless correct (e.g. safety circuits). In particular, after electrical installations and electrical service work, check the protective devices such as the grounding system and the protective ground leads. Only then may the TOPMET web coating system be restarted.

- **Media supply**

Water hoses, electrical power leads and gas lines must be conducted separately to the TOPMET web coating system. They must be protected against mechanical stress. They must be run without restricting the safety and reliability of the TOPMET web coating system in any way.

- **Accident prevention regulations of specific lines of business**

Additional accident prevention regulations for your line of business and for your local area always apply when operating the TOPMET web coating system.

- **Operating instructions**

These operating instructions must always be available at a set place near the TOPMET web coating system. The personnel must be informed about this location.



2.3 Structure of safety precautions

In the various chapters of this manual, the following safety precautions point out the different degrees of danger.

	⚠ DANGER Stands for work or operational procedures which directly cause serious injuries or death if not adhered to.
	⚠ Warning Stands for work or operational procedures which must be adhered to precisely to rule out possible serious hazards to persons and/or the environment.
	⚠ Caution Refers to work and operational procedures which must be adhered to precisely to avoid slight injuries, environmental harm or damage to equipment.
	Note! Is placed before technical requirements which must be especially noted.

2.4 Safety philosophy

2.4.1 General information

To ensure maximum safety during production and maintenance, this system was constructed in accordance with the guidelines of a safety philosophy. The latter is based on design and information techniques for eliminating safety-critical system areas, resulting from a safety analysis.

Dangerous areas are eliminated by:

- identifying and naming the dangerous area (warning symbols, documentation)
- design measures at the dangerous areas (protective covers, safety circuits, safety interlocks)

2.4.2 Access to the operating rooms

The system owner is responsible for ensuring that only authorized persons have access to the operating rooms of the TOPMET web coating system (see chapter 2.1).

	⚠ DANGER Forbidden for persons with pacemakers. Magnetic fields influence the functioning of pacemakers. This can disturb the heart beat or cause death. The system owner absolutely must ensure that no endangered persons enter the dangerous area near the cathodes. The minimum radius is 5 meters.
---	---

2.4.3 How to react to dangerous situations

	Note!
	The TOPMET system is equipped with a safety line at the winding system, above the coating drum. The function of the safety line is equivalent to that of an emergency-off switch. These instructions will treat the safety line as such a switch. The safety line is used by pulling on it (corresponds to pressing an emergency-off button).

If a dangerous situation occurs during operation of the TOPMET web coating system, the system must be stopped immediately by pressing one of the emergency-off buttons.

To avoid equipment damage, you are strongly recommended to press an emergency-off button and stop the TOPMET web coating system whenever any of the following events occur:

- you hear unusual noises (when the system is being pumped down, a change in the noises of the fore vacuum pumping systems is normal)
- you notice smoke or smell a fire

The functions of the emergency-off buttons are described in chapter 2.5.1.1.

The positions of the emergency-off buttons are shown in the following figure.

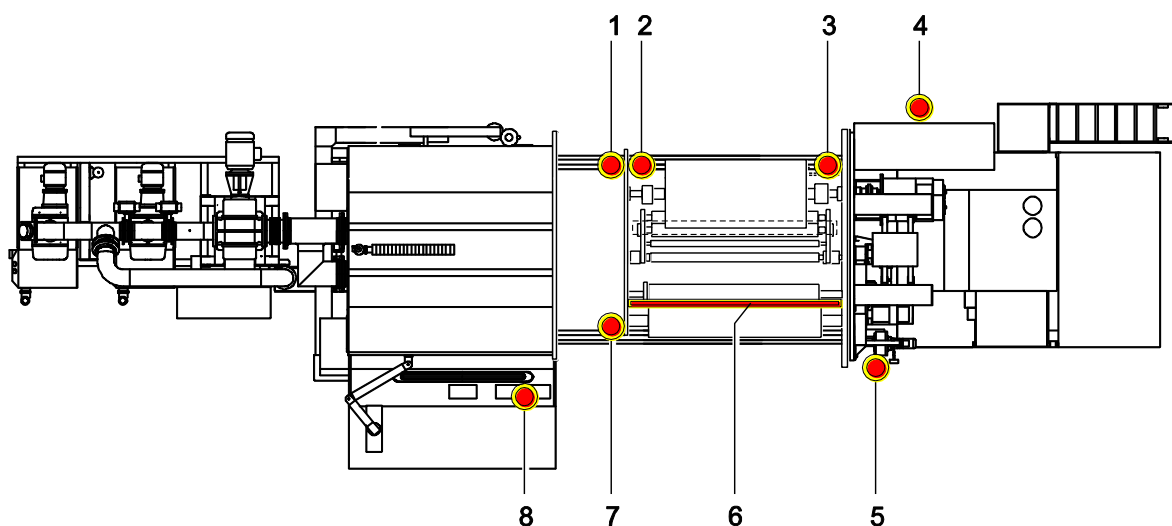


Fig. 2-1 Positions of the emergency-off buttons

- | | |
|--|---------------------------|
| 1 - Winding system (bearingplate) | 5 - Operating Panel +R23 |
| 2 - Winding system (rewinder at the right) | 6 - Safety line |
| 3 - Winding system (rewinder at the left) | 7 - Coating window (left) |
| 4 - Keypad +G21 | 8 - Keypad +R29 |

2.4.4 Definition of personnel

Only authorized, trained personnel are permitted to operate and maintain the system. The system personnel are grouped into three categories according to their training and access rights. Those assigned to a given group are required to have the knowledge and skills of the lower groups:

- Operating personnel
- Maintenance and service personnel
- Supervisors

The rights of personnel to access the various levels of the programmable system control are defined by assigning passwords. For details, please refer to your system control manual (user manual).

Access rights	Personnel		
	Operating personnel	Maintenance and service personnel	Supervisor
AUTOMATIC/ MANUAL MODE			
MAINTENANCE MODE			
ACCESS RIGHTS; RECIPES			

Fig. 2- 2 Access rights for system control

2.4.4.1 Operating personnel

Operating personnel is permitted to operate the TOPMET web coating system in normal operations (automatic coating). This comprises the following activities:

- Carrying out coating processes via the Graphic User Interface (GUI) and system control keypads
- Loading and unloading the system with rolls/films
- Providing the system with coating material
- Replacing the pressurized process gas bottles
- Cleaning the evaporation station, coating window and winding system (drums) between coating cycles

Only those persons are permitted to operate the TOPMET web coating system who have successfully completed system operation training. Employees of APPLIED MATERIALS or experienced supervisors of the system's owner must conduct the training.

2.4.4.2 Maintenance and Service Personnel

Maintenance and service personnel must have the system knowledge of operating personnel.

	⚠ Warning
	Reduced level of safety. Except for the emergency-stop buttons, none of the protective functions are enabled. Improper operation can cause serious injuries of the personnel and considerable equipment damage. Only authorized maintenance and service personnel is permitted to operate the TOPMET web coating system in maintenance mode.

The maintenance and service personnel may operate the TOPMET web coating system in normal operation, regularly perform necessary maintenance work and additionally perform all further service work which is necessary for proper operation. For this purpose, the TOPMET web coating system may be operated in manual and service mode.

Maintenance of the TOPMET web coating system comprises the following activities:

- Switching the system on and off.
- All work mentioned in the maintenance schedule (see chapter 7).

Service of the TOPMET web coating system comprises the following activities:

- Unpacking, setting up and installing the system.
- Initial start-up of the system.
- Shutting down and dismantling the system; packing the system components.
- Repair work on mechanical and electrical components.
- Configuring the system control and altering machine constants, e.g. after changes to the hardware configuration.

Maintenance and service of the TOPMET web coating system may only be carried out by trained employees of APPLIED MATERIALS or employees of the system owner with comparable training.

For work on electrical components, training as a master electrician or comparable technical training is necessary.

2.4.4.3 Supervisor

Supervisors must have the system knowledge of maintenance and service personnel.

The supervisor may operate the TOPMET web coating system and has access to all the functions of the operator station. The supervisor is entitled to create and store process recipes, assign access rights and change passwords.

Supervisors are employees of APPLIED MATERIALS or specially trained employees of the system owner.



2.5 Safety and protective interlocks

The TOPMET film system is equipped with numerous safety and protective interlocks, to protect the system personnel and avoid equipment damage. The safety and protective interlocks for process steps and operations are divided into two categories: hardware interlocks and software interlocks.

	Note!
	Only the original safety keys may be used. It is strictly forbidden to duplicate keys and to use such duplicates. Contact the manufacturer if you lose safety keys.

2.5.1 Hardware interlocks

Hardware interlocks are protective mechanisms consisting of limit switches and pushbuttons with direct contact actuation. They protect and ensure the active safety of the personnel.

2.5.1.1 Emergency-off pushbuttons

The positions of the emergency-off pushbuttons are shown in Fig. 2- 1.

The system's power supply and components remain on even after an emergency-off pushbutton is pressed. The system is still connected to the source of electrical power.

Pressing an emergency-off pushbutton stops all moving parts of the system. The following system functions are immediately interrupted and automatically locked to prevent them from being switched back on:

- Vacuum pumps (switch off).
- Winding system (is slowed to a stop).
- All motor-driven system parts are stopped (such as travelling part, coating window).
- Glow cathode (switches off).

You can unlock an emergency-off pushbutton by pulling it out.

2.5.1.2 Service key switch

The service key switch protects the personnel during maintenance and service work on motor-driven system parts (winding system, travelling part, coating window, etc.).

The safety interlock is enabled by rotating the key switch to the left position (service position). In the service position, the key can be removed.

The following are blocked and cannot be switched back on: the drives of the winding system, operation of the travelling part (chamber open/closed), the function of the pivot arm and the glow cathode.

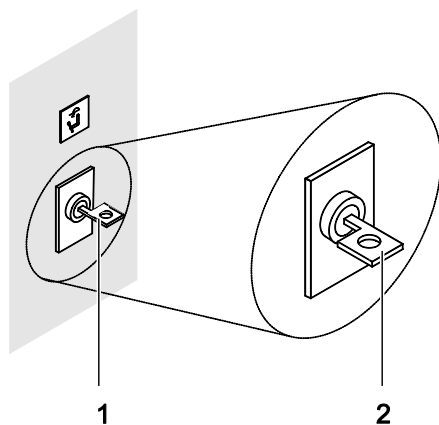


Fig. 2- 3 Key switch at the keypad +R02

1 - Position for normal mode

2 - Position for service mode

2.5.1.3 Chamber contact switch

The chamber contact switch installed on the top of the system prevents the following from being switched on:

- Evaporator station
- Pretreatment station (cathode)

When the vacuum chamber is open, the contact switch is opened, and the safety lock is enabled. The above components cannot be switched on.

2.5.2 Software interlocks

	Note!
	All software interlocks are displayed at the visualization system as warning and error messages.

Software interlocks are protective mechanisms consisting of PLC safety circuits, limit switch functions and programs for monitoring errors. These interlocks protect the system from damage and improper operation. They also protect the operating personnel.

Due to the large number of integrated software interlocks, only an overview of major interlock situations can be provided.

Interlock situations:

- Cooling water-driven system components are monitored by flow monitors. If insufficient cooling water flows, the affected system component is switched off.
- The system control monitors pressure switches and vacuum gauges. Via set switching points, it opens and closes pressure-dependent valves, and it connects or disconnects pumps.
- The vacuum chamber is only closed when
 - the pivot arm has moved to the rewinder.
 - the coating window is located at the work position.
- The coating window can only be moved when the vacuum chamber is open all the way (limit switch for travelling part).
- Sensors monitor the unwinder and rewinder couplings in the winding system. The winding motors (winding unit) can only be switched on when
 - the winding couplings are completely closed.
 - the mechanical safety interlocks are closed.
- A redundant pressure measurement system in the pretreatment station immediately stops the supply of oxygen (shut-off valves) when the chamber pressure exceeds 0.1 mbar.

2.5.3 Safety circuit (electrical safety chain)

	Note!
	You are only allowed to switch the system on and operate it if the interlock components are working properly. The system owner is obliged to inspect the following hardware interlock components and make sure they are working properly • Emergency-off pushbuttons • Chamber contact switch • Service key switch This must be done on a regular basis.

The system is monitored by an electrical safety circuit. This circuit consists of hardware interlocks placed in series, which form a safety chain.

If one of the interlock components trips, the safety chain is broken. All system functions connected with the safety chain are switched off and are prevented from being switched back on.

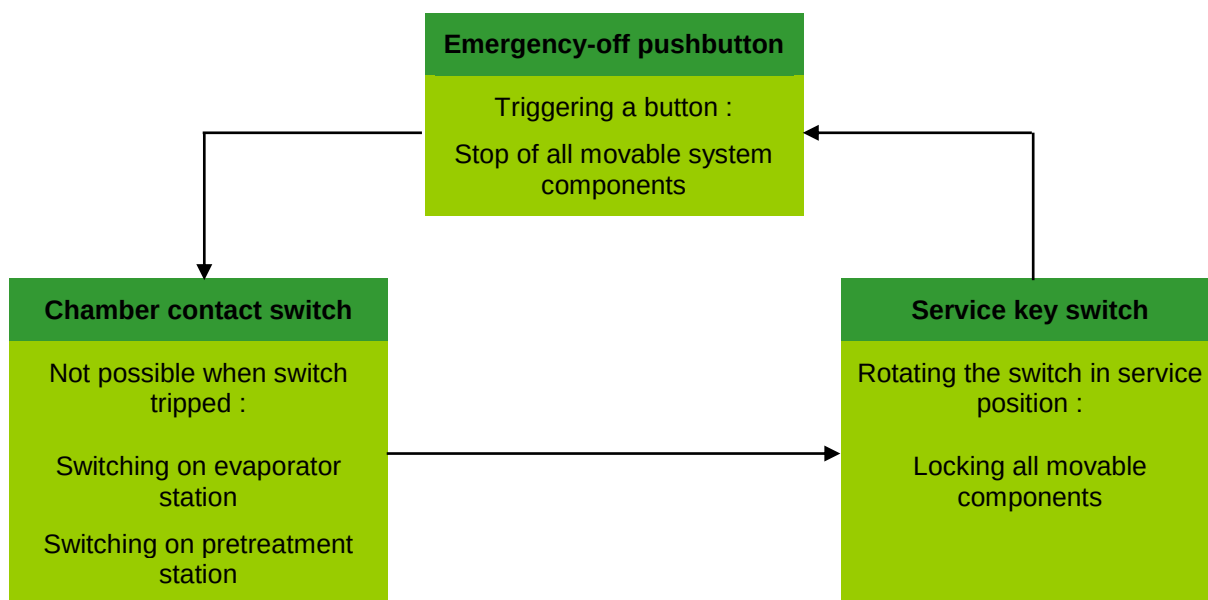


Fig. 2-4 Electrical safety chain



2.5.4 Grounding rod

	⚠ DANGER
	Danger of fatal electrical shock if the grounding rod is not used properly. The grounding rod may only be used after it is properly installed. Only service personnel with special electrical training is permitted to install and ground the grounding rod. Visually inspect grounding rods regularly and check them for proper functioning. Immediately eliminate any faults of the grounding system or insulation damage. Use only grounding rods which are not damaged and which work properly.

The grounding rod discharges residual voltage at the cathode of the pretreatment station before maintenance and service work is performed.

It is located at the pretreatment station at the chamber wall of the travelling part, to the right of the unwinder drum.

How to use grounding rods:

- Before touching any component with a grounding rod, disconnect it from the source of electrical power. Make sure it is voltage-free.
- When discharging equipment with the grounding rod, first discharge equipment in front (closest to the affected person), and then equipment further away.

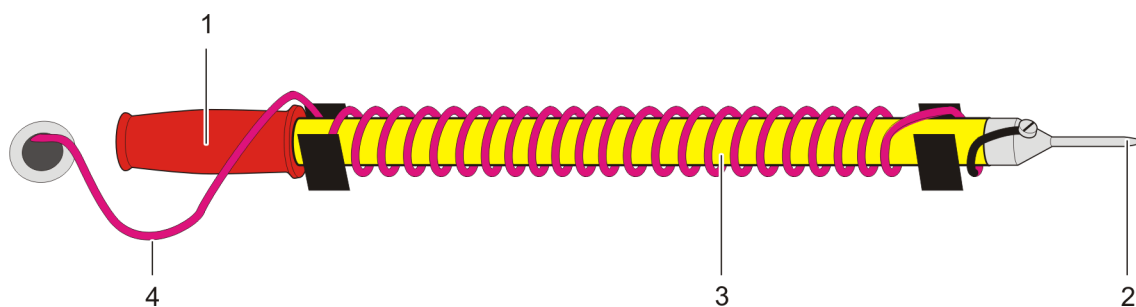


Fig. 2- 5

Grounding rod

1 - Handle

2 - Grounding cable

3 - Grounding rod

4 - Probe tip

2.6 Dangerous areas for operating personnel

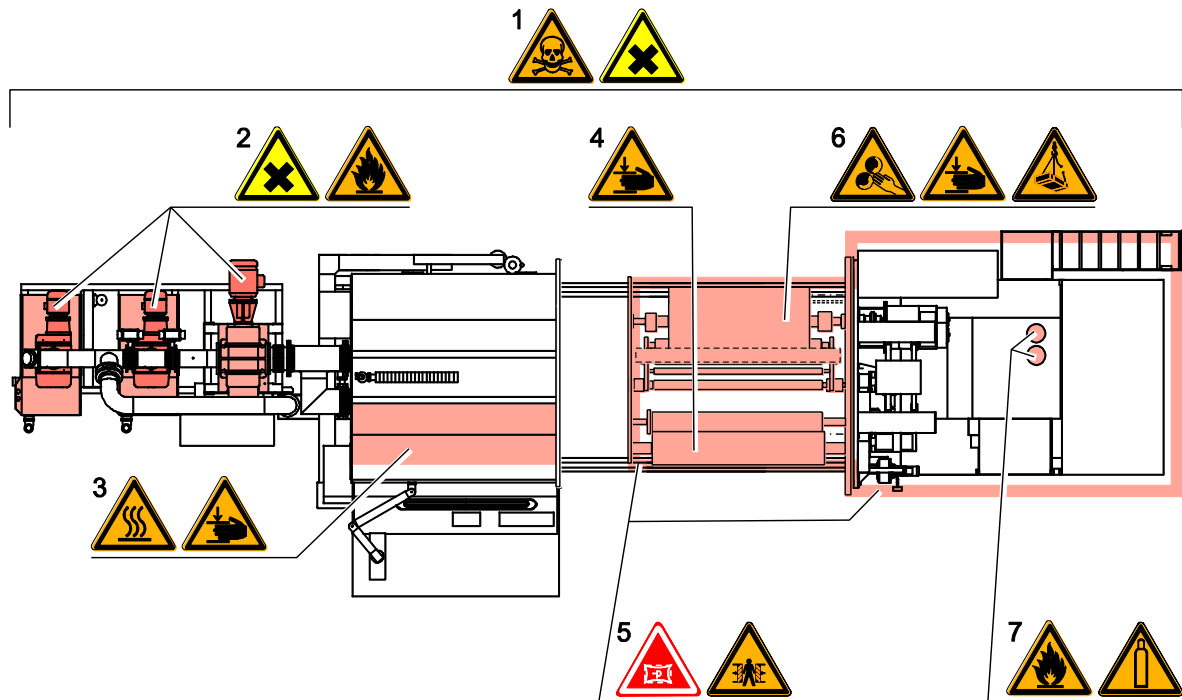


Fig. 2- 6 Dangerous areas for operating personnel

- 1 - Cleaning:
 - Health hazard from metal and metal oxide dusts
 - Health hazard when using cleansers
- 2 - Vacuum pumps:
 - Fire hazard from oxygen flowing through the fore pumps
 - Health hazard from inhaling oil mist and exhaust gases
- 3 - Evaporator station:
 - Danger of combustion from hot surfaces and molten metal
 - Danger of contusions from moving parts (evaporator flap, wire coil bobbin)
- 4 - Coating window:
 - Danger of contusions when lifting or lowering the coating window
- 5 - Travelling part:
 - Danger of suffocation if locked in the system
 - Danger of injuries when the travelling part is in motion
- 6 - Winding system:
 - Danger of injuries from the crane
 - Danger of injuries from rotating drums
 - Danger of contusions from adjustable drums (pivot arm, axial motion)
- 7 - Process gas supply:
 - Danger of combustion when dealing with oxygen
 - Danger of explosions from improper use of pressurized gas bottles

2.6.1 Cleaning

	⚠ Warning
	Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal and metal oxide dusts which are a hazard to your health. Always avoid creating dust during cleaning. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.

	⚠ Caution
	Health hazard Skin contact, eye contact and inhaling cleansers can be hazardous to your health. Use cleansers in accordance with international, national and local specifications.

2.6.2 Vacuum pumps

	⚠ Warning
	Fire and explosion hazard Pumping oxygen with the fore pumps can cause a fire. Operate the pumps only with a functioning gas ballast device. Make sure the redundant flow monitoring system always works properly. Carefully read the operating instructions for the pumps and process gases. Precisely adhere to the precautions specified in them.

	⚠ Caution
	Health hazard Oil vapors and exhaust gases can be inhaled if the exhaust filter at the exhaust gas side of the pumps is not installed correctly. Operate a pump only if its exhaust line is installed properly and leads to the outside air at a safe place.

2.6.3 Evaporator station

	⚠ Warning
	Danger of burns from hot surfaces Directly after the coating process and after the chamber is opened, the evaporator station and the evaporation material is still hot. Whenever you work at the evaporator station, wear the following protective gear as protection against hot parts and hot molten metal: • goggles • fire-resistant protective gloves • work garments which protect against molten metal (e.g. leather apron)

	⚠ Warning
	Contusion hazard The moving components • evaporator flap • wire roll frame can cause injuries. When working at the evaporator station, make sure these components do not move unintentionally. In particular, make sure you do not press the pushbutton <evaporator flap>. Never place your hands within the radius of motion of these components.

2.6.4 Coating window

	⚠ Warning
	Contusion hazard There is the danger of contusions within the radius of motion of the coating window! Before you press the pushbutton <coating window>, make sure no persons are located within the radius of motion of the coating window. Never place your hands within the radius of motion of the coating window.

2.6.5 Travelling part

	⚠ DANGER
	Danger of suffocation from vacuum When the system is being pumped down, the vacuum would kill any person who might be locked inside. Before you close the chamber, make sure no persons are located in front of, near or in it.

	⚠ Warning
	Injury and contusion hazard Moving the travelling part could injure persons standing nearby. Persons are not permitted to be located within the radius of motion of the winding machine before the chamber is opened or closed. When closing the chamber, do not reach between the chamber plate and flange.

2.6.6 Winding system

	⚠ Warning Rotating drums There is the danger that rotating drums could pull in body parts. When cleaning the winding system, especially the coating drum, wear only tight clothing. Persons are not permitted to be located directly at the winding system when the pushbutton <film feed> is pressed. Do not reach into the drum system while it is running. Another person must be on alert at the next emergency-off pushbutton.
	⚠ Warning Contusion hazard Injuries or contusions can be caused by • movements of the pivot arm. • axial movements of the rewinder and unwinder drum. Do not reach between the winding tube and the pivot arm. Do not reach between the bearing plate and the rewinder or unwinder. Persons are not permitted to be located directly at the winding system when the pushbuttons <pivot arm> and <axial motion> are pressed.
	⚠ Warning Danger of injuries from hovering loads There is the danger of injuries when inserting or removing a roll with the crane. Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use hoists.



2.6.7 Process gas supply

	⚠ Warning
	Fire hazard Oxygen promotes the spread of fires. There is a fire hazard if oxygen is mishandled. Make sure oxygen installations are sufficiently far away from sources of ignition. Do nothing which creates sparks when near oxygen installations.

	⚠ Warning
	Beware of gas bottles Pressurized gas bottles can explode and cause serious injuries and damage if mishandled. Gas installations must comply with international, national and local safety regulations for handling pressurized gas bottles.

2.6.8 Control desk

	⚠ DANGER
	Forbidden for persons with pacemakers. Magnetic fields influence the functioning of pacemakers. This can disturb the heart beat or cause death. The system owner absolutely must ensure that no endangered persons enter the dangerous area near the control desk. The minimum radius is 5 meters.

2.7 Dangerous areas for maintenance and service personnel

In addition to the dangers for operating personnel, maintenance and service personnel is exposed to further hazards described in chapters 2.7.1 and 2.7.2.

2.7.1 Electrical energy as a source of danger

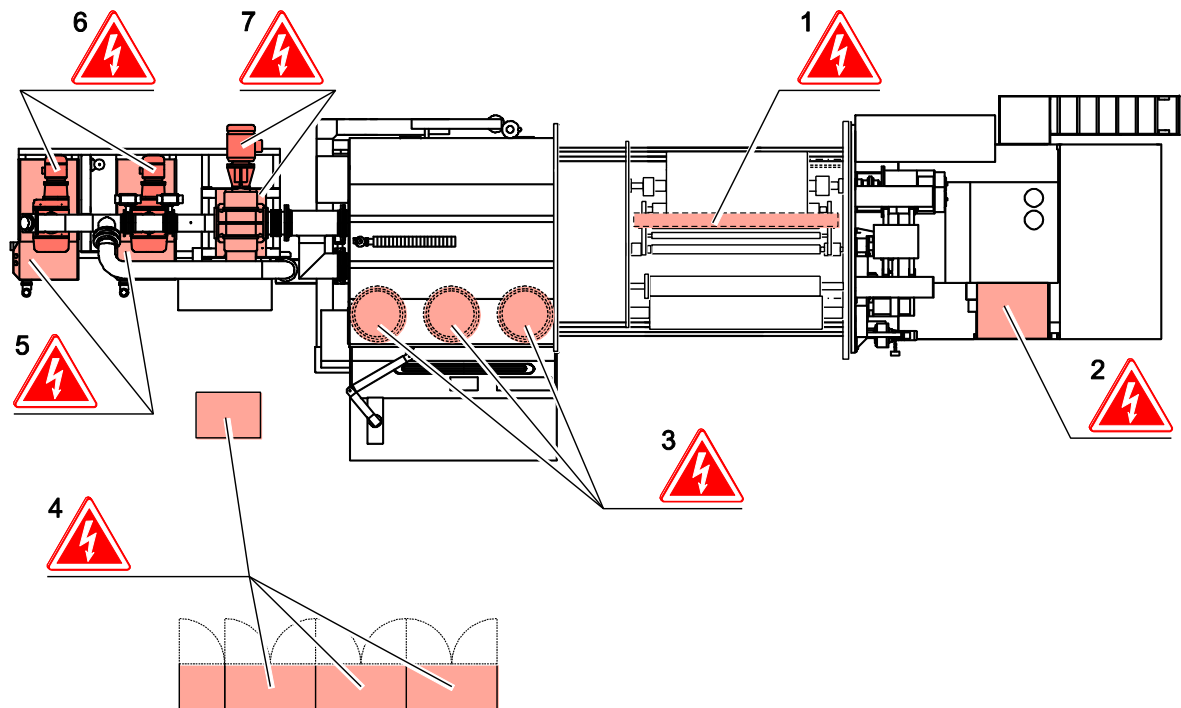


Fig. 2- 7 Areas dangerous to maintenance and service personnel due to electrical energy

- 1 - Pretreatment station (cathode)
- 2 - Polycold refrigeration units
- 3 - Oil diffusion pumps
- 4 - Electrical cabinets
- 5 - Fore vacuum pumps: rotary vane pumps
- 6 - Fore vacuum pumps: roots pumps I
- 7 - Fore vacuum pumps: roots pumps II

	⚠ DANGER
	Beware of high voltage parts. Danger of fatal electrical shock when touching high voltage parts. Service work may only be performed on the electrical equipment when it has been disconnected from the electrical power supply. Switch off the effected components and secure them against being accidentally switched on again. Always use the grounding rod to discharge those components which have a residual voltage after they are switched off. <u>Important:</u> Even with the main switch in the off position, certain system components can be live. In particular, this is true for devices connected to the maintenance outlets. The primary side of the main system switch always carries voltage. Only trained personnel with experience in high-voltage systems is permitted to perform service work on the electrical equipment.

2.7.2 Dangers from non-electrical energy

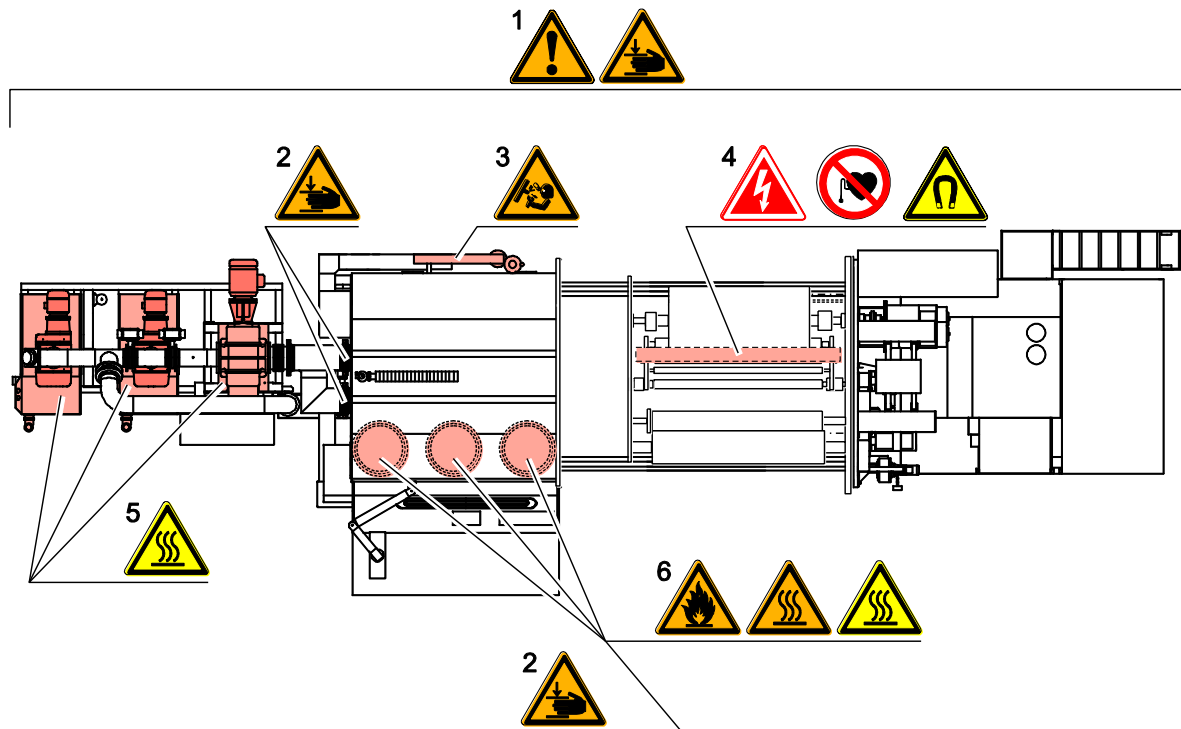


Fig. 2- 8 Dangerous areas for maintenance and service personnel

- 1 - Entire system area:
 - Reduced level of safety in service mode
 - Danger of injuries when heavy parts move
- 2 - Vacuum chamber:
 - Danger of contusions from vacuum flaps
 - Danger of contusions from high-vacuum valve lids
- 3 - Media supply:
 - Compressed air system: danger of injuries from flying parts
 - Water distribution unit: danger of injuries from flying parts
- 4 - Pretreatment station:
 - Lethal danger for persons with pacemakers
 - Lethal danger from electrical shock
 - Danger of contusions from strong magnetic fields
- 5 - Fore vacuum pumping station:
 - Danger of burns from hot surfaces
- 6 - Oil diffusion pumps:
 - Danger of explosions when venting hot oil diffusion pumps
 - Danger of burns from hot oil
 - Danger of burns from hot surfaces

2.7.2.1 Entire system area

	⚠ Warning
	Reduced level of safety When the system is operated in service mode, the level of safety is reduced, meaning a greater potential for injuries. Be especially careful when performing maintenance and service work. Adhere to the safety precautions for every job you do.

	⚠ Warning
	Danger of contusions There is the danger of contusions when dealing with heavy parts, especially: • Fore pumps and other components of the pumping stations. • Components of the process stations (cathode, wire roll frame) • Drums • Energy supply and control cabinets. You absolutely must adhere to the relevant safety precautions when dealing with heavy parts.

2.7.2.2 Vacuum chamber

	⚠ Warning
	Danger of contusions There is the danger of contusions: • When lowering the lids of the high-vacuum valves • At the vacuum flaps Perform cleaning and service work only when the relevant opening and closing control valves are secured against forbidden access. Before closing the lifting valve V1 and the valve flaps, make sure no persons are located near the open valve or the vacuum flaps.

2.7.2.3 Media supply

	⚠ Warning
	Overpressure, causing parts to fly There is the danger of injuries from flying parts when dismantling parts of the compressed air system or water distribution unit. Dismantle parts only when all pressure has been bled off the system. Before servicing the compressed air maintenance unit (e.g. replacing the filter cartridge): • Close the supply of compressed air (customer's side) and secure it against unintentional reopening. • Bleed the compressed air off the maintenance unit.

2.7.2.4 Pretreatment station

	⚠ DANGER
	Banned for persons with pacemakers Magnetic fields influence the functioning of pacemakers. This can disturb the heart beat or cause death. The system owner absolutely must ensure that no endangered persons enter the dangerous area near the cathodes. The minimum radius is 5 meters.

	⚠ DANGER
	Caution high voltage. There is lethal danger from residual voltage on the cathode. Before performing any work, discharge the cathode with the grounding rod, to eliminate residual voltage. Do not touch the cathode with your bare hands.

	⚠ Caution
	Strong magnetic fields The magnetic fields of the cathode at the pretreatment station can • Cause contusions when objects snap shut. • Make electronic equipment and magnetic cards useless. Do not use magnetized tools when working on the cathode. Lay credit cards, watches, floppy disks, etc., aside at a safe distance (approx. 5 m) from the cathode.

2.7.2.5 Fore vacuum pumping station

	⚠ Caution
	Beware of hot surfaces. There is the danger of burns when touching the housings of fore vacuum pumps and oil diffusion pumps. Let these components cool down. Wear gloves and work garments with long sleeves. Do not place any combustible textiles (work garments, cloths, etc.) on the housings. Fire hazard!

2.7.2.6 Oil diffusion pumps

	⚠ Warning
	Explosion hazard There is the danger of explosions when venting hot oil diffusion pumps. Before venting, allow the diffusion pump to cool down to less than 90° C. Check the diffusion pump's temperature on the corresponding monitor page.

	Warning Fire and explosion hazard The oil diffusion pump can explode or the area around catches fire. If the prevacuum pressure is higher than 5 mbar and the oil diffusion pump temperature is over 100°C an acoustic alarm and flash light appears. • Leave the vacuum chamber immediately! • Leave the area around diffusion pumps immediately! • Opening of the vacuum chamber is blocked!
	Warning Danger of burns There is the danger of skin burns from hot oil when working on a hot oil diffusion pump. Perform any and all work only when the pumps are cool, below 30° C.
	Caution Beware of hot surfaces. There is the danger of burns when touching the housings of fore vacuum pumps and oil diffusion pumps. Let these components cool down. Wear gloves and work garments with long sleeves. Do not place any combustible textiles (work garments, cloths, etc.) on the housings. Fire hazard!

2.7.2.7 Control Desk

	⚠ DANGER
	Forbidden for persons with pacemakers. Magnetic fields influence the functioning of pacemakers. This can disturb the heart beat or cause death. The system owner absolutely must ensure that no endangered persons enter the dangerous area near the control desk. The minimum radius is 5 meters.

2.8 Additional hazards

2.8.1 Assembling/disassembling heavy components

Adhere to the following safety precaution when handling heavy system components during installation, maintenance and service work:

	⚠ Warning
	Beware of hovering loads. Improper handling of heavy parts can cause serious contusions of body parts, fractures and death. In particular, this applies for: • Vacuum chambers and their attached modules • Components of the process stations • Components of the winding system • Components of the vacuum system, in particular vacuum pumps and valves • Electrical cabinets Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use carts, crane trucks and hoists.

2.8.2 Process gases

2.8.2.1 Argon (Ar)

Argon is:

- Colorless, odorless and tasteless.
- Non-toxic.
- Non-corrosive.
- Non-combustible.
- Heavier than air.

Argon does not bind chemically and is only minimally soluble in water.

Inhaling pure or highly concentrated argon can cause nausea and has the risk of death through suffocation.

	Note!
	Make sure there are no leaks in the process gas supply system. Before starting any work, close all gas line valves and vent the lines.

	Note!
	Carefully read the safety data sheets of the process gases in use. Strictly follow the local safety regulations.



2.8.2.2 Oxygen (O₂)

Oxygen is

- Colorless, odorless and tasteless.
- Non-toxic.

	⚠ Warning
	Combustion hazard Pure oxygen increases the rate of reaction with combustible materials. Keep oxygen away from combustible materials. Become familiar with suitable fire-fighting techniques, and have fire-extinguishing materials at hand.

Oxygen may react with the hot oil and the oil mist in the fore pumps. For continuous fore pump operation, oxygen concentration of less than 25% is uncritical. If higher concentrations are used in the system, pure oxygen must be thinned with argon. Always run fore pumps with gas ballast.

	Note!
	Make sure there are no leaks in the process gas supply system. Before starting any work, close all gas line valves and vent the lines.

	Note!
	Carefully read the safety data sheets of the process gases in use. Strictly follow the local safety regulations.

2.8.3 Bead blasting of coated components

Adhere to the following safety precaution when cleaning coated components by means of bead blasting:

	⚠ Warning
	Beware of possibly poisonous coating materials. Depending on the nature of the process, poisonous substances might be used as coating material. Follow these rules when blasting coated surfaces: • Carefully read the safety data sheets of the process materials in use. Precisely adhere to the precautions specified in them. • Always wear a P3 fine dust mask, goggles and gloves. • Avoid skin contact.

2.8.4 Measures against noise

After the TOPMET web coating system is set up, the system owner must perform a workplace-related noise measurement. As a result, he must take any action which might be necessary to protect operating personnel, such as requiring ear protection.

2.8.5 Exhaust gases

With regard to reactions with other gases, the exhaust gases of the vacuum pumps are harmless. However, in order to eliminate the possibility of unexpected reactions with the process gases of other systems, run the exhaust pipes of the fore vacuum pumping stations separately to the outside air. The exhaust gas must not be allowed to escape to the outside air in close vicinity to a fresh air inlet. Provide the usual equipment for disposing of condensation.

The exhaust air system for the TOPMET web coating system must fulfill the requirements described in chapter 1.6.8.

3

Design and Function

The chapter "Design and Function" provides an overview of the system's configuration and how the web coating system works. The significance and tasks of each component are explained.

The basic principles of coating technology and applied coating techniques are introduced, to promote understanding of the production process.

3.1 Basic principles of coating technology

3.1.1 Depositing

The TOPMET system metallizes films by means of the coating technique called "thermal depositing in a high vacuum".

The depositing technique is based on the principle that coating materials (e.g. aluminum) evaporate in a vacuum and condense on a relatively cold surface (coating drum) of the traversing film. This results in the following procedural steps for the coating process:

- Evaporation phase
- Transport phase
- Condensation phase (layer structure)

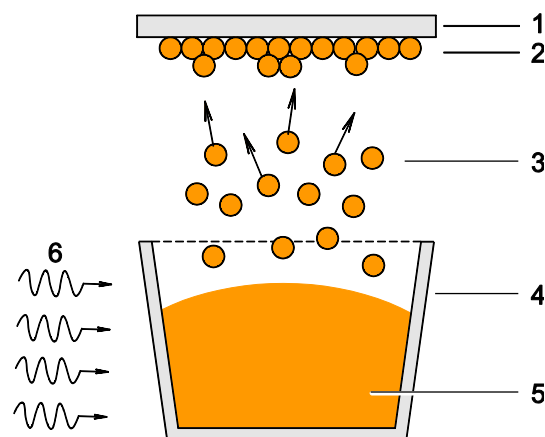


Fig. 3- 1 Schematic illustration of the depositing process

- | | |
|-------------------------|---|
| 1 - Substrate | 4 - Crucible (evaporator boats) |
| 2 - Deposited layer | 5 - Coating material (evaporation material) |
| 3 - Evaporated material | 6 - Heat |

Evaporation phase

Evaporator boats heated by electrical power and made of material with a high melting point are used as the source of evaporation. The aluminum is continuously fed to the evaporator boats, where it melts and evaporates.

When the evaporation material (aluminum) is hot enough, it evaporates, creating a club-shaped cloud. The rate of evaporation (evaporating mass over the evaporation area per unit time) depends on the amount of material in the evaporation boat and the set temperature.

Transport phase

The transport phase describes the path of the evaporated particles from the point of evaporation (evaporator boat) to the point of condensation (film).

The transport phase proceeds well if:

- the path of the evaporated particles is as short as possible.
- the chamber is as free as possible of residual gases (vacuum ranging from 10^{-3} to 10^{-6} mbar), to minimize collisions and reactive processes.

Condensation phase (layer structure)

The process of condensation of the evaporation particles at the substrate is very complex. It is influenced by:

- the condition of the substrate (nature of the surface, material type, temperature).
- the condition of the condensing particles (material type, energy, rate of collision).

3.1.2 Pretreatment**3.1.2.1 Process fundamentals**

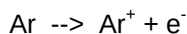
Film pretreatment in the glow station is based on the effect of sputtering of a target surface (sputter process).

A plasma is required to conduct a sputter process. A plasma is an ionized gas. It can be created by exposing a gas, under reduced pressure, to a static electrical field or a high-frequency alternating field.

Igniting the plasma

The following discussion assumes argon is used as the working gas, and is converted to a plasma by applying a static electrical field. In this case, the negative pole of the voltage is at the target, i.e. the target is the cathode. The positive pole of the voltage is generally connected with chassis ground.

Due to the ionizing radiation (cosmic radiation and natural radioactivity) which is always present, any gas has a certain, small ion content. That means that some atoms of the working gas are split into positive argon ions and into negative electrons, in accordance with the reaction



Due to the electrical field, these argon ions migrate to the cathode, and the free electrons migrate to the anode. The migrating particles collide with additional atoms and ionize them with their kinetic energy. This process repeats many times, causing an exponential increase in the number of argon ions and electrons. The plasma ignites.

Gas pressure

To maintain a plasma at a voltage of a few hundred volts, the pressure of the working gas must lie between around 10^{-4} and 10^{-1} mbar. If the pressure is set too high, then the migrating particles will collide with the gas atoms before reaching the minimum level of energy required for ionization. On the other hand, if the pressure is too low, the particle density will be low, reducing the probability of collisions.

The plasma process is supported by adding oxygen or oxygen-containing gas mixtures (argon + oxygen). The efficiency of the glow process depends mostly on the applied voltage, the target material in use, the nature of the gas and the pressure.

Sputtering

After the plasma ignites, many positive argon ions strike the target (see Fig. 3- 2). Due to their large mass, they have very great momentum. This momentum transfers to the atoms on the target's surface, ejecting individual or entire clusters of atoms.

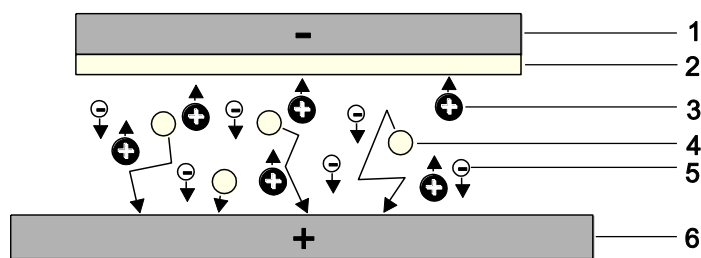


Fig. 3- 2 Principle of cathode sputtering

- | | |
|--------------------------|---|
| 1 - Cathode (minus pole) | 4 - Particles of the target material |
| 2 - Target | 5 - Free electrons |
| 3 - Gas ions | 6 - Substrate carrier and anode (plus pole) |

Magnetron sputtering

The magnets in the cathode support the process of electron discharge directly in front of the cathode, especially where the magnetic field lines run parallel to the target surface. The magnetic field focuses the ion blasting; eventually, an erosion profile (erosion groove) is formed, raising the temperature at the target. Circulating cooling water channels protect the cathode and the area around it from overheating.

3.1.2.2 Objectives and properties

The basic prerequisite for optimum adhesion of deposited layers is that the substrate surface be clean and absorbent.

All substrates are exposed to atmospheric conditions before the coating process. For this reason, a thin layer of water, oxides and other contaminants is spread over the substrate's surface. This reduces the adhesiveness of the deposited layers on the substrate.

Thus, for good adhesiveness, most substrates (films) must be pretreated (activated).

They are pretreated in the glow station directly before the coating process. An aluminum target is sputtered by glow discharge (plasma). The sputtered particles collect on the film's surface, forming small aluminum islands which are condensation "seeds" for the subsequent depositing process. Basically, pretreatment produces the following:

- Desorption of water components, oxides and contamination on the film's surface.
- Preseeding of the film's surface with aluminum, to improve the adhesiveness of the subsequent coating.

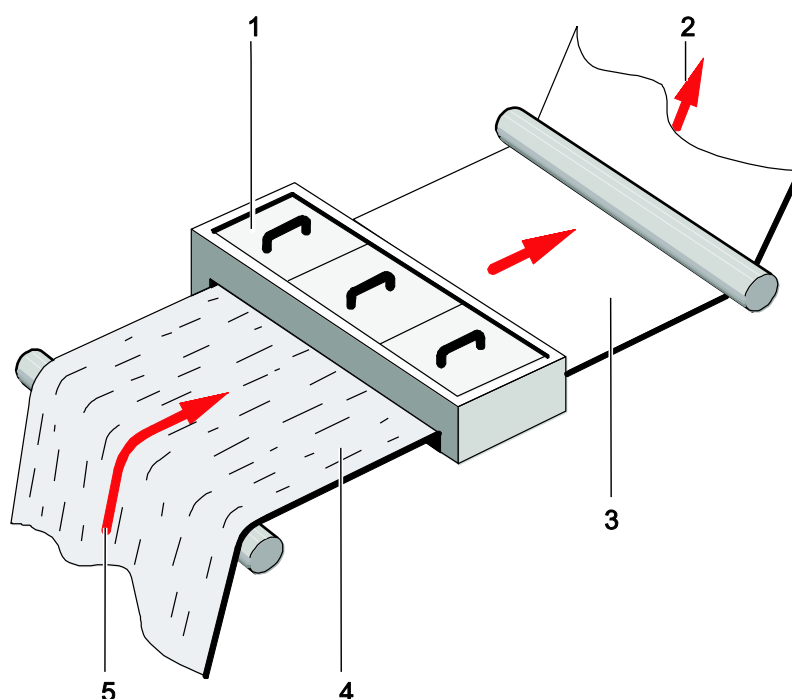


Fig. 3-3 Illustration of in-line plasma pretreatment

- 1 - Pretreatment station (glow station)
 2 - To the coating drum
 3 - Pretreated film

- 4 - Untreated film
 5 - From the unwinder

3.2 System overview

The following diagram illustrates the opened TOPMET film coating system, with the major components of the system's design.

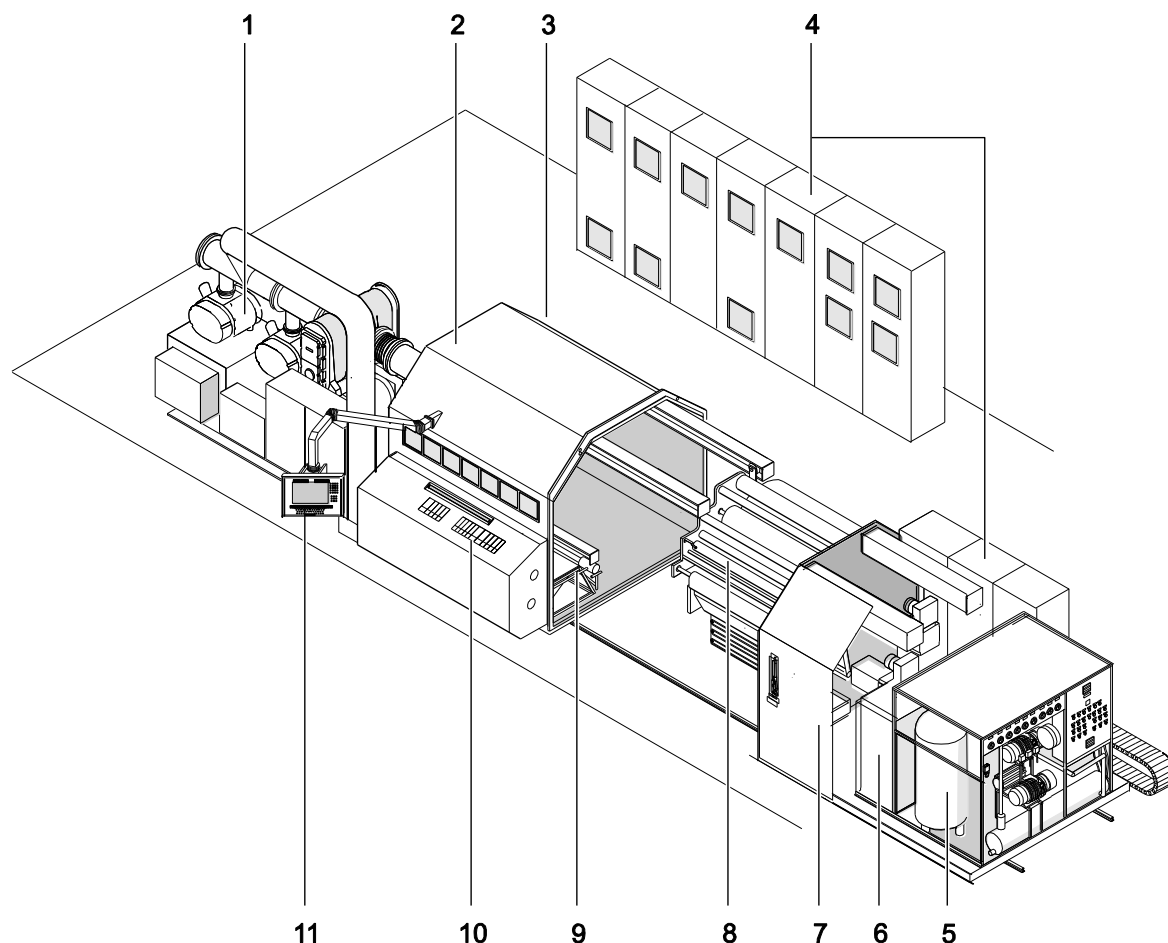


Fig. 3- 4 TOPMET plant (standard arrangement)

- | | |
|--------------------------------------|---|
| 1 - Fore vacuum pumping station | 7 - Travelling part |
| 2 - Vacuum chamber | 8 - Winding system and pretreatment station |
| 3 - Media supply (chamber rear side) | 9 - Evaporator station |
| 4 - Switching cabinets | 10 - Operating panels |
| 5 - Cooling and heating equipment | 11 - Graphic user interface with monitor |
| 6 - Polycold refrigerator | |

- Fore vacuum pumping station:
Rotary vane pump and roots pump for pumping down the vacuum chamber and high vacuum pumps, see chapter 3.4.1
- High vacuum pumping station:
Oil diffusion pumps (not shown in Fig. 3- 4) for generating the high vacuum in the coating chamber, see chapter 3.4.1
- Polycold refrigerator:
Cooling equipment and cooling trap for faster pumping down of water vapor, see chapter 3.4.4

- Vacuum chamber:
Pumped down chamber, consisting of the winding chamber (with winding system and pretreatment) and coating chamber (with evaporator station), see chapter 3.3.1 und 3.4
- Media supply:
Unit for supplying the system with water and compressed air, see chapter 3.6
- Switching cabinets:
Control and electrical supply of all system components, see chapter 3.5
- Cooling and heating equipment:
Temperature regulation of the coating drum and tension roll (cooling drum), see chapter 3.3.2
- Travelling part:
For opening and closing the vacuum chamber. The winding system and the cooling and heating equipment are moved during opening and closing.
- Winding system:
System of drums for transporting and guiding the film during the process, see chapter 3.3.2
- Pretreatment station:
Treatmag cathodes and gas supply for pretreating the film, see chapter 3.3.3 und 3.3.4
- Evaporator station:
Evaporation material (aluminum) and evaporator boats for coating the film, see chapter 3.3.5 und 3.3.6
- Operating panels:
Operating elements to control system functions, see chapter 4.3
- Graphic user interface with monitor:
Operation of the system control and visualization software (process control software), see chapter 4.2

3.3 Functional principle

3.3.1 Chamber cross-section and process

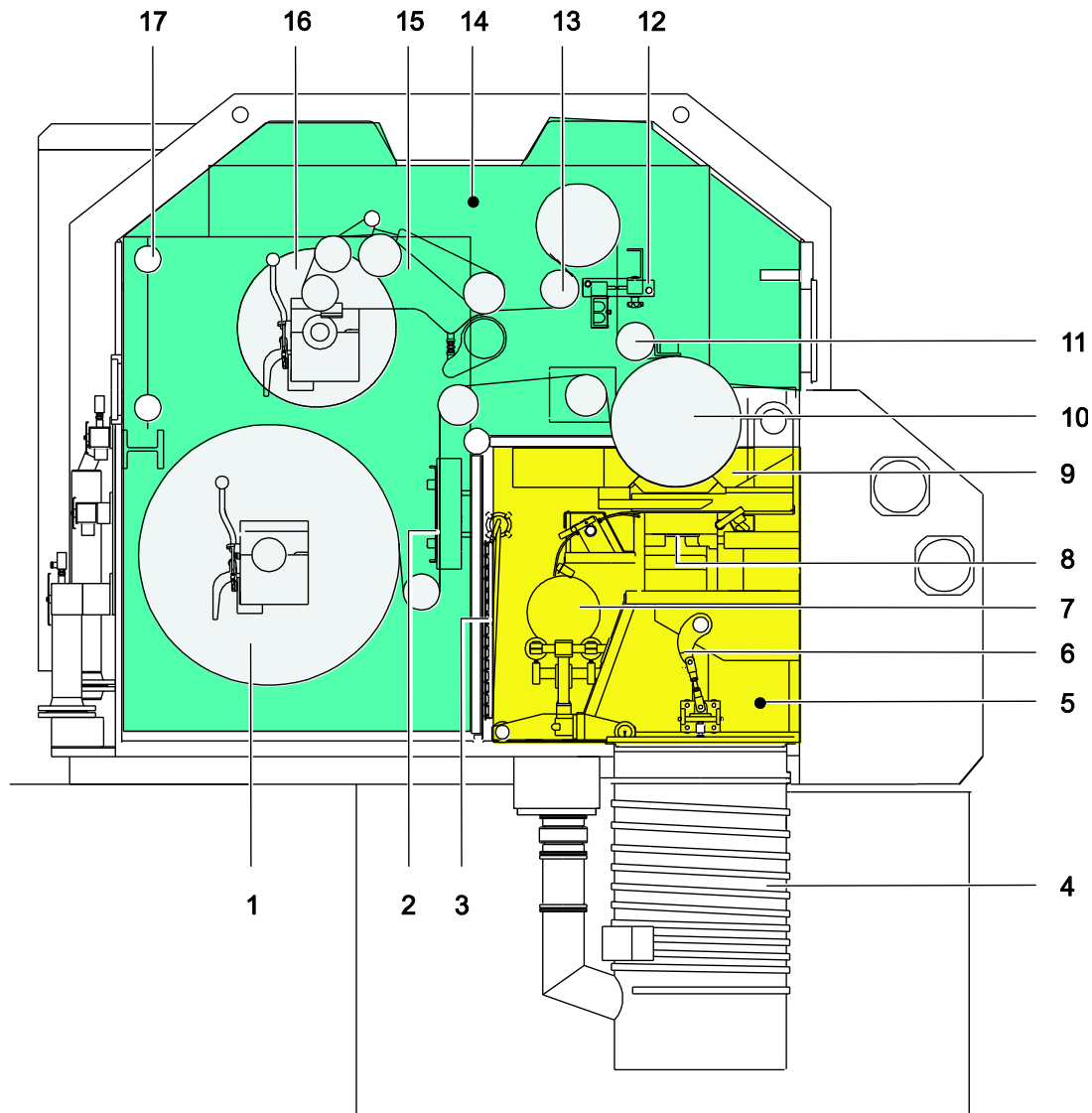


Fig. 3- 5 Functional principle TOPMET (chamber cross-section)

- | | |
|---|--|
| 1 - Unwinder | 10 - Coating drum |
| 2 - Pretreatment station (glow station) | 11 - Measuring roller A (LMS-C) (option) |
| 3 - Cooling trap | 12 - Layer thickness measurement unit |
| 4 - Pump system | 13 - Measuring roller B (LMS-C) (option) |
| 5 - Coating chamber | 14 - Winding chamber |
| 6 - Drive for high-vacuum valve V1 | 15 - Swivel arm (Pivot-Arm) |
| 7 - Wire coil | 16 - Rewinder |
| 8 - Evaporator boats | 17 - Cooling trap (optinal) |
| 9 - Coating window | |

The system consists of two chambers, the winding chamber and the coating chamber. Except for two slots through which the film runs, the chambers are separated by a chamber seal, and are pumped down by separate pump systems. That makes it possible to pump down the considerable gas loads from the film with a relatively small pumping station. The pressure in the winding chamber can be 100 times higher than the pressure in the coating chamber. Process control requires a pressure of approximate 2.5×10^{-4} mbar in the coating chamber.

To improve the vacuum and thus increase the product quality, a cooling trap can be optionally installed in the winding chamber as a water vapor pump.

The winding chamber contains the drum system which automatically guides the film, and the pretreatment station. To load and unload films, the travelling part with the winding system is moved out of the chamber.

The coating chamber contains the evaporator station.

After the vacuum chamber is closed, the pump system automatically pumps the winding and coating chambers down to the required pressures. When the process pressure is reached, the film is transported from the unwinder to the pretreatment station. Here, the film surface is preseeded by sputtering a target.

The film then runs via the coating drum directly into the coating chamber (process chamber). The film is coated there. An aluminum wire coil continually feeds evaporation material into the evaporator boats, where this material is heated and evaporated. The film's evaporation width can be set by means of a coating window. A cooling trap in the coating chamber binds the water vapor which is generated.

The heat loading from the evaporator boats, along with the heat of condensation released as the layer grows, are a considerable thermal stress for the film. During the process, the film is sufficiently cooled by means of the cooled coating drum and tension roll, so it is not damaged. The drums are cooled by the cooling and heating equipment at the travelling part.

The layer thickness is measured during the entire coating process. For this, the film is run over a layer thickness measurement unit (LMS). The measurement values are used for quality control and to control the thickness of the deposited layer (comparison of setpoint and actual values).

The film then runs over a tension roll. At the rewinder, it is wound into a ball; the winding tension can be adjusted.

A process control and monitoring system controls the entire operation. This system displays the events on a system monitor.



3.3.2 Drum system

Fig. 3- 6 shows the position of the drums in the winding system; the routing of the film is shown in red.

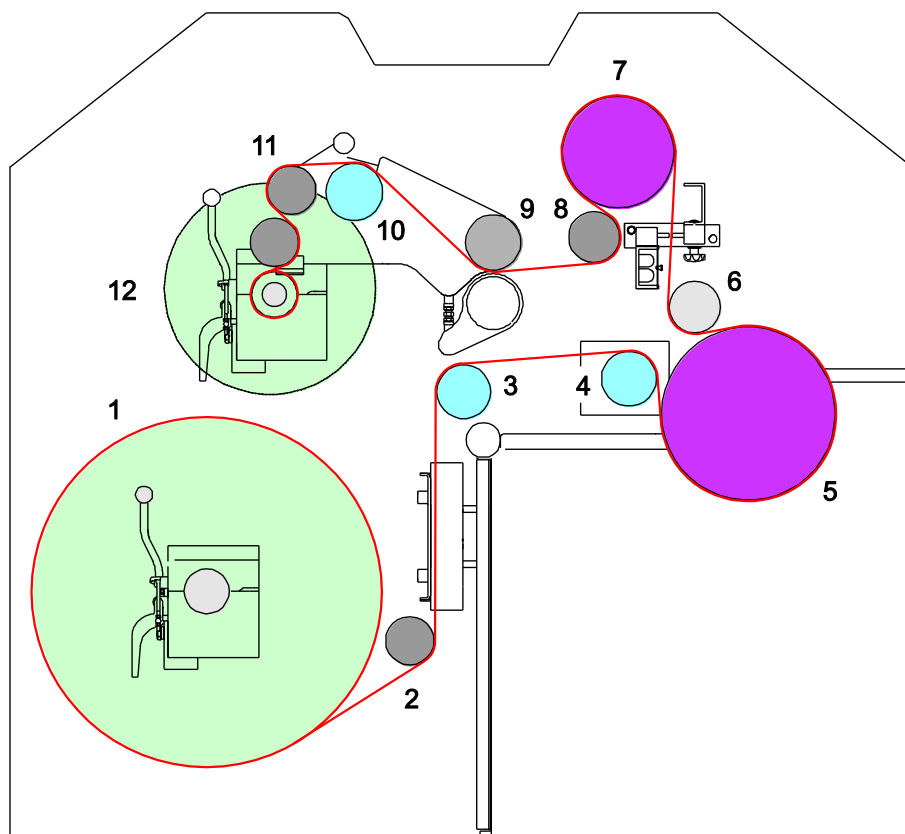


Fig. 3- 6 Drum system (chamber cross-section)

- | | |
|---------------------------------|---------------------------------------|
| 1 - Unwinder drum | 7 - Tension roll or Post cooling drum |
| 2 - Spreader roll spiral shaped | 8 - Guide roller 3 |
| 3 - Guide roller 1 | 9 - Guide roller 4 |
| 4 - Spreader roll, variable | 10 - Spreader roll "Fixbow" |
| 5 - Coating drum | 11 - Moving drum package (pivot arm) |
| 6 - Guide roller 2 (option) | 12 - Rewinder drum |

The various types of drums work as follows:

- **Unwinder:**
At the beginning of the process, the unwinder drum accepts the uncoated roll. During the process, the film unwinds from the unwinder. The drive motor of the unwinder drum places tension on the film, away from the coating drum (away from the direction of the film's motion).
- **Guide roller:**
The guide rollers ensure the film is routed evenly, and they bend the film in the required direction.

- **Spreader roll:**
When running, the film tends to wind to the middle.
A spreader roll consists of a bent shaft on which ball bearing and bobbin are mounted, forming a rotating drum part. This bent shaft axially stretches the film as it runs over the spreader roll.
- **Coating drum:**
The coating drum is the central drum of the system. It is driven by a motor; its speed (which controls the film's speed) is regulated.
The coating drum transports the film from the winding chamber into the coating chamber, to the evaporator station (evaporation boats), and back from the coating chamber to the winding chamber. This drum also serves as the vacuum isolation between the two chambers.
During coating, the film is exposed to considerable thermal stress; this is why the coating drum is cooled. For good thermal contact between the film and drum, the film must lie on the coating drum. This is accomplished by applying tension to the film (the unwinder "slows down", the rewinder and tension roll "pull").
- **Tension roll:**
The drive motor of the tension roll applies tension to the film, away from the coating drum (along the film's direction of motion).
The tension roll also decouples the tension from the rewinder, so that the film tension can be freely adjusted during rewinding.
Instead of the tension roll the plant can be equipped with a post cooling drum.
The post cooling drum is used to after cool the film.
- **Pivot arm:**
The pivot arm keeps the distance from the rewinder always stays the same.
- **Rewinder:**
The rewinder accepts the coated roll. During the process, the film is wound from the unwinder to the rewinder.
The drive motor of the rewinder drum places tension on the film, away from the coating drum (in the direction of the film's motion).

3.3.3 Pretreatment station (optional)

The pretreatment station (glow station) is used to preseed the film's surfaces, for better adhesion of the deposited layers.

The central component of the pretreatment station is the Magnetron glow cathode (TreatMag). The pretreatment station is placed before the coating station, allowing in-line film treatment.

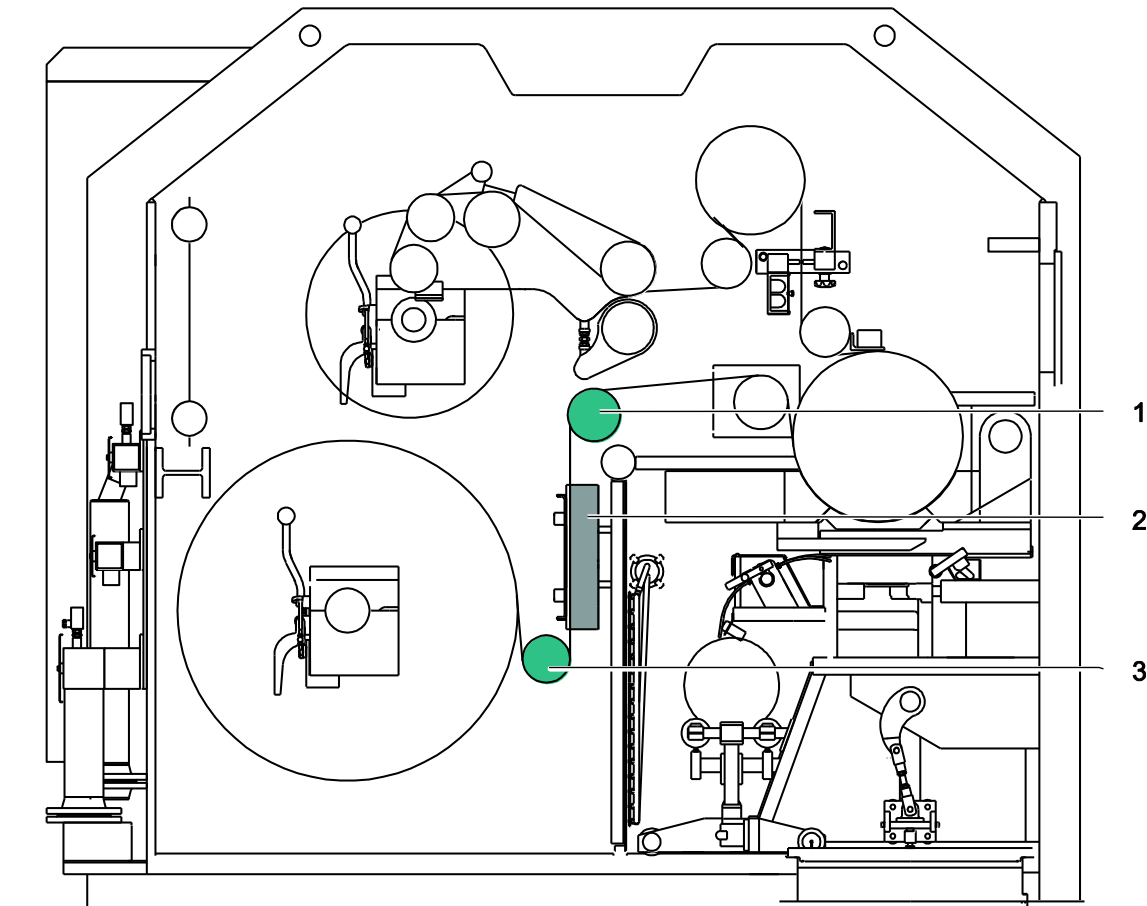


Fig. 3- 7 Pretreatment station

- 1 - Spreader roll Spiral shaped
- 2 - Glow cathode Treatmag

- 3 - Guide roller

The pretreatment station contains the following system components:

- Gas supply Provides and distributes the process gases required for glowing (Ar and O₂), see chapter 3.3.3.1 and 3.3.3.2
- Oxygen safety interlock Gasballast flow monitoring of the fore pumps and O₂ shut-off valves in the gas supply line, see chapter 3.3.3.3
- Glow cathode (TreatMag) Body of the cathode with stainless steel target, see chapter 3.3.4

- Power supply Built-in unit which supplies the glow station and controls its power, see chapter 3.3.4.3
- Guide rollers Ensures the film runs along the pretreatment station without folds or contact, see chapter 3.3.2
- Control Controls and regulates the glowing, via the screen pages of the process visualization software; refer to the Software Guide

3.3.3.1 Gas system

The process gases required for glowing are argon (Ar) and oxygen (O₂).

The gases can be used separately (only Ar or O₂), or as a gas mixture (Ar + O₂). These process gases are used as follows:

- Argon Required to condition the cathode.
- Oxygen Primarily used only to pretreat the film.
- Oxygen + Argon Primarily used to clean and pretreat the film.

The gases are supplied to the system from pressurized gas bottles located at the travelling part between the Polycold and the cooling and heating equipment.

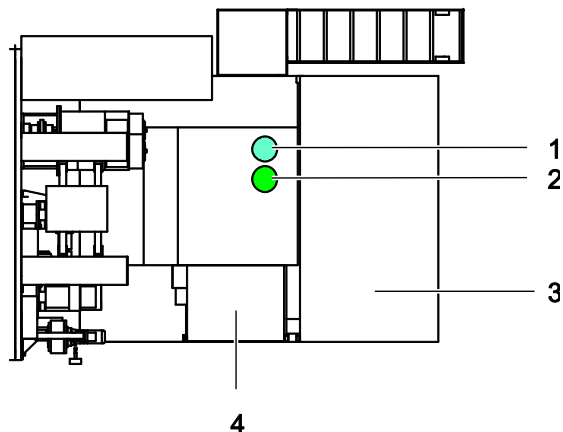


Fig. 3- 8 Position of the compressed gas bottles at the travelling part

- | | |
|--------------------------------------|-----------------------------------|
| 1 - Compressed gas bottle for oxygen | 3 - Cooling and heating equipment |
| 2 - Compressed gas bottle for argon | 4 - Polycold |

The process gases flow from the gas bottles via separate supply lines past a gas flow controller and then directly to the gas inlet of the glow cathode.

3.3.3.2 Gas distribution and flow control

Arrangement

Gas distribution is the arrangement for deciding which process gases and amounts are to be directed to the glow cathode. Gas flow controllers are used for this; there is a separate gas control unit for each gas connection. The system control determines the gas flow volumes and valve positions (open or closed) by controlling the gas flow control units.

Structure

A gas flow control unit essentially consists of an electro-magnetic shut-off valve and a flow controller. The process gas flows from the gas bottles via the shut-off valve to the flow control unit and then into a common line. Via a manual shut-off valve, the gas then enters the glow cathode.

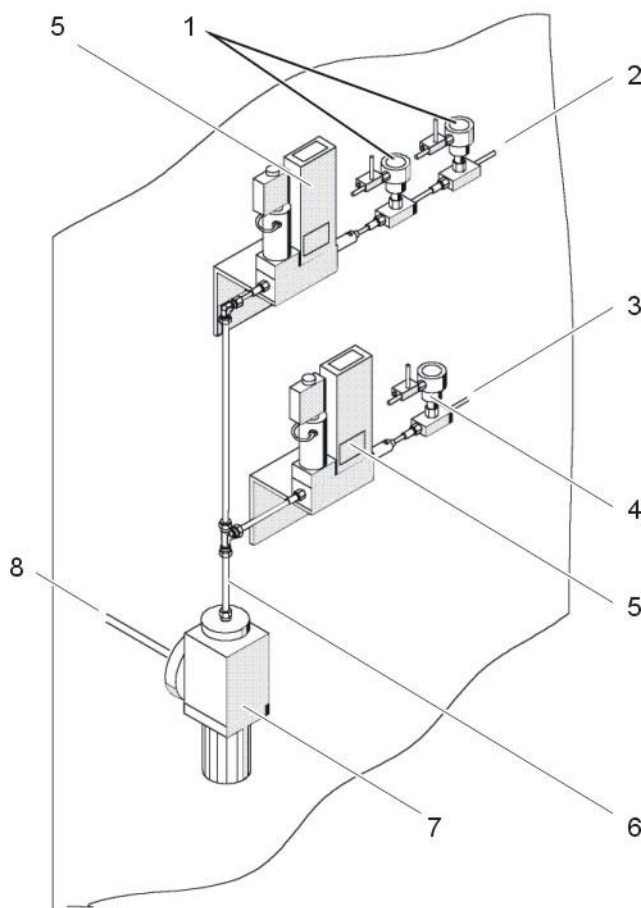


Fig. 3- 9 Structure of the gas control and distribution unit

- | | |
|--|------------------------------|
| 1 - Magnetic, redundant shut-off valves (oxygen) | 5 - Flow controller |
| 2 - Gas supply for oxygen | 6 - Common line |
| 3 - Gas supply for argon | 7 - Manual shut-off valve |
| 4 - Magnetic shut-off valve (Ar) | 8 - Gas inlet to the cathode |



	Note!
	Comply with the manufacturer's documentation for flow controllers.

Controlling and monitoring the gas supply

The process gases required for glowing are controlled, monitored and activated via the screen pages of the process visualization software.

- Data is entered and the process gas is selected from the screen page "Parameters", Pre-Treatment (see Software Guide)
- The value of the actual gas flow is displayed on the screen page "Process", Pre-Treatment (see Software Guide)
- A test function for gas valves is available on the screen page "General Parameters", Test Pre-treatment (see Software Guide)

System prerequisite to execute the test function:

- The chamber is closed and pumped down to a pressure of $p < 0.1$ mbar.

3.3.3.3 Safety system for the oxygen influx

	Note!
	To ensure proper functioning of the safety system for oxygen shut-off, the gas ballast devices and oxygen shut-off valves must be maintained and cleaned on a regular basis.

High oxygen content in the oil-filled (Anderol) fore vacuum pumps can result in combustible gas mixtures.

A redundant safety circuit monitors the oxygen influx. This way, a combustible oxygen/oil vapor mixture cannot be created in the fore pumps.

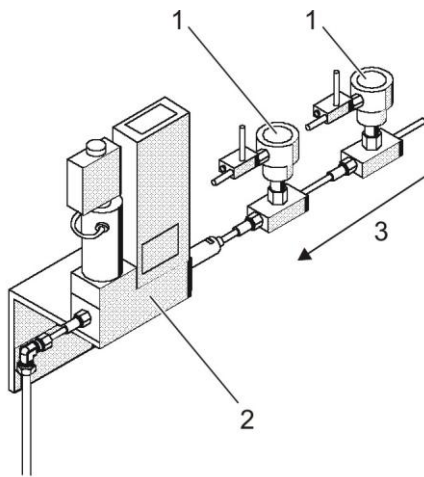


Fig. 3- 10 Safety system for the oxygen influx

- 1 - Oxygen shut-off valve
- 2 - Gas flow controller

- 3 - Oxygeninlet



Fig. 3- 11 Fore vacuum pump with gas flow sensor

Two series electro-magnetic shut-off valves in the oxygen inlet line are essential elements of the safety circuit. A safety circuit controls both shut-off valves, for redundant monitoring of the condition of system components placed after the valves. Each component condition is monitored by two independent hardware- and software-controlled circuits, resulting in redundant monitoring. In conjunction with two shut-off valves (which function simultaneously), this results in a duplicate safety chain.

System conditions monitored for excessive oxygen concentration:

- Oxygen return flow of the fore vacuum pumps
Two gas flow sensors mounted at the fore vacuum pump monitor the volume of oxygen. Both oxygen shut-off valves are immediately closed if excessive oxygen is measured or one of the installed gas ballast valves fails. The supply of O₂ for the pretreatment station is interrupted.
- Vacuum chamber pressure
When the pressure in the winding chamber increases to $p > 0,1$ mbar, both oxygen shut-off valves are closed immediately and the supply of O₂ is interrupted.

3.3.4 Magnetron cathode (optional)**3.3.4.1 Performance data and specifications for the glow cathode**

- Glow width 2,450 mm
- Working vacuum 10^{-3} to 10^{-2} mbar
- Maximum current 24 A
- Supply voltage 300 – 1,000 V
- Gas connection Oxygen and argon



3.3.4.2 Body of the cathode

The following diagram shows the mechanical structure and main parts of the Magnetron glow cathode.

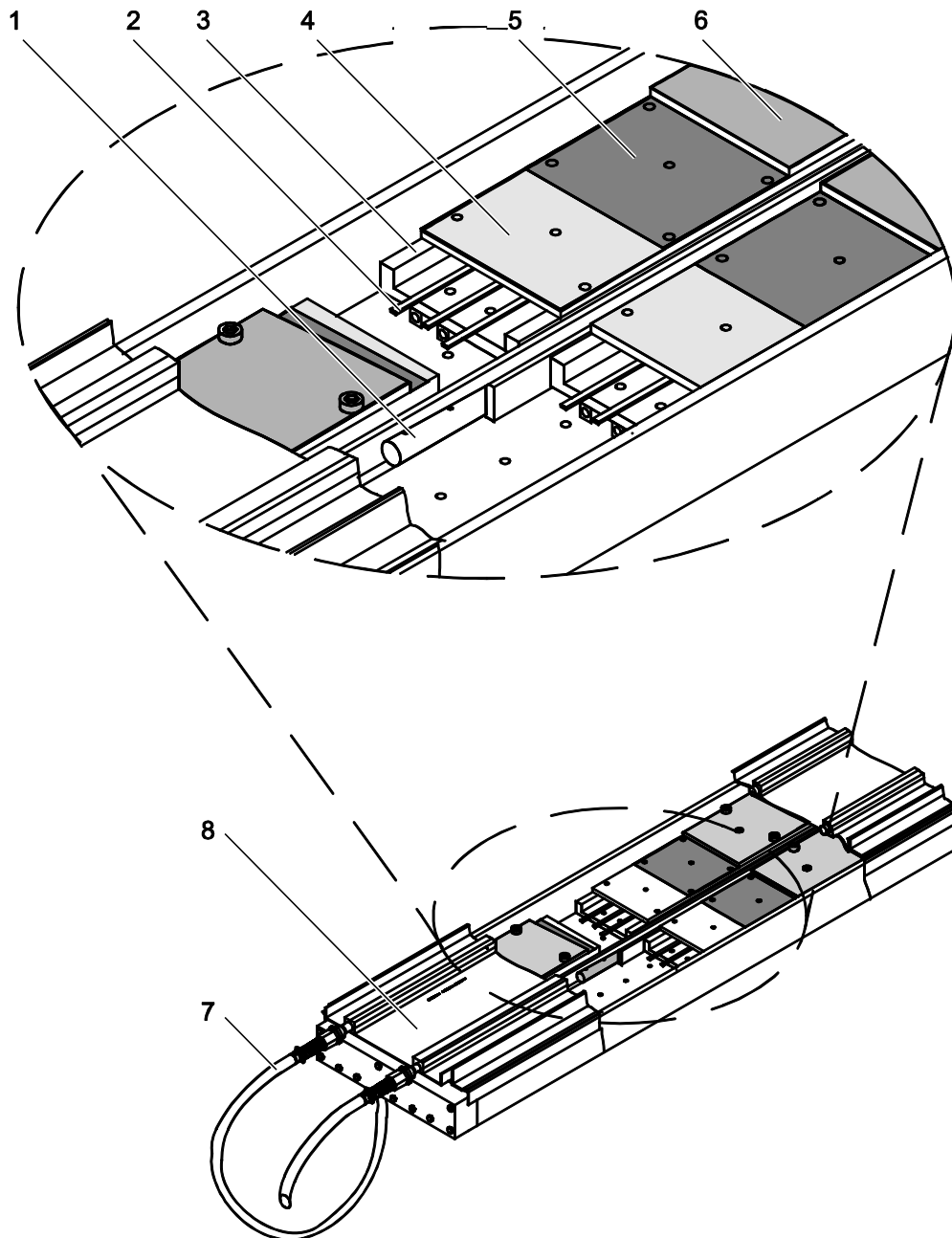


Fig. 3- 12 Body of the cathode

- 1 - Gas distribution tube
- 2 - Magnets
- 3 - Body of the cathode
- 4 - Cooling plate

- 5 - Graphite film
- 6 - Target
- 7 - Cooling water connection
- 8 - Cathode lid

3.3.4.3 Power supply

	Note!
	Comply with the manufacturer's documentation for the power supply.

The power supply supplies the glow cathode with electrical power. It converts the AC line voltage provided by the customer to the DC supply voltage required for glowing. This device is a separate built-in component at the cabinet +G3.

The power supply is connected with the system control for control purposes. The process visualization software is used to operate and communicate with the power supply.

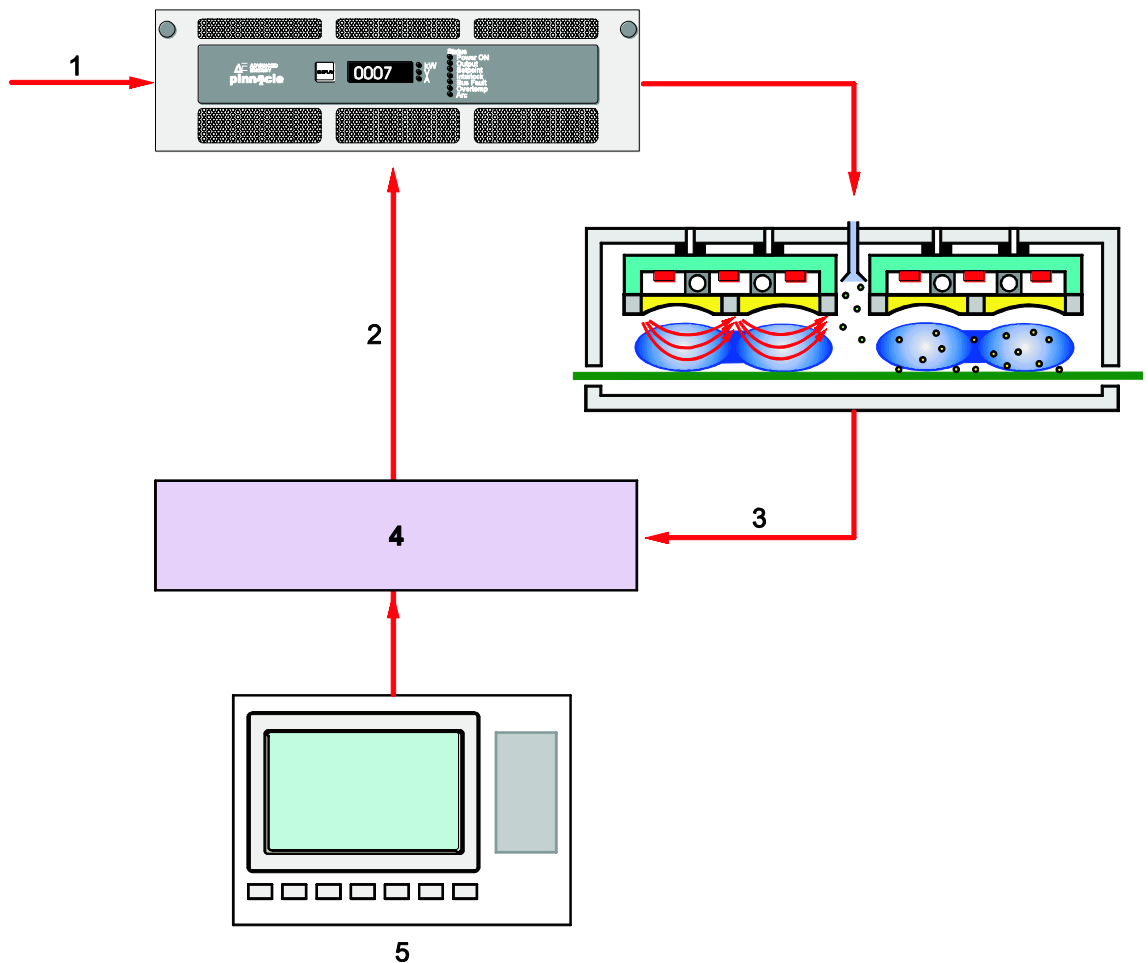


Fig. 3- 13 Block diagram of the power supply control

- 1 - Main Power in
- 2 - Power Control Setpoint
- 3 - Actual Values

- 4 - Process Control
- 5 - PLC

3.3.5 Evaporator station

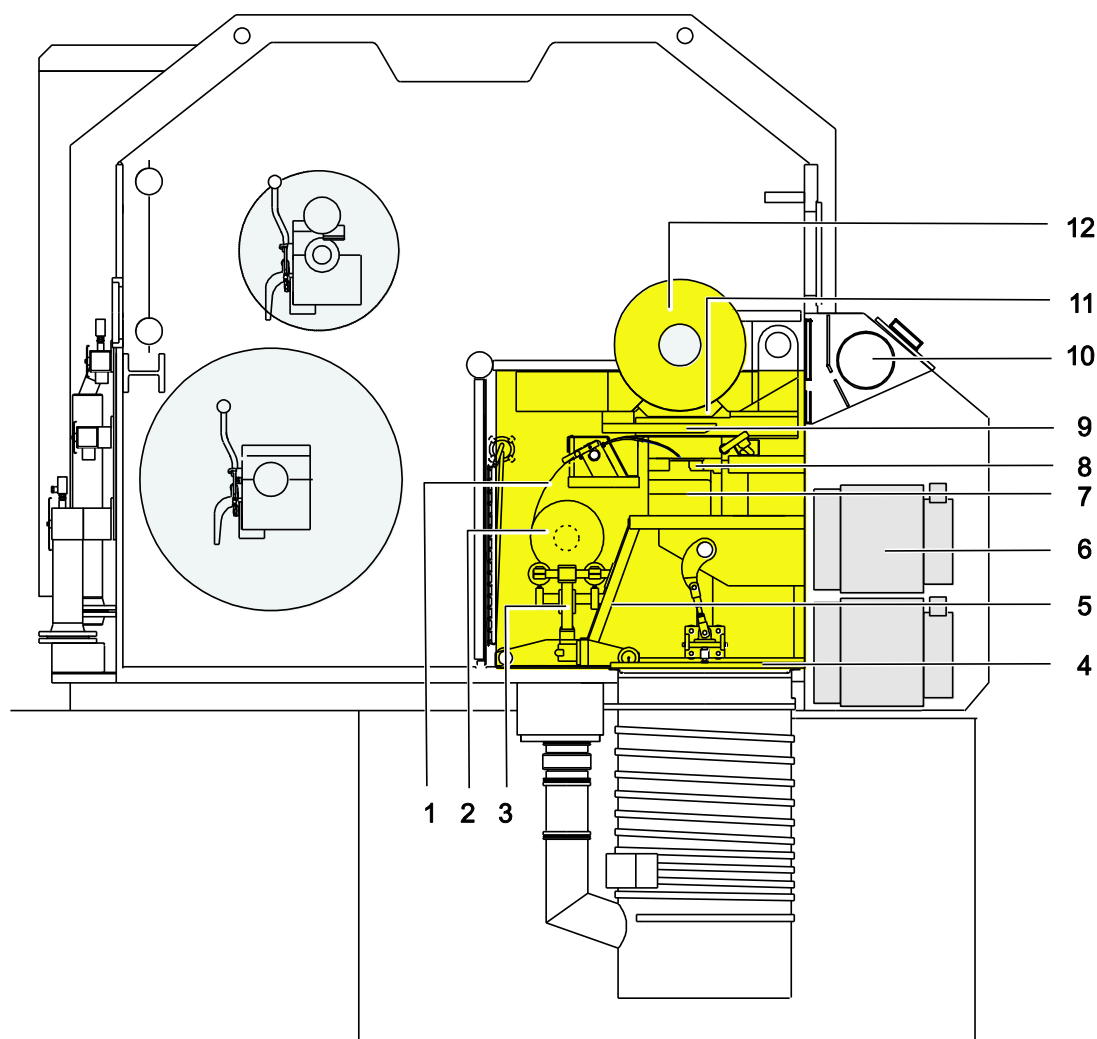


Fig. 3- 14 Evaporator station

- | | |
|------------------------------|---------------------------------------|
| 1 - Aluminum wire | 7 - Container for evaporation residue |
| 2 - Roll with aluminum wire | 8 - Evaporator boats |
| 3 - Portable wire coil frame | 9 - Shutter |
| 4 - High-vacuum valve V1 | 10 - Tubular stroboscope |
| 5 - Sieve | 11 - Coating window |
| 6 - Transformers | 12 - Coating drum |

Depending on the exact implementation of your system and the coating width, the TOPMET system is equipped with 14 - 56 (depending how your plant is implemented) evaporator boats made of special ceramics. The evaporator boats are patented staggered arranged.

On a portable frame, aluminum wire (evaporation material) is continuously fed from the aluminum wire coils (one for each evaporation boat) to the evaporator boats, where it is heated and evaporates. The feed for each wire coil can be adjusted and matched to the selected rate of coating.

After the coating chamber is pumped down to high vacuum, the evaporator boats will be heated up. Transformers supply the boats with electrical power, via a current feedthrough. This heats and evaporates the material in the evaporation boats above the melting point. Evaporation residue is collected by means of a removable container placed under the boats.

With the evaporator shutter open, the vapor cloud expands and condenses on the film as it runs by. The coating width of the film is adjusted via a coating window.

During the process, a tubular stroboscope can be used to observe the aluminum wires through a flap, to make sure they are fed and run correctly.

The area around the evaporator boats is exposed to high thermal stress. To avoid overheating, some components such as the current feedthrough are water-cooled and made of copper (excellent thermal conductivity), to quickly dissipate the heat.

As the film runs by, it is also exposed to great thermal stress. Therefore, it must firmly press against the cooled coating drum (cooling and heating equipment).

High vacuum is required for the coating process. Since the pressure in the pumps is different than that in the coating chamber, high-vacuum valves can separate them. Sieves protect the valves and pumps from damage caused by coarse dirt from the evaporator station.

3.3.6 Layer thickness measurement system

The layer thickness is measured continuously during the entire coating process. For this, the film passes over a layer thickness measurement unit (LMS) with several sensors. These sensors are arranged perpendicular to the direction of the film's motion, along the film's width.

The measurement values are used for quality control and to control the thickness of the deposited layer (comparison of setpoint and actual value).

The layer's thickness can be controlled in two ways:

- Control along the film's width (cross control): Coating is controlled such that each aluminum wire feed unit is adjusted to the setpoint for the layer thicknesses. This results in a homogeneous thickness profile along the film's width.
- Control along the film's direction of motion (length control): Coating is controlled such that all single-signals of each measuring sensor are averaged. This average value is compared with the adjusted setpoint for the layer thickness. The film speed is regulated. This results in a homogeneous thickness profile along the film's direction of motion.

The following chapters briefly describe the measurement principle of the layer thickness measurement systems. For a detailed explanation, please refer to the manufacturer's documentation.

3.3.6.1 Layer thickness measurement system LMS-OD

The layer thickness measurement system LMS-OD is based on an optical measurement method.

A transmitter (IR LED) emits a very stable, clocked beam of infrared light. This beam is directed through the coated film and is partially absorbed or transmitted. The weakened beam is detected by an ultra red-sensitive photo diode. The absorbed component can be determined by comparing the intensity of the emitted and reflected beam. The layer thickness can then be determined.

The layer thickness is specified in units of OD (Optical Density).

The system must be calibrated before each coating process. For this, the infrared absorption of an uncoated film is determined.

For flawless measurements, the surfaces of all components of the layer thickness measurement system (transmitter LEDs, lenses) must be clean.

3.3.6.2 Layer thickness measurement system LMS-R (option)

The layer thickness measurement system LMS-R is based on contactless measurement of the electrical resistance by means of eddy currents.

The layer thickness is specified in units of resistance per square.

For flawless measurements, the surfaces of all components of the layer thickness measurement system (sensors) must be clean.



3.3.6.3 Layer thickness measurement system LMS-C (option)

The layer thickness measurement system LMS-C is based on current and voltage measurement between the measuring rollers A and B (see Fig. 3- 5). The analogue measured values are transferred to the PLC. There the resistance calculation is carried out.

	Note!
	For the cross control of the layer thickness the measurement system LMS-C cannot be used.

3.3.7 Gasinlet

The gasinlet system uses the property of gas to cool down during expansion. In the course of the coating process the film is cooled down by the gas (air) let in the vacuum chamber (see Fig. 3- 15). Additionally a smoothing of the film at the coating drum is achieved, which improves heat transmission from the film to the cooled coating drum.

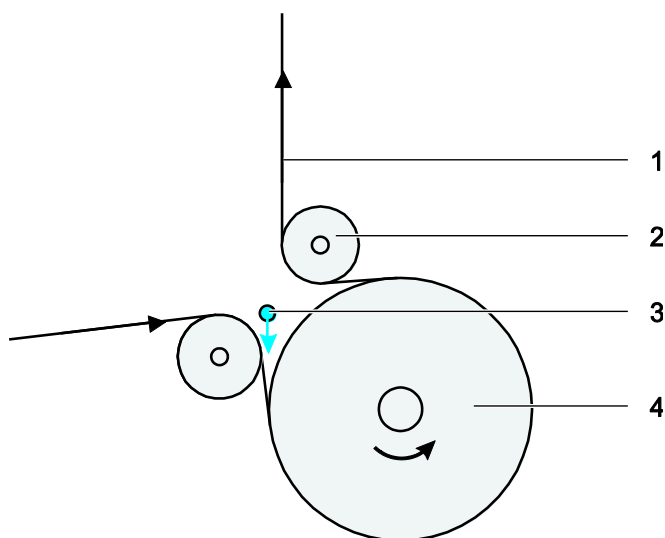


Fig. 3- 15 Gasinlet before coating drum

1 - Film

2 - Guide roller

3 - Gasinlet tube

4 - Coating drum

3.4 Vacuum system

The vacuum system:

- Pumps down the vacuum chamber until the process pressure is reached. The pump-down time depends on the chamber volume (depends on the specific implementation of your system) and the condition of the chamber (clean and dry system).
- Monitors and maintains the vacuum during the coating phase.
- Vents the system after the coating process.

3.4.1 Pumping stations

The pumping station of the TOPMET system comprises:

- Fore vacuum pumping station, with:
 - Rotary vane pumps
 - Roots pumps
- High-vacuum pumping station, with:
 - Oil diffusion pumps

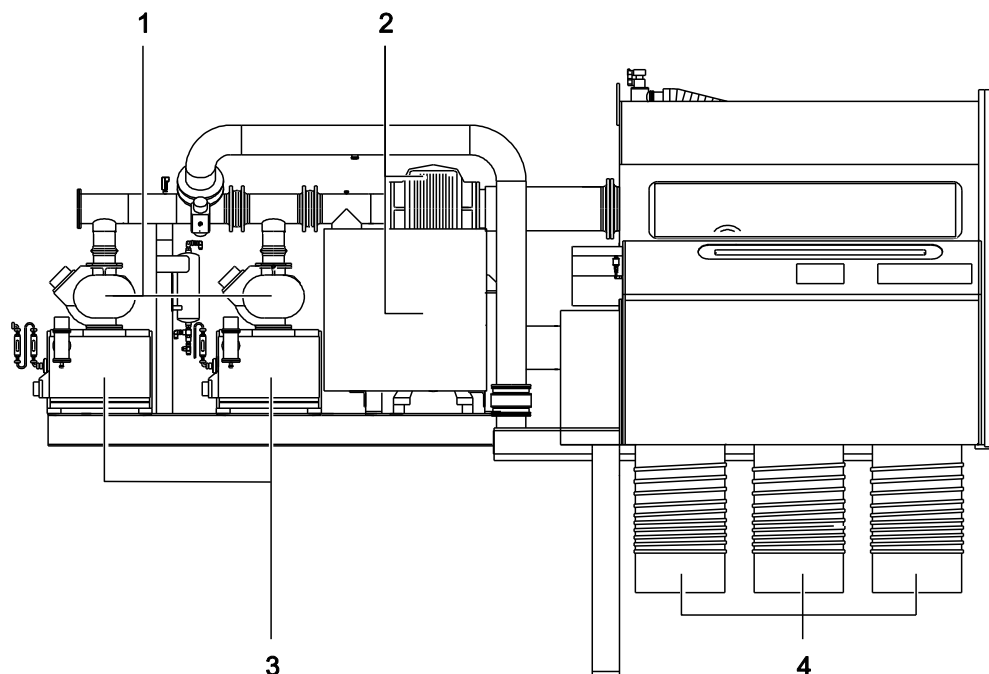


Fig. 3- 16 Vacuum pumping station

1 - Roots pump WAU 2001 / RA 3001
2 - Roots pump RA 3001/7001/9001

3 - Rotary vane pump SV 750 B (F)
4 - Oil diffusion pump DIP 20000

3.4.2 Vacuum diagram

The vacuum diagram shows the structure of the vacuum system. A fore and a high-vacuum pumping station pumps down the vacuum chamber. The latter is divided into a winding chamber and a coating chamber. The functioning principle of the pump system is described in detail in chapter 3.4.5.

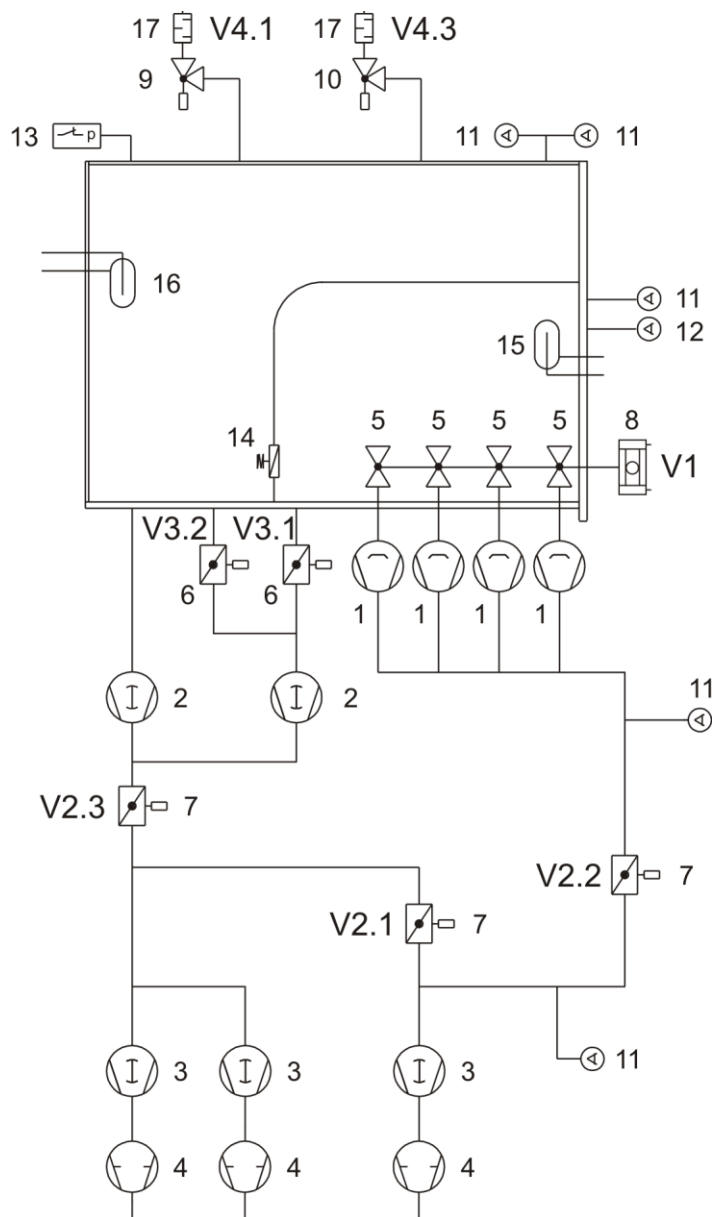


Fig. 3- 17 Vacuum diagram

- | | |
|------------------------------------|--|
| 1 - Oil diffusion pump DIP 20000 | 10 - Pneumatic angle valve DN 16 |
| 2 - Roots pump RA 7000/9001 | 11 - Pirani transmitter PSG 500-S |
| 3 - Roots pump WAU 2001/RA3001 | 12 - Penningvac transmitter IKR 251 |
| 4 - Rotary vane pump SV 750 B (F) | 13 - Difference pressure transducer |
| 5 - High-vacuum valve V1 | 14 - Non-return valve |
| 6 - Pneumatic shut-off flap DN 300 | 15 - Coldtrap for coating chamber |
| 7 - Pneumatic shut-off flap DN 200 | 16 - Coldtrap for winding chamber (optional) |
| 8 - Pneumatic drive for valve V1 | 17 - Muffler |
| 9 - Pneumatic shut-off flap DN 63 | |

3.4.3 Vacuum valves

The functioning principle of the pump system is described in detail in chapter 3.4.5.

Valve	Function
• V1	High-vacuum valve for DIP 20000
• V2.1	Fore vacuum valve
• V2.2	Fore vacuum valve
• V2.3	Fore vacuum valve
• V3.1, V3.2	Fore vacuum valve
• V4.1, V4.3	Venting valve

3.4.4 Coldtrap

The coldtrap is a special kind of pump. The pumping effect is based on the principle of water vapor condensation on cold surfaces. The coldtrap is used to quickly or continually pump the water vapor (from material degassing) out of the vacuum chamber.

The coldtrap consists of a cooling plate (cooling coil) in conjunction with an external refrigerator. The cooling plate, refrigerator and circulating cooling medium form a closed cooling circuit.

3.4.4.1 Cooling plate



Fig. 3- 18 Cooling plate (cooling coil) with collecting tray underneath

The cooling plate is formed as a cooling coil. The refrigerator keeps the cooling coil's temperature so low (-70°C to -140°C) that the water vapor in the chamber's air condenses on the coil's surface as ice. The condensation on the cooling plate relieves the entire pump system during the phase of pumping down and coating. The vacuuming of the coldtrap lessens as the condensation layer (ice) on the cooling plate increases.

After the coating process, in **Venting on** mode the cooling coils are automatically defrosted. The condensation layer (ice) melts and drops down into the collecting tray.

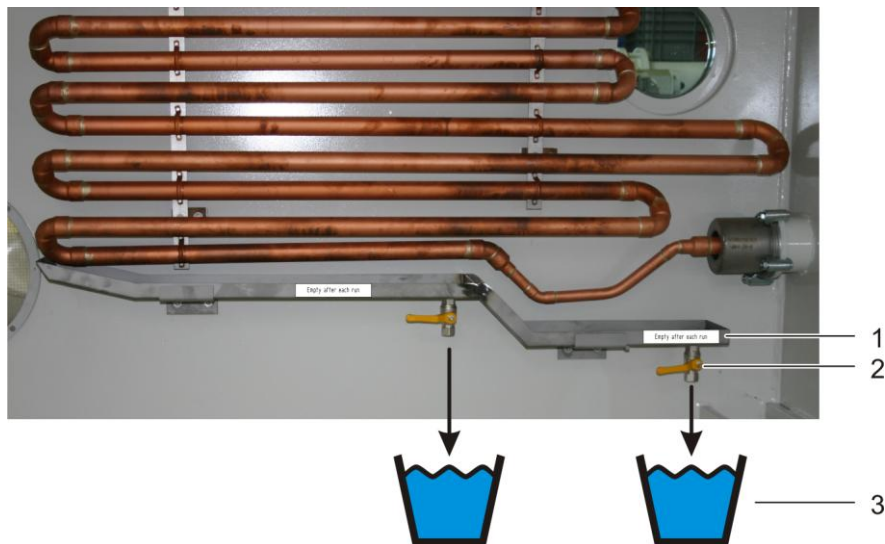
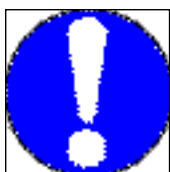


Fig. 3- 19 Collecting tray with ball valves

- 1 - Collecting tray
2 - Ball valve

- 3 - Container



Note!

The collecting tray must be emptied after each coating run. Place a container (e.g. bucket) underneath each ball valve, open the two ball valves and let the condensate flow off. Close the ball valves and move the container out off the vaccum chamber.

3.4.4.2 Polycold refrigerator

During normal operation, the refrigeration unit is switched on and off with the system's power supply. For this, the Polycold's main switch must be turned to ON.



Fig. 3- 20 Polycold refrigerator

1 - Operating panels

2 - Compressor pressure display

3 - Device's main switch

The coldtrap - i.e. heating and cooling the connected cooling plate – is controlled directly by the refrigerator. It is part of the pump system's process control:

- During the pumping down and coating phase, the connected cooling plate is chilled to the lowest temperature.
- During the venting phase of the vacuum chamber, the coldtrap is heated and the cooling plate is defrosted.

The temperature setpoints and actual values are set and monitored at the system monitor, in the corresponding pages of the visualization software.

	Note!
	For a detailed operation description, refer to the manufacturer's documentation for the Polycold refrigeration unit.

3.4.5 Pump cycle

A pump cycle is the fully automatic control of valves during a complete coating cycle, after the chamber is closed until the coated films are removed.

During loading and unloading (when the chamber is open), the high-vacuum pumps remain in operation.

A pump cycle consists of four steps (see also Fig. 3- 17):

- Evacuate
1000 mbar > p > 60mbar

V2.1, V2.3, V3.1 and V3.2 open, V2.2 closes.
At a pressure of p = 60 mbar the roots pumps VC31 and VC32 automatically start to pump.
The roots pump RA 9001 without frequency converter is switched on at a pressure of p = 60 mbar.
- $p < 4,0 \times 10^{-2}$ mbar

V1.1 and V1.2 open.
The coating chamber is pumped down to high vacuum.
At a pressure of p = $1,0 \times 10^{-2}$ mbar in the screen page "Vaccum" the pressure indication in the coating chamber switches from VM15 (Pirani) to VM21 (Pennigvac). The Pirani gauge (VM15) is still working and supplies the switching point for valve V3.1.
- $p < 8,0 \times 10^{-3}$ mbar

V3.1 closes.
- $p < 8,0 \times 10^{-4}$ mbar

At a pressure of p < $8,0 \times 10^{-4}$ mbar in the coating chamber, the interlock of the evaporator station is released.
- Venting

Before venting, the roots pumps (2) are switched off and valves V1, V2.1 and V2.3 are closed. The rotary vane pumps now run in gas ballast mode.
The coldtrap and the cooling and heating unit switch to "Heating" mode, the coating drum is heated to ambient temperature.
Valve V2.1 and V2.3 close, V2.2 opens. The roots pumps switch off.

The vent valves V4.1 opens.
The interlock for opening the chamber is released at a pressure of p > 980 mbar.

The set pressure switching points are shown on the screen page "Vaccum Parameters".

3.4.6 Vacuum monitoring

The vacuum must be monitored continuously, in order to assess the current status of the pump system (in other words, the vacuum). The TOPMET's vacuum system uses two different pressure sensors:

- Pirani-type heat conductivity vacuum meter (Pirani PSG 500S)
- Ionization vacuum meter (Penningvac IKR 251)

The following methods provide an overview of the possible test procedures for monitoring the vacuum at the system.

Pump test

Pressure changes over time can be easily recognized by regularly comparing the pump times during production. The pump times are indicated on the screen page "Vacuum" (see the Software Guide).

For the pump time test, the value when the chamber is dry, clean and empty (without substrate or degassing evaporation material) is special. APPLIED MATERIALS personnel will measure this value for new systems, after pumping down for at least 12 hours*, with the chamber opened for 5 minutes* thereafter. This pump time * is a reference parameter; it is the shortest possible pump time of the system.

Measurement of final pressure

Pumping the system down for more than 12 hours makes it possible to determine if the required final vacuum* was reached.

Measurement of pressure increase

At regular intervals or whenever you suspect leaks (pumping takes too long), determine the leak rate Q. This is done by measuring the pressure increase (see Software Guide).

The leak rate Q is a parameter for the tightness of the entire system. When the system is clean and empty, the leak rate should be in agreement with the vacuum test log. High leak rates indicate dirt or leaks in the system.

* In accordance with the specification for your specific system or the vacuum log.



3.5 Electrical supply and control

3.5.1 Control and power cabinets

The control and power cabinets contain all electrical connections and equipment for supplying electrical power (feeding and distributing) and controlling the system. All components relevant to system operation (such as motor protective switches, relays, transformers, converters) are identified and noted in the wiring diagram. The component identifiers are part of the error message texts shown by the visualization system. This makes it easy to find faulty electrical parts in the switching cabinets.

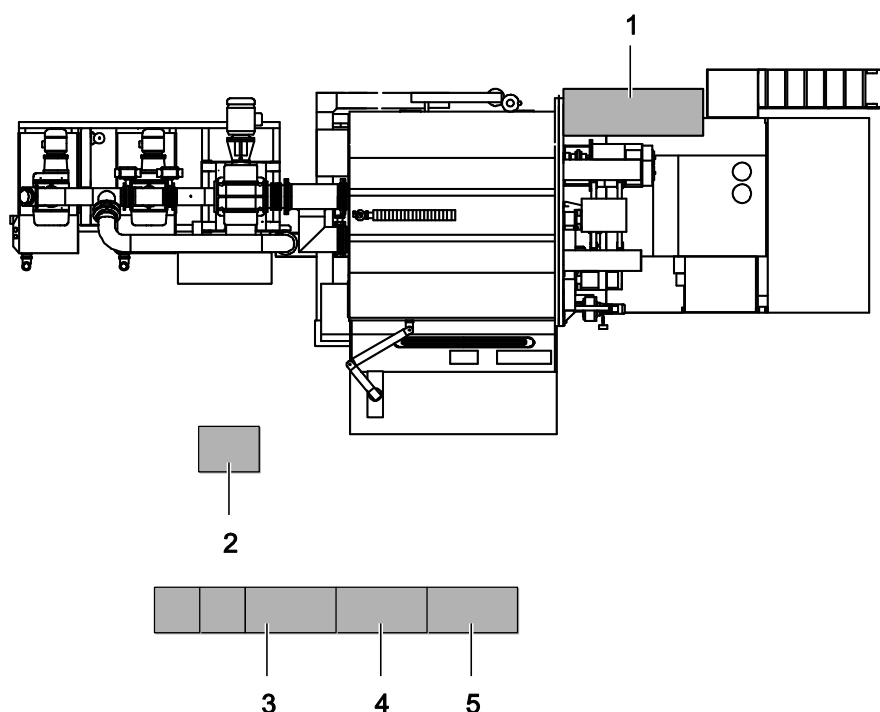


Fig. 3- 21 Arrangement of the electrical cabinets (top view), standard layout

- | | |
|---|------------------------------|
| 1 - Winding cabinet (+G6) | 4 - Evaporator cabinet (+G1) |
| 2 - Pretreatment cabinet (+G3) (option) | 5 - Wire units cabinet (+G5) |
| 3 - Power cabinet (+E1) | |

The installed power and control cabinets contain:

- +E1 Power cabinet: mains power supply and pumps
- +G1 Evaporator cabinet: power supply for evaporator boats
- +G3 Glow cabinet: power supply for Treatmag cathodes
- +G5 Wire units cabinet: drives for wire coils
- +G6 Winding cabinet: drives for drum drive motors

3.5.2 System control

3.5.2.1 Overview

The TOPMET system uses a control system based on standard hardware and software, but the process software was designed specifically for TOPMET. The control software has three main functions:

- Starting and booting the system, such as the control of the vacuum system.
- Controlling and monitoring system functions during operation, such as the tension of the winding system.
- Controlling and monitoring the coating process, such as layer thickness.

The control system controls auxiliary current switches, monitoring systems, safety interlocks and alarm messages for the entire system in the modes Automatic, Manual and Service. The user operates the control system via graphic user interfaces and operating panels. Process information (such as recipes and setpoints) is entered in the process computer. The operator uses operating panels in manual mode for certain functions, such as opening and closing the chamber.

Three modes are supported – User, Service and Supervisor.

During production, the system runs in Automatic mode. The system is booted up in Manual mode. All required control commands are available in both modes.

During operation, the control system always allows programmed system control, for safe operation. In Service mode, however, some interlocks are disabled.

3.5.2.2 SIMATIC S7-400

The SIMATIC S7 controls and monitors operation of the entire system. All system devices and components for drive control, regulation, pressure measurement, etc., are connected with the Simatic via L2 bus lines. This bus is used to exchange data and information between the system components and the control. The system monitor and an optional network connection (Ethernet) are connected to a central processor in the S7.

Within the framework of the operational modes, the control system provides the following main features:

- Fully automatic control and monitoring of the overall system and of the vacuum and winding systems.
- Calculation of setpoints.
- Status display of system components, vacuum pumps, chamber pressures and valve positions.
- Automatic process control, using recipes and process parameters.
- Display of setpoints and actual values for the process and winding systems.
- Error recognition and alarm displays.
- Storing (hard disk) and outputting (hard copy) of process and alarm messages.
- Software and hardware interlocks and limit switches ensure the system can only be operated within safe limits. An acoustic signal sounds in case of a fault.
- Safety system with emergency-stop pushbuttons.



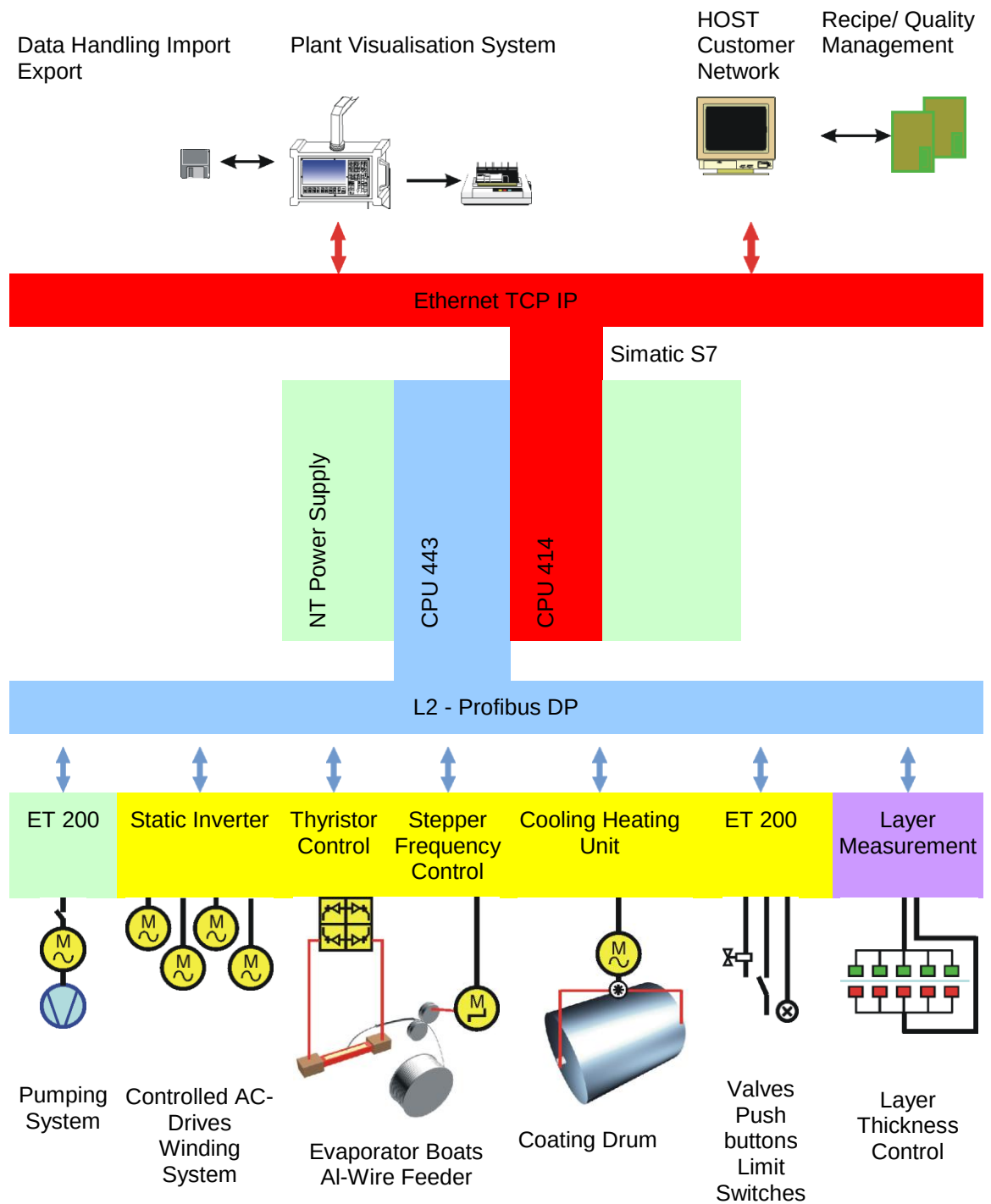


Fig. 3-22 System control diagram

3.6 Media supply

The media supply for the TOPMET film coating system encompasses the supply of cooling water (see chapter 3.6.1) and compressed air (see chapter 3.6.2).

3.6.1 Cooling water supply

The system components are supplied with cooling water from the customer's water source and a water distribution system installed at the system.

	Note! The cooling water provided from the customer's water source must comply with the specification stated in the technical specifications for the system.
---	--

3.6.1.1 Water distribution system

The following schematic diagram shows the water distribution system of the TOPMET. The water distribution unit is located at the system on the rear side of the chamber.

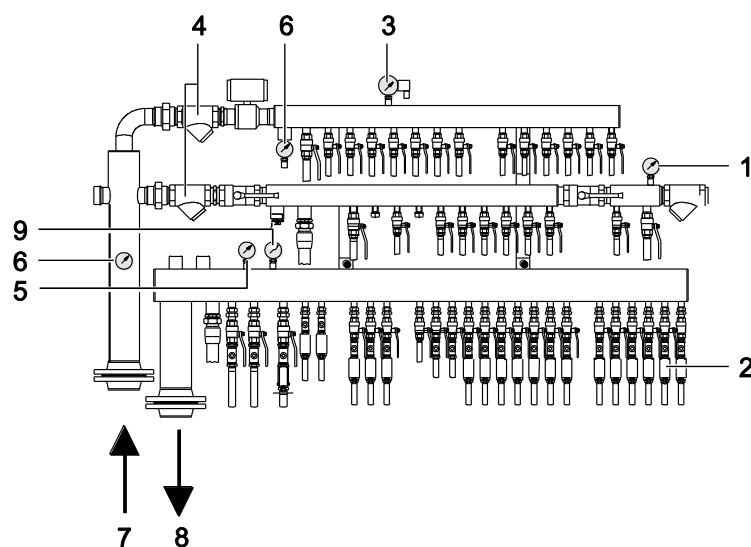


Fig. 3- 23 View of the water distribution system

- | | |
|---|------------------------------|
| 1 - Inlet pressure for cooling devices | 6 - Water inlet temperature |
| 2 - Flow monitor | 7 - Water inlet line |
| 3 - Inlet pressure for evaporator and pressure switch | 8 - Water return line |
| 4 - Dirt filter | 9 - Water return temperature |
| 5 - Return line for differential pressure | |

All system components supplied with cooling water are monitored via the screen page "Water Scheme" (see Software Guide).

3.6.1.2 Water diagram

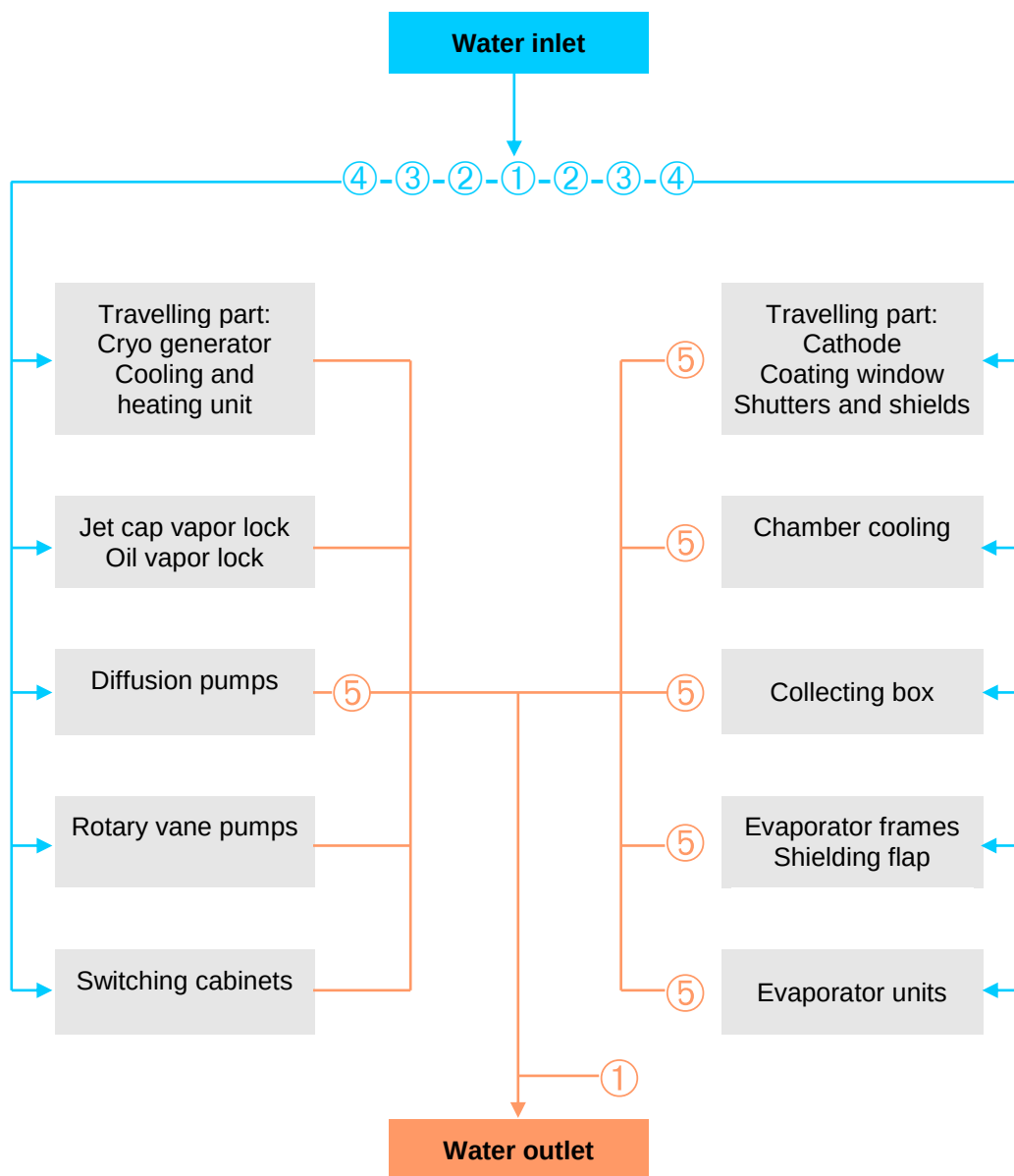


Fig. 3- 24 Water diagram

- 1 - Thermometer
- 2 - Filter
- 3 - Ball valve

- 4 - Pressure gauge
- 5 - Flow monitor

3.6.2 Compressed air supply

The system's supply of compressed air is implemented with a compressed air maintenance unit plus following distribution units. The compressed air maintenance unit is located on the reverse side of the vacuum chamber, next to the water distribution system.

	Note!
	The compressed air provided from the customer's air source must comply with the specification stated in the technical specifications for the system.

3.6.2.1 Compressed air maintenance unit

A compressed air maintenance unit installed at the system processes the compressed air provided by the customer's source (water separation, filtering, pressure regulation). The compressed air then flows to the main distribution units (pilot valve units) of the chamber and travelling part, as well as to the following loads.

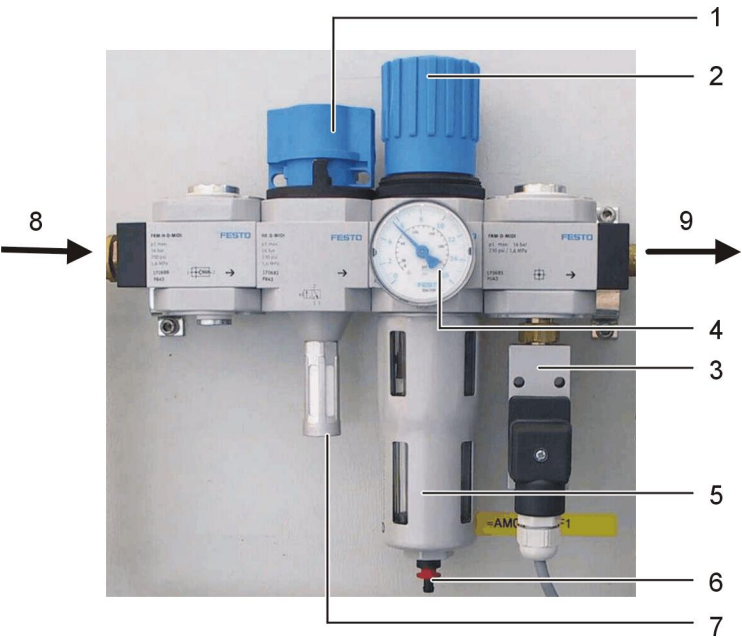


Fig. 3- 25 Compressed air maintenance unit

- | | |
|--------------------------------|---|
| 1 - Manual shut-off valve | 6 - Drain button |
| 2 - Control valve | 7 - Muffler |
| 3 - Pressure monitoring switch | 8 - Compressed air connection (customer's side) |
| 4 - Pressure gauge | 9 - Compressed air distribution system (system) |
| 5 - Water separator | |

3.6.2.2 Compressed air storage tank

A compressed air storage tank with spare compressed air is part of the compressed air supply. If the compressed air fails, this tank ensures that all pneumatically operated components come to a final safe position.

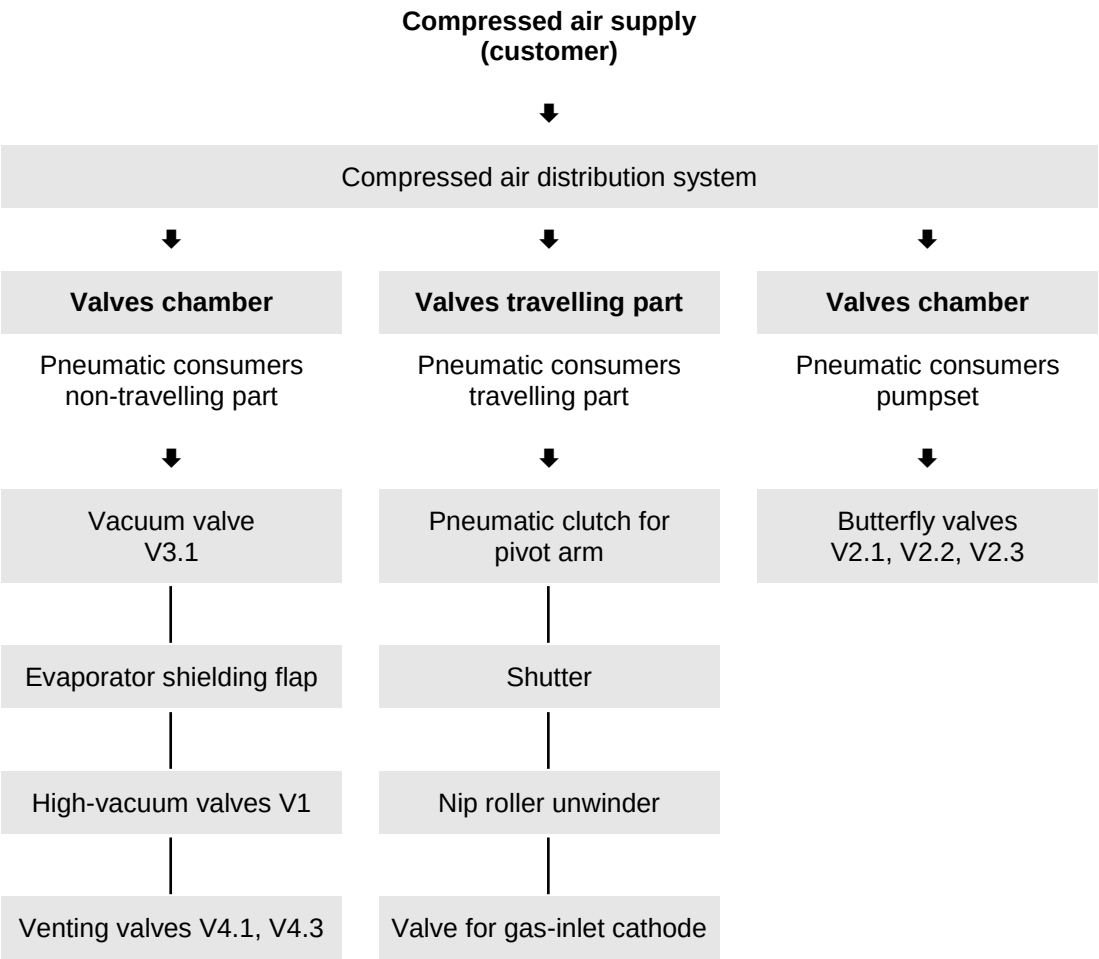
The compressed air storage tank is attached to the support frame of the fore pumping station.



Fig. 3- 26 Compressed air storage tank

3.6.2.3 Compressed air diagram

The schematic diagram shows the compressed air distribution system and the supplied loads of the TOPMET system. The compressed air flows from a central compressed air maintenance unit. The compressed air then branches to two main distribution units for the chamber and the travelling part. The main distribution units then supply all compressed air-controlled components.



4

Operating Elements

The chapter "Operating Elements" describes all operating panels, switches and key elements of the TOPMET system.

4.1 Overview

Die TOPMET film coating system is controlled by means of various operating elements:

- Graphic user interface for system control and visualization, e.g. entering parameters, process monitoring, see chapter 4.2.
- Operating panels to control system functions, see chapter 4.3.
- Main switch and other switches for supplying power to the system, see chapter 4.4.
- Safety-related switches, such as key switches and emergency-stop pushbuttons, see chapter 4.5.



4.2 System control and visualization

4.2.1 System configuration

The PC monitor panel PC is part of the system control PLC (SIMATIC S7). An Ethernet bus enables data communication between the PC monitor panel PC and the S7 system control.

The system configuration of the hardware and software is described in your Software Guide.

4.2.2 Graphic user interface

The system's visualization system features a monitor panel with a PC and visualization software (InTouch™) tailored specifically to operation of this system. The color monitor is the visual interface between the operational personnel and the system control. There are screen pages for all operational steps required to monitor and control the production process and for service work. The monitor is conveniently placed on a movable pivot arm in front of the system.

For detailed information about the system refer to the suppliers documentation.

During system operation, the monitor panel is responsible for the following tasks:

- User guidance, consisting of screen pages with graphic process displays, diagrams and data fields for controlling and operating the system.
- It is the interface for data entry (setpoints) and data output (display of actual values).
- It allows the user to output reports and screen settings (hard copies) for process documentation.
- It allows the user to acquire data (data logging) and manage recipes.
- It includes an error message system for the system and its components.

The visualization system is based on menus. Data is selected, used and entered via function keys.

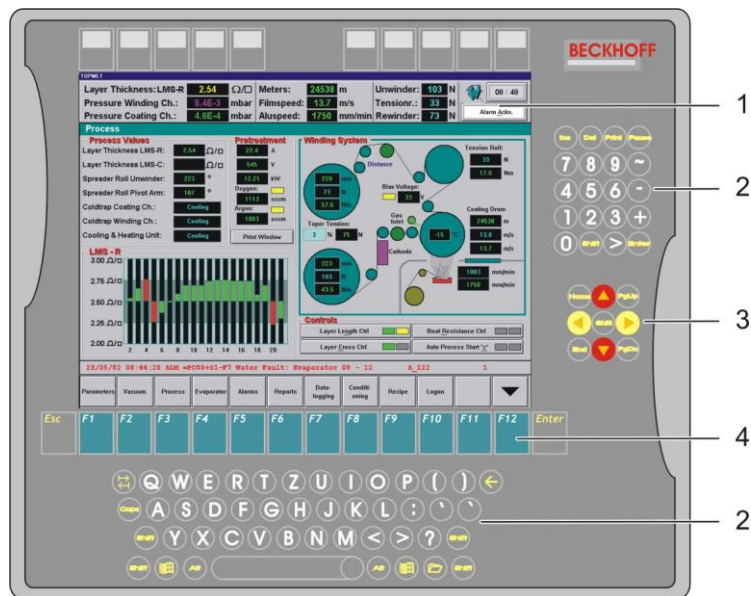


Fig. 4-1 Graphical user interface

- | | |
|-----------------------|----------------------|
| 1 - Screen page | 4 - Function keys |
| 2 - Operating keypad | 5 - Operating keypad |
| 3 - Special functions | |

- Screen page See chapter 4.2.3
- Operating keypad See chapter 4.2.4
- Special functions See chapter 4.2.4
- Function keys See chapter 4.2.4

4.2.3 Structure of screen pages

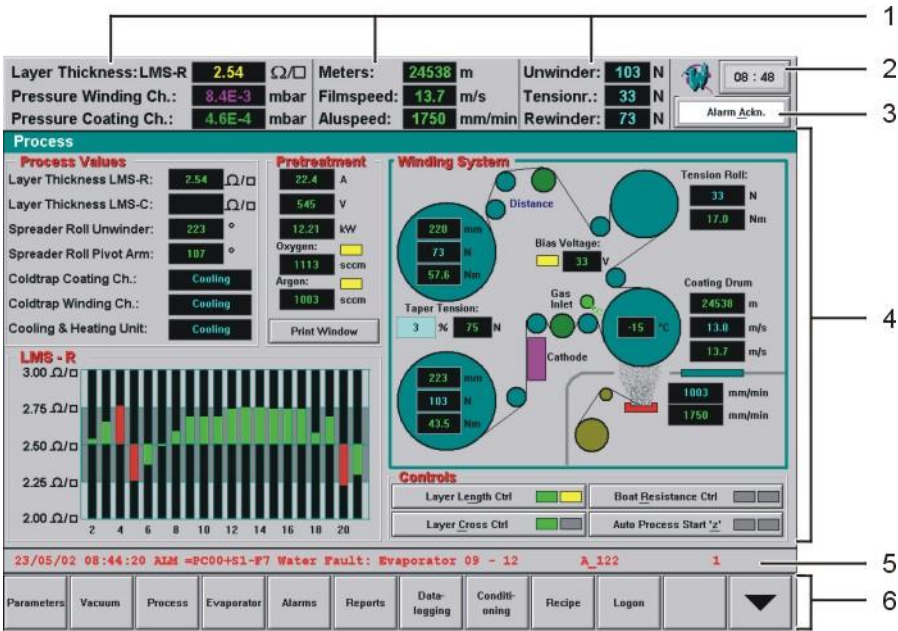


Fig. 4- 2 Structure of a screen page

- 1 - Process data

2 - Time of day

3 - Button Alarm Ackn.
- 4 - Visualization

5 - Alarm bar

6 - Function buttons

The screen pages are divided as follows:

Screen element	Description
• Permanent display: • Process data • Time • Button Alarm Ackn.	These display fields are always shown, even when you branch to another page. You can observe the process data, independent of the selected screen page (see Software Guide). Displays the current process data. Displays the system's current time of day. Button for acknowledging alarm messages.
• Visualization	The currently selected screen page is shown. The name of the current page is shown in the header. The screen page contains graphic displays of process events, diagrams, buttons, data entry fields, and displays of setpoints and actual values (see chapter 4.2.3.1 and 4.2.3.2).
• Alarm bar	This bar shows the most recent current error message, or the status of alarms (see Software Guide).



Screen element	Description
Function buttons	Allows you to control machine and process functions, with the buttons F1...F12 (see Software Guide). A new set of buttons appears when you change to a different screen page. The new set contains all functions required to use this screen page.

4.2.3.1 Input/output fields and buttons

Screen pages for setting process parameters and using operating functions are divided into sections, as shown below.

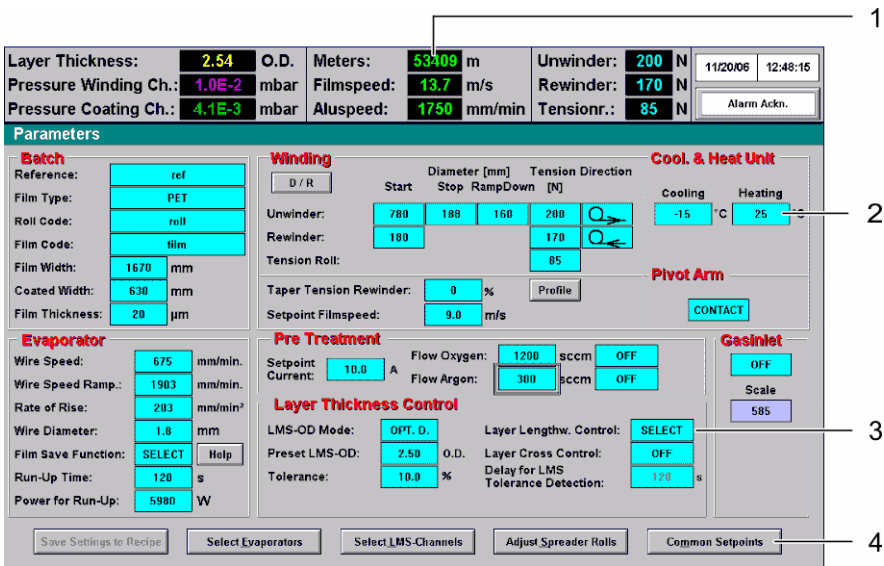


Fig. 4- 3 Entry fields and buttons

- | | |
|------------------|-------------------|
| 1 - Output field | 3 - Status button |
| 2 - Entry field | 4 - Menu button |

The meaning of the fields is as follows:

- | | |
|-----------------|--|
| • Output field | An output field displays setpoints or actual values. |
| • Entry field | Click on the entry field and enter the values or parameters directly. |
| • Status button | Click on the button to toggle between predefined system functions. |
| • Menu button | Click on the button to activate the corresponding (branching) menu or the system function. |



Hot keys

Some buttons are hot keys. You can immediately execute the button operation by entering the proper identification letter at the keypad (e.g. "S" for <Adjust Spreader Rolls ...>).

Except for a few exceptions, you can execute hot key functions no matter what screen page is currently selected. A few hot key functions can only be executed if the corresponding screen page is selected.

Clicking means the following:

With the cursor keys, place the cursor in the desired field. Then press the ENTER key (see chapter 4.2.4) (touchscreen: directly press on the field).

Always follow your entries by pressing the ENTER key. Only then does the program accept your entries.

4.2.3.2 Status colors

The process status and parameters are color-coded.

- Red Text: Error message, alarm condition.
 Icon: Defective or nonfunctioning system component.
- Yellow Icon: Status display for a currently active operational function.
- Green Text: Displays the current operational status (actual value). Status display for a preselected function.
 Icon: Status display for the active process.
- Blue Text: Displays the setpoint for a parameter.
 (background)
- White or gray Icon: Neutral status. No processes or operational functions are active.



4.2.4 Keypad

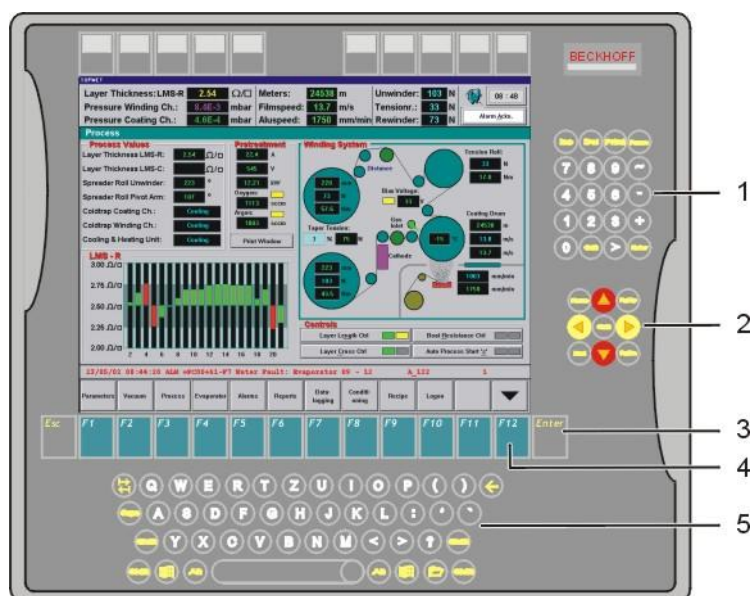


Fig. 4- 4 Operating keypad

- | | |
|---|----------------------------|
| 1 - Numeric entry field | 4 - Function keys |
| 2 - Special function keys / Cursor keys | 5 - Alphabetic entry field |
| 3 - Enter | |

- Numeric entry field** Use this portion of the keyboard to enter numeric values in screen entry fields. Use the alphabetic keys to enter letters.
- Special function keys** These keys are used to branch between pages or the begin and the end of a page.
- Cursor keys** These keys control the movement of the cursor on the screen. Use them to place the cursor on buttons or in entry fields.
- Alphabetic entry field** Use this part of the keyboard to enter letters and characters in screen entry fields. Use the numeric keys to enter numbers.
- Enter** Activates selected buttons, confirms and accepts entered parameter values.
- Function keys** The function keys F1 – F12 are used to directly select screen pages. These function keys correspond to the function buttons displayed in the screen pages (see Software Guide).
- Delete / Insert** Delete values in data fields.
Insert new values in data fields.



4.3 Operating panels for system control

4.3.1 Overview

The pushbuttons and switches for system operation are placed on operating panels. The following illustration shows the positions and designations of the operating panels at the TOPMET film coating system.

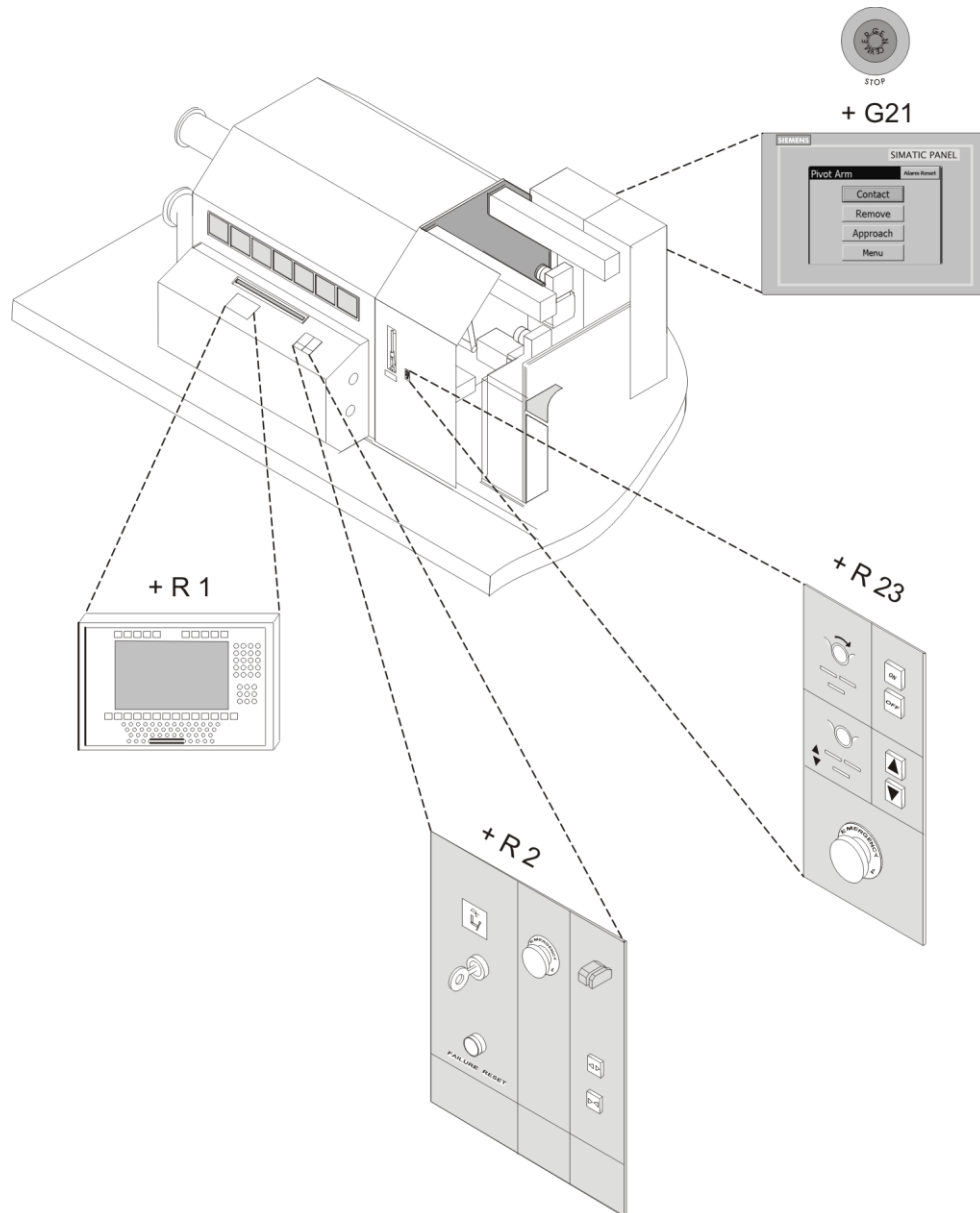


Fig. 4-5 Arrangement of the operating panels

4.3.2 Operating panel +R01

The operating panel is used to operate

- operate and set the winding and pump system
- control and monitor the evaporator station

For detailed information refer to “Software Guide”.

4.3.3 Operating panel +R02

The operating elements on operating panel + R02 are used for general system control. The position of operating panel +R02 is shown in Fig. 4- 5.

The settings for the operating elements of this operating panel affect the parameters for system control (see Software Guide) and overwrite them.

Element	Function
Service key switch	Safety switch for locking motor-driven system parts. The safety interlock protects personnel during system maintenance and service work. The key can be removed when in the vertical position.
Emergency-stop pushbutton	Pressing the emergency-stop pushbutton stops all motor-driven system parts (see chapter 2.5.1.1). Press this pushbutton immediately in dangerous situations (when persons could be injured or equipment could be damaged).
Open/close chamber	Pushbutton function for opening and closing the vacuum chamber. When opening the chamber, always move the travelling part to its final position (limit switch position).
FAILURE/RESET	When the lamp is lit, this indicates a current error message. The signal lamp remains lit as long as the error message is active, i.e. as long as the cause of the fault has not been eliminated. The fault is acknowledged by pressing the button (signal lamp turns off).



4.3.4 Operating panel +R23

The operating elements at operating panel +R23 are used to clean the coating drum during service work. The position of operating panel +R23 is shown in Fig. 4- 5.

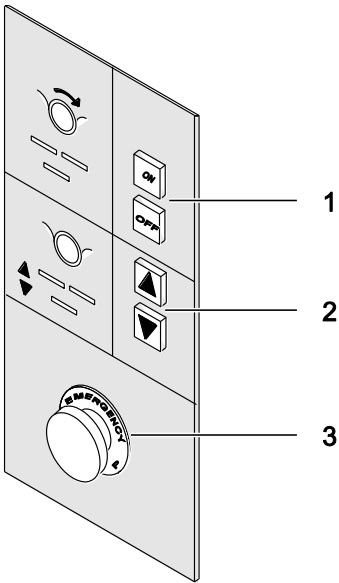


Fig. 4- 6 Operating panel +R23

- 1 - Coating drum service mode
- 2 - Raise/lower coating window

- 3 - Emergency-stop button

Element	Function
Coating drum service mode	This pushbutton function is used to start and stop the coating drum for servicing. This function can only be executed when the chamber is open. - ON: Only the coating drum rotates and accelerates to the assigned film speed. The drum rotates opposite to the normal direction of travel. - OFF: The coating drum slows to a stop.
Raise/lower coating window	- Raising: The coating window to raised to the work position. - Lowering: The coating window to lowered to the cleaning position. The shutter automatically opens before lowering.
Emergency-stop button	Pressing the emergency-stop pushbutton stops all motor-driven system parts (see chapter 2.5.1.1). Press this pushbutton immediately in dangerous situations (when persons could be injured or equipment could be damaged).



4.3.5 Operating panel +G21

The operating elements at operating panel +G21 are used primarily to thread and align a new film. The position of operating panel +G21 is shown in Fig. 4- 5.

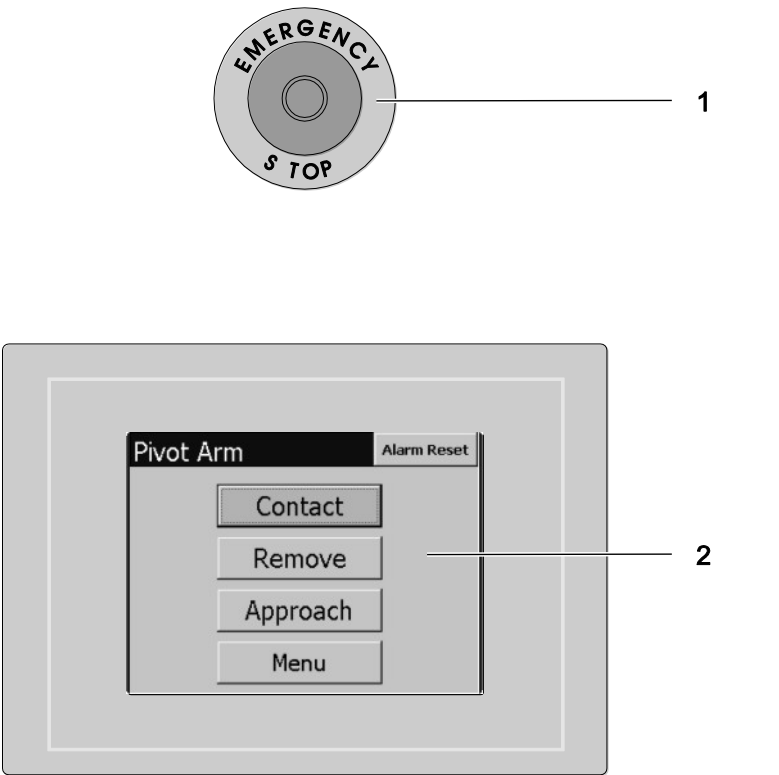


Fig. 4- 7 Operating panel +G21

- 1 - Emergency-stop button
- 2 - Touchscreen

The settings for the operating elements of this operating panel affect the parameters for system control and overwrite them.

For a detailed description of the screenshots refer to your Software Guide, chapter 7 “Operating Panel at Cabinet +G21”.

Element	Function
Emergency-stop button	Pressing the emergency-stop pushbutton stops all motor-driven system parts (see chapter 2.5.1.1). Press this pushbutton immediately in dangerous situations (when persons could be injured or equipment could be damaged).
Film tension	This function is used to thread a film placed in the winding system from the unwinder side to the rewinder side. - ON: After this pushbutton is pressed, only the coating drum and tension roll are driven, at a speed of < 0.3 m/s. All other drums of the winding system are idle. - OFF: The function "Film tension" is stopped. The coating drum and tension roll drives are switched off.

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Element	Function
Axial adjustment (unwinder) (optional)	Function for shifting the axial position of the unwinder shaft from the center position. The axial position of the unwinder can be manually shifted ± 15 mm from the center position. This function is used to correct the setting for film travel from the coating drum towards the rewinder.
Spreader roll Fixbow (unwinder) (rewinder)	Pushbutton function for motor-driven rotating of the roll position relative to the web. - Rotate the spreader roll's bow into the web: middle of film is stretched (button "+"). - Rotate the spreader roll's bow away from the web: edge of film is stretched (button "-"). The set bow amount is shown on the screen page "Process" under Process Values.
Pivot arm	Pushbutton function for motor-driven shifting of the pivot arm position: - Move the pivot arm away from the rewinder. Press button "Remove". The pivot arm moves upwards as long as the pushbutton is pressed. - Move the pivot arm towards the rewinder. Press button "Approach". The pivotarm moves automatically by pressing the pushbutton once to the limit switch.

4.4 Main switch

4.4.1 System main switch

The system main switch is located on the front door of the power cabinet +E1 (for the location, see chapter 3.5.1). It is used to switch the system's electrical power on and off. The function of the main switch is determined by three predefined switch positions.

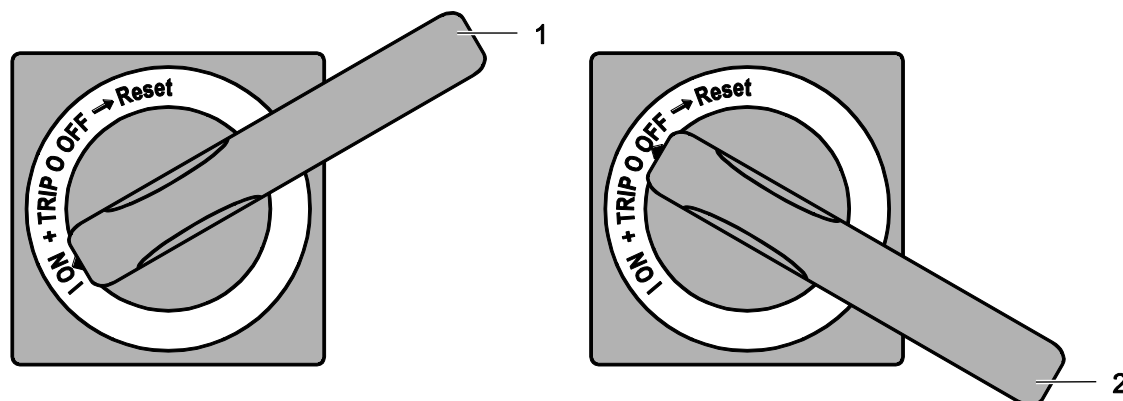


Fig. 4- 8 Main switch

1 - Position ON

2 - Position OFF

I Position "ON"

The system's power supply is switched on. The switching cabinets and modules are connected to the source of electrical power.

+ Position "Trip"

When the line current fluctuates (power failure, overload current, short circuit), the power supply automatically shuts down. The main switch automatically switches from the position "ON" to the position "Trip".

Switching the electrical power back on again:

- Rotate the switch from the position "Trip" to the position "OFF".
- Rotate the switch from the position "OFF" to the position "ON".

O Position "OFF"

The system's power supply is switched off. The front door of the power cabinet can only be opened when the main switch is turned off.

4.4.2 Polycold

The position of the main switch of the Polycold refrigerator is shown in Fig. 3- 20.

A detailed description of its operation can be found in the manufacturer's documentation for the Polycold refrigerator.

4.5 Safety-related switches

4.5.1 Key switch

	Note!
	Only the original safety keys may be used. It is strictly forbidden to duplicate keys and to use such duplicates. Contact the manufacturer if you lose safety keys.

Key switch	Position	Comments
Service key switch	Operating panel +R02	See chapter 2.5.1.2

4.5.2 Emergency-stop pushbuttons

The positions of the emergency-stop pushbuttons and their function are described in chapter 2.4.3 or in chapter 2.5.1.1.

4.5.3 Signal lamp and horn

Inside the vacuum chamber a signal lamp and horn indicates a hazard of the diffusion pump.

	Warning
	Fire and explosion hazard The oil diffusion pump can explode or the area around catches fire. If the prevacuum pressure is higher than 5 mbar and the oil diffusion pump temperature is over 100°C an acoustic alarm and flash light appears. • Leave the vacuum chamber immediately! • Leave the area around diffusion pumps immediately! • Opening of the vacuum chamber is blocked!



Fig. 4- 9 *Signal lamp and horn*

5

Installation

The chapter "Installation" provides information on setting up and connecting the TOPMET web coating system.

5.1 General information

This chapter contains general guidelines on installing the TOPMET film coating system and provides information on how to set up, connect and check the TOPMET film coating system for proper functioning.

	Note!
	Only duly trained service technicians of APPLIED MATERIALS are permitted to connect, install and test the system. Other persons are not permitted to perform this work.

The lines of responsibility for the installation are as follows, unless the purchase contract contains other provisions:

The end user of the TOPMET film coating system is responsible for:

- Providing the supply connections (water, electrical power, compressed air, etc.) and disposal facilities (exhaust gases) in accordance with specifications made in the quotation and order confirmation.
- In-house transportation of the TOPMET film coating system to the installation site.
- Unpacking the TOPMET film coating system.

By contract, APPLIED MATERIALS is responsible for:

- Setting up and commissioning the TOPMET film coating system.
- Testing the TOPMET film coating system, including the acceptance test protocol.

APPLIED MATERIALS will supply the following documents to the end user of the TOPMET film coating system:

- With the purchase contract:
 - layout drawing, media documentation and specifications for the electrical connections.
- With delivery of the system:
 - delivery documents, machine documentation, end user-specific documents.



5.2 Installation site requirements

At the installation site, ambient conditions must be created as specified in chapter 1.6.4 of these instructions.

5.2.1 Space required for the TOPMET film coating system

For the end user, the **layout drawing specific to each order** applies. The safety distances for evacuation routes must be adapted to local safety regulations.

5.2.2 Preparing the Installation

The preparatory work which the end user must perform for the installation is described in the chapter of these operating instructions.



5.3 Setting up and connecting the TOPMET film coating system

5.3.1 Setting up the TOPMET film coating system

	Note!
	The TOPMET film coating system will be set up and installed by service technicians of APPLIED MATERIALS. The end user is only permitted to set up the TOPMET film coating system if supervised by APPLIED MATERIALS.

	⚠ Warning
	Beware of hovering loads. Improper handling of heavy parts can cause serious contusions of body parts, fractures and death. In particular, this applies for: • Vacuum chambers and their add-ons. • Process station components (drums, glow station, etc.). • Pumping system components, in particular vacuum pumps and valves. • Cabinets. Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use carts, crane trucks and hoists.

The structural components are set up and connected in accordance with the layout drawing. Adhere to the installation notes for the structural components (see separate folder with OEM operating instructions).

5.3.2 Media Supply

The end user is responsible for connecting the TOPMET film coating system to the media supplies, such as water, compressed air, etc. Adhere to the media schematic diagrams provided by APPLIED MATERIALS.

There are no provisions for APPLIED MATERIALS to make the initial connection of the TOPMET film coating system to the media supply.



	Note !
	The end user is responsible for fulfillment of the specifications for the supply media, as stated in the chapter Use and technical data.

5.3.3 Electrical connection

The connection values to be provided by the end user are stated in the chapter 1.6.5.

	Note!
	Only properly trained service technicians of APPLIED MATERIALS are permitted to cable the system and to connect it to the source of electrical power, in order to avoid damage from improper implementation and the resulting exemption of liability on the part of the supplier.

	⚠ DANGER
	Beware of high voltage parts. Danger of fatal electrical shock when touching high voltage parts. Service work may only be performed on the electrical equipment if it has been removed from the source of electrical power. Switch off the affected components, secure them against being accidentally switched on again, and discharge them with the grounding rod. Important: Some parts of the system might be connected to electrical power even when the main switches are turned to the Off position. This applies in particular to devices connected to the maintenance power outlets. The primary side of the main switch always carries voltage. Only trained personnel with experience in high-voltage systems is permitted to perform service work on the electrical equipment.

5.4 Testing/repeating the tests

5.4.1 Functional test after the installation

Only service technicians of APPLIED MATERIALS will perform the functional test after the TOPMET film coating system is set up and connected to the supply of electrical power. Under no circumstances is the end user permitted to perform the functional test. The result of the functional test will be documented in the acceptance test protocol.

5.4.1.1 Checking the electrical connections

	⚠ DANGER Beware of high voltage parts. Danger of fatal electrical shock when touching high voltage parts. Service work may only be performed on the electrical equipment if it has been removed from the source of electrical power. Switch off the affected components and secure them against being accidentally switched on again. Important: Some parts of the system might be connected to electrical power even when the main switches are turned to the Off position. This applies in particular to devices connected to the maintenance power outlets. The primary side of the main switch always carries voltage. Only trained personnel with experience in high-voltage systems is permitted to perform service work on the electrical equipment.
--	--

Carefully check all connections for leads conducting the line voltage, as well as those signal ground and protective ground leads which must be connected for installation of the equipment.

Primarily make sure,

- all protective equipment such as lids, covers and interlocks are properly in place.
- the cables and ground leads are connected to the proper terminals.
- the terminals are isolated from one another. This means no strands of wire protrude and no wires are pinched.
- the cable connections are tight enough. Stranded wire cables should always be provided with cable casings or crimped-on pins.
- there is strain relief for the cables.

5.4.1.2 Checking the emergency-stop circuit

Check safe functioning of the emergency-stop circuit as follows:

- Turn the main system switch on.
- Press one of the emergency-stop buttons.
- Unlock the emergency-stop button by hand.
- Repeat the test for all other emergency-stop buttons.

5.4.1.3 Checking the safety chain

	Note!
	This check must be based on the wiring diagram for your specific system. The safety-related sheets of the wiring diagram are marked with the note "Safety circuit".

5.4.2 Repeating the tests

Repeat the tests of the electrical and mechanical equipment every **six months** and in accordance with applicable national safety regulations.

	Note!
	Reference is made here only to the overall system. For devices with discharging resistors, refer to their operating instructions for the required information on repeating the tests. Record the results in the record book which the end user must keep.

5.4.2.1 Checking the safety circuits after service work

Check the safety circuits after any and all service work. Perform the check in accordance with chapters 5.4.1.2 and 5.4.1.3 of these operating instructions.

5.4.2.2 Six-month test repetitions

Electrical tests

- Check the safety circuits in accordance with chapters 5.4.1.2 and 5.4.1.3 of these operating instructions. Actuate the safety locks several times (at least five times).
- Check all high voltage conductors, including the related return leads and the signal and protective ground lead connections. Check the connection points by wiggling them and visual inspection.
- The cooling water monitors (water flow monitors) must actuate whenever the pressure or flow volume drops below the specified minimum. Check this at least five times.
- Check the entire water system for leaks and for damaged or broken hoses.
- Check the contacts of the connection points of all high-current cables (e.g. heating equipment).

Mechanical tests

- Visually inspect all parts which are subject to wear and tear. See the specific operating instructions for details.
- Make sure all protective devices (e.g. lids, covers, locks, etc.) are properly in place.
- Whenever the system is modified mechanically or statically (e.g. drilling in reinforcement ribs), check it for mechanical strength. Reinforce it if necessary.
- Also check the entire cooling water system for leaks and for damaged or bent hoses.

The customer and the service technician of APPLIED MATERIALS must make a record of the test repetitions in the acceptance test protocol.

5.4.3 Knowledge of the safety locks

When the TOPMET film coating system is handed over to the end user, the function of the safety locks and interlock circuits must be demonstrated. For this, the TOPMET film coating system must be completely ready to operate and tested for proper functioning.

Refer to the chapters 5.4.1.2 and 5.4.1.3 for information on the functions to be demonstrated and tested.

The customer and the service technician of APPLIED MATERIALS must confirm in the acceptance test protocol that these safety features are functioning properly.



6

Operation

The chapter "Operation" explains how to operate the TOPMET web coating system. All operational steps important for production are explained.

6.1 Introduction

6.1.1 Personnel qualification

Only authorized personnel with the corresponding rights is permitted to switch the system on and operate it. The end user of the system is responsible for ensuring that only duly trained personnel operates the system.

You are only permitted to operate the TOPMET system after having read and completely understood chapter 2. In particular, note the safety precautions in chapter 2.6.

	Note!
	APPLIED MATERIALS shall assume no liability for damage caused by operation of the system by nonauthorized personnel. The safety regulations in the chapter Safety and access rights must always be adhered to.

6.1.2 Operation

The user operates the TOPMET system by means of the keyboard, the screen of the operating terminal and the operating panels. The PLC controls and monitors all system functions.

6.1.3 Cleanliness

	Note!
	The process unit can only ensure a high throughput if it is clean. Therefore, adhere to the following rules: • Put on clean room gloves before you touch parts in the process unit. • Make sure only clean, dry objects are placed in the process unit. • Restrict use of vacuum grease to a minimum.

You should clean the evaporator station and the coating window after each coating cycle. This work is described in chapter 7.3.1.

6.2 Modes

The system can be operated in two modes:

- Automatic coating cycle (Auto-Process-Start)
- Manual mode

Both modes are selected or deselected from the screen page "Process" (see Fig. 6- 11). The control has an interlock which prevents both modes from being selected simultaneously. In addition, numerous protective and safety interlocks prevent improper system operation in the selected mode.

6.2.1 Manual mode

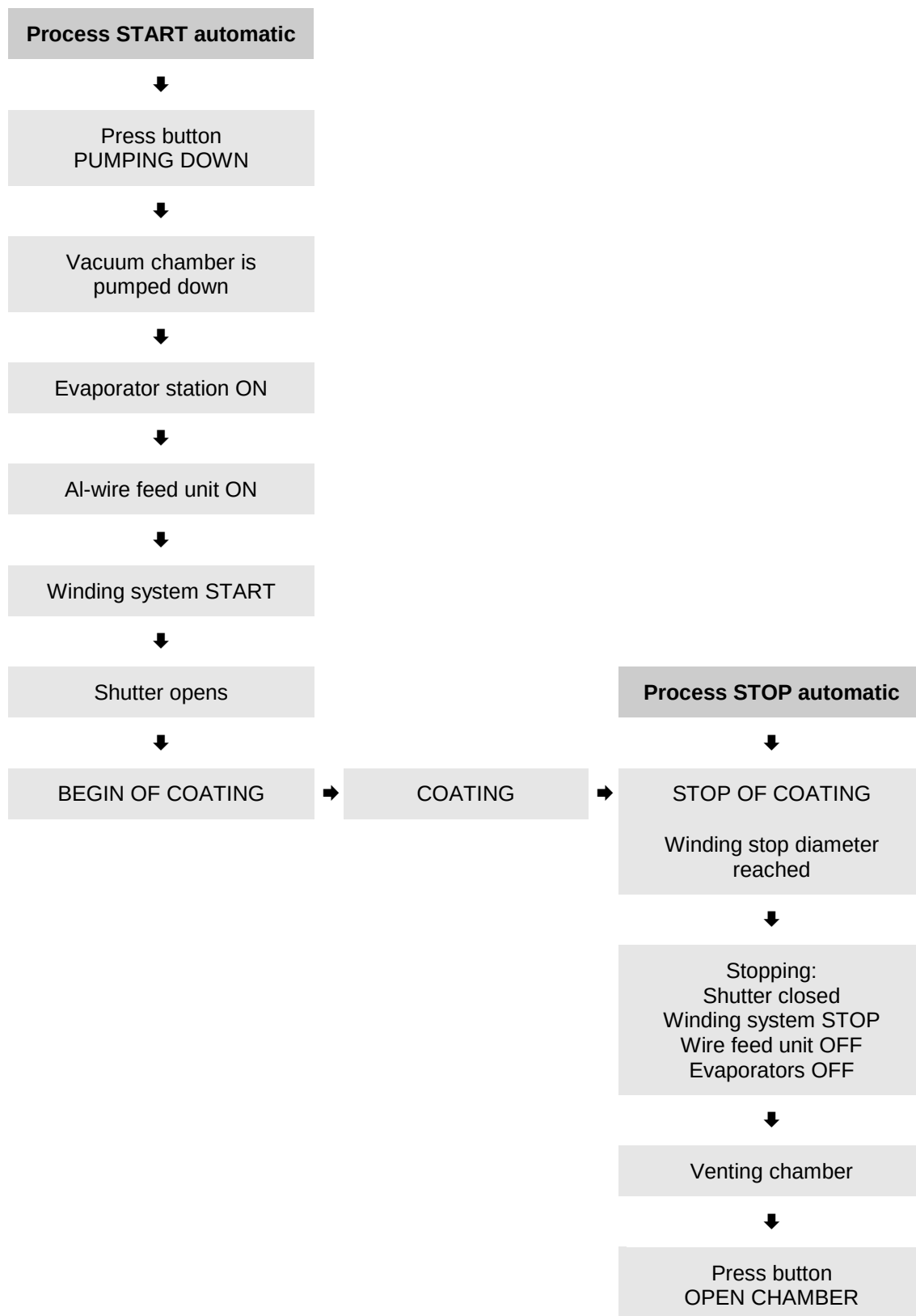
Manual mode is used to condition and optimize the system and its components after maintenance and repair work. It is not appropriate for regular production operation.

In this mode, the automatic process start-up feature is disabled. You must perform all steps for system operation manually (using pushbuttons).

6.2.2 Automatic coating cycle

This mode is for regular production operation of the system. In this mode, most process steps for the entire coating cycle are automated. In particular, the automatic process start-up feature during start-up and an automatic shut-down feature at the end of processes reduce the workload for the operational personnel.

The following schematic diagram illustrates the role which automatic features play during coating cycles.

Course of a coating cycle

6.3 Prerequisites for operation

Before you switch the system on and start operation, make sure the following components are functioning properly:

- | | |
|-----------------------------|--|
| Installation work: | <ul style="list-style-type: none">• Finish installation work and inspect the system, see chapter 5. |
| Emergency-off pushbuttons: | <ul style="list-style-type: none">• Make sure all emergency-off pushbuttons are reset, see chapter 4.5.2. |
| Compressed air supply: | <ul style="list-style-type: none">• Proper working pressure (6 - 7 bar), see chapter 1.6.3. |
| Water distribution system: | <ul style="list-style-type: none">• Check the inlet pressure (4 - 6 bar), see chapter 1.6.2.• Check the inlet temperature (20 - 25°C), see chapter 1.6.2. |
| Vacuum pumps: | <ul style="list-style-type: none">• Check the oil level in the appropriate sight glass, see chapter 7.2.1. |
| Vacuum rotary feedthroughs: | <ul style="list-style-type: none">• Check the oil level in the appropriate sight glass, see chapter 7.5.9.1. |
| Polycold: | <ul style="list-style-type: none">• Switch the main switch of the Polycold on, see chapter 4.4.2.• Select REMOTE mode, see chapter 3.4.4.• Switch the 24-V power supply to I . |



6.4 Switching the TOPMET system on and off

	⚠ Warning
	Hazard of contusions. When the travelling part moves, there is the hazard of serious injuries or contusions. Before opening the chamber, make sure no persons are located in the travelling part's path.

	⚠ Caution
	Hazard of damage to the system. The system can be damaged if any objects are located in the travelling part's path. Before opening the chamber, make sure no objects are located in the travelling part's path.

6.4.1 Switching the system on

	Note!
	When the main switch is turned to ON, an electric interlock prevents mechanical movement of the system. A warning based on this emergency-stop function indicates when the system is ready.

- Switch the power supply on: Turn the main switch to ON.
- The line voltage power is available. The system and its components are connected to electrical power.
- The system control (PLC) begins the automatic start-up procedure. The operating system Windows boots.
- Log onto Windows as "Operator". A password is not required.
- The visualization system InTouch starts automatically (auto-start function of Windows).
- After around two minutes, the log-on window appears at the system control monitor. You must enter your password to access the levels User, Service and Administrator.

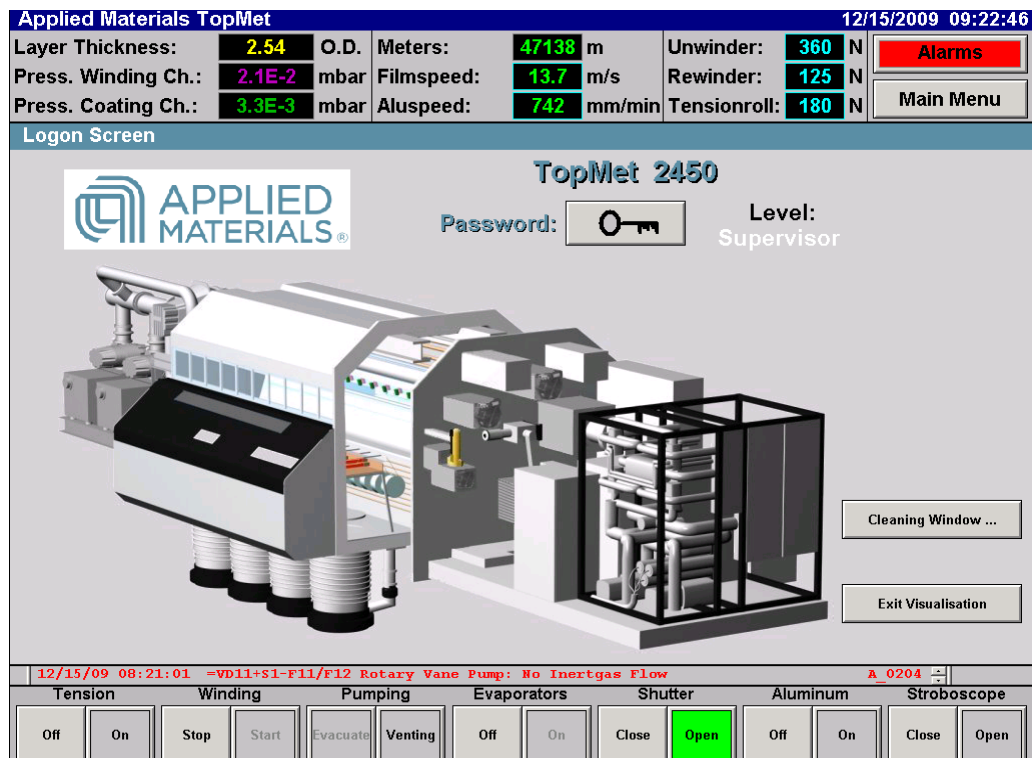


Fig. 6-1 Screen page "Logon"

- Acknowledge the warnings (hooter signal). For this, press the pushbutton FAILURE/RESET at operating panel +R02. The system is now completely ready.
- Switch the vacuum pump system on:
 - Select the screen page "Vacuum".

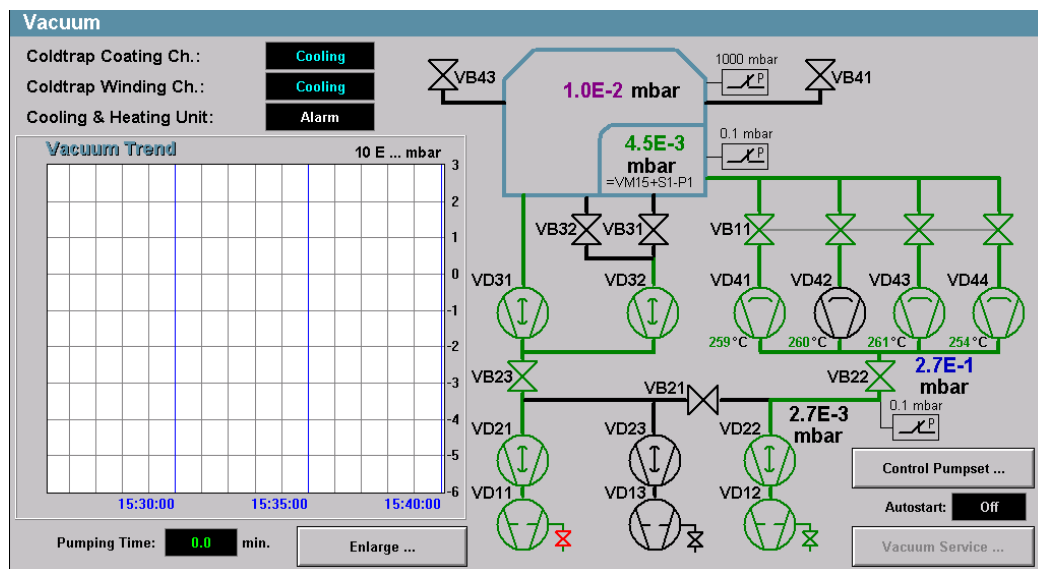


Fig. 6-2 Screen page "Vacuum"

- Click on the button <Control Pumpset>.
- In the function box, click on the mode switch, to switch the pump system on:
The fore vacuum pumps and diffusion pumps are switched on.
The diffusion pumps are heated to their operating temperature (>240°C).

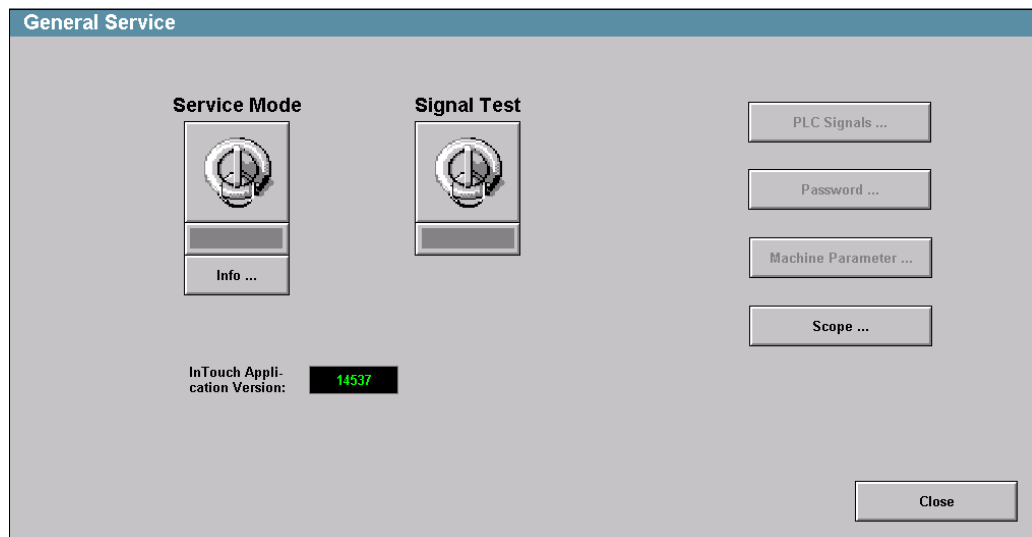


Fig. 6-3 Screen page "General Parameters"

- Perform the function test for the signal lamps:
 - Select the screen page "General Parameters".
 - Perform the function **Signal Test**.
- Open the vacuum chamber by pressing the pushbutton OPEN CHAMBER at operating panel +R02.
 - The hooter sounds while the travelling part moves.
 - The travelling part moves as long as you keep the pushbutton pressed.
 - Always move the travelling part to its final position. Only then does a limit switch activate other system functions.

6.4.2 Switching the system off

6.4.2.1 Shutting the system down

The following text describes how to switch the system off at the end of coating or for a temporary production interruption (next day). In these cases, the system's power supply is not disconnected from the line voltage (the main switch remains on).

- Clean the system:
 - Discharge the winding system and remove any remaining film from the system.
 - Clean the coating window.
 - Clean the evaporator station.
 - Clean the interior of the chamber.
- Close the chamber by pressing the pushbutton CLOSE CHAMBER at operating panel +R02. The hooter sounds while the travelling part moves.
- Pump the chamber down by pressing the pushbutton PUMP CHAMBER DOWN at operating panel +R02.
- Switch the vacuum pump system off:
 - Select the screen page "Vacuum" (see Fig. 6- 2).
 - Click on the button <Control Pumpset>.
 - In the function box, click on the mode switch to switch the pump system off:
While the diffusion pumps are cooling, the fore vacuum pumping station remains in operation. It switches off automatically when the diffusion pumps have reached the assigned trigger temperature of < 90°C.

6.4.2.2 Switching off the entire system

In the following cases, switch the system off by isolating it from the main power supply (main switch off):

- For a long-term production interruption (weeks, months)
- For major service, repair or maintenance work

Proceed as follows:

- Perform the steps described in chapter 6.4.2.1.
- Close the following supply lines:
 - cooling water
 - compressed air
 - process gas(es)



	⚠ Caution Hazard of damage to the system. Hot diffusion pumps could be damaged if the system's power supply is switched off prematurely. Turn the main switch off only when: • The diffusion pump temperature is below 90°C. • The fore vacuum pumping station has switched itself off automatically.
---	--

- Turn the main switch off.
 - The system and its components are disconnected from electrical power.
 - You will be automatically logged off the application software and the computer will shut down.

6.5 Coating cycle

6.5.1 Loading the winding system

	⚠ Warning Hazard of injuries from hovering loads There is the danger of injuries when inserting and removing rolls with the crane. Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use hoists.
	⚠ Caution Hazard of damage to the system from improper use of the crane, lifting gear or hoisting equipment. Each time the system is loaded or unloaded, lift the hoisting equipment all the way to the top, and remove it from the radius of motion of the winding machine.

6.5.1.1 Inserting a new material roll

- Open the winding bearing of the unwinder:
 - Open both folding bearings/bearing clips (to the left and right of the drum).
 - Open the bearing coupling. For this, take the retaining rod from the chamber plate, and insert it in the foreseen bore at the bearing coupling. Unlock the bearing coupling by rotating the locking rod. Then press the coupling lever to the front, to open the toothed wheel coupling.
- Insert the new material roll:
 - Hook on the hoisting equipment. Insert the new material roll in the system.
 - Slowly lower the material roll into the open unwinder bearing. Unhook the hoisting equipment and remove it from the vicinity of the system.
 - Close both folding bearings.
 - Close the winder coupling. For this, press the hand lever back into the toothed wheel coupling until it snaps into place.
 - Lock the winder coupling into place with the retaining rod.

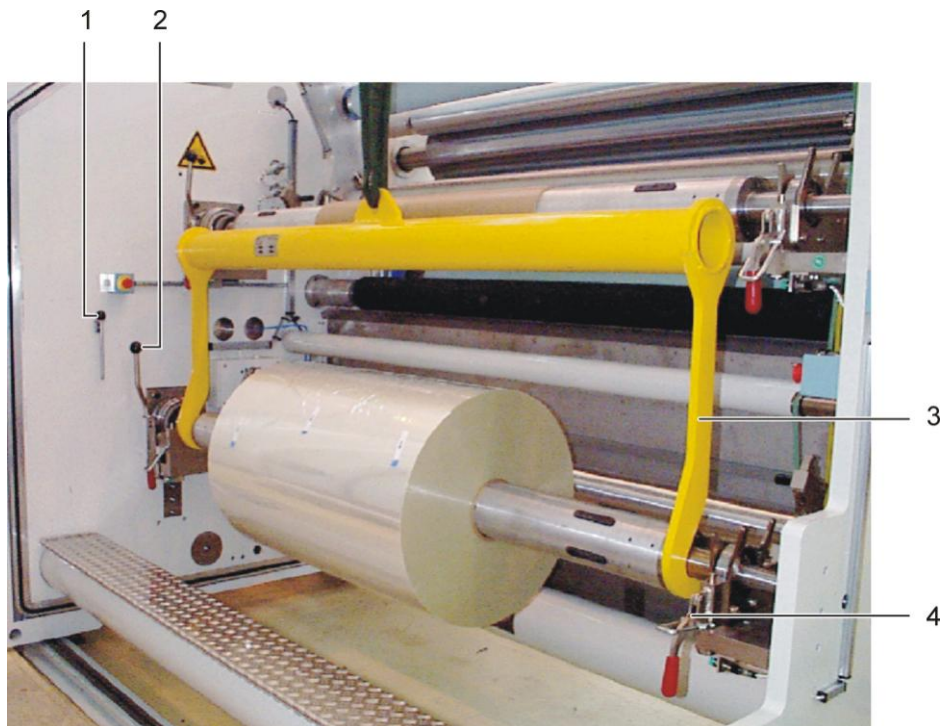


Fig. 6- 4 Inserting the material roll

1 - Retaining rod

2 - Hand level for the bearing coupling

3 - Hoisting equipment

4 - Folding bearings/bearing clips

6.5.1.2 Inserting the rewinder shaft

Fitting the tube

- Place the empty tube on the shaft by hand.
- Pick up the shaft with the crane and hoisting equipment.
- Align the shaft to the tube such that the distance between the tube edge and shaft edge is equal on both sides. Measure the distance with a tape measure.

- Tense the tube on the shaft.
For mechanically tensing shafts, use a clamping device to tighten the tube and shaft.
For pneumatically tensing shafts, insert a compressed air pistol at the compressed-air supply of the shaft and tense the tube at a pressure of $p = 4\text{-}5$ bar (see Fig. 6- 5).

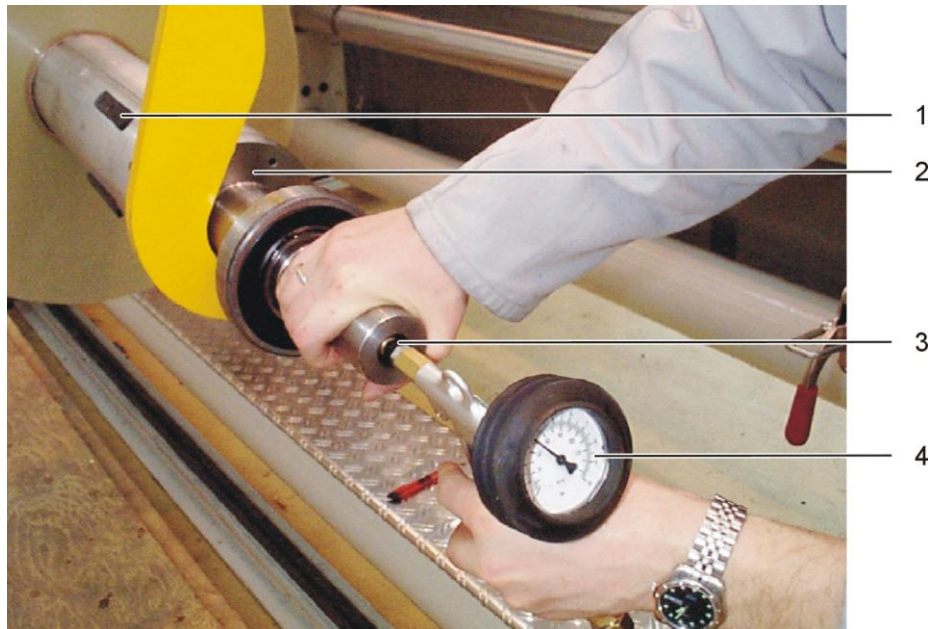


Fig. 6- 5 Tensioning the tube

- | | |
|----------------------|---------------------------|
| 1 - Clamping profile | 3 - Compressed-air supply |
| 2 - Shaft | 4 - Compressed-air pistol |

Inserting the rewinder shaft

- Insert the rewinder shaft in the open rewinder coupling.
- Remove the crane and hoisting equipment.
- Close both folding bearings.
- Close the rewinder coupling until the toothed wheel coupling snaps into place. Lock the coupling into place with the retaining rod.

6.5.2 Threading the film

	⚠ Warning
	Rotating drums There is the hazard that body parts can be pulled in by the rotating drums. Wear only tight clothes when cleaning the winding system, especially the coating drum. No persons may be located in the direct vicinity of the winding system while the pushbutton function < film feed > is pressed. Do not place your hands or arms in the drum system while it is running. Another person must be on alert at the next emergency-off pushbutton.

	⚠ Caution
	Before and during work on the winding system, make sure no unauthorized persons are located at the system.

After the material roll is inserted, the film is threaded through the drum system.

Threading includes inserting and threading the film from the unwinder side to the rewinder side.

6.5.2.1 Trimming the film ends

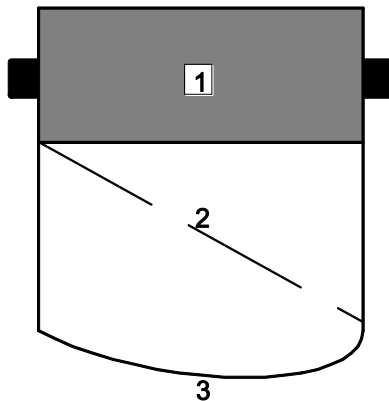


Fig. 6- 6 Trimming the film ends

1 - Material roll
2 - Knife cut

3 - Film end

- Pull the end of the film from the bale and cut it off diagonally.
- Cutting the film diagonally makes it easier to feed it through the winding system.

6.5.2.2 Threading the film

- Pull the film from the bale, and thread it through the drum system in the selected direction of rotation (UP/DOWN).
 - Winding direction UP:
Thread the film end from above via the first guide roller into the winding system. The interior side of the film will be coated.
 - Winding direction DOWN:
Thread the film end from below via the first guide roller into the winding system. The exterior side of the film will be coated.
 - Set the direction of rotation for the winder on the screen page "Parameters".
- Fasten a piece of adhesive tape on the film end, in the film's direction of travel.
- Fasten the film end at the threading belt.

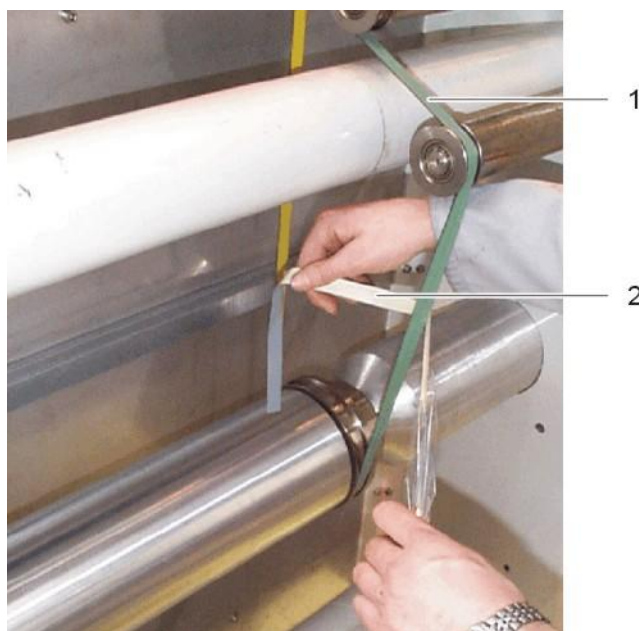


Fig. 6-7 Fastening a piece of adhesive tape on the film end

1 - Threading belt

2 - Adhesive tape

6.5.2.3 Feeding the film

The film feed system automatically moves the film to the rewinder side.

- Press the pushbutton THREADING ON at operating panel +R24.
The coating drum and tension roll rotate. The film is moved to the rewinder side at a slow speed (< 0.2 m/s).
- Assist the unwinding of the film by hand. The threading belt only guides the film through the winding system.
- Press the pushbutton THREADING OFF, when the film end has reached the rewinder side (one revolution of the threading belt). The coating drum stops.

6.5.2.4 Fastening the film end at the rewinder tube

- Remove the adhesive tape from the threading belt, and remove the film end from the winding system.
- Fasten a piece of double-sided adhesive tape onto the rewinder tube.
- Thread the film end through the pivot arm, depending on the selected rewinding mode (up/down).
 - Rewinding direction down:
After the film end has passed the final guide roller, place the film end around the winding core from below and attach it. The coated film side is wound to the inside.
 - Rewinding direction up:
Thread the film end between the final two guide rollers. Thread it directly around the winding core from above and attach it. The coated film side is wound to the outside.
 - Set the direction of rotation for the winder on the screen page "Parameters" (see Fig. 6- 10).

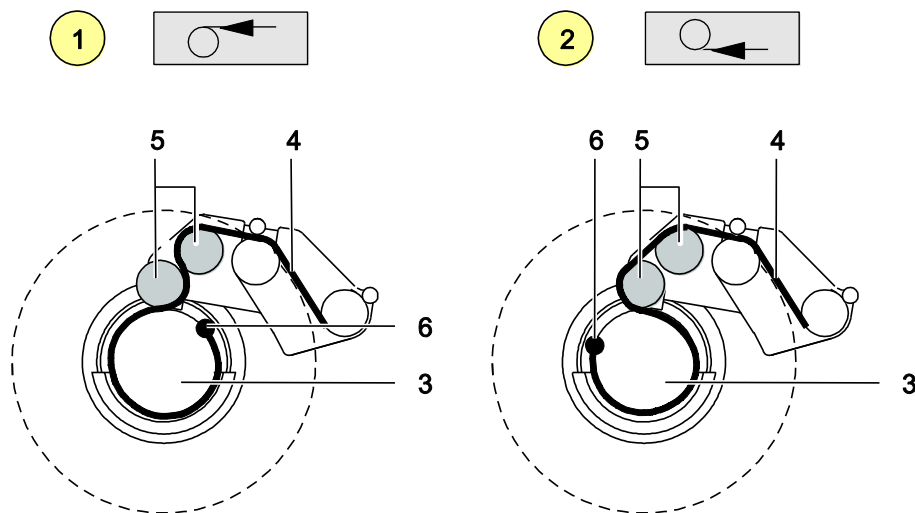


Fig. 6- 8 Threading the film through the pivot arm

- 1 - Rewinding direction UP
- 2 - Rewinding direction DOWN
- 3 - Winding Core

- 4 - Film travel
- 5 - Guide rollers
- 6 - Point of attachment

- After the film end has passed the final guide roller, fasten the film end to the double-sided adhesive tape of the winding core.

6.5.2.5 Initially stressing the film

⚠ Warning	
	Hazard of contusions Injuries or contusions are possible from • Motion of the pivot arm • Axial motion of the rewinder and unwinder drums Do not place your hands or arms between the tube and pivot arm. Do not place your hands or arms between the bearing plate and rewinder or unwinder. No persons may be located in the direct vicinity of the winding system while the pushbutton function <pivot arm> or <axial motion> is pressed.

	Note!
	It is very important that you initially stress the film by hand before the system's winding tension is activated with the pushbutton function FILM TENSION ON. If the film is loose when the tension starts, it can tear.

- Press the pushbutton PIVOT ARM APPROACH at operating panel + G21. The pivot arm moves to the rewinder tube.
- Rotate the tube at the rewinder by hand, until the film between unwinder, pivot arm and rewinder is free of folds and tightly tensed.



Fig. 6-9 Initially tensing the film

1 - Tube

2 - Pivot arm

6.5.2.6 Setting winding and coating parameters

Select the screen page "Parameters". Check or update all winding and coating parameters relevant to the next coating process.

Parameters											
Batch			Winding				Cool. & Heat Unit				
Reference:	Ref001		Unwinder:	160	680	188	220	360	Direction	Cooling: -8 °C	Heating: 26 °C
Film Type:	PET		Rewinder:	160	186			125		Coldtrap Wind.Ch. Off	
Roll Code:	Roll 0001		Tension Roll:				180				
Film Code:	Film 0002										
Film Width:	2375	mm									
Coated Width:	2300	mm									
Film Thickness:	18	µm									
Evaporator			Layer Thickness Control				Pivot Arm				
Wire Speed:	1500	mm/min.	LMS-OD Mode:	Opt. Density		Layer Lengthw. Control:	Select		Distance		
Wire Speed Start:	1100	mm/min.	Preset LMS-OD:	2.00	O.D.	Layer Cross Control:	Off		Off		
Rate of Rise:	10	mm/min²	Tolerance:	20.0	%	Delay for LMS	20 s		Scale 100		
Wire Diameter:	1.6	mm									
Film Save Function:	Select	Help									
Run-Up Time:	180	s									
Power for Run-Up:	6318	W									
Pre Treatment											
			Current:	10.0	A	Flow Oxygen:	1200	scm	Select		
						Flow Argon:	300	scm	Off		
Save Settings to Recipe ...											

Fig. 6- 10 Screen page "Parameters"

6.5.2.7 Switching the film tension on

Note!	
	Before you execute the pushbutton function TENSION ON, make sure: • Bearings and couplings are closed tight/locked. • Film tension values for the inserted film are set correctly. • Unwinder and Rewinder Diameter are set correctly. Make sure no persons are located in or at the winding system.

- Press the pushbutton TENSION ON at operating panel +R02. The film is tensed to the set value.

6.5.3 Preparing system components

	⚠ Warning
	Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal dusts and metal oxide dusts which are hazardous to your health. During all cleaning work, avoid creating dust. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.

Preparing the system includes all measures and steps required to prepare the system for the next coating process.

6.5.3.1 Evaporator station

	⚠ Warning
	Hazard of burns from hot surfaces Directly after a coating process and after opening the chamber, the evaporator station and the evaporation material inside are still hot. During all work on the evaporator station, protect yourself from hot parts and hot molten metal. Wear the following protective gear: • Goggles • Fire-resistant protective gloves • Appropriate work garments offering protection from molten metal (e.g. leather apron)

	⚠ Warning
	Contusion hazard The movable components • Evaporator flap • Wire roll frame can cause injuries. When working at the evaporator station, make sure you do not move these components by mistake. In particular, do not press the pushbutton <evaporator flap> by mistake. Never place your hands or arms in the radius of motion of these components.

- Clean the aluminum evaporator station (see chapter 7.3.1).
- Check the number of required evaporator boats with regard to the assigned coating width. Switch the wire feed off for all nonactive evaporator boats, depending on the evaporator configuration (drive with stepper motors or continuous shaft).
 - If the aluminum wire units are driven by stepper motors, then the number of required wire feed units is automatically controlled by the assigned coating width.
 - If the wire units are driven by a continuous shaft drive, then you must switch off all wire feed units not required for the assigned coating width by hand (turn the locking lever up).
- Check the amount of aluminum wire (see chapter 6.6.3.1).
- Check the rate of wear of the evaporator boats (see chapter 6.6.3.2).

6.5.3.2 Coating window

	⚠ Warning
	Contusion hazard There is the danger of contusions in the radius of motion of the coating window! Before you press the pushbutton function <coating window>, make sure no persons are located in the radius of motion of the coating window. Do not place your hands or arms in the radius of motion of the coating window.

When the chamber is open (travelling part at its final position), the coating window can be placed anywhere between the cleaning position and the work position. Perform the following work:

- Clean the coating window before each new coating cycle (see chapter 7.3.1).

	Note!
	Note the following when working with the limit shields: • The screws for clamping the limit shields must always be tightened, even if no limit shields are in place.

- Whenever you change the coating width, set the limit shields to the proper coating width (screen page "Parameters" under Batch, see Fig. 6- 10), see chapter 6.6.4.2.
- Place the coating window in the work position.

6.5.4 Starting production (beginning of coating)

	⚠ DANGER
	Beware of the danger of being locked in the coating system. Pumping down the system would kill anyone locked inside. Before you close the chamber, make sure no persons are located in front of, near or in the chamber.

	⚠ Warning
	Hazard of injury and contusion Persons near the travelling part could be injured when it moves. No persons may be located within the radius of motion of the winding machine before the chamber is opened or closed. When closing the chamber, do not place your hands or arms between the chamber plate and flange.

	Note!
	Due to the complexity of the monitoring and control work during the coating cycle, at least two persons are required to operate the TOPMET system. Never operate the system alone and without supervision.

Starting production includes all steps from the status "Chamber closed" until coating begins. Production is started when all steps required to prepare the system for the next coating cycle have been completed. You can start production in manual mode or as an automatic coating cycle. Select the corresponding mode with the button <Auto Process Start> on the screen page "Process" (see Fig. 6- 11).

6.5.4.1 Checking the system's status

Before you close the chamber, work through the following checklist on the system's status:

- Supply of aluminum wire checked.
- Coating window placed in the work position and shutter closed.
- Remaining film, tools and aids removed from the chamber and the winding system.
- Chamber seal checked for dirt (clean if necessary).
- Stroboscope checked for dirt (clean if necessary).
- Lateral chamber sight glasses checked for dirt (clean if necessary).
- Coating drum checked for cleanliness (clean if necessary).
- Predefined and assigned winding and coating parameters checked on the screen pages.

6.5.4.2 Preselecting a mode of operation

Note!	
	The button <Auto Process Start> is used only to preselect the mode. Switching modes during the coating phase is improper operation of the system. Do not change the mode from automatic coating cycle to manual mode, or vice versa, during the coating phase (chamber closed and evaporator on).

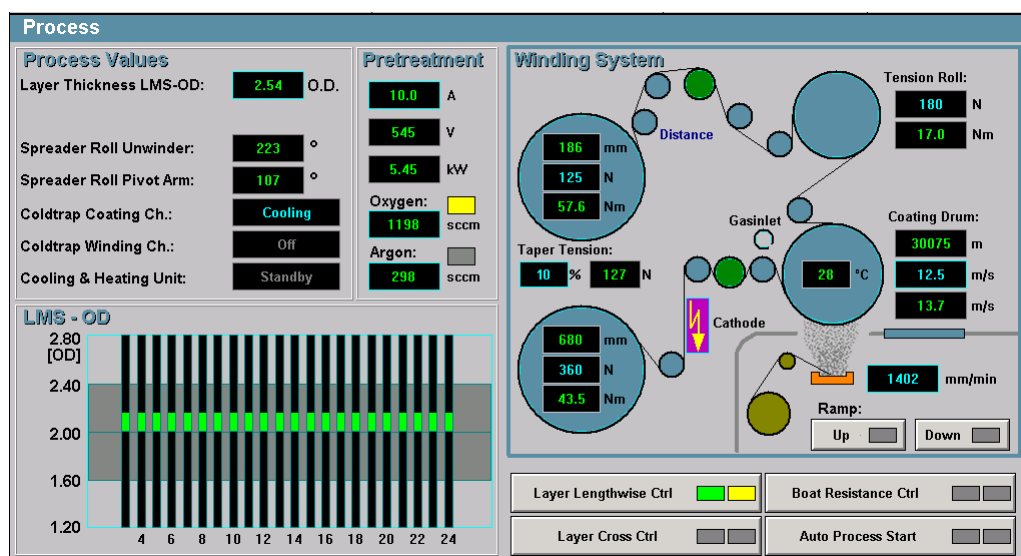


Fig. 6- 11 Screen page "Process"

- Select the screen page "Process".
- Setting the mode of operation:
 - Automatic coating cycle
Click the button <Auto Process Start> to ON.
 - Manual mode:
Click the button <Auto Process Start> to OFF.

6.5.4.3 Automatic coating mode

	Note!
	Compare the recipe parameters with the system configuration, when a new coating recipe was loaded or when the recipe is changed directly before starting production.

In automatic coating mode, the automatic process start-up feature is in charge of most steps for starting production.

Step	Pushbutton	Comments
Closing the vacuum chamber	CLOSE CHAMBER, operating panel +R02	The hooter sounds while the travelling part moves.
Pumping the vacuum chamber down	PUMPING DOWN ON, operating panel +R02	Prerequisites for switching on: • The pushbutton TENSION ON is switched on. • The operating temperature of the diffusion pumps is > 240° C.

The steps of the automatic process feature then follow:

- The aluminum evaporators switch on.
- The wire feed switches on.
- The winding system starts.
- The evaporator shutter opens.

6.5.4.4 Manual mode

In manual mode, you must perform all steps for starting production, by pressing pushbuttons.

Step	Pushbutton	Comments
Closing the vacuum chamber	CLOSE CHAMBER, operating panel +R02	The hooter sounds while the travelling part moves.
Pumping the vacuum chamber down	PUMPING DOWN ON, operating panel +R02	Prerequisites for switching on: • The pushbutton TENSION ON is switched on. • The operating temperature of the diffusion pumps is > 240° C.
Switching the aluminum evaporator station on	EVAPORATOR ON, operating panel +R10	Prerequisites for switching on: • The vacuum switching point (8×10^{-4} mbar) for releasing the evaporator interlock must be reached. The evaporator boats are heated in accordance with the assigned run-up time.
Switching the wire feed unit on	WIRE FEED ON, operating panel +R10	Prerequisites for switching on: • Wait for a time equivalent to the assigned run-up time for the evaporator boats.
Starting the winding system	WINDING ON, operating panel +R02	Prerequisites for switching on: • Wait until the aluminum in the boats is completely melted.
Opening the shutter	SHUTTER OPEN, operating panel +R10	Prerequisites for switching on: • Open only when the setpoint for film speed is reached.



6.5.5 Stopping coating

6.5.5.1 Automatic coating mode

If coating was performed in automatic coating mode, then an automatic feature will automatically perform all steps required to end the process and to stop the system.

When the roll diameter approaches the final winding diameter, the following procedure takes place:

- The evaporator shutter closes.
- The winding drives slow to a stop.
- The aluminum wire feed switches off.
- The evaporator station switches off.
- The cooling and heating equipment switches to the status "HEAT".
- The coldtrap (Polycold) switches to the status "DEFROST".
- The venting valves open.

When the final winding diameter is reached at the end of coating, the winding system is idle and the hooter sounds.

Step	Pushbutton	Comments
Acknowledge the hooter	FAILURE/RESET, operating panel +R02	The hooter turns off.
Opening the vacuum chamber	OPEN CHAMBER, operating panel +R02	Prerequisites for switching on: • The pressure in the vacuum chamber is > 950 mbar (safety interlock through PLC). The hooter sounds while the travelling part moves. Always move the travelling part to its final position (limit switch).



6.5.5.2 Manual operation

If you performed coating in manual mode, then you must perform all steps required to end the process and to stop the system by hand.

When the roll diameter approaches the final winding diameter, the following procedure takes place:

- The evaporator shutter closes.
- The winding drives slow to a stop.
- The aluminum wire feed switches off.
- The evaporator station switches off.
- The cooling and heating equipment switches to the status "HEAT".

When the final winding diameter is reached at the end of coating, the winding system is idle and the hooter sounds.

Step	Pushbutton	Comments
Acknowledge the hooter	FAILURE/RESET, operating panel +R02	The hooter turns off.
Venting the vacuum chamber	VENT CHAMBER, operating panel +R02	Prerequisites for switching on: • The venting valves can only be opened when the temperature switching point for heating the coating drum is reached (safety interlock through PLC).
Opening the vacuum chamber	OPEN CHAMBER, operating panel +R02	Prerequisites for switching on: • The pressure in the vacuum chamber is > 950 mbar (safety interlock through PLC). The hooter sounds while the travelling part moves. Always move the travelling part to its final position (limit switch).



6.5.6 Unloading the winding system

	⚠ Warning Rotating drums There is the hazard that body parts can be pulled in by the rotating drums. Wear only tight clothes when cleaning the winding system, especially the coating drum. No persons may be located in the direct vicinity of the winding system while the pushbutton function < film feed > is pressed. Do not place your hands or arms in the drum system while it is running. Another person must be on alert at the next emergency-off pushbutton.
	⚠ Warning Hazard of contusions Injuries or contusions are possible from • Motion of the pivot arm • Axial motion of the rewinder and unwinder drums Do not place your hands or arms between the tube and pivot arm. Do not place your hands or arms between the rewinder or unwinder chucking system. No persons may be located in the direct vicinity of the winding system while the pushbutton function < pivot arm > or < axial motion > is pressed.
	⚠ Warning Danger of injuries from hovering loads There is the danger of injuries when inserting or removing a roll with the crane. Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use hoists.

You can remove the coated film after the chamber opens and the travelling part moves to its final position. Perform the following steps.

- Switching the film tension off:
 - Press the pushbutton TENSION OFF at operating panel +R02.
 - The film tension is released. The film becomes loose.
 - The current roll diameters (which are stored in the programmable control) are replaced by the initial diameters.
- Place the pivot arm in the position "Remove":
 - Press the pushbutton PIVOT ARM DISTANCE at operating panel + G21.
 - Remove the pivot arm to its final position.
- Remove the material roll:
 - Cut the film off at the rewinder side.
 - Open the rewinder bearing.
 - Open both folding bearings.
 - Hook on the crane and hoisting equipment, and lift the coated roll out of the winding system.
- By hand, remove the remaining coated film between the unwinder and rewinder.



Fig. 6- 12 Unloading the winding system

1 - Separating the film

2 - Rewinder bearing

6.6 Operating the process stations

6.6.1 Pretreatment station

The system's automatic process feature carries out glowing (switching the cathode on and off).

	⚠ Caution
	Perform a glow process only when the cathode is clean and conditioned. If the glow cathode was not used for a long time, then you must condition it before use. Instructions on conditioning the cathode can be found in the chapter Maintenance instructions.

6.6.1.1 Prerequisites for glowing

	Note!
	Glowing is performed only in automatic coating mode. You are not permitted to switch the glow cathode on when a coating process is in progress.

Adhere to the following rules when pretreating plasma with the TreatMag glow cathode:

- Perform a glow process only if the cathode is clean and conditioned. Worn and very dirty cathodes
 - have reduced performance.
 - increase the time required to pump down the vacuum chamber.
 - encourage arcing in the plasma.
- Perform a glow process only when film has been inserted.
- For the glow process, use only the specified process gases (argon and oxygen):
 - Argon primarily to clean the film and condition the cathode
 - Oxygen for the process, i.e. to pretreat the film
- Clean the glow station each time the film tears. Remove remaining film from the body of the cathode.

6.6.1.2 Preparing glowing

	Note!
	Starting glowing depends on selection or deselection of the process gases on the screen page "Parameter". If you have not preselected any process gas, then glowing is disabled and will not be part of the process. The cathode remains switched off!

Checking the status of the system:

- System opened
- Film inserted in the winding system
- Cathode cleaned and conditioned (see chapter 7.5.11.4)
- Level of the pressurized gas bottles (oxygen and argon): maintain a minimum pressure of 10 bar
- System mode <Auto-Process-Start> preselected (screen page "Process", see Fig. 6- 11)

Setting the glow parameters

- Select the screen page "Parameter" (see Fig. 6- 10).
- Select the gas (Ar or O₂) appropriate for the procedure. Click on the buttons (turn from OFF to SELECT). Glowing is disabled if both buttons stay at OFF.
- Set an initial setpoint of approx. 800 -1200 sccm for the gas flow.
- Set an initial current setpoint of approx. 12 A for the cathode.



6.6.1.3 Starting glowing

- Close the chamber and pump it down (see chapter 6.5.4.3).

The system's automatic process start-up feature is activated. The following takes place after the vacuum switching point is reached:

- The winding system starts.
- The gas valves switch from SELECT to ON . The gas flow controllers open.
- Electrical power is then applied to the cathode.
The plasma is ignited. Within 15 seconds, the cathode current ramps up to the assigned setpoint. The plasma is visible in the vacuum chamber as a bluish glow.

6.6.1.4 Regulating the glow process

In essence, this procedure includes all steps required to set the operational values for the glow power and gas flow. There is a common data field for entering setpoints and displaying actual values. You can switch from setpoints to actual values.

- Select the screen page "Process", Pre-Treatment (see Fig. 6- 11).
- With the winding chamber pressure at 10^{-3} mbar, set the current setpoint to the maximum setpoint of 24 A, in 5 A increments. The winding speed must be set to > 0,5 m/s.
- The actual value must flawlessly follow the assigned setpoint, meaning the plasma is stable without arcing.
- At the maximum setpoint of 24 A, the operating voltage must settle at approx. 350 - 600 V (actual value). For a constant current value, the voltage depends on the pressure and process gas flow:
 - High gas flow means low voltage.
 - Low gas flow means high voltage.
- Approximate values for operational parameters (depends on the film):
 - Pressure $p = 2 \times 10^{-2}$ mbar
 - Current (setpoint) 12 A
 - Voltage (actual value) 400 V
 - Gas flow 1,500 sccm
 - Winding speed 7 m/s

	Note!
	Excessive voltage discharges (frequent, strong arcing in the plasma) results in the following: • The assigned cathode current must be reduced until the plasma becomes stable. • In case of overloading, the source of electrical power is automatically disconnected from the cathode. The error message "Pretreatment: not ready" will appear on the screen.

The glowing process is flawless when no voltage discharges (arcing) are shown in the plasma. Plasma arcing is indicated in three ways:

- Flickering, bluish plasma light in the vacuum chamber
- Display of actual voltage instable
- The LED "ARC" at the cathode's power supply will flash during arcing.

6.6.1.5 Resuming operation after a shutdown caused by arcing

The cathode's power supply can become overloaded if voltage discharges are too strong (frequent, strong arcing in the plasma). The power supply will then automatically switch off. The error message "Pretreatment: not ready" will appear on the screen.

Proceed as follows:

- Check the LEDs at the power supply. Eliminate the cause of the fault based on the manufacturer's documentation.
- Reset the power supply. On the screen page "Process", switch the button for the selected process gas from ON to OFF and then back to ON (or SELECT) (see Fig. 6- 11).



6.6.2 Gasinlet

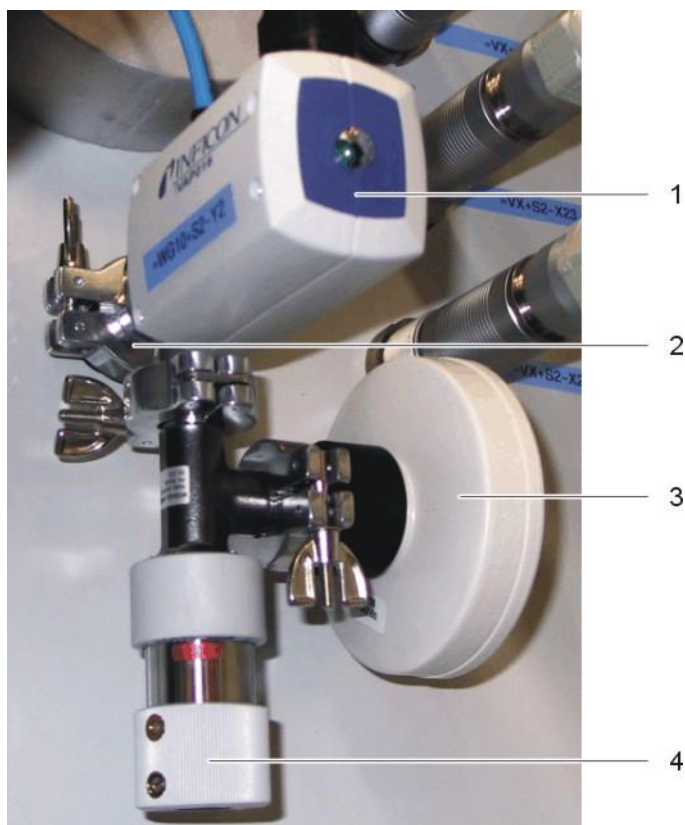


Fig. 6-13 Gasinletunit

- 1 - Angle valve
2 - Chamberduct

- 3 - Airfilter
4 - Needle valve

Adjusting the gasflow:

- Select the gasinlet at the screenpage "Parameters" (see Fig. 6- 10).
- The angle valve is triggered, as soon as the coating window opens. After that the gasflow must be set with the needle valve at the outside of the recipient plate (see Fig. 6- 13).
- The process gas pressure must not exceed $p = 6,0 \times 10^{-4}$ mbar.

	Note !
	Open the needle valve as far as a smoothing effect of the film at the coating drum is visual. If "Tramlines" appear on the film during coating process, the gas flow must be manually readjusted with the needle valve.

6.6.3 Evaporator station

⚠ Warning	
	Danger of burns from hot surfaces Directly after the coating process and after the chamber is opened, the evaporator station and the evaporation material is still hot. Whenever you work at the evaporator station, wear the following protective gear as protection against hot parts and hot molten metal: • goggles • fire-resistant protective gloves • work garments which protect against molten metal (e.g. leather apron)

⚠ Warning	
	Contusion hazard The moving components • evaporator flap • wire roll frame can cause injuries. When working at the evaporator station, make sure these components do not move unintentionally. In particular, make sure you do not press the pushbutton <evaporator flap>. Never place your hands within the radius of motion of these components.

6.6.3.1 Loading and adjusting the wire feed unit

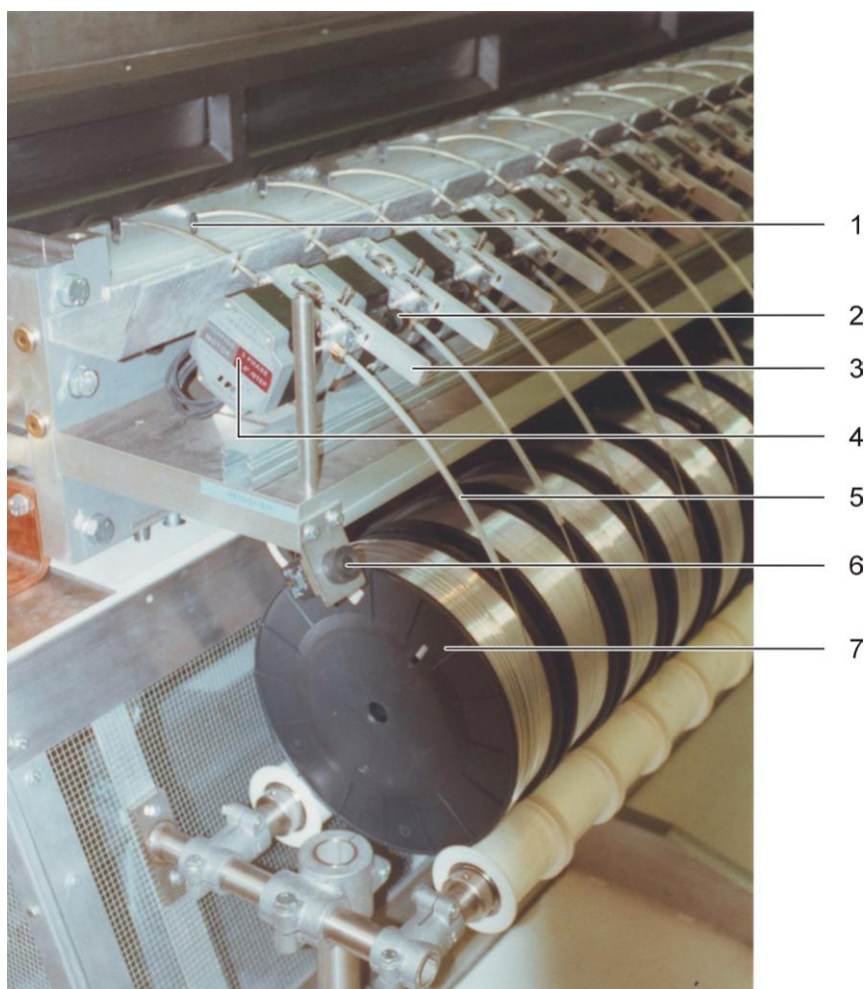


Fig. 6- 14 Loading the wire feed unit

- | | |
|--|---|
| 1 - Feedthrough to evaporator box
(wire feed pipette) | 5 - Guide pipette |
| 2 - Locking pin | 6 - Pushbutton for operating all feed units |
| 3 - Lever | 7 - Wire coil |
| 4 - Wire feed unit | |

- Remove the covers. For this, swing the handles out. When removing and replacing the covers, make sure you do not bend the feed pipettes.
- Unthread the aluminum wire:
 - Raise the lever, and snap the locking pin into place. This releases the wire tension.
 - Pull the wire out of the wire feed pipette.
- Remove the old aluminum wire coil and insert a new one.

- Clean the wire feed pipettes (see chapter 7.3.1):
- Thread the aluminum wire:
 - Thread the wire into the feed system, and shove it to up the feedthrough to the evaporator box (wire feed pipette).
 - Release the locking pin and lower the lever. The wire is tensed.
- Check the wire's travel. The aluminum wire should make contact between the middle and first third of the evaporator boat cavity. Readjust the wire feed pipettes if necessary.
- The pushbutton for all feed units is used to switch all wire feed units on and off:
 - Press this pushbutton and keep it pressed.
 - The aluminum wire of all feed units is transported with the preselected speed (screen page "Parameters", see Fig. 6- 10). The wire feed stops as soon as you release the pushbutton.

	Note!
	Automatic evaporator boat shut-off: The number of required evaporator boats is automatically preselected, as a function of the set film width. Manual boat shut-off: You can switch off individual evaporator boats with the button <Select Evaporators> on the screen page "Parameters". In both cases, it is important that you disable the wire feed units of the inactive evaporator boats: • Raise the lever, and use the locking screw to lock it into the upper position. • Pull the aluminum wire out of the guide pipette. If stepper motors are used, the wire shut-off is automatically coupled with evaporator boat shut-off.

6.6.3.2 Replacing evaporator boats

	Note!
	Always replace the entire set of evaporator boats, even if the degree of wear and tear varies from boat to boat.

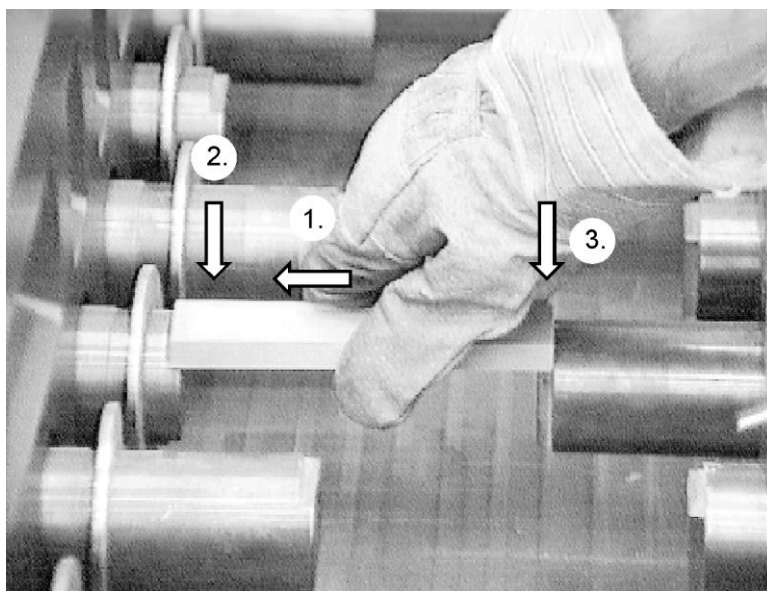


Fig. 6- 15 Replacing evaporator boats

- Replace the evaporator boats:
 - Remove worn boats from the holder.
 - Clean boat holder and seat.
 - Coat the cleaned parts with a thin layer of graphite paste.
 - Insert new boats. Place them in the center of the holder. At the front sides, place graphite foil underneath them.
- Condition the new set of boats:
 - Select the screen page "Boat Conditioning".

	Note!
	The new evaporator boats should not visual glow in the first 5 minutes when heated-up.

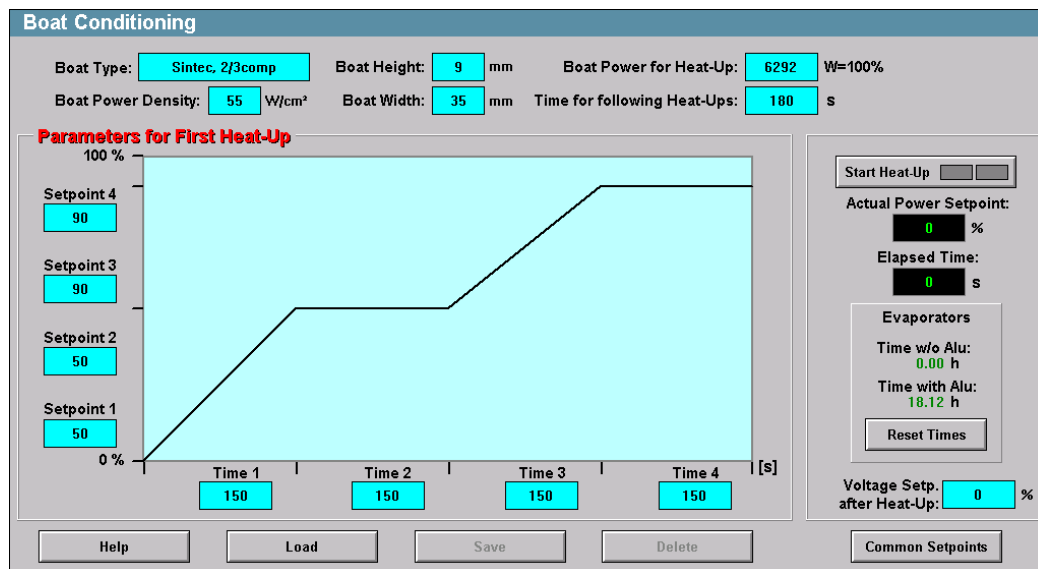


Fig. 6-16 Screen page "Boat Conditioning"

- Reset the operating times for the new boats to 0 (button <Reset Times>).
- Enter all conditioning parameters for the new boats in the proper data fields.
- Pump the system down.
- Activate the button <Start Heat-Up>.



6.6.4 Coating window

	⚠ Warning
	Contusion hazard There is the danger of contusions within the radius of motion of the coating window! Before you press the pushbutton <coating window>, make sure no persons are located within the radius of motion of the coating window. Never place your hands or arms within the radius of motion of the coating window.

During system operation, perform the following work at the coating window:

- Clean the coating window before each new coating cycle (see chapter 7.3.1).
- Each time the coating width changes, reset the limit shields to the new width (see chapter 6.6.4.2).

6.6.4.1 Operation

When the chamber is open (travelling part at its final position), the coating window can be placed anywhere between the cleaning position and the work position.

- Press the pushbutton LOWER COATING WINDOW (operating panel +R23) and keep it pressed. The coating window swivels down to the cleaning position. The evaporator shutter is automatically opened.
The following safety interlocks are active:
 - The shutter function (open/closed) is locked (operating panel +R10).
 - The travelling part function (open/closed) is locked (operating panel +R02).
- Press the pushbutton LIFT COATING WINDOW and keep it pressed. The coating window swivels back to the work position. In the work position, a limit switch checks the final position. The function CLOSE CHAMBER is only enabled when the coating window is completely raised.

6.6.4.2 Adjusting the coating width

	Note!
	Note the following when working with the limit shields: • The screws for clamping the limit shields must always be tightened, even if no limit shields are in place.

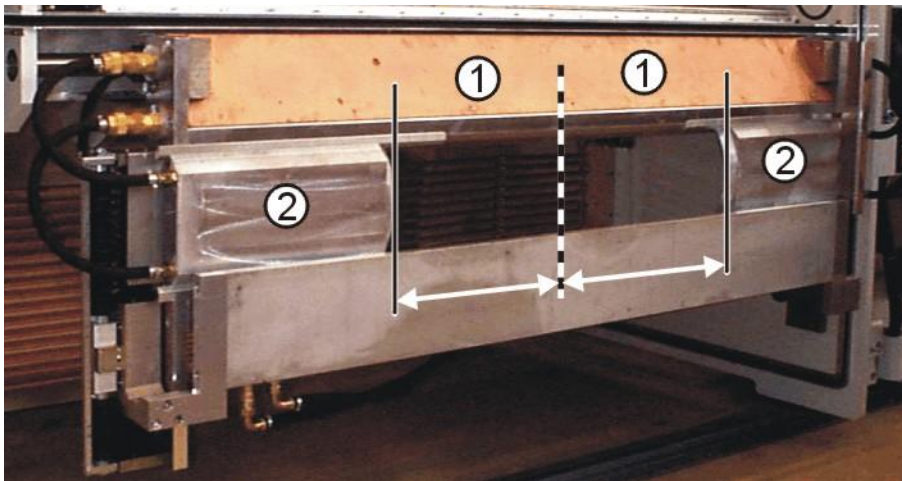


Fig. 6-17 Adjusting the coating width

- 1 - Half the coating width
(to the middle of the coating window)
- 2 - Limit shield

Depending on the assigned coating width (screen page "Parameters", see Fig. 6-10), you must align the lateral limit shields to the middle relative to the coating window:

- Remove the screws from the clamping bar of the limit shields.
- Measure the required coating width to the middle, relative to the coating window.
- Select the appropriate limit shields. Connect the cooling water hoses.
- Shift the limit shields accordingly.
- After you adjust the coating width, tighten all screws of the clamping bar.

6.7 Disturbances and resuming operation of the system

6.7.1 Pressed emergency-stop pushbutton

	⚠ DANGER
	Hazard of lethal electrical shock when touching parts and components connected to electrical power. The system is still connected to electrical power after an emergency-stop pushbutton is pressed. Trouble-shoot the system only if it has been thoroughly secured.

	⚠ Warning
	Contusion hazard. After an emergency-stop pushbutton has been pressed, there is the hazard of injuries from system parts which still move. Wait for all moving parts to come to a complete stop before starting to work on the system.

6.7.1.1 Procedure

Pressing an emergency-stop pushbutton automatically activates the following system functions:

- All moving system parts are stopped immediately.
- The signal horn sounds.
- An error message appears at the monitor. The message text is printed out.
- The signal lamp FAILURE/RESET at operating panel +R02 turns on.
- The pump system switches to the mode PUMPS OFF.

6.7.1.2 Resuming operation

- Acknowledge the error message:
 - Press the pushbutton FAILURE/RESET at operating panel + R09.
 - The color of the message text on the monitor changes from red to dark red.
 - The signal lamp FAILURE/RESET turns on, the tooter turns off.
- Eliminate the source of hazards and eliminate the disturbances.
- Unlock the emergency-stop pushbutton.
- Reset the error message:
 - After eliminating the cause of the error, press the pushbutton FAILURE/RESET.
 - The message text is no longer shown on the monitor.
 - The signal lamp FAILURE/RESET turns off, and the emergency-stop relay is reset.
- Pump the chamber down.

6.7.2 Power failure

If the power fails during the coating process, the entire system is automatically switched off.

6.7.2.1 Procedure

- The system control (PLC) is stopped. All inputs and outputs are reset to their default settings.
- The evaporator system switches off (shutter closed, aluminum wire feed off).
- All water valves close. Thus, the cooling circuits for diffusion pump and evaporator are no longer active.
- The pump system switches off.
- The winding system switches off. The rewinder and unwinder are not slowed.



6.7.2.2 Resuming operation

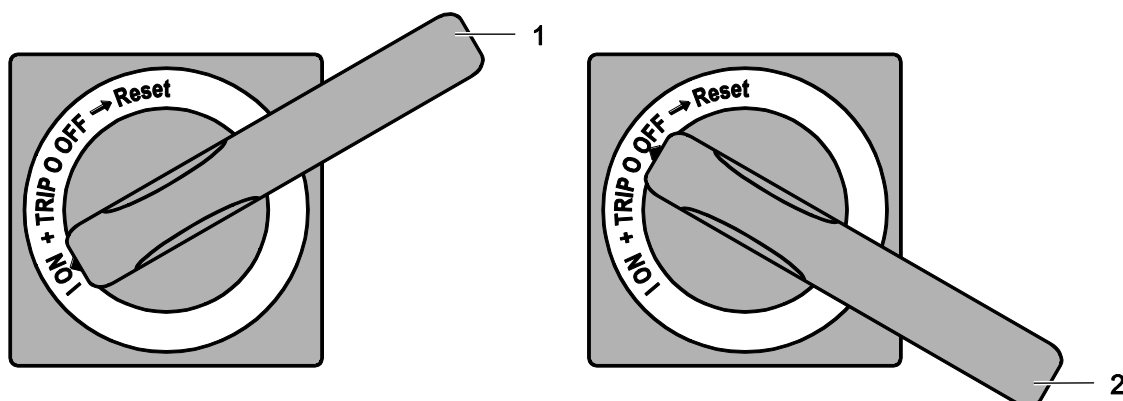


Fig. 6- 18 Main switch

1 - Position ON

2 - Position OFF

- Switch the power supply (main switch) on again (see chapter 4.4.1).
- Switch the pump system on:
 - Select the screen page "Vacuum" (see Fig. 6- 2).
 - Click on the button <Control Pumpset>, and switch the pumps on again.
 - Or: Press the hotkey <P>, and switch the pumps on again.
- Vent and open the vacuum chamber.
- Clean the system:
 - Remove remaining film from the winding system.
 - Check the coating drum for dirt (see chapter 7.5.9.2).
 - Check the gap lock (see chapter 7.5.9.3).
- Restart the coating cycle (see chapter 6.5):
 - Reenter the roll diameters.
 - Rethread the film.
 - Close the chamber and pump it down.

6.7.3 Film tears

If the film tears during the coating process, the automatic system for detecting film tears is activated.

6.7.3.1 Procedure

- The rewinder and unwinder are slowed to a stop.
- The evaporators switch off.
- The shutter closes.
- The hooter sounds.
- An error message appears on the monitor. A message text is printed out.
- The signal lamp FAILURE/RESET turns on.

6.7.3.2 Resuming operation

	Note!
	If the film tears, the system control automatically activates the function "Stop winding". The current winding diameters are then automatically reset to the initial diameters. Measure the current winding diameters of the rewinder and unwinder with a tape measure. Reenter the measured values in the screen page "Parameters".

Resume system operation as follows:

- Acknowledge the error message:
 - Press the pushbutton FAILURE/RESET at operating panel + R09. The error message is acknowledged and reset.
 - The signal lamp FAILURE/RESET turns off.
- Vent and open the vacuum chamber.
- Clean the system:
 - Remove remaining film from the winding system.
 - Check the coating drum for dirt (see chapter 7.5.9.2).
 - Check the gap lock (see chapter 7.5.9.3).
- Restart the coating cycle (see chapter 6.5):
 - Reenter the roll diameters.
 - Rethread the film.
 - Close the chamber and pump it down.



6.7.4 PLC failure/restarting the CPU

Failure of the system control relates primarily to communications disturbances in the central processor (CPU).

The central processor is implemented in a modular compact assembly as part of the SIMATIC S7-400 system control in the switching cabinet + E1.

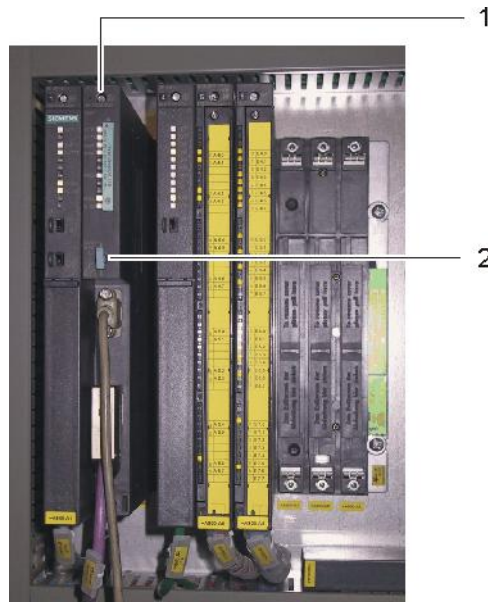


Fig. 6- 19 Compact assembly for the CPU

1 - CPU unit

2 - Selection switch

Error displays or LEDs at each assembly (CPU) indicate the nature and seriousness of any disturbances.

For trouble-shooting, always consult the manufacturer's documentation for the SIMATIC S7-400.

6.7.4.1 Restarting the CPU

	Note! Perform an OVERALL reset only if the LED STOP of the CPU flashes rapidly. An overall reset copies the contents of the CPU's EPROM chip to RAM. Current process data can be overwritten when this occurs.
---	--

- Check the error status of the CPU:
When the CPU is stopped, the LED STOP flashes slowly (0.5 Hz). The CPU must be restarted.

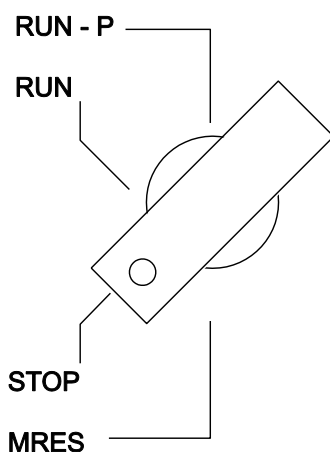


Fig. 6- 20 Selection switch

- Restarting the CPU:
 - Turn the selection switch from the position *MRES* to the position *STOP* and back again.
 - Result: The LED STOP flashes quickly (at 2 Hz) for approx. 3 s, indicating the PLC is rebooting. The LED is then lit continuously (booting is finished).



6.7.4.2 Disturbance of the PAC-F frequency control unit

The drive system for individual control of the aluminum wire feed basically consists of:

- Stepper motors directly at the evaporator.
- Frequency control unit (PAC-F)

The frequency control unit (PAC-F) contains its own program for separately clocking the individual stepper motors. An Interbus interface connects the internal PAC-F program with the higher-level PLC system control. The control pulses of the frequency control unit which clock individual stepper motors are sent directly to each stepper motor.

Errors or disturbances of the PAC-F program are shown as error numbers in a segment display.

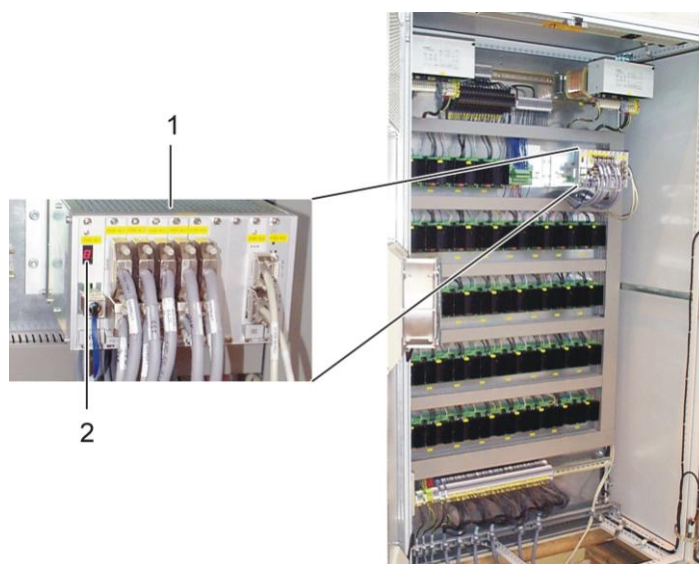


Fig. 6- 21 Frequency control unit (in the switching cabinet +G5)

1 - Frequency control unit PAC-F

2 - Segment display of error messages

Errors or disturbances of the drive system for the stepper motors are shown as error messages on the monitor (communication PLC-PA control error).

Resuming operation

- Trouble-shoot the frequency control unit (PAC-F):
 - The cause of the disturbance is shown on the display of the frequency control unit PAC-F.
 - Read the error number off the display. Look it up in the manufacturer's documentation to determine the cause of the disturbance.
- After the problem has been solved, restart the CPU of frequency control unit (PAC-F).



7

Maintenance

The chapter "Maintenance" contains the maintenance schedules for the TOPMET web coating system and describes all maintenance work to be performed on the system. Instructions on how to perform the work are provided.

7.1 Introduction

7.1.1 Safety precautions

	Note!
	During maintenance work, the personnel is sometimes located in hazardous zones specific to your system. It is imperative that you adhere to the safety precautions listed in the section Safety. Additional safety precautions can be found in the section Maintenance instructions directly before each maintenance step.

	Note!
	Before beginning maintenance work • at electrical equipment and the pump system: Disconnect the system from the electrical power, and secure it against being switched on again unintentionally. • at the winding system and in the radius of motion of the system: Turn the service key switch at control cabinet +H1 to SERVICE. Remove the service key and store it somewhere.

Operational prerequisites and requirements for maintenance personnel

- System maintenance work may only be performed
 - by technically qualified and specifically authorized maintenance personnel.
 - in agreement with the system's maintenance schedule.
- The end user is responsible for ensuring that the maintenance personnel has received all of the training and instructions required for these tasks.

APPLIED MATERIALS shall assume absolutely no liability for any damage and injuries caused by not adhering to these prerequisites and requirements.



7.1.2 Using the maintenance documents

The maintenance documents for the TOPMET system consist of the sections

- Maintenance schedules, see chapter 7.2
- Cleaning the system, see chapter 7.3
- Cleaning sealing surfaces and sealing rings, see chapter 7.4
- Maintenance instructions, see chapter 7.5

For all components, maintenance is performed on an assembly basis. Each maintenance schedule consists of a table listing the component to be maintained and the activity to be done, with the appropriate maintenance interval. A column Info links the maintenance schedule with maintenance instructions, when applicable.

Performing the maintenance intervals

d	=	daily
w	=	weekly
m	=	monthly
3m	=	quarterly
6m	=	semi-annually
a	=	annually
1000 h	=	operating hours
*)	=	depending on wear and tear

7.2 Maintenance schedules

7.2.1 Vacuum pumps

	Note!
	Comply with the manufacturer's documentation for the vacuum pumps and accessories. Perform the maintenance and cleaning work only in conjunction with the manufacturer's documentation. Comply with the safety precautions stated in the documentation.

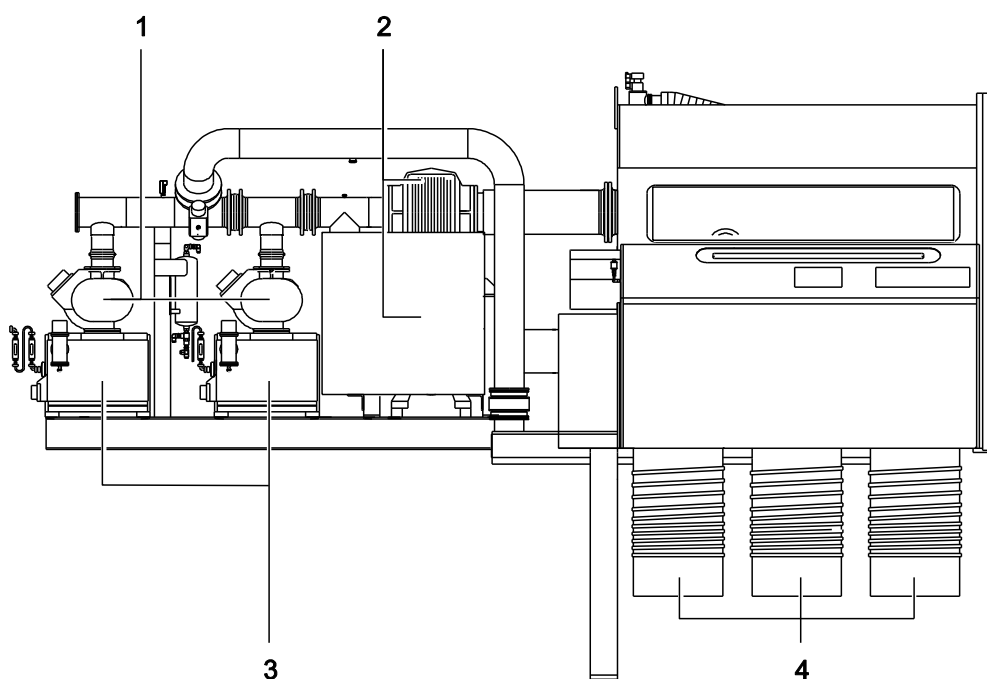


Fig. 7- 1 Vacuum pumps (standard layout')

- 1 - Roots pump RUVAC WAU 2001/RA 3001
 2 - Roots pump RUVAC RA 7001/9001

- 3 - Rotary vane pump SOGEVAC SV 750F
 4 - Oil diffusion pump DIP 20000

7.2.1.1 Rotary vane pump SOGEVAC SV 750 F

Note: Use the operating instructions SOGEVAC SV 750 F (OEM-Documentation).

Component/designation	Work to be done	Interval	Info
Exhaust filter	- Check the exhaust filter for dirt. Replace filter if very dirty. - Replace the filter before the specified maintenance interval if the pressure relief valve opens due to filter clogging.	6000 h	
Oil level/oil quality	At the oil sight glass, check the oil level while the pump is running. The oil level must be within the maximum/minimum range of the sight glass. Add oil only when the pump is switched off. When adding oil, use only oil type ANDEROL (mixable).	d	
	Visually inspect the oil quality at the sight glass. The oil is normally clear and transparent. Change the oil if it is cloudy.	d	
	Change oil and replace oil filter. Change the oil before the specified maintenance interval if it is very dirty. IMPORTANT: After each oil change, the hour counter of the pump SV 750F must be reset to zero. For this, select screen page "Vacuum Parameters" and click on the button <Reset> for the proper pump.	1000 h	
Pump motor	Check the motor fan and cooling fins for dirt. Remove dust and deposits with a brush and compressed air.	a	
	Check the tension of the V-belt by hand. Increase the tension if it is loose. A loose belt will cause slippage.	6 m	
Gas ballast device	Clean the gas ballast filter with compressed air and cleanser	m	



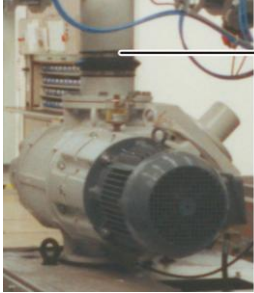
Component/designation	Work to be done	Interval	Info
	- Visually inspect the flow monitor: Cloudiness at the sight glass indicates dirt. - In case of dirt: Open device, remove and clean inner parts.	m	
Intake port and intake port valve	Clean the dirt trap in the intake port and intake port valve. Use suitable cleanser and compressed air.	6000 h	

7.2.1.2 Roots pump RUVAC WAU 1001/2001

Note: Use the operating instructions RUVAC WAU 1001/2001 (OEM-Dokumentation).

Component/designation	Work to be done	Interval	Info
Oil level/oil quality	Check the oil level of the bearings when the pump is not in operation. When adding oil, use only oil type N 62.	d	
	Check the oil level of the shaft sealing ring housing. The oil level must be visible in the oiler. When adding oil, use only oil type N 62.	d	
	Visually inspect the oil quality at the sight glass. The oil is normally clear and transparent. Change the oil if it is cloudy	d	
	Change oil and replace oil filter. Change the oil before the specified maintenance interval if it is very dirty. IMPORTANT: After each oil change, the hour counter of the pump WAU 1001/2001 must be reset to zero. For this, select screen page "Vacuum Parameters" and click on the button <Reset> for the proper pump.	3000 h	
Fan hood and cooling fins	Clean	m	



Component/designation	Work to be done	Interval	Info
Dirt trap 	- Remove dirt trap (1) and clean it IMPORTANT: The dirt trap sieve protects the pump from process residue and dirt particles. The sieve must be cleaned regularly to ensure reliable pump performance. Procedure: - Switch the pumping station off and let it cool down - Vent the vacuum pump. Open blanking flange lines and allow air to slowly flow in. - Remove the seal and dismantle the vacuum flange - Remove the sieve and clean it. Wash out oil residue and solvents. Then thoroughly dry the sieve with compressed air. Replace the sieve if it is defective. - Reassemble it by reversing the above steps	m	
Pump area	Clean	a	
Valve of the pressure equalization line	Clean	a	

7.2.1.3 Roots pump GM/GL or RUVAC RA

Note: Use the operating instructions GM/GL or RUVAC RA (OEM-Dokumentation).

Component/designation	Work to be done	Interval	Info
Oil level/oil quality	Check the oil level of the bearings when the pump is not in operation. When adding oil, use only oil type N 62 or equivalent oil qualities.	d	See OEM-Dokumentation
	Check the oil level of the shaft sealing ring housing. The oil level must be visible in the oiler. When adding oil, use only oil type N 62 or equivalent oil qualities.	d	See OEM-Dokumentation
	Visually inspect the oil quality at the sight glass. The oil is normally clear and transparent. Change the oil if it is cloudy.	d	See OEM-Dokumentation
	Change oil. Change the oil before the specified maintenance interval if it is very dirty. IMPORTANT: After each oil change, the hour counter of the roots pump must be reset to zero. For this, select screen page "Vacuum Parameters" and click on the button <Reset> for the proper pump.	3000 h	See OEM-Dokumentation
Fan hood and cooling fins	Clean	m	
Dirt trap	Clean	m	
Pump area	Check for dirt and deposits.	a	
Shaft sealing rings	Replace	*)	



7.2.1.4 Oil diffusion pump DIP 20000

Note: Use the operating instructions for Oil diffusion pump DIP 20000 (OEM-Documentation).

Component/designation	Work to be done	Interval	Info
Pump fluid	Check the pump fluid level. The pump fluid level is read directly off the oil sight glass. The sight glass has markings for the minimum and maximum pump fluid level. When the pump is switched off, the pump fluid level must be in the middle of the sight glass. Add pump fluid if the pump fluid level is wrong. IMPORTANT: Only fill the diffusion pump with original silicon oil from the original filling. Note the information printed on the pump. If alternative pump fluid is used, then thoroughly clean the pump before filling it.	w	
	Check the quality of the pump fluid. Pump fluid is normally clear and transparent. Change the pump fluid if it is cloudy.	w	
	Change pump fluid/clean pump Change pump fluid only when the pump is switched off, cooled down and vented to atmosphere. Clean the pump before filling it with new pump fluid. NOTE: Clean the pump before the specified maintenance interval in case of very long pumping times. IMPORTANT: After each oil change, the hour counter of the pump DIP 20000 must be reset to zero. For this, select screen page "Vacuum Parameters" and click on the button <Reset> for the proper pump.	2500 h	



Component/designation	Work to be done	Interval	Info
Heating rods	Visually inspect the fuses for the heating rods. The fuses are located in a fuse box, with a transparent cover. If a fuse blows, determine the reason. Then reinsert a new fuse. NOTE: If a power fuse drops out again, then the corresponding heating rod is defective (short circuit). Replace the defective heating rod. IMPORTANT: Perform work on the heating rods only when the pump is disconnected from electrical power. Turn the power switch in the switching cabinet off!	w	



7.2.2 Media supply (water/compressed air/gas)

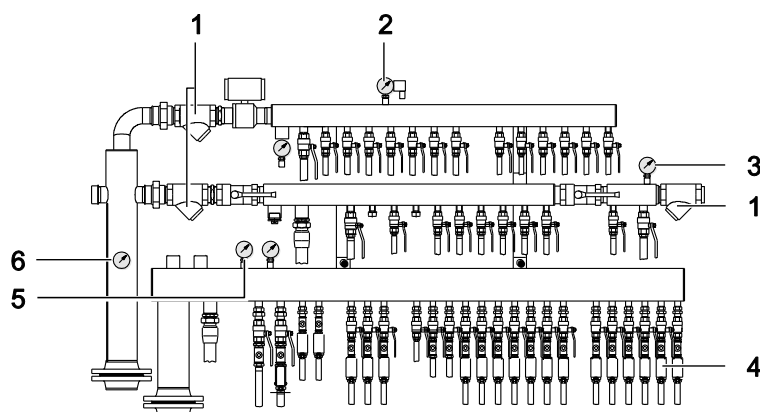


Fig. 7-2 Cooling water supply

- | | |
|--|--------------------------------------|
| 1 - Cooling water: filter cartridge (dirt sieve) | 4 - Cooling water: flow monitor |
| 2 - Cooling water: evaporator - inlet pressure | 5 - Cooling water: return pressure |
| 3 - Cooling water: cooling system - inlet pressure | 6 - Cooling water: inlet temperature |

Component/designation	Work to be done	Interval	Info
Cooling water supply	Check water pressure and temperature at the water distribution system.	d	Chapter 7.5.1
	Close the shut-off valve. Make sure that gauge 2 and 4 (see Fig. 7- 2) have the same value. Shut-off power and water supply, release water pressure and clean solenoid valve.	d	
	Check water ph value, using ph test strips: Make sure the ph value is between 8 - 10.	m	
	Check the water specification	m	Chapter 1.6.2
Cooling water supply (continued)	Check the entire system for leaks, visually inspect pipe couplings and hoses.	m	
	Clean flow monitors and check for proper functioning.	6 m	Chapter 7.5.1
	Check/clean the filter cartridges (dirt sieves) of the water distribution system. The specified maintenance interval depends on the quality of the cooling water.	6 m	Chapter 7.5.1

Component/designation	Work to be done	Interval	Info
Compressed air supply	Check operating conditions at the compressed air maintenance unit (see Fig. 7- 3). Set the operating pressure to 6 - 7 bar.	d	Chapter 7.5.2
	Check the compressed air specification.	m	Chapter 1.6.3
	Check the entire system for leaks. Visually inspect the compressed air connections and hoses. Listen for the sounds of leaks.	m	
Gas system (oxygen/argon)	Check the entire system for leaks. Check lines and connections for leaks.	m	Chapter 7.5.3
	Check the oxygen shut-off valves for proper functioning. Check the valve function for leaks.	m	Chapter 7.5.3

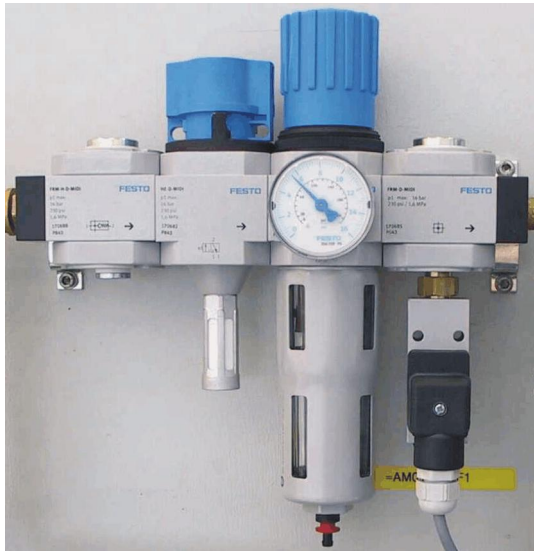


Fig. 7- 3 Compressed air supply



7.2.3 Function test of the safety equipment

Component/designation	Work to be done	Interval	Info
Emergency-stop pushbuttons	- Check the function of the EMERGENCY-STOP PUSHBUTTON When you press the EMERGENCY-STOP PUSHBUTTON, all moving parts of the system are stopped and locked. Resetting the interlock: - Unlock the EMERGENCY-STOP PUSHBUTTON by hand. - Press the pushbutton FAILURE/RESET at operating panel +R02. IMPORTANT: After you reset the interlock, no automatic movements of system parts are permitted.	3 m	Chapter 4.5.2
System main switch	Check the function of the main switch: Turn the main switch off and on again. When it is turned off, the system is disconnected from the electrical power; the system is not under voltage. IMPORTANT: After the main switch is turned back on, no automatic movements of system parts are permitted.	3 m	Chapter 4.4.1
Service safety switch	Check the function: When you turn the key switch to the service position, the travelling part, rewinder, unwinder and pivot arm are locked. These parts will not function.	3 m	Chapter 4.5.1

Component/designation	Work to be done	Interval	Info
Grounding rod	- Check the function and visually inspect: Visually inspect: - Check the insulation of the ground wire for mechanical damage. - Make sure the ground wire is not broken. Check the function: - Test with conductivity of the grounding rod with an external source of electrical power.	3 m	

7.2.4 Electrical equipment

Component/designation	Work to be done	Interval	Info
Automation device SIMATIC S7-400	- Check the battery level indicator. - Only replace the buffer battery when a warning is displayed. *)	d *)	Chapter 7.5.4.1 and 7.5.4.2
Switching cabinet	Clean the fan: use a vacuum cleaner.	3 m	
Check the overall electrical system	Check the electrical system for safety and proper functioning.	a	Chapter 7.5.4.3
Signal lamps	Check the signal lamps for proper functioning.	w	Chapter 7.5.4.4



7.2.5 Vacuum system

7.2.5.1 Vacuum valves

Note: Use the operating instructions for chamber flaps.

Component/designation	Work to be done	Interval	Info
High-vacuum valve V1	Clean the area around the valve and the sealing surface. Visually inspect the lip-seal.	3 m	Chapter 7.5.5
Chamber flaps V3	Clean the valve flap and seat.	m	Chapter 7.5.6

7.2.5.2 Vacuum chamber

Component/designation	Work to be done	Interval	Info
Chamber sight glasses	- Clean the sight glasses Remove deposits with common cleansers. Do not use aggressive solvents. - Immediately replace defective sight glasses. When replacing them, use only the glasses specified for the system.	w	
Stroboscope	- Clean the cylinder stroboscope Make sure it runs freely. - Check the oil level of the vacuum rotary feedthrough: the oil level must be in the middle of the sight glass	*)	Chapter 7.5.7
Vacuum test for leaks	Perform a pressure increase test. Enter the result in the test log.	3 m	Chapter 7.5.8



7.2.5.3 Vacuum measuring equipment

	Note!
	Comply with the manufacturer's documentation for the vacuum measuring equipment. Perform the maintenance and cleaning work only in conjunction with the manufacturer's documentation. Strictly comply with the safety precautions stated in the documentation.

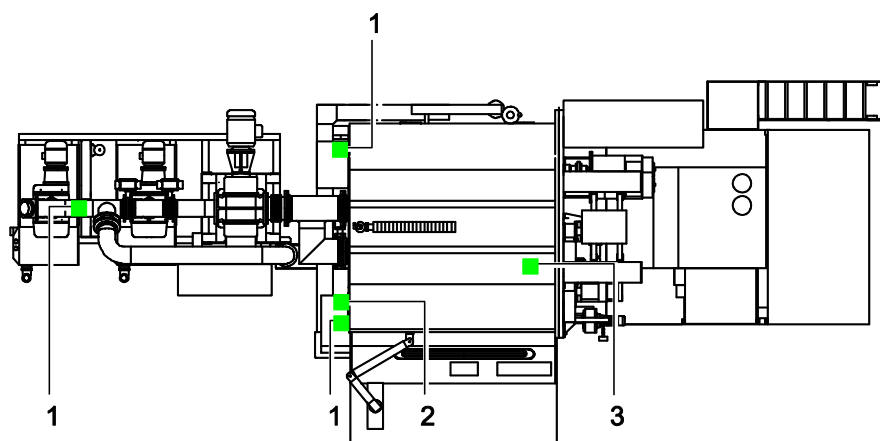


Fig. 7- 4 Vacuum measuring equipment

1 - Pirani PSG 500S

2 - Penningvac IKR 251

3 - Pirani PSG 500S (located at diffusion pump)

Component/designation	Work to be done	Interval	Info
Pirani transmitter PSG 500S	- Inspect, clean and replace the measuring equipment unit. *) depending on the amount of dirt - Check plugs and connection cables. Make sure the connectors are seated properly (cables, plugs).	*) 6 m	
Penningvac IKR 251	- Inspect, clean and replace the measuring equipment unit. *) depending on the amount of dirt - Check plugs and connection cables. Make sure the connectors are seated properly (cables, plugs).	*) 6 m	



7.2.6 Winding unit

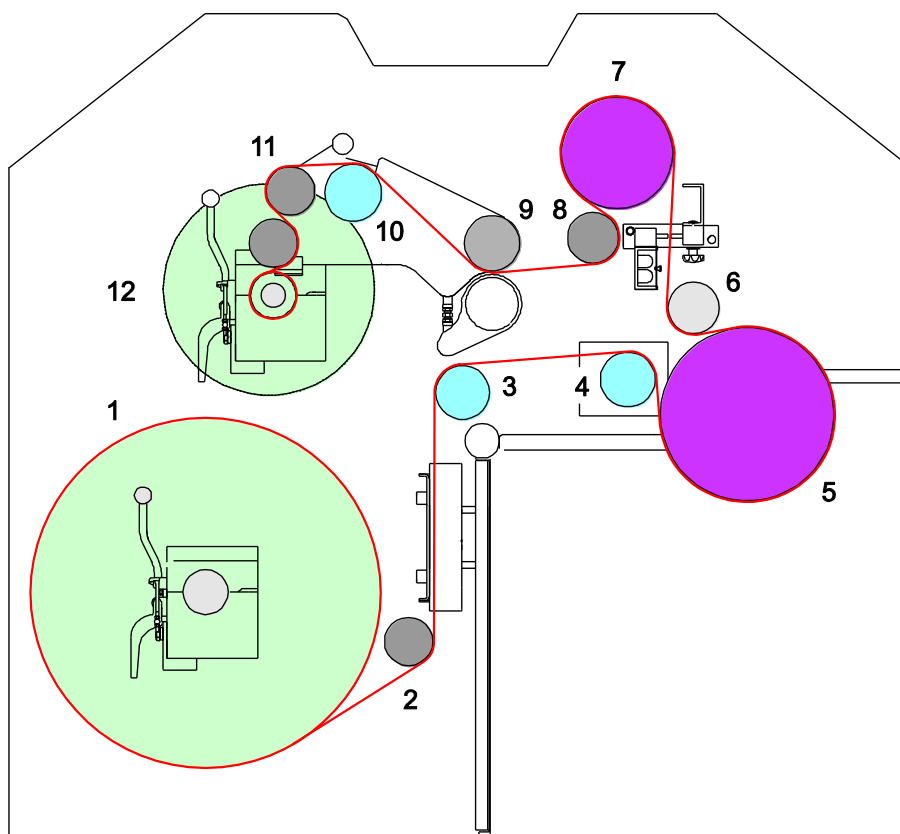


Fig. 7- 5 TOPMET chamber cross-section

- | | |
|---|---------------------------------------|
| 1 - Unwinder drum | 7 - Tension roll or Post cooling drum |
| 2 - Spreader roll spiral shaped | 8 - Guide roller 3 |
| 3 - Guide roller 1 | 9 - Guide roller 4 |
| 4 - Spreader roll "Varibow" | 10 - Spreader roll "Fixbow" |
| 5 - Coating drum | 11 - Pivot arm |
| 6 - Guide roller 2 (with tension sensors) | 12 - Rewinder drum |

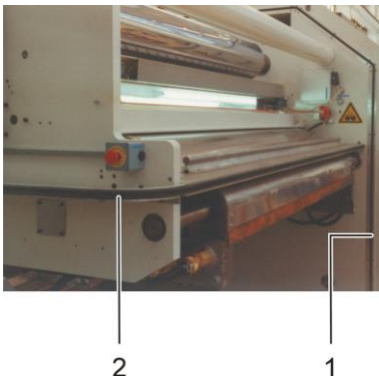
Component/designation	Work to be done	Interval	Info
Vacuum rotary feedthrough	- Check the oil level in the sight glass. The level must be in the middle of the sight glass. Check the oil level only when the system is switched off. - Check the shaft sealing ring for leaks. Replace the shaft sealing ring in case of oil leaks.	w w	Chapter 7.5.9.1
Coating drum	- Clean the drum's surface *) depending on the amount of dirt - Replace the drum bearings If the bearings make loud noises, they must be replaced before the specified maintenance cycle.	*) a	Chapter 7.5.9.2
Tension roll or Post cooling drum	- Clean the drum's surface. Rotate the drum by hand and rub with alcohol the steel surface. - Replace the drum bearings. If the bearings make loud noises, they must be replaced before the specified maintenance cycle.	w a	
Rewinder/unwinder	Replace the bearings of the rewinder/unwinder. If the bearings make loud noises, they must be replaced before the specified maintenance cycle.	a	
Spreader roll - Fixbow (rewinder side)	- Clean the roll's surface. Rotate the drum by hand and rub with alcohol. - Replace the roll. Remove the entire roll and perform a general overhaul NOTE: APPLIED MATERIALS recommends that you send it to the manufacturer.	m 2 a	
Spreader roll - Varibow (unwinder side)	Clean the roll's surface. Rotate the drum by hand and rub with alcohol.	m	



Component/designation	Work to be done	Interval	Info
	Replace the roll. Remove the entire roll and perform a general overhaul NOTE: APPLIED MATERIALS recommends that you send it to the manufacturer.	2 a	
	Check the hydraulic lines of the bow motion Visually inspect all lines and connectors. In case of leaks, remove hydraulic oil and service the system. When filling oil, use oil type ENERPACK, AF 95 Y	m	
Rewinder/unwinder coupling 	- Check the function of the bearing coupling. The bearing coupling must open and close easily by hand. Otherwise, replace the plastic toothed gear (Item 2) - Make sure the coupling lever (Item 1) is parallel. When the bearing coupling is open, the coupling lever must remain in a vertical position. There must not be any play in the lever.	m	
V-belts (spreader rolls)	Check the V-belt tension for spreader rolls (Fixbow und Varibow). Check the belt tension by hand. The belt must be tight. If it is loose, there will be slippage and the V-belts must be tensed using the cam.	m	Chapter 7.5.9.5
Guide rollers	Replace the roller bearings. If the bearings make loud noises, they must be replaced before the specified maintenance cycle.	a	
Characteristic of friction	Enter the characteristic of friction in the PLC	6 m	Chapter 7.5.9.4

Component/designation	Work to be done	Interval	Info
Threading belt (flat belt) 	Check the belt tension by hand; adjust if necessary	m	Chapter 7.5.9.6

7.2.7 Travelling part

Component/designation	Work to be done	Interval	Info
Lip-seal (Item 1) 	- Cleaning: Clean the seal only with solvent-free cleansers. Slightly grease the cleaned seal with vacuum grease. - Visual inspection: Check the seal for mechanical damage. Replace the seal if defective.	3 m	Chapter 7.4
Chamber isolation (inflatable seal, Item 2)	- Cleaning: Remove deposits and dirt from the seal groove. Clean the seal only with solvent-free cleansers. - Visual inspection: Check the seal for mechanical damage. Replace the seal if defective.	3 m	Chapter 7.4



7.2.8 Layer thickness measurement systems

	Note!
	Comply with the manufacturer's documentation for the layer thickness measurement systems. Perform the maintenance and cleaning work only in conjunction with the manufacturer's documentation. Strictly comply with the safety precautions stated in the documentation.

7.2.8.1 Optical layer thickness measurement system LMS-OD

Note: Use the operating instructions for LMS-OD.

Component/designation	Work to be done	Interval	Info
Probes	Clean the probe lens	3 m	
	- Check the intensity of the light in the measurement channels: On the screen page "LMS-Service", the bar graph value for each measurement channel must be greater than 5,000 units. - Replace any defective transmitter and receiver diodes.	3 m	Chapter 7.5.10.1

7.2.8.2 Resistance measurement system LMS-C (option)

Note: Use the operating instructions for LMS-C.

Component/designation	Work to be done	Interval	Info
Measurement drums	- Clean the measurement drums: Rub the drum with a soft, lint-free cloth. Remove stubborn dirt only with ethyl alcohol or ethanol. - Check the slip rings of the carbon brushes: Replace any carbon brushes which might be too short.	m	Chapter 7.5.10.2

7.2.8.3 Resistance measurement system LMS-R (option)

Note: Use the operating instructions for LMS-R.

7.2.9 Pretreatment station (option)

Component/designation	Work to be done	Interval	Info
Magnetron cathode	• Visually inspect the cathode Check the cathode's parts: • dark room shielding • target surface • gas tube • Clean the cathode and the area around it • Replace worn and consumed targets. Condition new targets.	w	Chapter 7.5.11



7.2.10 Cooling and heating equipment

	Note!
	Comply with the manufacturer's documentation for the cooling and heating equipment. Perform the maintenance and cleaning work only in conjunction with the manufacturer's documentation. Strictly comply with the safety precautions stated in the supplier's documentation.

Component/designation	Work to be done	Interval	Info
Refrigerant circuit	- Check the sight glass in the refrigerant circuit. - Humidity indicator green: refrigerant normal - Humidity indicator yellow: Too much water content in refrigerant. Inform the APPLIED MATERIALS service department. Bubbles in the sight glass indicate there is not enough refrigerant in the circuit: Add refrigerant.	w	
	Check the temperature of the refrigerant During operation, read off the temperature value for the R404A circuit at the high-pressure pressure gauge: Normal temperature approx. 40° C (R404A) In case of deviations from normal: Adjust the temperature for the refrigerant to normal, using the control valve. The scale value at the control valve should be approx. 4 – 4.5.	w	
Compressor (refrigeration circuit)	Check the oil level Check the oil level only when the compressor is switched off.	w	

Component/designation	Work to be done	Interval	Info
Temperature circuit (coating drum)	- Check the anti-freeze content of the brine. IMPORTANT: The proper brine mixture (water/anti-freeze) must be maintained at all times. Refer to the manufacturer's specifications (see the labeling on the system) for the specified brine ratio. For frost protection and optimum utilization of the system, the correct mixture must be maintained. Add only original anti-freeze in accordance with the manufacturer's specifications. The proper brine mixture can be measured in two ways: - Measurement with a Leica Refractometer: Take a sample from the brine tank and perform the measurement. The measuring equipment indicates the anti-freeze content as a percentage. Maintain a value between 52% and 55%. - Measurement with an acid siphon Place the measuring equipment in the brine tank and read off the temperature. The measurement is based on the temperature of the brine. The equipment shows the current temperature of the anti-freeze in the brine. Maintain a value between -35 ° C and -40° C.	w	
	Take a brine sample (250 ml) and analyze it. Send the brine sample to the next service facility of the company Höchst, and have it analyzed for residue (abrasive materials, retarding agents).	a	



7.2.11 Polycold refrigerator

	Note!
	Comply with the manufacturer's documentation for the Polycold refrigerator. Perform the maintenance and cleaning work only in conjunction with the manufacturer's documentation. Strictly comply with the safety precautions stated in the supplier's documentation.

Component/designation	Work to be done	Interval	Info
Oil level/oil quality	Oil level during operation: 1/8 to 1/4 of the sight glass's height Maximum oil level: not higher than the middle of the sight glass Make sure the oil is not dirty. If it is, contact Polycold service.	m	
Oil change	It might be necessary to change the oil ahead of schedule. This depends on the operating conditions and the oil quality. Contact Polycold service.	5 a	
Cooling water – heat exchanger	Test criteria Use the manufacturer's documentation. If the heat exchanger is full of lime, delime it with a deliming agent.	m	
Compressor	Check the equalization pressure Before performing the check, switch the refrigerator off and wait 48 hours. After this time, the equalization pressure (suction/discharge) at the pressure gauges should be for PFC 550 110psi, PFC 650 130psi, PFC 1100 150psi. Note: If this equalization level is not reached, this indicates that there is not enough refrigerant. Action: Leakcheck with soap bubbles. Only special trained personnel is allowed to add refrigerant. Contact Polycold service.	6 m	

7.3 Cleaning the system

	⚠ Warning Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal dusts and metal oxide dusts which are hazardous to your health. During all cleaning work, avoid creating dust. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.
---	---

	⚠ Caution Health hazard Skin contact or eye contact with cleansers or inhaling them can be hazardous to your health. Use cleansers in accordance with international, national and regional specifications.
---	--

	Note! The specified pump-down time is only possible if the vacuum chamber is clean and free of metal deposits, and if the coating components (evaporator station, coating window, shutter, etc.) are clean. The pump-down time will be longer if the vacuum chamber is dirty.
---	---

	Note! For cleaning, the vacuum chamber must be open. Before you enter the chamber, activate the system's safety interlock. Turn the service key switch at the operating panel to the Service position and remove it.
---	---



	Note!
	The use of compressed air for cleaning the system and its components is not allowed. There is a danger of parts flying around and distributing particles throughout the entire chamber. This increases the particle contamination and deteriorates the pump-down time for the vacuum chamber. Use a vacuum cleaner to remove scraped off or peeled off material and dust or dirt at the coating components (evaporator station, coating window, shutter, etc.).

When cleaning the system and its components, use only the following:

- spatula
- wire brush
- alcohol

Never clean system parts with water or aggressive solvents.

With regard to the work to be done, cleaning the system is divided into two phases:

- Cleaning between the coating cycles (see chapter 7.3.1) includes cleaning system parts directly after each coating process or before beginning a new coating cycle.
- Overall cleaning (see chapter 7.3.2) is an extended cleaning program, and is performed after long production phases.

7.3.1 Cleaning between coating cycles

7.3.1.1 Coating window

	⚠ Warning Contusion hazard There is the danger of contusions within the radius of motion of the coating window! Before you press the pushbutton <coating window>, make sure no persons are located within the radius of motion of the coating window. Never place your hands or arms within the radius of motion of the coating window.
---	--



Fig. 7- 6 Coating window (cleaning position)

- Lower the coating window to the cleaning position: Press the pushbutton LOWER COATING WINDOW (operating panel +R23) and keep it pressed. The evaporator shutter automatically opens.
- Remove coating residue from the shutter, guide rail and limit shields. For cleaning, use a spatula and wire brush.
- Coat all cleaned parts with a thin layer of graphite paste.
- Raise the coating window to the coating position: Press the pushbutton RAISE COATING WINDOW (operating panel +R23) and keep it pressed. The evaporator shutter automatically closes.

7.3.1.2 Evaporator station

	⚠ Warning Danger of burns from hot surfaces Directly after the coating process and after the chamber is opened, the evaporator station and the evaporation material is still hot. Whenever you work at the evaporator station, wear the following protective gear as protection against hot parts and hot molten metal: • goggles • fire-resistant protective gloves • work garments which protect against molten metal (e.g. leather apron)
	⚠ Warning Contusion hazard The moving components • evaporator flap • wire roll frame can cause injuries. When working at the evaporator station, make sure these components do not move unintentionally. In particular, make sure you do not press the pushbutton <evaporator flap>. Never place your hands within the radius of motion of these components.

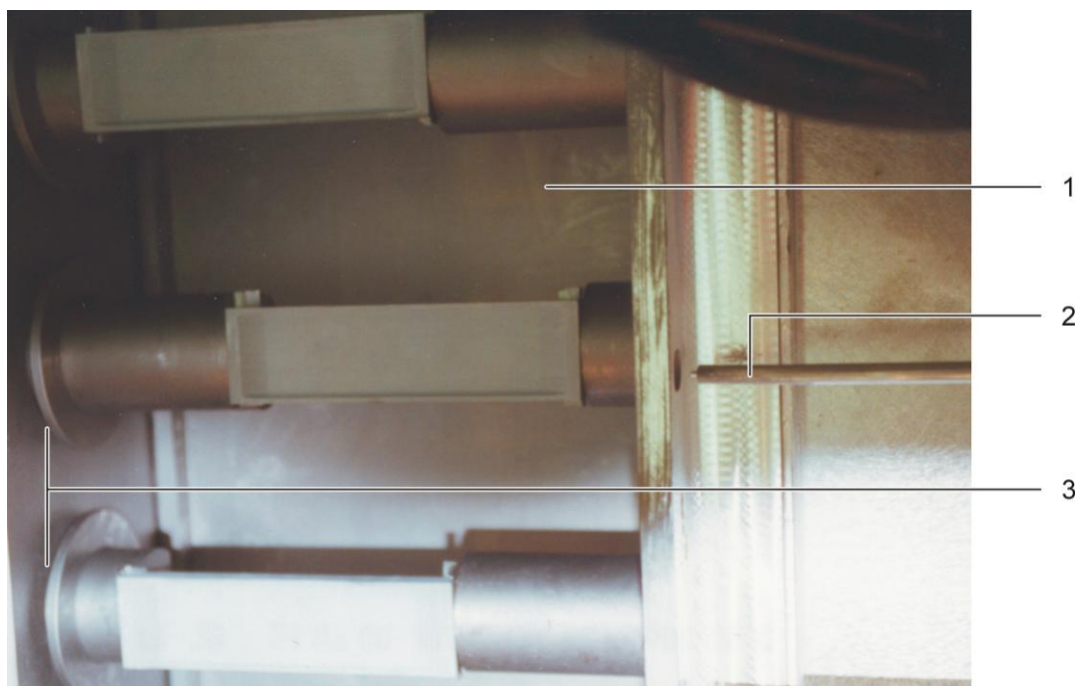


Fig. 7-7 Evaporator station

1 - Collector

2 - Wire feed pipette

3 - Gap between current bolt and cover

Wire feed pipettes

	Note!
	Thoroughly clean the gap between the current bolt and the cover. Deposits here can cause short circuits during operation.

- Clean the openings of the wire feed pipettes. Remove aluminum deposits from the evaporator box feedthrough. Material here can cause short circuits.
- Coat all cleaned parts with a thin layer of graphite paste.

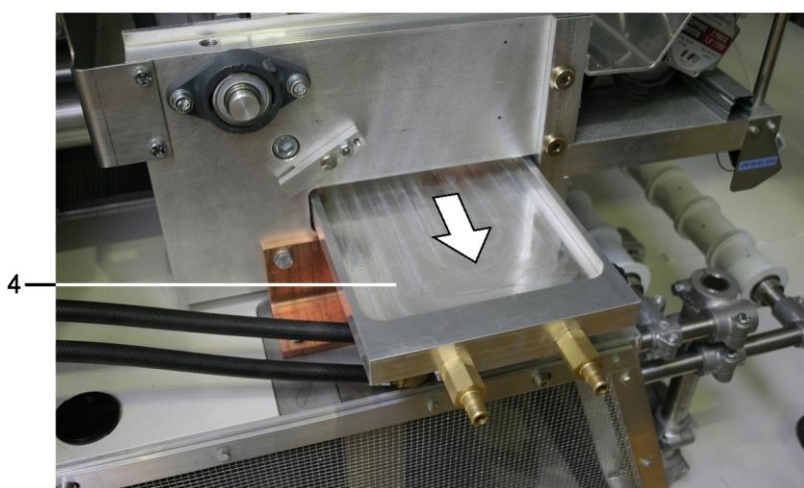


Fig. 7-8 Collector

- | | |
|----------------------------------|---------------------|
| 1 - Lock bolt | 3 - Water couplings |
| 2 - Connectors for cooling water | 4 - Collector |

Collector

- Loosen both cooling water couplings and remove them.
- Swivel the lock bolt in the shown direction
- Pull the collector out and empty it.
- Remove aluminum residue from the box. Use a wire brush and spatula to remove stubborn aluminum deposits.
- Coat the cleaned box with graphite paste.
- Put the collector back in place, and mount the cooling water couplings.

Evaporator box and area around it

Use an industrial vacuum cleaner to remove all remaining aluminum, residue and dirt remaining from cleaning.

7.3.2 Overall cleaning

Overall cleaning of the system includes cleaning specific system components as stated in the maintenance schedules, see chapter 7.2.



7.4 Cleaning sealing surfaces and sealing rings

7.4.1 Prerequisites

Adhere to the following rules:

- During all cleaning work, comply with all local, national and international laws (respiratory protection, ventilation etc.).
- Wear clean, lint free gloves when cleaning the sealing surfaces and sealing rings.
- In clean rooms, wear clean room gloves. Do not allow particles to form.

7.4.2 Permitted Cleaning Materials

Permitted are:

- Sandpaper with a graining of 400 or more.
- Scotch Brite.
- IPA (Isopropyl alcohol, 2-Propanol).
- Ethanol (Ethyl alcohol) and Acetone (Propanone) (do **not** use for O-rings!)

	Note!
	Ethanol (Ethyl alcohol) and Acetone (Propanone) must not be used for O-rings. These alcohols destroy the rings!

- Lint free cloths, e.g. Dastex Series 400.

7.4.3 Cleaning sealing surfaces

Stainless steel and aluminum sealing surfaces must be scratch free. For all vacuum ranges, the tolerable surface roughness is $R_z = 6.3 \mu\text{m}$.

Remove scratches using sandpaper (graining 400 or more) and Scotch Brite. The polishing movements must be **parallel to the sealing groove**. Do not deepen the sealing surface. Finally clean the surface with IPA (Isopropyl alcohol) using a lint free cloth.

7.4.3.1 Cleaning round sealing rings

Types of sealing rings:

- Sealing rings are ring shaped, made by a cord through vulcanizing or gluing.
- O-rings are endless sealing rings which are pressed as one piece.

The used sealing rings are made either of:

Material	Abbreviation	Trade name
Fluor caoutchouc	FKM, FPM	Viton
Acrylnitril-butadien-caoutchouc	NBR	Perbunan

Both materials are chemically resistant to IPA, but will be destroyed by Ethanol or Acetone (see chapter 7.4.2).

Flawless rings have the following characteristics (see Fig. 7- 9):

- equal in diameter.
- free from noticeable gluing joints.
- free from any damage (tears, cuts, burrs, porous areas).
- clean surface.

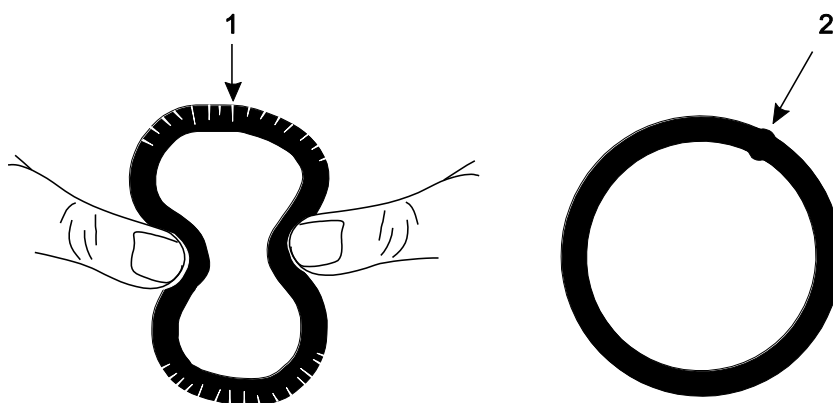


Fig. 7- 9 Checking O-rings

1 - Cuts, porous areas

2 - Enlargement, swelling

Proceed as follows to clean the sealing from oil, grease, particles, dust, talc or other soiling:

- Pull the ring through a dry cloth to remove coarse soiling.
- Pull the ring through a cloth drenched with IPA to remove slight soiling.

Note!	
	
	The following two steps are prohibited if the ring temperature will exceed 100°C. The material temperature must not exceed the following limits: • O-ring NBR: 100° C. • O-ring FPM: 200° C. • Vacuum grease Leybold LITHILEN: 150° C.

- Sparingly apply vacuum grease to the ring (very thin film of Leybold LITHILEN).
- Pull the ring through a dry cloth, so that the film of grease on the surface and pores is very thin.

	Note!
	Slightly greased O-rings collect dust particles. Assemble the ring at once, rather than storing it somewhere.

- In the groove, the ring should extend beyond the sealing surface by a uniform amount. Depending on the ring's thickness, the excess should be as follows:

Ring thickness	Excess (Fig. 7- 10)
5 mm	0.8 mm
6 mm	1.1 mm

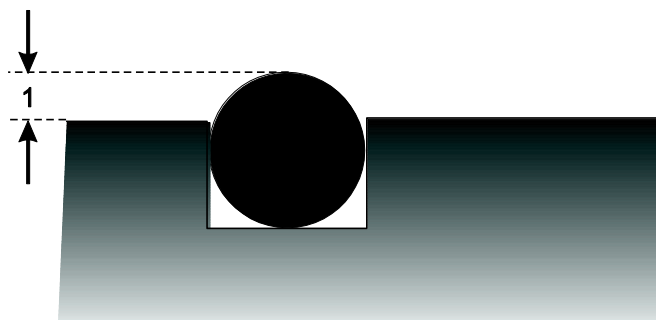


Fig. 7- 10 Excess of a sealing ring beyond the sealing ring

1 - Excess of the ring

7.4.3.2 Cleaning the groove

The groove must be clean, too.

Procedure:

- Remove the O-ring. Be careful not to damage or scratch the sealing surface, the groove or the O-ring itself. Never use sharp or pointed tools (e.g. screwdrivers).
- Before inserting the cleaned O-ring, clean the groove with an IPA drenched cloth. Always wipe in one direction parallel to the groove, never back and forth (Fig. 7- 11).

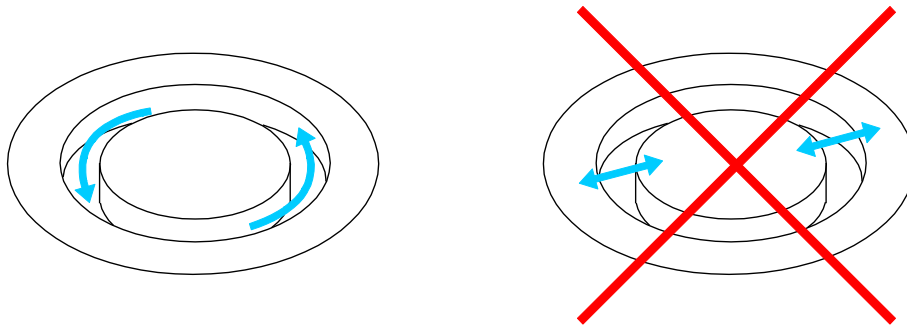


Fig. 7- 11 Direction of wiping in the groove

7.4.3.3 Cleaning the mounted O-ring

Procedure:

- Use a vacuum cleaner to remove particles and dust from the area around the ring.
- Wipe the ring with an cloth drenched in IPA.
- After a ring has been cleaned several times, you are recommended to remove it and thoroughly clean the ring and groove (see chapter 7.4.3.1 and chapter 7.4.3.2). Replace the ring with a new one if necessary.

7.5 Maintenance instructions

7.5.1 Cooling water supply

	⚠ Warning
	Overpressure, causing parts to fly There is the danger of injuries from flying parts when dismantling parts of the compressed air system or water distribution unit. Dismantle parts only when all pressure has been bled off the system. Before servicing the compressed air maintenance unit (e.g. replacing the filter cartridge): • Close the supply of compressed air (customer's side) and secure it against unintentional reopening. • Bleed the compressed air off the maintenance unit.

7.5.1.1 Checking the operating conditions (water pressure and temperature)

	Note!
	Make sure there is a pressure difference of at least 3 bar between the cooling water inlet and return, to ensure the system is completely cooled.

The operating values for the system's cooling water supply are displayed directly at the water distribution unit, with pressure gauges and thermometers. Check the following operating values and conditions:

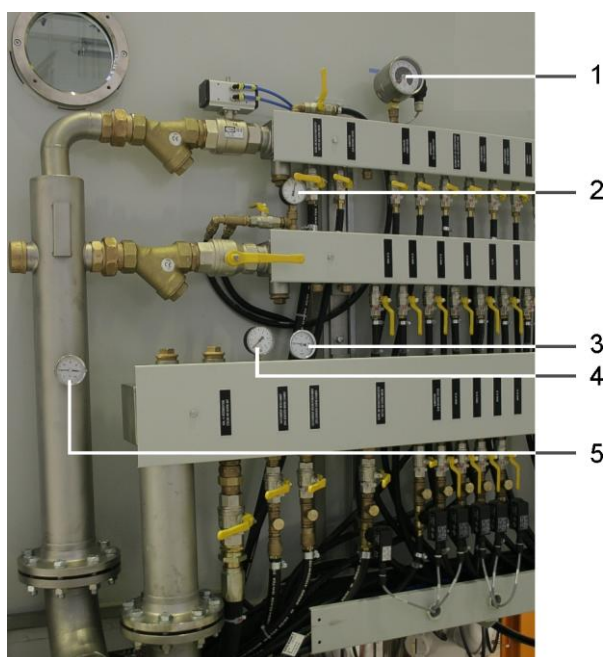


Fig. 7- 12 Checking the operating conditions for the cooling water

- | | |
|-------------------------------------|-----------------------|
| 1 - Inlet pressure (pressure gauge) | 4 - Return pressure |
| 2 - Inlet pressure | 5 - Inlet temperature |
| 3 - Return temperature | |

Cooling water - inlet pressure

- Maintain a pressure difference of 3-4 bar.
- During system operation, check the cooling water flow at the pressure gauge.

Cooling water - inlet temperature

The permissible range is 20 – 25° C.

Cooling water - return pressure

Maintain a maximum value of < 1 bar.

7.5.1.2 Checking the cooling water monitors for proper functioning

The monitors installed in the cooling water return circuit monitor the cooling water flow for all hot system components (pumps, evaporator systems).

All monitors must be checked regularly for proper functioning, to avoid monitor failure or erratic behavior.

Electrical switching contacts connect the monitors with the system control. When the cooling water fails, the corresponding monitor reports this. The failure is shown on the monitor page "Water Scheme"; the corresponding system component flashes red.

Possible causes for defective monitors:

- Dirty cooling water can cause deposits in the monitor. The measurement lug strip cannot trip.
- Mispositioning of the measurement lug strip. When the system is operated a long time, the measurement lug strip can become mispositioned due to deposits.

Procedure

- Isolate the system from the supply of cooling water: Close the main valve (customer's side) of the water supply.
- Relieve the water pressure from the cooling water system:
 - Slightly loosen the coupling at the monitor, until cooling water escapes. A small amount of water escapes, and can be neglected.
 - Observe the pressure reduction at the pressure gauge of the cooling water distribution unit. Wait until the pressure gauge shows 0 bar.

	Note!
	Turn the monitor outwards only when there is no pressure in the cooling water system.

- Turn the monitor outwards: Open the coupling completely and remove the monitor.

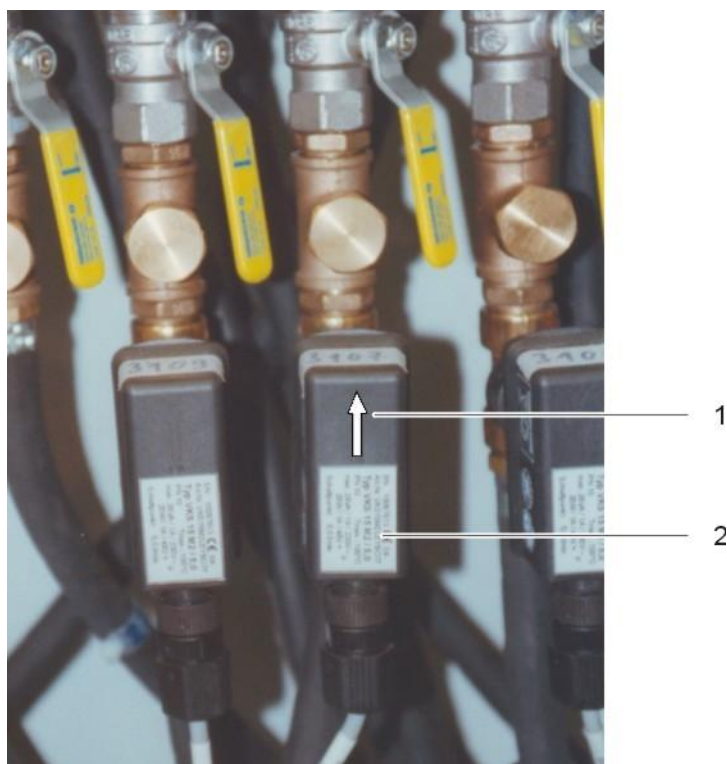


Fig. 7- 13 Monitor

1 - Direction of flow

2 - Electrical switching contact

- Clean and check the monitor:
 - Cleaning: Remove deposits and contaminations of the measurement lug strip with common cleansers.
 - Checking: Deflect the measurement lug strip by hand. It must then automatically return to its normal position. Replace the monitor if it is defective or does not work properly.

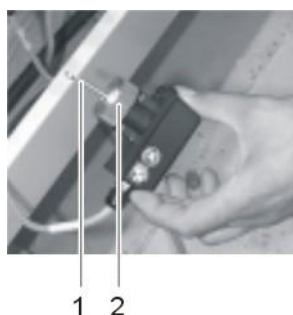


Fig. 7- 14 Measurement lug strip

1 - Measurement lug strip

2 - Coupling

- Mounting and connecting:
 - After the maintenance work, mount the monitor again. Note the indicated direction of flow.
 - Reopen the main shut-off valve of the cooling water supply.



7.5.1.3 Checking and cleaning the dirt sieves in the cooling water inlet lines

The inlet lines of the cooling water supply are equipped with dirt sieves. These sieves prevent contaminants and sediments from flowing into the system components. Depending on the water quality, matter will collect in the sieves. They must be checked and cleaned regularly to avoid cooling water failures.

	Note!
	During operation, the cooling water system is pressurized. Before opening screw couplings and connections at the water distribution unit, perform the following: • Switch the pump system off. • Close the main spigot of the cooling water supply. • Relieve the pressure from the cooling water system.

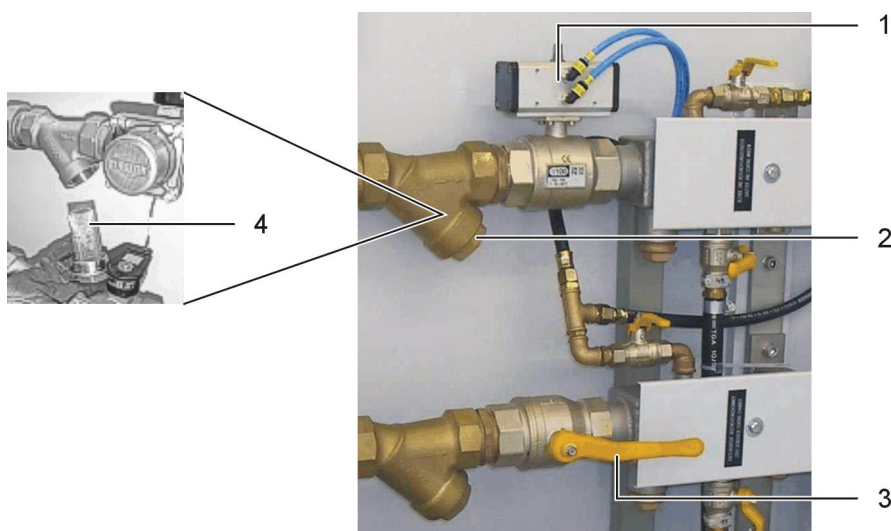


Fig. 7-15 Dirt sieve

- | | |
|---|---------------------------|
| 1 - Shut-off valve (controlled automatically) | 3 - Manual shut-off valve |
| 2 - Screw coupling | 4 - Dirt sieve |

- Isolate the system from the supply of cooling water:
 - Switch the pump system off. Switching the fore vacuum pumps off closes the automatically controlled manual shut-off valves at the cooling water inlet.
 - Close the main spigot (customer's side) of the cooling water supply.
 - Close all manual shut-off valves at the cooling water inlet.
- Removing, cleaning and remounting the dirt sieves:
 - Open the screw coupling and remove the dirt sieve.
 - Clean the sieve; replace it if it is defective.
 - Remount the dirt sieve. Close the screw coupling.
- Open the cooling water supply again: open all manual shut-off valves and the main spigot.

7.5.2 Checking the compressed air supply (operating conditions)

	⚠ Warning
	Overpressure, causing parts to fly There is the danger of injuries from flying parts when dismantling parts of the compressed air system or water distribution unit. Dismantle parts only when all pressure has been bled off the system. Before servicing the compressed air maintenance unit (e.g. replacing the filter cartridge): • Close the supply of compressed air (customer's side) and secure it against unintentional reopening. • Bleed the compressed air off the maintenance unit. • Released the compressed air at the compressed air storage tank.

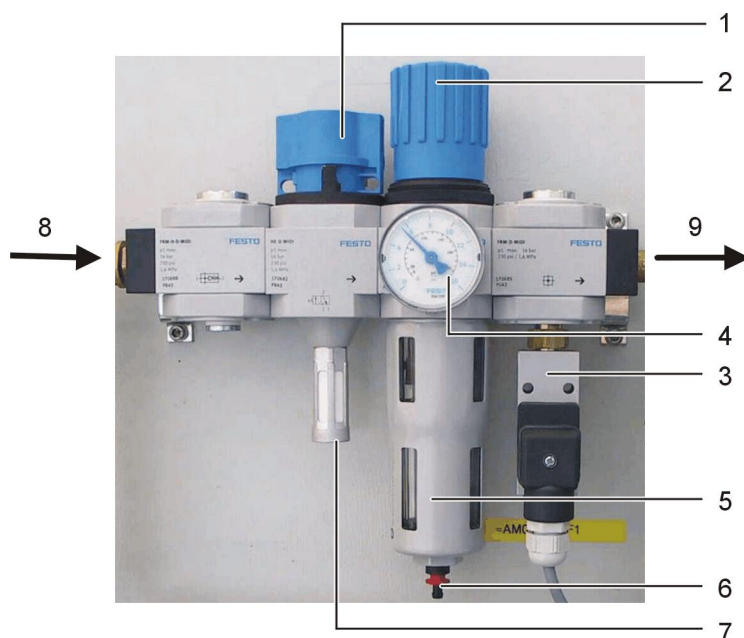


Fig. 7- 16 Compressed air maintenance unit

- | | |
|--------------------------------|---|
| 1 - Manual shut-off valve | 6 - Drain button |
| 2 - Control valve | 7 - Muffler |
| 3 - Pressure monitoring switch | 8 - Compressed air connection (customer's side) |
| 4 - Pressure gauge | 9 - Compressed air distribution system (system) |
| 5 - Water separator | |

- Check the pressure of the compressed air supply at the maintenance unit. For system operation, a pressure of 6 - 7 bar is required at the pressure gauge (Item 4 in Fig. 7- 16). If the pressure deviates from this, it must be adjusted at the control valve (2).
- Check the level of condensation in the water separator (5). If it is too high, press the drain button (6).

7.5.3 Process gas supply

	⚠ Warning Fire hazard Oxygen encourages the spread of fire. Fires can be caused if oxygen is used improperly. Make sure the oxygen installations are sufficiently far away from sources of ignition. Do not do anything which could create sparks near oxygen installations.
---	--

	⚠ Warning Beware of gas bottles Pressurized gas bottles can explode and cause serious injuries and damage if mishandled. Gas installations must comply with international, national and local safety regulations for handling pressurized gas bottles.
---	--

7.5.3.1 Checking the oxygen shut-off valves for leaks

The following safety system monitors the system's supply of oxygen:

- Redundant pressure measurement in the coating chamber.
- O₂ shut-off valves in the oxygen influx line.
- Redundant monitoring of gas flow (air) of the gas ballast device at the fore pumps.

The safety system prevents excessive concentrations of oxygen in the fore pumps. This prevents formation of combustible and explosive oil vapor/oxygen gas mixtures in the fore pumps.

You must regularly check the valves for leaks, to ensure that the safety function of the O₂ shut-off valves is working properly.

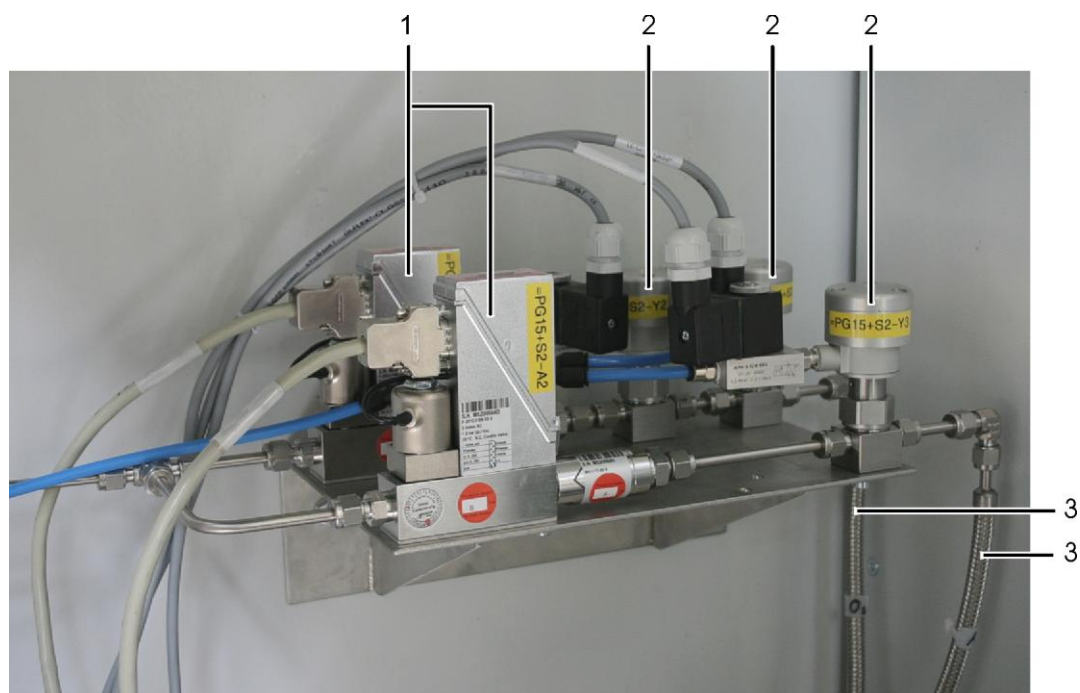


Fig. 7-17 Gas supply unit

- 1 - Flow controller
2 - Shut-off valves

- 3 - Gas influx

	Note!
	Before you begin the leak test, the following aids must be at hand: • Helium (gas bottle) as a test gas • Leak detector UL 200 (test equipment)

Procedure

- Switch the pump system off.
- Connect the leak detector UL 200:
 - Connect the leak detector to the intermediate vacuum system of one of the two fore vacuum pumps WAU 2001.
 - Switch the leak detector on and adjust it.
- Switch the pump system on (pump the chamber down): The required vacuum pressure for starting the test is approx. 1×10^{-4} mbar.
- Connect the helium gas supply (bottle) to the oxygen shut-off valves:
 - Connect the main oxygen supply (gas bottle).
 - Remove the oxygen connection at the shut-off valve, and connect the helium gas supply (gas bottle) directly to the oxygen shut-off valves.
 - Set the helium pressure to approx. 2-3 bar at the pressure reducer.
- Read the test result off the UL-200 display:
 - If the test value is $< 10^{-5}$ mbar l/s, a oxygen shut-off valve is tight.
 - If the test value is $> 10^{-5}$ mbar l/s, a oxygen shut-off valve is defective and must be replaced.
- Finishing the test:
 - Vent the chamber.
 - Switch the pump system off.
 - Remove the leak detector UL-200, close the intermediate vacuum system.
 - Remove the helium supply from the shut-off valves, and reinstall the oxygen supply.

7.5.3.2 Checking the gas supply line for leaks (oxygen/argon)

This test is a quick leak check of the gas supply line (argon/oxygen). The following describes the procedure for testing the gas influx line. This procedure can be used to test the oxygen and argon influx lines.

Procedure

- Setting and noting the default setting:
 - Open the vacuum chamber, and switch the vacuum pump system on.
 - Select the screen page "Process". Note the actual gas flow for the oxygen or argon influx, displayed under "Pre-Treatment",.

When the vacuum chamber is open, the gas flow controller is at the zero position. A residual flow of approx. 3 to 10 sccm is usually shown. However, the actual gas flow is 0 sccm.



- Pumping the chamber down:
 - Close the vacuum chamber, and pump it down to a vacuum of approx. 8×10^{-4} mbar.
- Switching the oxygen gas flow on:
 - After the required vacuum has been reached, at the screen page "General Parameters" under "Test Pre-treatment" click on the button <Oxygen Flow> or <Argon Flow>.
 - Branch to the screen page "Process", and click on the entry field Oxygen or Argon. Enter a gas flow setpoint of 2000 sccm.
 - Observe the display of the actual gas flow. Wait until the setpoint 2000 sccm is reached.
- Start the leak test:
 - When the gas flow is approx. 2000 sccm, close the manual shut-off valve at the gas influx line (pressurized gas bottle).
 - Then note the display of the actual gas flow.
The gas influx is closed, so the actual flow of 2000 sccm must steadily decrease to the noted initial value of approx. 3 - 10 sccm.
- Result:
 - If the actual flow is less than 10 sccm, the line is OK.
 - If the actual flow is greater than 10 sccm, the line has a leak.
In this case, thoroughly check the oxygen influx line with the leak detector. Find the exact location of the leak and replace the defective parts of the influx line.
- Finishing the test:
 - In "General Parameters", terminate the gas influx (reset the button <Oxygen Flow> or <Argon Flow> from ON to OFF).
 - Reopen the manual shut-off valve of the pressurized gas bottle.
 - Vent the vacuum chamber.



7.5.4 Electrical equipment

	⚠ DANGER
	Beware of high voltage parts. Danger of fatal electrical shock when touching high voltage parts. Service work may only be performed on the electrical equipment when it has been disconnected from the electrical power supply. Switch off the effected components and secure them against being accidentally switched on again. Always use the grounding rod to discharge those components which have a residual voltage after they are switched off. <u>Important:</u> Even with the main switch in the off position, certain system components can be live. In particular, this is true for devices connected to the maintenance outlets. The primary side of the main system switch always carries voltage. Only trained personnel with experience in high-voltage systems is permitted to perform service work on the electrical equipment.

7.5.4.1 Checking the charge of the buffer batteries

Two buffer batteries back up the data and programs stored in the system's PLC. If the power fails, the stored data and programs are not lost.

A safety circuit monitors the charge of the buffer batteries. If the charge decreases, the following warning will appear at the monitor:

Warning: PLC S7-400 Buffer Battery low

Replace the batteries if this warning appears. In any case, they must be replaced every year. The buffer batteries are part of the automation device SIMATIC S7-400 in switching cabinet + E1. Refer to the manufacturer's documentation when replacing these batteries.

7.5.4.2 Replacing a buffer battery

	Note!
	When replacing the buffer battery, follow the instructions given in the OEM SIMATIC S7-400. Replace the batteries only when the system's power supply is switched on. If the power connection is interrupted, then a supply voltage must be applied to the battery supply before replacing the battery. Otherwise, the PLC's data will be lost.

7.5.4.3 Overall electrical inspection

The regular inspection of the electrical system and equipment

- is to be performed only by trained electricians.
- is to be repeated at regular intervals.
- is to be performed before commissioning (e.g. after enhancements or repairs).
- is to be performed only with appropriate measurement equipment and tools.

	Note!
	The end user has the entire responsibility for performing repeated tests of the electrical systems and equipment. The following tests are recommendations. They do not release the end user from his responsibility to perform the overall test. The results of the tests shall be documented in a test log.

Inspections

The electrical system and equipment are to be inspected, to determine if for example:

- there is obviously recognizable damage and deficiencies.
- the electrical system and equipment are capable of withstanding external influences at the installation site.
- the protective mechanisms preventing direct contact with electrical equipment are still in place.
- the protective mechanisms still comply with the set-up standards when touched indirectly. Note protective measures with and without ground wires.
- the overcurrent protective mechanisms are properly arranged, in accordance with the wire diameter.
- the overcurrent protective mechanisms are still in place and adjusted properly.
- required wiring diagrams, circuit labels, Software Guides, warning signs and other signs are still in place and accurate.
- changes comply with established standards, and are documented.

Trial testing and function testing

The electrical system and equipment are to be trial tested as follows:

- Check isolation monitoring equipment for proper functioning.
- Check protective equipment for proper functioning:
 - Emergency-stop/emergency-off pushbuttons and switches
 - Temperature monitoring systems
 - safety switches and current fault/ground fault protective switches

Measuring and determining measurement values

- Check equipment with ground wires for low-resistance continuity and the proper connection.
- For equipment without ground wires, measure the isolation resistance of the operating circuit relative to nearby metal parts.
- Check the isolation resistance of the electrical system and equipment.

7.5.4.4 Performing a function test of the signal lamps

Many process statuses and system conditions are indicated by signal lamps. For flawless system operation, all signal lamps must be checked regularly for proper functioning.

For the lamp test, refer to screen page "General Parameters" and make sure the system is connected to electrical power.

	Note!
	Defective signal lamps may only be repaired or replaced when the system is disconnected from electrical power. Only a trained electrician is permitted to perform the work.

Performing the lamp test

- Select the screen page "General Parameters".
- Click on the button <Signal Test>. The color of the control field changes from gray to yellow (from OFF to ON).
- Visually inspect all signal lamps at the operating panels. All signal lamps at the panels must turn on.
- Terminating the lamp test: Reset <Signal Test> from ON to OFF.

7.5.5 Cleaning the high-vacuum valve - V1

	⚠ Warning
	Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal and metal oxide dusts which are a hazard to your health. Always avoid creating dust during cleaning. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.

	⚠ Warning
	Danger of contusions There is the danger of contusions: • When lowering the caps of the high-vacuum valves • At the vacuum flaps Perform cleaning and service work only when the relevant opening and closing control valves are secured against forbidden access. Before closing the lifting valve V1 and the valve flaps, make sure no persons are located near the open valve or the vacuum flaps.

	⚠ Warning
	Explosion hazard There is the danger of explosions when venting hot oil diffusion pumps. Before venting, allow the diffusion pump to cool down to less than 90° C. Check the diffusion pump's temperature on the corresponding monitor page.

	⚠ Warning
	Danger of burns There is the danger of skin burns from hot oil when working on a hot oil diffusion pump. Perform any and all work only when the pumps are cool, below 30° C.

	⚠ Caution
	Beware of hot surfaces. There is the danger of burns when touching the housings of fore vacuum pumps and oil diffusion pumps. Let these components cool down. Wear gloves and work garments with long sleeves. Do not place any combustible textiles (work garments, cloths, etc.) on the housings. Fire hazard!

	⚠ Caution
	Danger of damage to the oil diffusion pumps. Hard objects in the oil diffusion pumps can damage them. During cleaning, make sure no objects fall into the open valve duct of the diffusion pump.

Cleaning the vacuum valves refers to removing metal deposits and dust residue from the valve seats and lid seals of the high-vacuum valves V1 (diffusion pumps).

Prerequisite for cleaning the high-vacuum valve V1:

- Vent the diffusion pumps
- Open the lifting valves V1 via a manual control valve in an external service box (lifting the valve caps)

	Note!
	Strictly adhere to the following safety precautions and system conditions when cleaning the valves: • The pump system is switched off and the diffusion pumps are cold (room temperature). • The entire system is disconnected from electrical power (main switch in position OFF). • The system is secured against being unintentionally reconnected with electrical power. • The vacuum chamber is open (chamber vented). • The compressed air supply is not interrupted (required to open the vacuum valves).

Strictly adhere to the following cleaning procedure. Only qualified personnel is permitted to perform the cleaning.

7.5.5.1 Clearing and cleaning the area around the valve

All aluminum wire coils are placed on a portable wire coil frame. To access the V1 valves, you must first pull the wire coil frame away. The wire coils will remain on the frame. The valve caps can now be accessed.

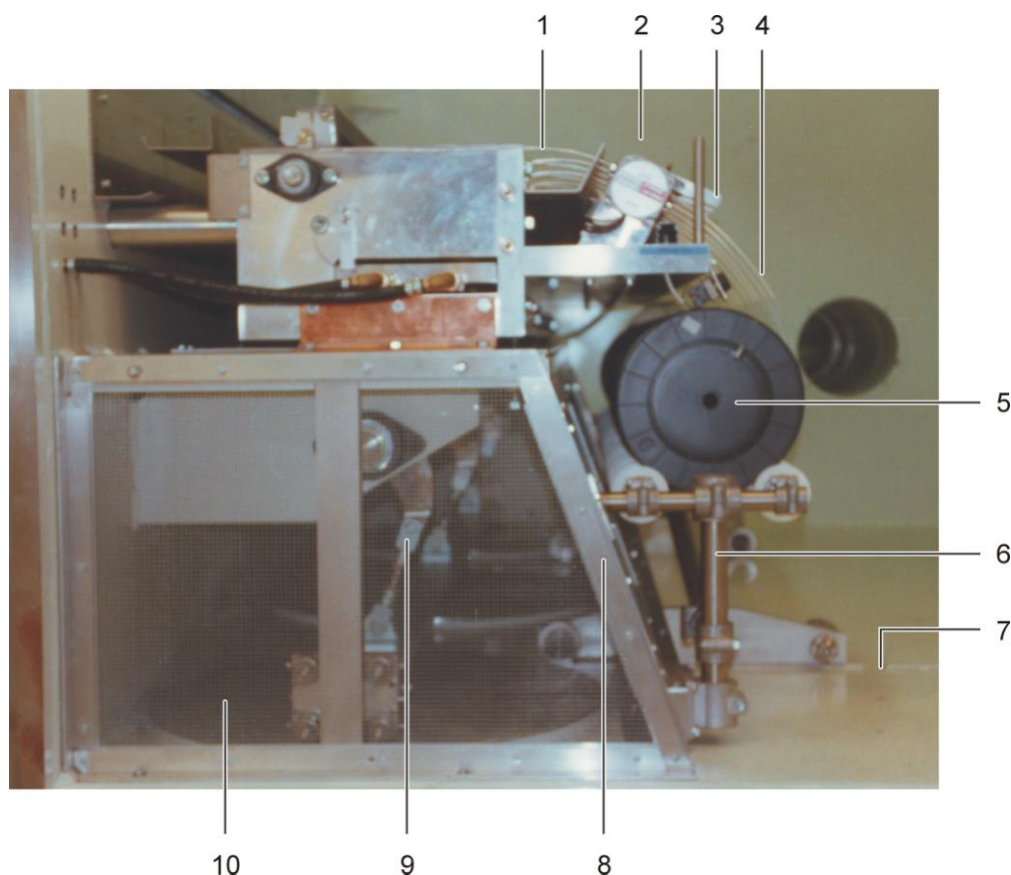


Fig. 7- 18 The area around high-vacuum valves V1

- | | |
|--------------------------------------|------------------------------------|
| 1 - Wire feed pipette | 6 - Wire coil frame |
| 2 - Protective lid (already removed) | 7 - Guide rail |
| 3 - Lever | 8 - Wire grid |
| 4 - Aluminum wire | 9 - Pneumatic driven lifting drive |
| 5 - Wire coil | 10 - Valve cap |

- Remove the aluminum wire from the evaporator boats:
Remove the protective lid, raise the lever for the wire feed pipettes, and pull wire out of the feed pipettes.
- Lock wire end:
Attach the wire end to the wire coil in such a way that the roll tension is maintained.
- Remove roll wire frame:
Unscrew all screws around the wire coil frame and grid.
- Swivel the wire coil frame away:
By hand, pull the wire coil frame on the guide rail away far enough to access the valve caps.
- Clean the area around valve V1:
The area around the valve and the valve caps are usually very dirty. For initial cleaning, keep the valve closed. This way, dirt particles will not enter the inside of the valves and pumps when you open the valve caps (V1). With the valve caps closed, use an industrial vacuum cleaner to remove deposits from the valve and area around it. Then clean all parts thoroughly, using a cloth and some cleanser.

7.5.5.2 Venting the diffusion pump

	Note!
	The nozzle system of the diffusion pumps can be shifted or damaged by abrupt venting of the pipe. Always vent the pipe slowly, in a controlled manner.

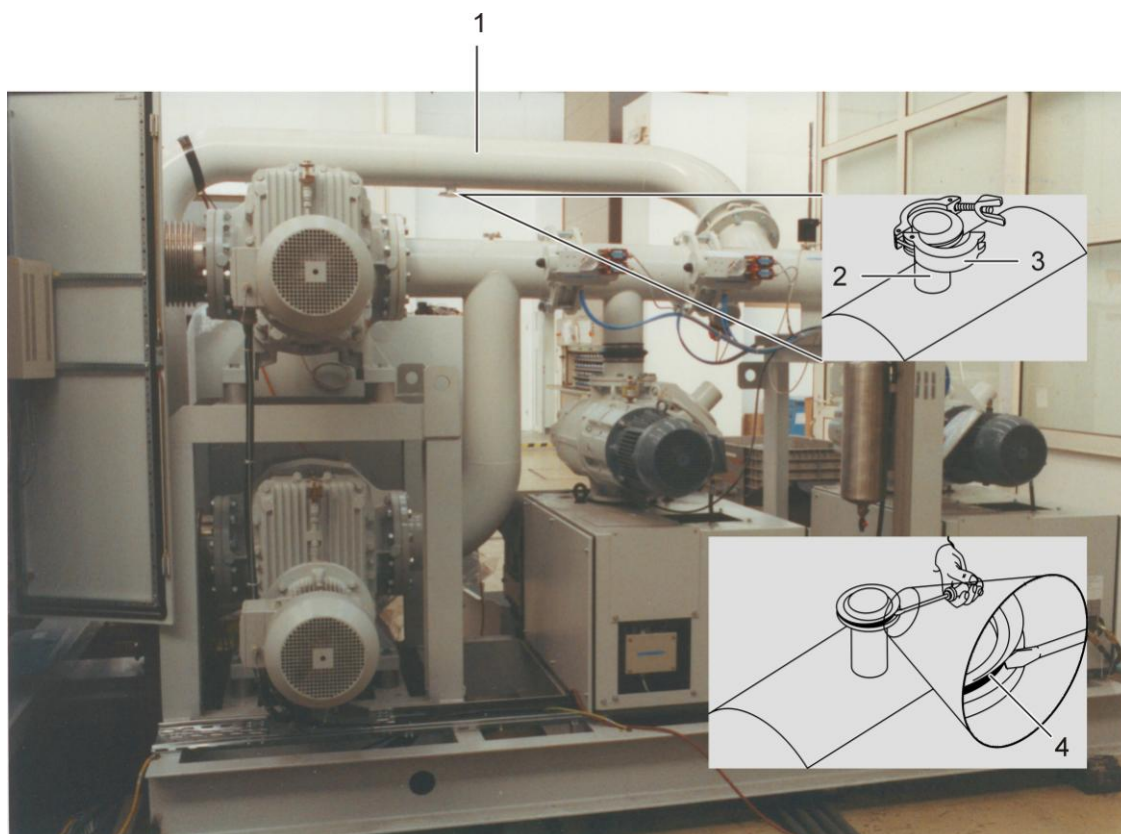


Fig. 7- 19 Venting the diffusion pumps

1 - Vacuum line
2 - Blanking flange

3 - Clamp
4 - O-ring

- Open one of the blanking flanges at the vacuum line to the diffusion pump. Loosen the clamp and remove it.

	Note!
	When you lift a blanking flange with a screwdriver, make sure you do not damage the sealing surface and the O-ring.

- After removing the clamp, slightly lift the blanking flange with a screwdriver, allowing air to slowly flow into the pipe.
- Continue venting until the sound of the entering air stops and the diffusion pumps are completely vented.
- After venting the diffusion pumps, remount the blanking flange tightly and close the clamp.

7.5.5.3 Open and clean the V1 valves

The V1 valves can be open and closed from a central location, using a manual control valve. The control valves for raising and lowering the valve caps are placed in a lockable service box.

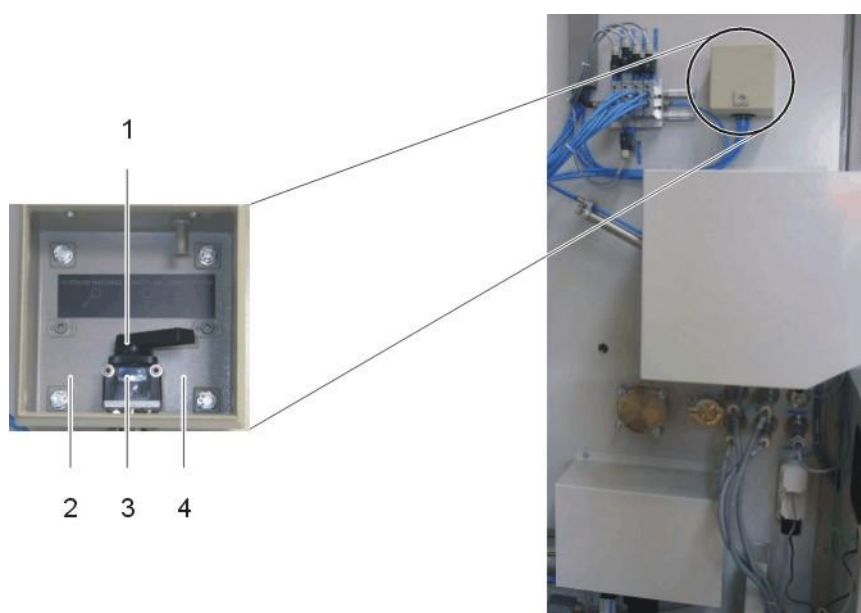


Fig. 7-20 Manual control valves in the service box

1 - Manual control valve
2 - Position OPEN

3 - Position SECURE
4 - Position CLOSED

- Lifting and securing valve cap V1:
 - Change the manual control valve directly from the position CLOSED to OPEN. All valve caps are pneumatically lifted.
 - Change the manual control valve from the position OPEN back to SECURE (center position). In case of a loss of pressure, the valve caps are secured against automatic closing.



Fig. 7-21 Opening high-vacuum valve V1

1 - Valve cap with lip-seal
2 - Valve plate seat

3 - Valve duct

- Cleaning valve V1:
 - Wipe the lip-seal at the valve cap with a dry cloth. Do not use aggressive solvents when cleaning the valve seal (see chapter 7.4). Do not rotate the valve cap when cleaning the lip-seal.
 - Check the seal for damage and make sure it is seated properly. Straighten it if it is bent or twisted. Replace it immediately if it is defective or damaged (see chapter 7.4).
 - Clean the valve plate seat at the bottom of the chamber.

7.5.5.4 Assembly

	Note!
	When the valve cap is lowered, parts lying around at the valve duct can damage the sealing surface. Make sure no objects are lying around the valve duct.

- Close high-vacuum valve V1:
Change the manual control valve from the position SECURE back to CLOSED. The valve caps are lowered.
- Close the service box for the manual control valves again.
- Remount the wire coil frame.
- Rethread the wire coils.

7.5.6 Cleaning the chamber flaps V3

	⚠ Warning Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal and metal oxide dusts which are a hazard to your health. Always avoid creating dust during cleaning. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.
	⚠ Warning Danger of contusions There is the danger of contusions: • When lowering the caps of the high-vacuum valves • At the vacuum flaps Perform cleaning and service work only when the relevant opening and closing control valves are secured against forbidden access. Before closing the lifting valve V1 and the valve flaps, make sure no persons are located near the open valve or the vacuum flaps.

Cleaning chamber flaps consists of removing metal deposits and dust residue from the vacuum flaps V3 located in the chamber wall.

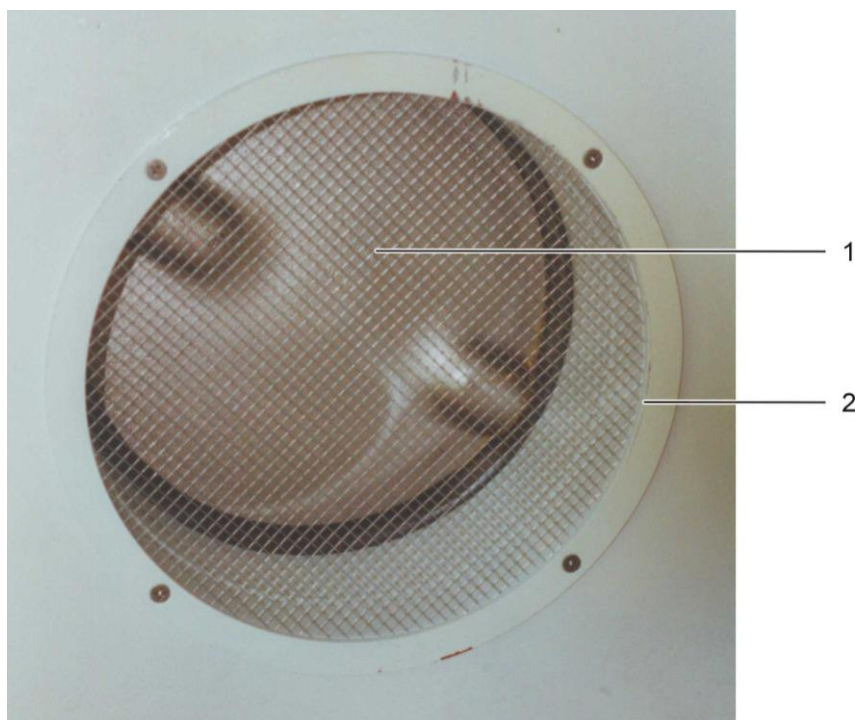


Fig. 7-22 Chamber flaps

1 - Chamber flaps

2 - Wire grid

Prerequisite for cleaning the chamber flaps V3:

- Venting the fore vacuum pumping station (fore vacuum line)
- Opening the chamber flaps V3 by means of manual control valves directly at the chamber flaps.

	Note!
	Strictly adhere to the following safety precautions and system conditions when cleaning the valves: • The pump system is switched off and the diffusion pumps are cold (room temperature). • The entire system is disconnected from electrical power (main switch in position OFF). • The system is secured against being unintentionally reconnected with electrical power. • The vacuum chamber is open (chamber vented). • The compressed air supply is not interrupted (required to open the vacuum valves).

The following cleaning steps absolutely must be adhered to. Only qualified personnel is permitted to perform the work.

7.5.6.1 Venting the vacuum pipe (fore vacuum pumps)

- Remove one of the blanking flange clamps at the fore vacuum line.
- Carefully open the blanking flange, so that air slowly flows into the pipe. Continue venting until the sound of the entering air stops.
- After venting, remount the blanking flange tightly (tighten the clamp).

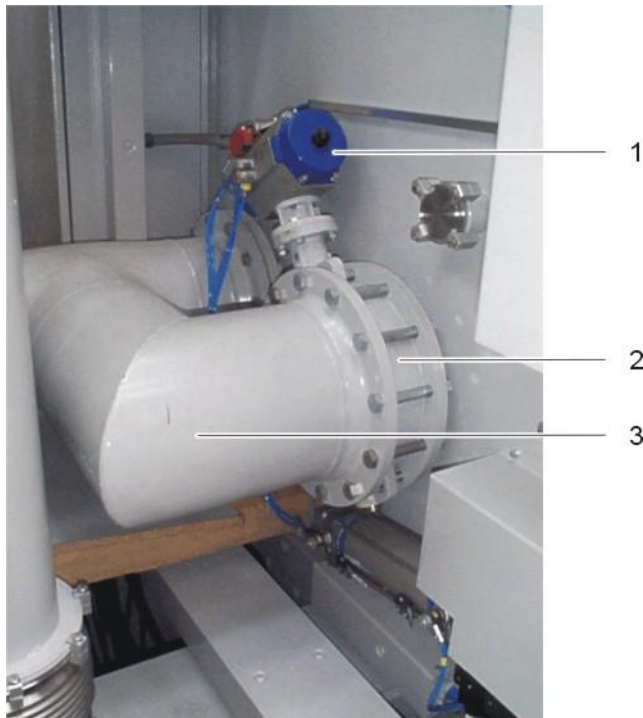
7.5.6.2 Opening chamber flaps V3

Fig. 7-23 Opening chamber flaps

1 - Pneumatic Drive

2 - Chamber flap

3 - Vacuum line

A manual control valve actuates the pneumatic drive which opens and closes the chamber flap.

- Press the adjusting lever at the manual control valve with a screwdriver.
- The chamber flap (V3) opens.

	⚠ Warning
	Overpressure, causing parts to fly There is the danger of injuries from flying parts when dismantling parts of the compressed air system or water distribution unit. Dismantle parts only when all pressure has been bled off the system. Before servicing the compressed air maintenance unit (e.g. replacing the filter cartridge): • Close the supply of compressed air (customer's side) and secure it against unintentional reopening. • Bleed the compressed air off the maintenance unit. • Released the compressed air at the compressed air storage tank.

- Remove both pneumatic hoses at the magnetic valve. Keep the screwdriver pressed until both hoses are removed.
- Close the compressed air valve at the compressed air supply unit.

7.5.6.3 Cleaning chamber flap V3

- Remove the protective grid from the interior of the chamber, and clean it (see Fig. 7- 22).
- Remove remaining film from the grid.
- Clean the area around the chamber flap with an industrial vacuum cleaner.
- Wipe the seal with a dry cloth and check it for damage. Note the cleaning instructions in chapter 7.4.

7.5.6.4 Assembly

- Mount the protective grid at the chamber flap.
- Close the chamber flap.
- Connect both pneumatic hoses. Then switch on the compressed air.
- The chamber flap automatically closes.



	Note !
	If the alarm message “Fault at vaccum valve V3” occurs at the operating monitor: • Change both pneumatic hoses at the magnetic valve.

7.5.7 Stroboscope service

	Note!
	Maintain the following system conditions when cleaning or opening the stroboscope flange: • Switch the pump system off and secure it against being switched on again. • Vent the vacuum chamber, and place the travelling part in its final position. • Protect the operating panels from dirt (e.g. cover them with foil).

A stroboscope window seals the tubular stroboscope in a vacuum. The chamber must be vented and open when you open the stroboscope window and clean the inside.



Fig. 7-24 Stroboscope window

1 - Plastic pane

2 - Hexagon socket screw

- Opening the stroboscope window:
Unscrew all hexagon socket screws, and remove the window.

	Note!
	The plastic stroboscope window is not scratch-resistant. The plastic pane could be scratched if you use mechanical aids, such as spatulas or wire brushes, or aggressive solvents. If the pane is scratched, you can polish it with grinding paste and a polishing wheel.

- Cleaning the tubular stroboscope:
 - Remove dirt and coating residue from the stroboscope with a wire brush.
 - Clean the O-ring and stroboscope flange. Note the information contained in chapter 7.4.
 - Clean both sides of the stroboscope window (plastic pane) with a common cleanser.
- Before you close the stroboscope window, check the sealing ring for damage. Note the information contained in chapter 7.4.
- Mounting the stroboscope window:

Put the stroboscope window in place and tighten the hexagon socket screws. Tighten them only until the sealing ring is slightly distorted. If you tighten them too much, the plastic pane could break.

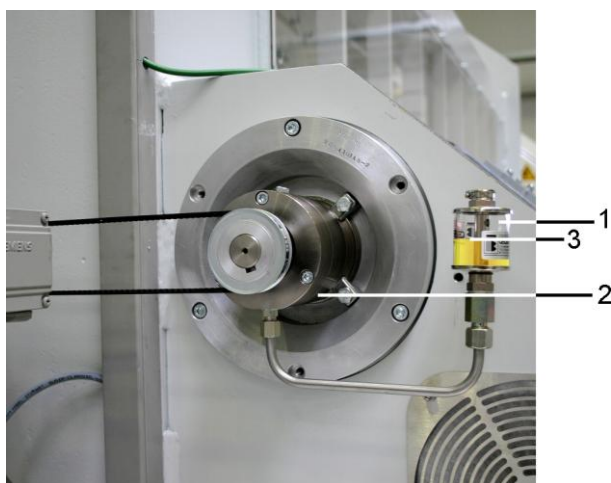


Fig. 7- 25 Oil sight glass for the vacuum rotary feedthrough of the stroboscope

- 1 - Oil sight glass
2 - Vacuum rotary feedthrough

- Check the vacuum rotary feedthrough:
Check the oil level in the sight glass only when the stroboscope is not moving. The oil level must be in the middle of the sight glass (use oil type: N 62).



Fig. 7-26 Stroboscope duct

1 - Covers

2 - Evaporator station

- Cleaning the stroboscope duct:
 - Remove the covers from the stroboscope duct.
 - Use a spatula to remove the aluminum deposits from the covers and in the duct.
 - Clean the duct with a vacuum cleaner. Grease all cleaned parts with graphite paste.
 - After cleaning, mount all covers.

7.5.8 Vacuum test

7.5.8.1 Preparing the system

- Clean the vacuum chamber and coating system parts. Do not leave any degassing materials in the chamber, such as films, hard paper cores etc.
- Close the chamber.
- Switch the pump system on and pump the system down.
Note: The system must be pumped down for at least 12 hours.
- Select screen page "Vacuum". Observe and note the reached pressure level during the pump-down phase for the winding and coating chambers.
- Note the final pressure after pumping down for at least 12 hours.

7.5.8.2 Starting the pressure-increase measurement

- Select screen page "Leak Rate Test".
- Press function pushbutton <Start Leak Rate Test>.
 - All chamber valves close automatically. The pumping station and chamber are functionally separated.
 - The pumping station and cooling trap remain in operation.
 - The pressure begins to increase in the chamber.
 - The test time is set at 30 minutes. The PLC then automatically calculates the leak rate Q .
 - If you terminate the leak test prematurely by pressing **Stop Leak Rate Test**, the leak rate will be calculated.
 Note that the test is less precise if you shorten the leak test.

7.5.9 Winding system

	⚠ Warning Health hazard After a coating process, the interior walls of the vacuum system are contaminated with metal and metal oxide dusts which are a hazard to your health. Always avoid creating dust during cleaning. Always wear a type P2 fine dust mask and appropriate protective garments. Carefully read the safety data sheets of the process materials being used. Precisely adhere to the precautions specified in them.
	⚠ Warning Rotating drums There is the danger that rotating drums could pull in body parts. When cleaning the winding system, especially the coating drum, wear only tight clothing. Persons are not permitted to be located directly at the winding system when the pushbutton <film feed> is pressed. Do not place your hands or arms in the drum system while it is running. Another person must be on alert at the next emergency-off pushbutton.
	⚠ Warning Contusion hazard Injuries or contusions can be caused by • movements of the castor device. • axial movements of the rewinder and unwinder drums. Do not place your hands or arms between the winding tube and the pivot arm. Do not place your hands or arms between the bearing plate and the rewinder or unwinder. Persons are not permitted to be located directly at the winding system when the pushbuttons <pivot arm> and <axial motion> are pressed.



7.5.9.1 Checking and adding oil to the vacuum rotary feedthroughs

The oil level in the rotary feedthroughs is only accurate if the system is idling. During system operation, the oil level sinks.

Thus, only check add oil when the system is idling.

- Check the oil levels in the sight glasses of the vacuum rotary feedthroughs (see e.g. Fig. 7- 25): The oil level must be in the middle of the sight glass.
- When adding oil, use only the following types.

Sight glass positions at the system	Oil type
Unwinder	N 62
Rewinder	N 62
Stroboscope	N 62
Tension roll (cooling drum)	N 62 (Reniso KM)
Coating drum	Reniso KM (low temperature stable)

7.5.9.2 Cleaning the coating drum

	Note!
	When cleaning the coating drum, adhere to the following: • Clean the drum's surface only with glass paper with a graining of 1200 or Scotch Brite. Mechanical aids such as spatulas or wire brushes damage the drum's surface.

- Preselect the drum speed: Select screen page "Parameters". Under Winding, set the coating drum setpoint to at most 5 m/s.

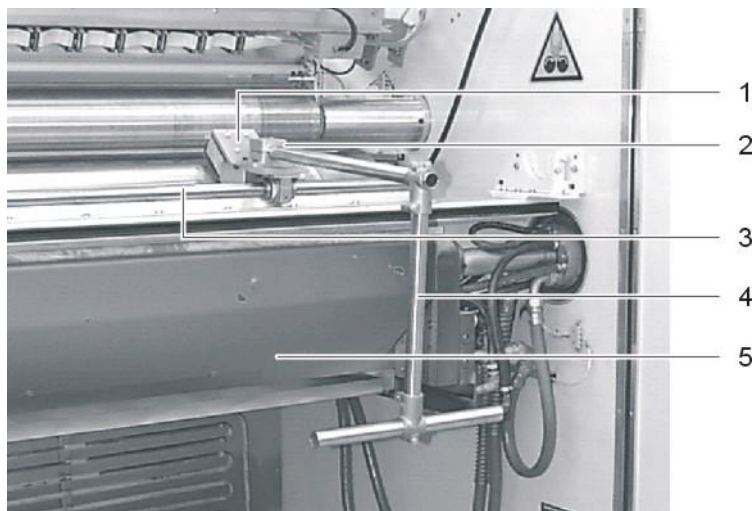


Fig. 7- 27 Cleaning the coating drum

- | | |
|-----------------------------------|---------------------|
| 1 - Cleaning holder | 4 - Cleaning device |
| 2 - Connection for vacuum cleaner | 5 - Coating window |
| 3 - Guide rail | |

- Preparing the cleaning device:
 - Clamp the polishing material or glass paper into the cleaning holder
 - Connect a powerful industrial vacuum cleaner to the tube of the cleaning device
 - Place the cleaning device in the stop and guide rail
- Start the coating drum in service mode:
 - Press the pushbutton ON at operating panel + R23. The coating drum accelerates to the set drum speed. The drum rotates opposite to the process travel direction (i.e. opposite direction of rotation).
 - Service mode of the coating drum can be stopped at any time by pressing the pushbutton OFF.
 - In order to revert to the proper direction of rotation for the next process, stop service mode by pressing the pushbutton OFF.
- Cleaning the drum's surface:

Press polishing material against the coating drum while it is rotating, from below. Run the polishing material over the rotating drum (entire width of the drum).

7.5.9.3 Cleaning the gap lock

	Note!
	Before you clean the gap lock, moving parts of the system must be made immobile. Clean and service the gap lock only in Service mode. Before you begin work, turn the service key switch to the Service position and remove the key.

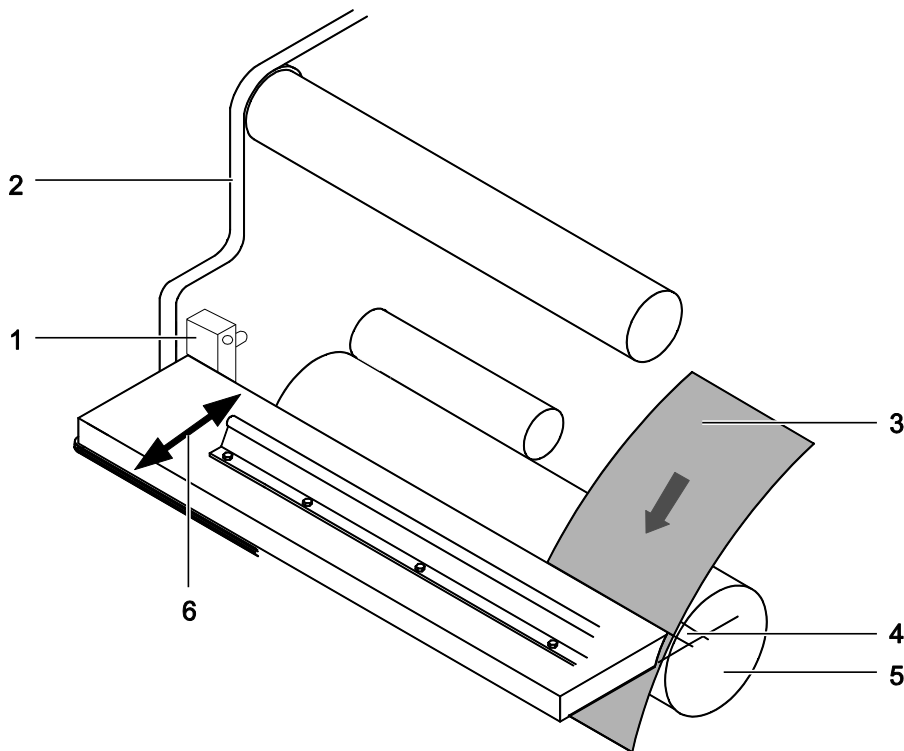


Fig. 7- 28 Gap lock

- 1 - Adjustment unit
- 2 - Bearing plate
- 3 - Plastic strip

- 4 - Gap lock
- 5 - Coating drum
- 6 - Lock jaw

The gap lock is the distance between the coating drum and a movable lock jaw, which is specified as part of the system's design. The gap (gap lock) is approx. 2 mm wide, and creates a vacuum gap between the coating chamber and winding chamber during operation. For cleaning, the gap can be opened to approx. 15 mm by using a movable lock jaw. The lock jaw is moved by means of adjustment screws at both the bearing plate and vacuum chamber plate. Once the gap lock is wide enough to insert a plastic strip, deposits and remaining film can be removed.

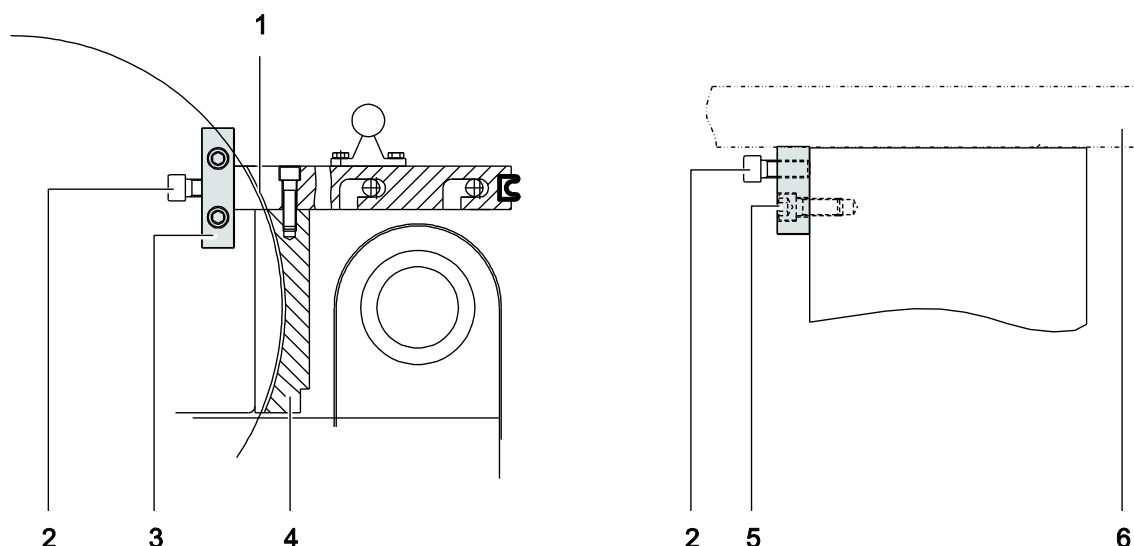


Fig. 7-29 Opening the lock jaw

- | | |
|---------------------|-------------------|
| 1 - Gap lock | 4 - Lock jaw |
| 2 - Forcing screw | 5 - Set screw |
| 3 - Adjustment unit | 6 - Bearing plate |

- Unscrew the set screw (5) on both sides.
- Screw in the forcing screw (2) with equal force on both sides. Press the lock jaw away from the coating drum, in parallel and uniformly. The maximum path is 15 mm.
- Insert a flexible plastic strip in the gap (approx. 15 mm). Move the strip back and forth in the gap, removing any residual film and deposits.
- Unscrew the forcing screw (2) on both sides. Use the set screw (5) to tighten the lock jaw to the coating drum with uniform pressure. Screw the set screw in until it stops, making the gap is narrow (2 mm).

7.5.9.4 Recording the friction characteristics

	⚠ Caution During operation, there is the hazard of damage to the chamber when uncovered clamping pieces in the winding shaft fly out. Make sure all clamping pieces of the shaft are completely covered by the tube.
---	---

- Preparing the winding shaft:
 - Place empty tubes on the rewinder and unwinder shafts.
 - Insert the shafts in the winding system, and close the couplings.
- Close the chamber and pump it down.
- Start the function **Friction recording**:
 - Select the screen page "Winding Parameters" (see Software Guide).
 - Click on the START button to start recording the friction characteristics.
Recording of the friction characteristics is interlocked with a pressure switching point. The buttons can only be activated when the chamber pressure is < 0.1 mbar. An error message is issued if the buttons are switched too early.
 - The friction characteristics for unwinder, rewinder and tension roll can be recorded together.
 - For rewinder and unwinder, you must record the friction characteristics for two directions of winding (counter-clockwise and clockwise).
 - The buttons are automatically reset when the recording is finished.

7.5.9.5 Checking and adjusting the V-belt tension (spreader rolls)

	Note!
	The V-belts are checked and adjusted with the chamber open, in the immediate vicinity of the winding system. The service personnel is thus located in the dangerous area of moving system parts (travelling part and winding system). Before working on the spreader rolls, turn the service key switch to the Service position. Remove the service key.

The coating and tension rolls indirectly drive the spreader rolls Fixbow and Varibow, by means of the V-belt system. The V-belts must maintain proper tension to avoid slippage. The drums would not be synchronized.

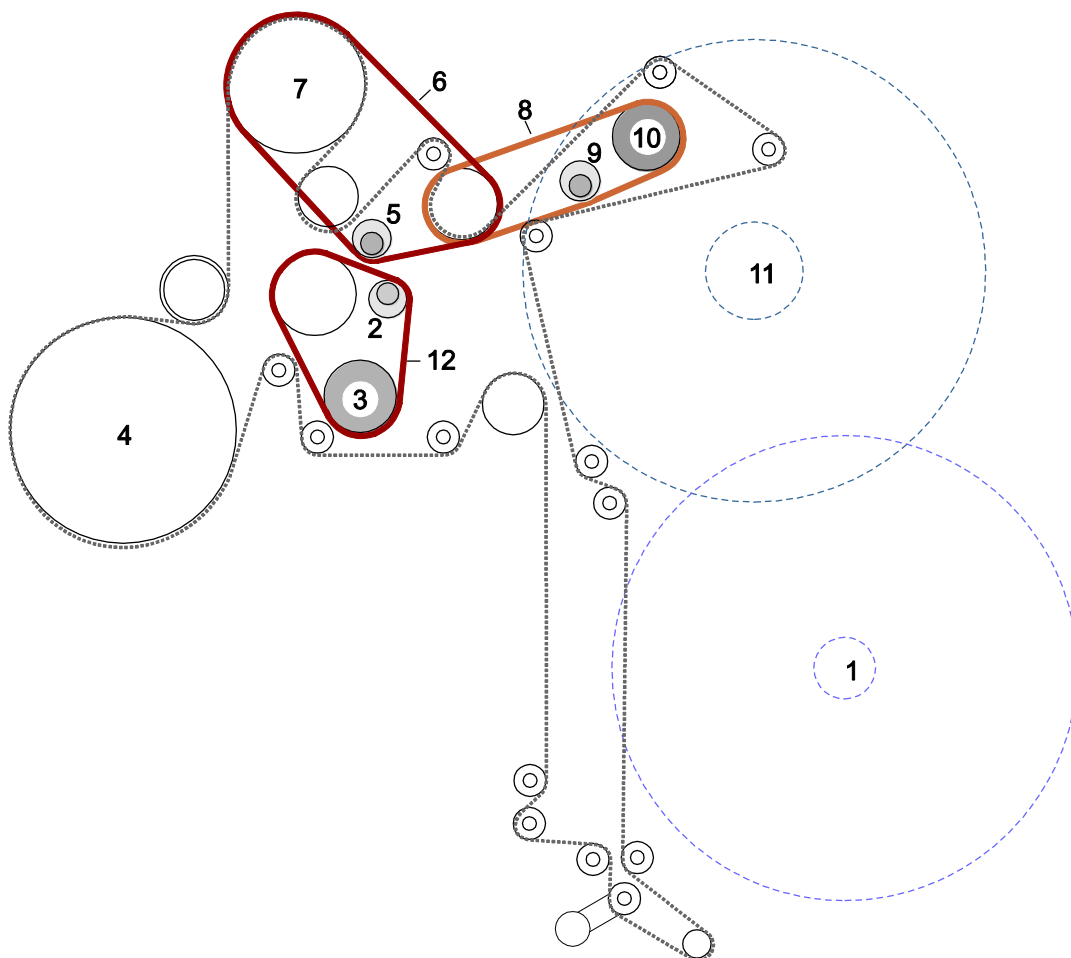


Fig. 7- 30 Arrangement of V-belts in the winding system

- | | |
|---------------------------|---------------------------------------|
| 1 - Unwinder | 7 - Tension roll or Post cooling drum |
| 2 - Cam tension roll 1 | 8 - V-belt |
| 3 - Spreader roll Varibow | 9 - Cam tension roll 3 |
| 4 - Coating drum | 10 - Spreader roll Fixbow |
| 5 - Cam tension roll 2 | 11 - Rewinder |
| 6 - V-belt | 12 - V-belt |



- Checking the V-belts:
 - Check all V-belts for damage and wear. Replace any defective belts.
 - By hand, check the tension of the V-belts. When moderate force is applied, the V-belt may only give way at most 1 cm. Otherwise, adjust the tension.
- Tensing the V-belt for the spreader roll Varibow:
 - Loosen the set screw for the cam tension roll (1) from the bearing plate.
 - Tense the V-belt:
Place the cotter pin in the proper bore and rotate the cam clockwise. The tensioning distance of the cam is limited. If the belt is too loose, remove it and take out some links.
 - Tighten the set screw for the cam tension roll (1) at the bearing plate.

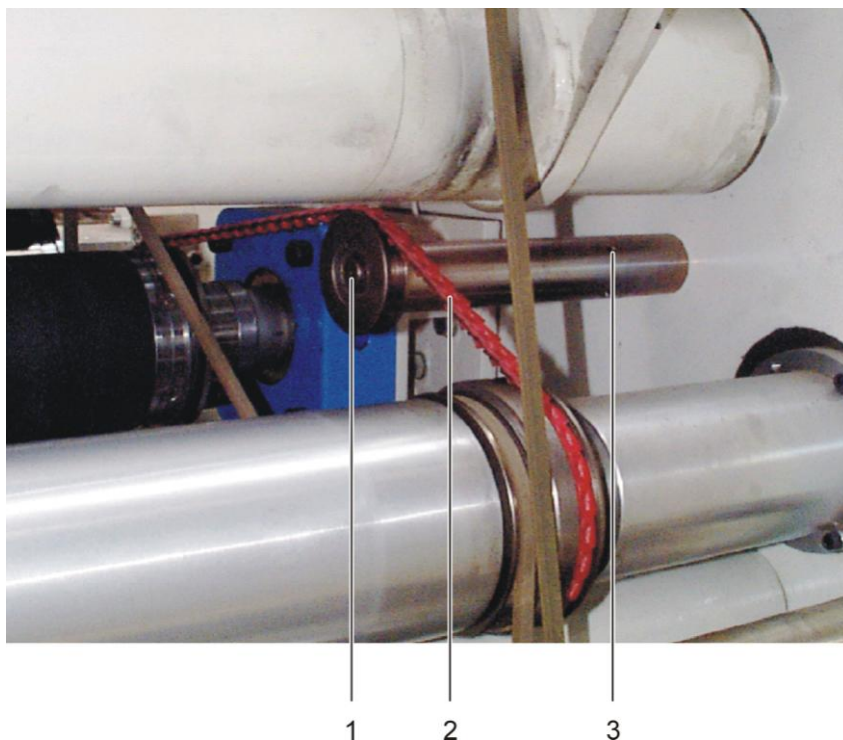


Fig. 7- 31 Tensing the V-belt

1 - Cam

2 - V-belt

3 - Bore for cotter pin

- Tensing the V-belts for the spreader roll Fixbow:
 - Loosen the cam tension roll: Loosen the set screws for the cam tension roll (2)/(3) from the pivot arm and bearing plate.
 - Tensing the V-belt:
Place the cotter pin in the proper bore and rotate the cam clockwise. The tensioning distance of the cam is limited. If the belt is too loose, remove it and take out some links (a detailed description is included in the OEM-folders: Fenner Drives, Power Twist Plus V-Belts).
 - Tighten the set screws for the cam tension roll (2)/(3) at the pivot arm and the bearing plate.

7.5.9.6 Checking and adjusting the threading belt tension

Check and adjust the tension of the threading belt only with the pivot arm in position *Contact*.

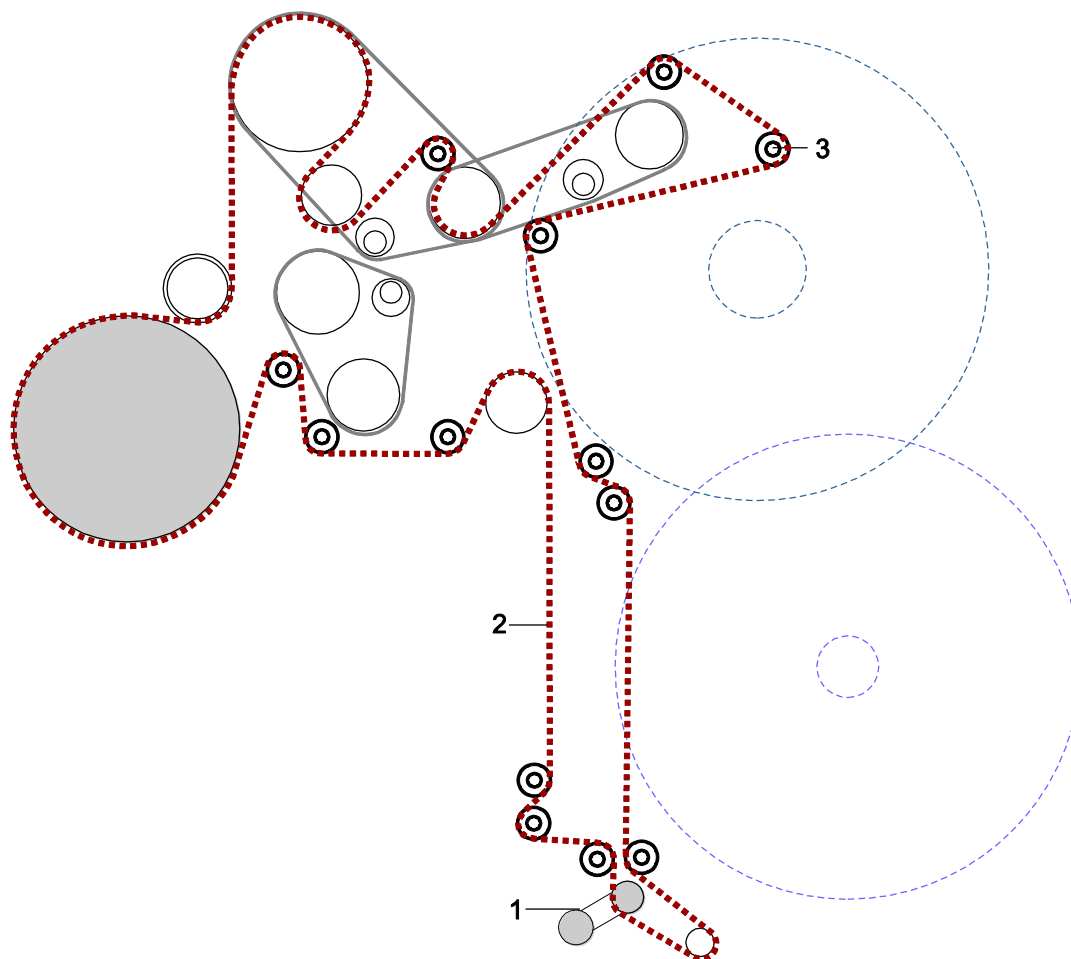


Fig. 7- 32 Course of the threading belt in the winding system

1 - Prestressing unit
2 - Threading belt

3 - Guide roller

- Checking the threading belt:
 - Check the threading belt for damage and wear. Replace any defective threading belts.
 - By hand, check the tension of the threading belt. When moderate force is applied, the threading belt may only give way at most 1 cm. Otherwise, adjust the tension.
- Tensing the threading belt
 - Check and adjust the prestressing unit:
The tension roll must always apply sufficient prestress to the threading belt.

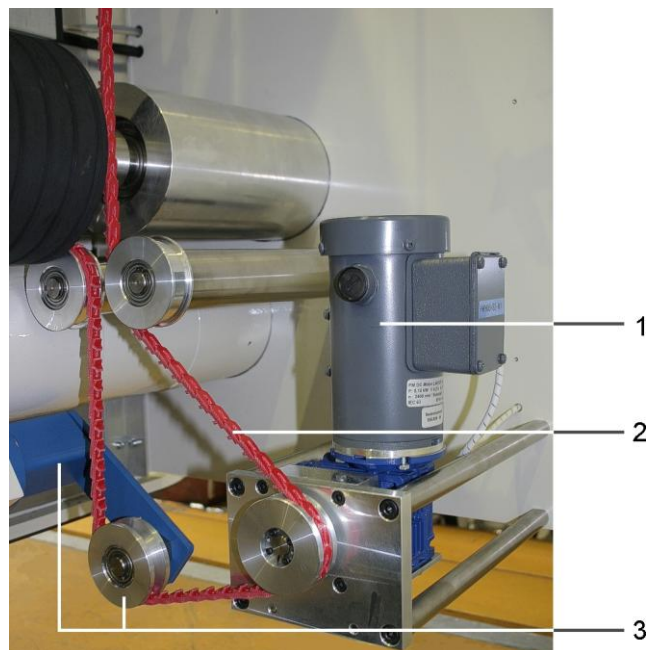


Fig. 7- 33 Threading belt

- 1 - Drive motor
- 2 - Threading belt

- 3 - Prestressing unit

7.5.10 Layer thickness measurement systems

	Note!
	For major maintenance and repair work, always use the OEM documentation.

7.5.10.1 Layer thickness measurement unit LMS-OD

For most process conditions, the layer thickness measurement system is basically maintenance-free. Particles and dirt on the probe lenses can reduce the intensity of the light transferred from the transmitter to the receiver. This would deteriorate the measurement results in the affected measurement channel.

You are therefore recommended to regularly clean the probe lenses and check the intensity of the transmitted light.

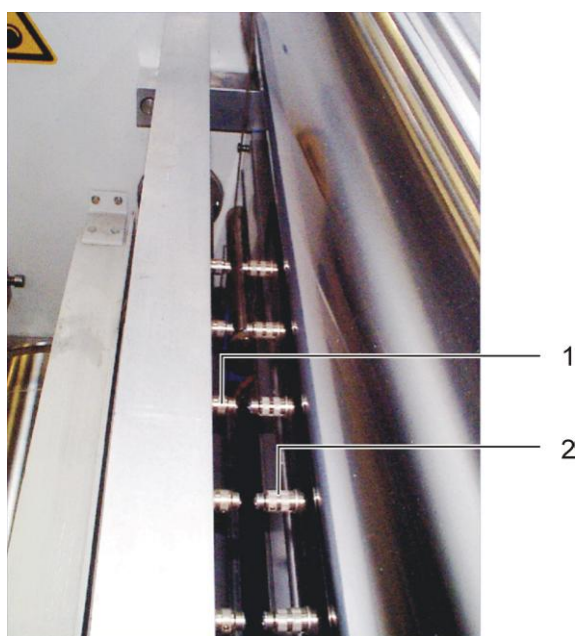


Fig. 7- 34 Layer thickness measurement unit

1 - Transmitter probe lens

2 - Receiver probe lens

Cleaning the probe lens

- Clean the probe lenses (transmitter and receiver) with soft, lint-free wool cloths. Remove stubborn dirt particles with spirits.

Checking the light intensity

	Note!
	Prerequisite for service work: • The system remains connected to electrical power. • The chamber is open, with the travelling part in its final position. • The service key switch of the system is activated.

The light intensity specifies how much of the light was transferred from the transmitter to the receiver. It can be reduced by dirty probe lenses or defective probe diodes.

- Select the screen page "LMS- Service". The diagram shows the actual light intensity per measurement channel as a bar graph.
- The optimum light intensity per measurement channel is between 5,000 and 20,000 counts. If the value is less than 5,000 counts, the measurement is imprecise. Possible causes are:
 - The probe lenses are dirty: clean them.
 - Transmitter or receiver diodes are defective: replace them.

Calibrating measurement systems

- The measurement system LMS-OD is calibrated automatically during operation, when the evaporator station is switched on.
However, you can calibrate it manually: select screen page "LMS-Service", button <LMS-Calibration>.

7.5.10.2 Resistance measurement system LMS-C

	Note!
	Do not damage the polished surface of the measurement drum. For cleaning, do not use mechanical aids such as spatulas or wire brushes, or aggressive solvents.

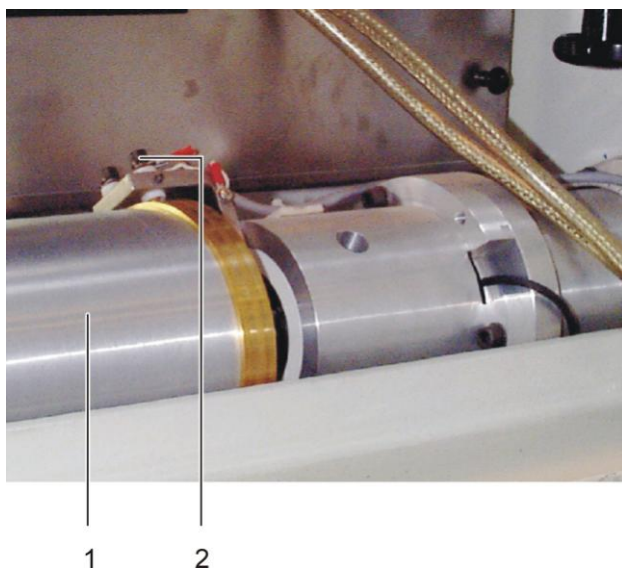


Fig. 7- 35 Layer thickness measurement unit

1 - Measurement drum

2 - Carbon contacts

- Cleaning the measurement drum:
Rotate the measurement drum by hand. Wipe dirt off with a soft wool cloth or Scotch Brite.
- Check and replace carbon contacts:
 - Unscrew the contact screw and carefully remove the carbon brush
 - Check the length and wear of the brush
 - Replace the carbon brush if it is very worn

7.5.11 Pretreatment station (TreatMag cathode)

	⚠ DANGER Forbidden for persons with pacemakers. Magnetic fields influence the functioning of pacemakers. This can disturb the heart beat or cause death. The system owner absolutely must ensure that no endangered persons enter the dangerous area near the cathodes. The minimum radius is 5 meters.
	⚠ DANGER Caution high voltage. There is lethal danger from residual voltage on the cathode. Before performing any work, discharge the cathode with the grounding rod, to eliminate residual voltage. Do not touch the cathode with your bare hands.
	⚠ Caution Strong magnetic fields The magnetic fields of the cathode can • Cause contusions when objects snap shut. • Make electronic equipment and magnetic cards useless. Do not use magnetized tools when working on the cathode. Lay credit cards, watches, floppy disks, etc., aside at a safe distance (approx. 5 m) from the cathode.

For disturbance-free glowing, you must maintain the cathode regularly.

Cathode maintenance includes:

- Cleaning and visually inspecting the cathode and the area around it.
- Replacing the target.
- Conditioning new targets.

7.5.11.1 Opening the cathode

- Secure the system functions:
 - Open the vacuum chamber, and move the travelling part to its final position.
 - Turn the service key switch at the operating panel to the Service position. Remove the key.

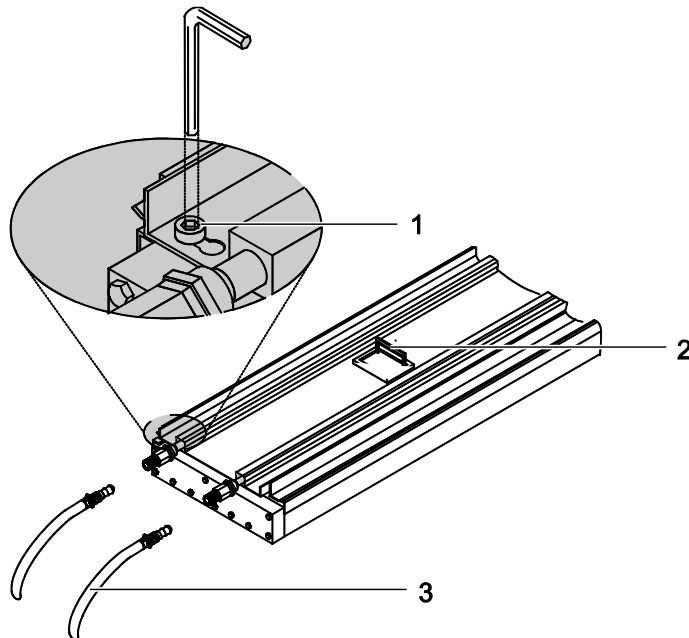


Fig. 7- 36 Opening the cathode

1 - Screw
2 - Handle

3 - Cooling water connection

- Open the lid of the cathode. Due to the size of the cathode, two persons are required.
 - Remove the couplings of the cooling water connection from the cathode.
 - Open the attachment screws from the lid of the cathode.
 - Remove the lid (grasp at the handles).



7.5.11.2 Cleaning the cathode parts

The following illustration shows the maintenance points at the glow cathode TREATMAG.

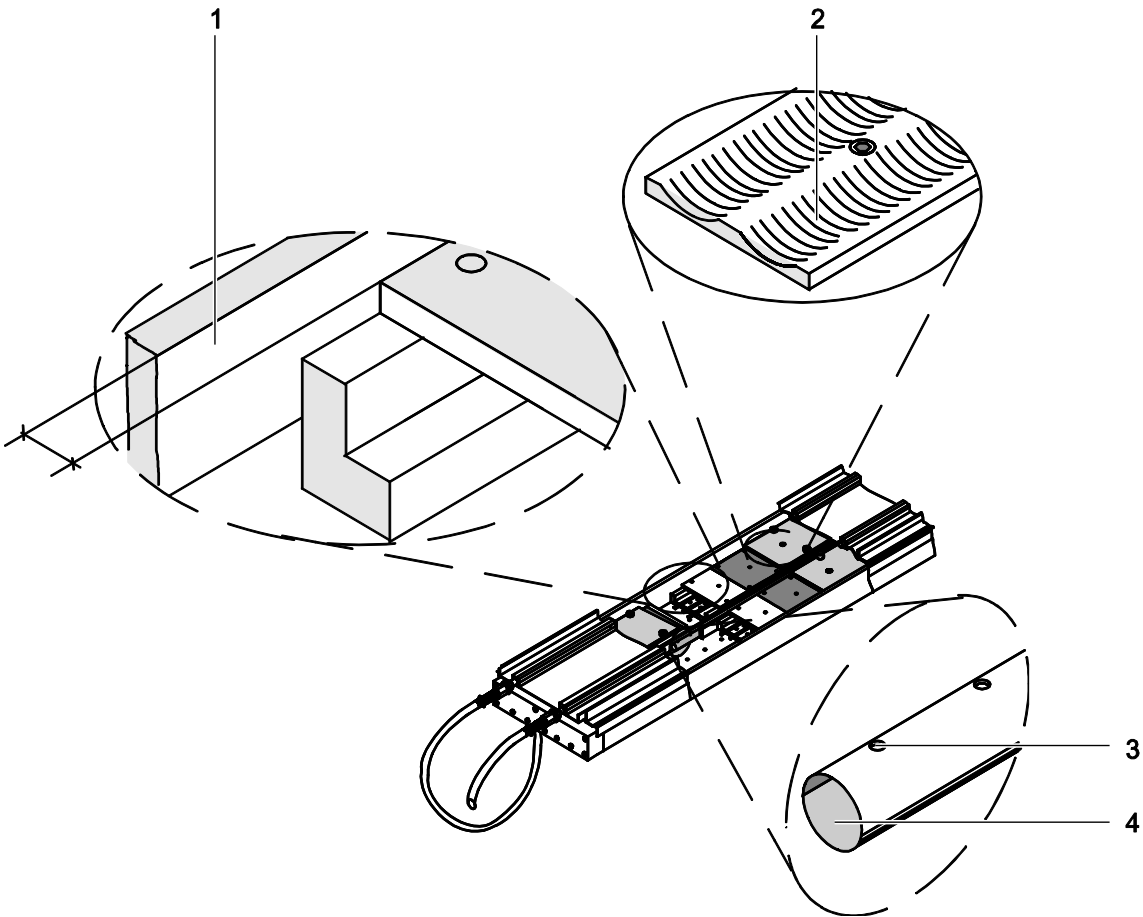


Fig. 7- 37 Maintenance points of the cathode

- | | |
|---|---------------------------|
| 1 - Dark room shielding | 3 - Gas inlet |
| 2 - Target surface (with sputter grooves) | 4 - Gas distribution tube |



Note!

- Do not touch the target with your bare hands.
When cleaning the target:
- Wear lint-free gloves.
 - Do not use aggressive solvents or cleansers.

Dark room shielding

- Burn-out and film residue in the dark room shielding can cause plasma arcing.
- Always remove dirt particles with an industrial vacuum cleaner.

Target surface

The target is consumed during glowing. A sputter groove is formed on the target's surface. You must continually check the depth of this groove and the number of burn-out spots around the cathode.

- Use a spatula or wire brush to remove burn-out and residue from the target's surface.
- Then clean the cathode and target thoroughly with an industrial vacuum cleaner.
- Measure the depth of the sputter groove. The maximum burn-in depth at the target's surface is 5 mm. Replace targets which are burned out or whose burn-in depth is > 5 mm.

Gas distribution tube

The openings (bores) at the gas tube distribute the process gases over the entire length of the cathode. Do not allow dirt to clog the openings; this would prevent the gas from flowing.

- Remove dirt and residue from the gas tube.
- Clean the openings in the gas tube with a needle.
- Clean the entire area around the cathode with an industrial vacuum cleaner.



7.5.11.3 Replacing the target

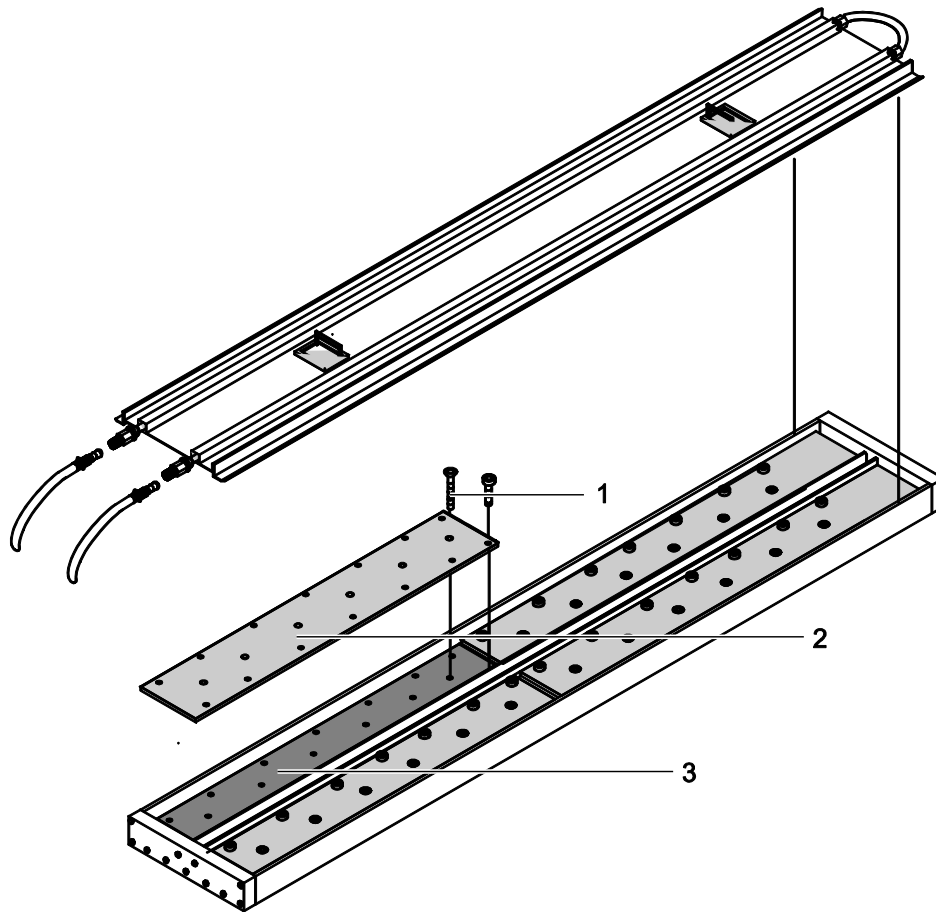


Fig. 7- 38 Replacing the target

1 - Attachment screw
2 - Target

3 - Graphite film

- Open the cathode, as described in chapter 7.5.11.1.
- Replace the target:
 - Unscrew the attachment screws from the target.
 - Remove a target segment.
 - Check the graphite film inside. Replace it if it is damaged.
 - Insert a new target and fasten it in place with the screws.
- Condition the new target (see chapter 7.5.11.4).

7.5.11.4 Instructions on conditioning a target

During glowing, oxidation on the target's surface will cause voltage discharges (plasma arcing).

Cathode conditioning consists of removing oxidation from the target's surface by incrementally increasing the cathode power. This is necessary in particular:

- When using new targets.
- After cleaning a cathode.
- When reusing a cathode after a long pause.

Basic system setting

Adhere to the following conditions for conditioning:

- Condition targets only in **Manual** mode.
- For the process gas, use only argon.
- Load the winding system with film.
Inserting the film is important, because enabling of power to the cathode is interlocked with the function **Start winding**. If the winding system is started without film, then the system for detecting film tears will issue an error message.

Procedure

- Select Manual mode (see chapter 6.2.1).
- Load the winding system and set the start parameters:
 - Insert foil and thread it.
 - Select the screen page "Parameters".
 - Under "Winding", set the film speed to 0.7 m/s.
 - Under "Pre-Treatment", set the cathode current setpoint to 1 A.
 - For the process gas, select argon (switch the button to <Select>).
 - Select the flow of argon to 300 sccm.
- Start conditioning:
 - Close the vacuum chamber.
 - Press pushbutton PUMP CHAMBER DOWN at operating panel +R02
 - Press pushbutton START WINDING at operating panel +R02.

Power is enabled to the cathode when the winding chamber pressure is < 1 mbar. Glowing starts, and the plasma is visible in the vacuum chamber as a bluish glow.



- Adjusting the conditioning parameters:
The goal of the adjustment is to incrementally increase the cathode current and gas flow. Based on the start parameters, the operating parameters must be reached with the plasma stabile. There should not be any voltage discharges (plasma arcing). Perform the next step only after the plasma has stabilized. The power must be reduced if there is too much arcing.
 - Increase the current setpoint in power increments: 1A/3A/5A/8A/10 A ... 24 A
 - Increase the gas flow in power increments: 300 sccm/400 sccm/ ... 1200 sccm

Length of time and end of conditioning

The length of time required for cathode conditioning depends on how conditioning proceeds. It depends on whether or not the plasma remains stabile during the power increases.

The following times are based on practical experience:

- Brand new targets: approx. 30 - 60 minutes.
- Used, cleaned targets approx. 10 - 30 minutes.

When the conditioning parameters are reached, the procedure can be terminated:

- Press the pushbutton STOP WINDING.
- Press the pushbutton VENT CHAMBER.
- Open the vacuum chamber.



8

Transport and Storage

The chapter Transport and Storage contains general guidelines on how to properly ship and store the TOPMET film coating system. It provides information on suitable ambient conditions whenever the system is to be stored for more than one half year. This chapter also includes important information on how to properly dispose the parts of the TOPMET film coating system and its operating media.

8.1 Storage

8.1.1 Shutdown

	Note!
	Only authorized, trained personnel with the maintenance or service qualification (cf. Chapter 2, Safety) is permitted to switch off the TOPMET film coating system.

The system should only be shut down in case of extended production interruptions (longer than one week), or to perform service or maintenance work.

Before an extended production interruption, the vacuum chamber should always be pumped down, to avoid contaminating the process chamber (mostly water vapor on the entire winding system).

Shutting down the system:

- Finish production
- Completely empty the system
- Pump for ten minutes
- Switch the system off, as described in chapter 6.4.2

Close the following supply lines:

- cooling water
- compressed air
- process gas(es)

The TOPMET film coating system is now shut down.

8.1.2 Packing

The packaging for the TOPMET film coating system depends on the way of transportation and the contract.

The system is partitioned into groups as follows:

- Media supply (water battery, etc.) packaged on a forklift-capable transport floor
- Travelling part and drum system packaged on a forklift-capable transport floor
- Process stations (evaporator station, glow station) packaged on a forklift-capable transport floor
- Vacuum chamber packaged on a forklift-capable transport floor
- Vacuum pumping stations packaged on a forklift-capable transport floor
- Electrical cabinets packaged on a forklift-capable transport floor
- System accessories sealed in plastic foil and packed in crates

The required packing symbols as shown in Fig. 8- 1 are printed on the packing material, and must be complied with during shipping.

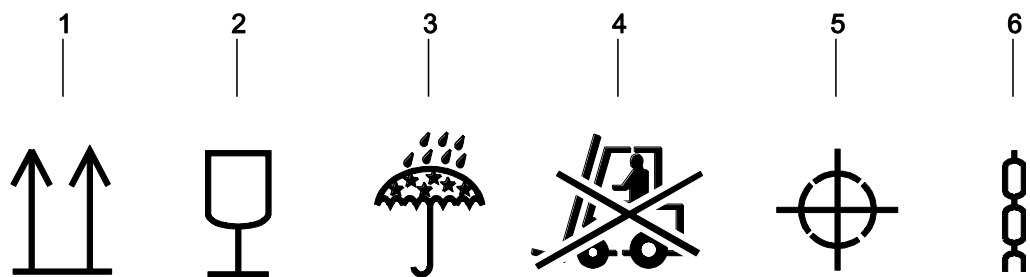


Fig. 8- 1

Packing symbols

- 1 - This side up
- 2 - Fragile, handle with care
- 3 - Keep dry

- 4 - Do not use fork lift truck here
- 5 - Center of gravity
- 6 - Lifting point

8.1.3 Storage conditions

The TOPMET film coating system may only be stored in a dry room with the following ambient conditions:

- Minimum storage temperature: 10 °C
- Maximum storage temperature: 70 °C
- Minimum humidity: 0 %
- Maximum humidity: 60 % (noncondensing)

8.1.4 Temporary Storage

The TOPMET film coating system may only be stored in the original packaging materials. The symbols in Fig. 8- 1 and the storage conditions stipulated in chapter 8.1.3 must be complied with.

8.1.5 Long-term storage

The TOPMET film coating system may only be stored in the original packaging materials. The symbols in Fig. 8- 1 and the storage conditions stipulated in chapter 8.1.3 must be complied with.

8.2 Transporting

 Warning	
	Beware of hovering loads. Improper handling of heavy parts can cause serious contusions of body parts, fractures and death. In particular, this applies for: • vacuum chambers and attached modules. • cathodes. • doors of the vacuum chambers. • pumping station components, in particular vacuum pumps and valves. • gate valves. • electrical cabinets. Persons are not permitted to stand under hovering loads. Only specially trained and authorized personnel is permitted to use carts, crane trucks and hoists.

The TOPMET film coating system may only be shipped in the original packaging materials. The shipping company must be specialized in hauling heavy and sensitive loads.

In-house transportation:

The TOPMET film coating system is designed such that the system modules can be transported to their destination. The modules contain suspension devices for transportation by crane.

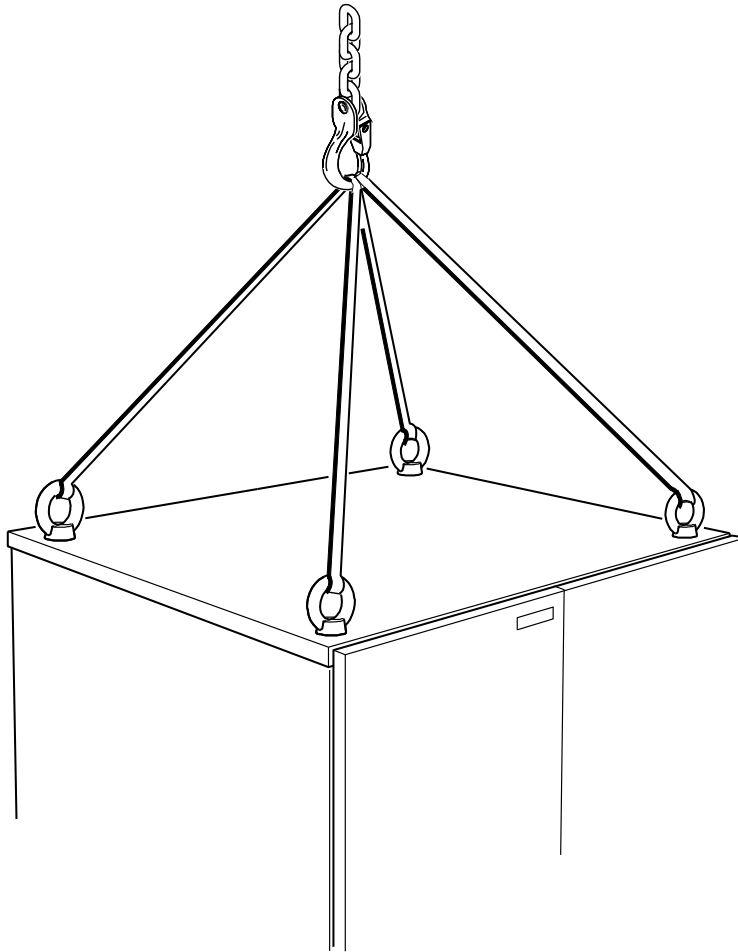


Fig. 8- 2 *Transportation by crane (example: electrical cabinet)*

8.3 Disposal

8.3.1 General Notes

	Note!
	Components and operating media must be disposed of in accordance with local regulations where the system is operated.

After dismantling the TOPMET film coating system, for disposal the components and operating media must be separated in the following categories:

- Operating media.
- Components filled with operating media or which had contact with operating media.
- Other components.

8.3.2 Operating media

	⚠ Warning
	Pump oil and oil mists contaminated with hazardous substances can be hazardous to your health. Skin or eye contact with pump oil and inhaling oil mists can be hazardous to your health. Wear protective clothing, oil-resistant gloves, goggles and a P3 respiratory protection mask when opening vacuum lines, vacuum valves or vacuum pumps containing harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.

	⚠ Caution The operating media of the vacuum pumps are harmful to the environment. Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil (even small amounts!) must be disposed of as hazardous waste.
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Operating media for the TOPMET film coating system can be returned to the supplier in unopened original containers.

Operating media in opened original containers and used operating media must be disposed of as hazardous waste in accordance with local regulations.

Consumables containing operating media (filter cartridges, cleaning rags, etc.) must also be disposed of as hazardous waste.

8.3.3 Components

All components of the TOPMET film coating system must be separated in the following classes, and recycled or disposed of, depending on the material:

- Aluminum
- Steel
- Stainless steel
- Precious metal
- Copper
- Glass
- Plastic
- Electronic or electrical parts
- Batteries

Components contaminated with operating media must be thoroughly cleaned before further processing.

8.3.4 Target materials

Precious metal or aluminum target materials can be recycled.

Other target materials must be disposed of in accordance with the safety data sheets and local regulations.



9

Alarm Messages and Warnings

The chapter "Alarm Messages and Warnings" contains a summary of alarms and warnings that may appear on the screen of the process monitoring system.

9.1 Dealing with alarm messages

Alarm messages and warnings indicate operational disturbances as well as critical process and operating situations.

Automatic reporting

A current disturbance and the corresponding alarm message are indicated as follows:

- The message appears on the monitor.
- The signal lamp FAILURE/RESET is lit.

	Note!
	When a new fault occurs at an electrical module, any existing alarm message is not automatically reset. After eliminating a fault at a module, press the pushbutton FAILURE/RESET again, to reset the alarm.

How to proceed when an alarm message appears:

- Select in the screenshot “Alarms” the alarm and press the button “Alarm Description” to get further information about trouble-shooting (see Fig. 9- 1).
 - Press button “Alarm Notices” to enter your own experiences according to an alarm.
 - To see alarm descriptions of alarm messages which are actually not in the alarm window use input field “Alarm No.”

Notice : Not all alarm numbers are used and not all alarms described are relevant for your plant.

[illegible]

Fig. 9- 3 Screenshot “Alarms”

- Acknowledge the alarm message by pressing the pushbutton FAILURE/RESET at operating panel +R02.
 - The color of the message text changes from red to dark red.
 - The signal lamp FAILURE/RESET turns on, because the error is still current.
- At the system control, select screen page "Alarms". The current alarm messages are listed in a table.
- Trouble-shoot the system and eliminate the alarm.
- Reset the error message:
 - After eliminating the error, press the pushbutton FAILURE/RESET:
 - The message disappears from the monitor.
 - The signal lamp FAILURE/RESET turns off.

9.2 Warnings and alarm messages

The control system includes an alarm reporting system for reliable operation of the TOPMET system. All main system components include a self-diagnostic function. Sensors and safety circuits sense critical conditions and operational irregularities. They are displayed on the screen as alarm messages and warnings.

The message texts for alarm messages and warnings are shown in red. There are two kinds of messages:

- Alarm messages:
System operation is interrupted. Operation can only resume after the cause of the disturbance is eliminated. The hooter sounds.
- Warnings:
System operation is not interrupted. The message is provided as information for operating personnel. The hooter sounds:
 - to take preventive action.
 - as an indication of possible improper operation.

Ignoring a warning may lead to an alarm message, stopping system operation.

9.3 Alarm list

The following is a list of all message texts for alarm messages and warnings.

No	Alarm Message	Alarm Description
A_0000	=AN01+E1-K12 Emergency Stop Button actuated	EMERGENCY-OFF button was pushed. Further plant operating: • Unlock EMERGENCY-OFF button • Press button "FAILURE/RESET" on operating field Important ! Take note of the security notes in plant documentation (user handbook) under section "Troubleshooting".
A_0001	=AN01+E1-K13 Safety Switch (Key) locked	The Service Key switch on the operating panel is switched to "SERVICE" position and is withdrawable. The interlock for moving plant parts is set to active. • Turn the key clockwise to deactivate the interlock
A_0002	=AN04+S2-S11 Rip Cord actuated	Reset with button on rip cord switch, press button "FAILURE/RESET" on operating panel.
A_0003	No Data from PLC S7-400 !	Possible reason: • No communication between monitor panel PC and PLC (S7). • Check cable connection • Check Hub =AS00+N11-A5, the green LINK LEDs of the connected ports must be on, the yellow DATA LEDs must flash if data is transmitted. (Compare LEDs TXD and RXD of CP443 in PLC rack) Switch power supply off and on • Restart PC • PLC is not running. • Restart the PLC system.
A_0004	=AM02+S1-F1 Compressed Air: Pressure is Low	Fluctuating air pressure supply or operating pressure not available • Check or set up operating pressure of at least 4 bar on the air pressure maintenance unit.
A_0005	PLC I/O Access Fault	Diagnostic with Step 7 possible, normally problem with ET200S module.
A_0006	PLC Programming Fault	Diagnostic with Step 7 possible.



No	Alarm Message	Alarm Description
A_0007	=AS21+G6 Touch panel: Fault Communication	No refresh of Profibus Watchdog signal from touch panel. Compare A_0840 To control the Pivot Arm use auxiliary buttons in Spreader Window of Visualization.
A_0010	After 30 min. Evacuating still Pressure Coating Chamber > 1mbar	Evacuating not possible, big leak or chamber not closed. Evacuating will be stopped.
A_0011	Pretreatment stopped! Pressure Winding Chamber > 0.1 mbar	All gas valves will be closed and Pinnacle switched off. • Check if material is much degassing • Check of dirt in Winding Chamber • Check for leaks
A_0012	Process stopped, Pressure higher than Evaporator Trigger	• Check Evaporator Trigger in Vacuum Parameters • Check High vacuum gauge function / dirt • Check water cycles for leaks • Check temperature of cold trap in coating chamber • Clean evaporator station and coating chamber • Check amount of the Gas inlet • Pay attention of outgassing materials
A_0020	=AA00+E1-Q2 Circuit Breaker for +N12 tripped	see fault number A_0200
A_0021	=AD01+E1-Fxx 230V-AC: Fuses are tripped	see fault number A_0031
A_0022	=AD02+E1-Fx 24V-DC: Fuses are tripped	see fault number A_0031
A_0023	=AD02+G6-F21/F22 24V Power Supply Spreader Rolls	see fault number A_0031
A_0024	=AD02+G6-Fxx 24VDC Fuse tripped	see fault number A_0031
A_0025	=AD02+G6-Q1 24V Power Supply Circuit Breaker tripped	see fault number A_0200
A_0026	=AD02+N11-Fxx 24V-DC: Fuses are tripped	see fault number A_0031
A_0027	=AD02+N12-Fxx 24V-DC: Fuses are tripped	see fault number A_0031
A_0030	=AD03+E1-Q1 24V-DC for Aluminum Feeder: Circuit Breaker tripped	see fault number A_0200
A_0031	=AD03+G4/5-Fxx 24VDC for Aluminum Feeder: Fuses tripped	• Switch on fuse. • If the breakdown recurs, check electrical circuit for a short circuit



No	Alarm Message	Alarm Description
A_0032	=AD03+E1-Q2 24V-DC for Aluminum Feeder #2: Circuit Breaker tripped	see fault number A_0200
A_0037	=AD03+E1-K2 Aluminum Feeder #2 24V-DC: Fault Contactor	see fault number A_0040
A_0040	=AD03+E1-K1 Aluminum Feeder 24V-DC: Fault Contactor	Contactor is in emergency off circuit. Check contactor: energized -> on, not energized -> off
A_0041	=WH05+G6-K21 Nip / Threading: Fault Contactor	see fault number A_0040
A_0050	=GA30+E1-E1 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0051	=GA30+G1-E1/2 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0052	=GA30+G3-E1 Overtemperature Cabinet / Condensate	• Temperature of cabinet is hotter than 7°K over the setpoint of cabinet cooling device (35°C) • Close cabinet door • Check cooling device setpoint is 35°C • Water leakage in cooling device • Check free condensate drain
A_0053	=GA30+G5-E1 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0054	=GA30+G6-E1/2 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0055	=GA30+N11-E1 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0056	=GA30+N12-E1 Overtemperature Cabinet / Condensate	see fault number A_0052
A_0060	=HB20+S1-F1 Water Fault: Coating Window + Shutter	see fault number A_0085
A_0061	=PC00+S1-F1 Water Fault: Ground bus bar #1	see fault number A_0065
A_0062	=PC00+S1-F2 Water Fault: Ground bus bar #2	see fault number A_0065
A_0063	=PC00+S1-F3 Water Fault: Ground bus bar #3	see fault number A_0065
A_0064	=PC00+S1-F4 Water Fault: Ground bus bar #4	see fault number A_0065



No	Alarm Message	Alarm Description
A_0065	=PC00+S1-F5 Water Fault: Evaporator 01 - 04	Not enough coolant supply is guaranteed. Water flow switch (WFS) indicates a defect. • Open window "Cooling Water" and locate defect. • Check differential pressure for coolant supply • Check dirt sieve in water inlet line • Check cool water connection for leakage • Check function of water flow switches. Use description in section MAINTENANCE INSTRUCTIONS.
A_0066	=PC00+S1-F6 Water Fault: Evaporator 05 - 08	see fault number A_0065
A_0067	=PC00+S1-F7 Water Fault: Evaporator 09 - 12	see fault number A_0065
A_0070	=PC00+S1-F8 Water Fault: Evaporator 13 - 16	see fault number A_0065
A_0071	=PC00+S1-F9 Water Fault: Evaporator 17 - 20	see fault number A_0065
A_0072	=PC00+S1-F10 Water Fault: Evaporator 21 - 24	see fault number A_0065
A_0073	=PC00+S1-F11 Water Fault: Evaporator 25 - 28	see fault number A_0065
A_0074	=PC00+S1-F12 Water Fault: Evaporator 29 - 32	see fault number A_0065
A_0075	=PC00+S1-F13 Water Fault: Evaporator 33 - 36	see fault number A_0065
A_0076	=PC00+S1-F14 Water Fault: Evaporator 37 - 40	see fault number A_0065
A_0077	=PC00+S1-F15 Water Fault: Evaporator 41 - 44	see fault number A_0065
A_0080	=PC00+S1-F16 Water Fault: Evaporator 45 - 48	see fault number A_0065
A_0081	=PC00+S1-F17 Water Fault: Evaporator 49 - 52	see fault number A_0065
A_0082	=PC00+S1-F18 Water Fault: Evaporator 53 - 56	see fault number A_0065
A_0083	=PC00+S1-F19 Water Fault: Evaporator Housing	see fault number A_0065

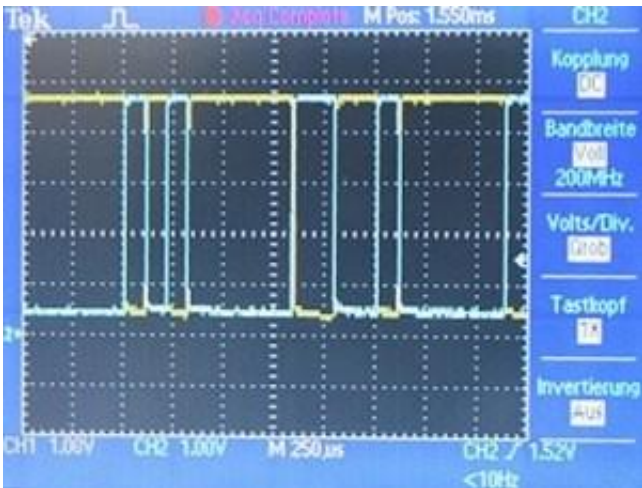


No	Alarm Message	Alarm Description
A_0084	=PC00+S1-F20 Water Fault: Pressure to low	- Check main water inlet - Check cooling water solenoid - Compare A_0065
A_0085	=PC00+S1-F21 Water Fault: Collecting Box for Debris #1	see fault number A_0065 Check water coupling inside chamber
A_0086	=PC00+S1-F22 Water Fault: Collecting Box for Debris #2	see fault number A_0085
A_0087	=PC00+S1-F23 Water Fault: Evaporator Frame/ Shielding Flap	see fault number A_0065
A_0090	=PC00+S1-F24 Water Fault: Chamber Cooling	see fault number A_0065
A_0091	=PC00+S1-F25 Water Fault: Edge Shielding #1	see fault number A_0085
A_0092	=PC00+S1-F26 Water Fault: Edge Shielding #2	see fault number A_0085
A_0093	=PC00+S1-F27 Water Fault: Frontal Shielding	see fault number A_0085
A_0094	=PC00+S1-F28 Water Fault:	see fault number A_0065
A_0100	=VM11+S1-P1 1000mbar-Switch Fault	Alarm is generated if chamber is 10 minutes vented and Switch is off or if pressure, measured with gauge =VM14, is smaller than 60 mbar and Switch is on. - Check function of 1000mbar Switch - Check function of gauge =VM14
A_0102	=VM12+S1-P1 Pirani Pressure Gauge Prevacuum	see fault number A_0104
A_0103	=VM12+S1-P1 MKS Pressure Gauge Prevacuum	see fault number A_0105
A_0104	=VM13+S1-P1 Pirani Pressure Gauge Diff Pump	Output Voltage of Pressure Gauge < 0.5 V - Check Filament Fracture - Check Cable Fracture - Check Connectors - Check Power Supply - Change Pressure Gauge



No	Alarm Message	Alarm Description
A_0105	=VM13+S1-P1 MKS Pressure Gauge Diffusion Pump	Three times no response from Pressure Gauge within 2 sec. • Check Cable Fracture • Check Connectors • Check Power Supply • Check addressing of gauge, see Window "MKS 901 Vacuum Gauge Info Screen" • Change Pressure Gauge
A_0106	=VM14+S1-P1 Pirani Pressure Gauge Winding Chamber	see fault number A_0104
A_0110	=VM15+S1-P1 Pirani Pressure Gauge Coating Chamber	see fault number A_0104
A_0111	=VM15+S1-P1 MKS Pressure Gauge Coating Chamber	see fault number A_0105
A_0112	=VM16+S1-P1 Pirani Pressure Gauge Pretreatment	see fault number A_0104
A_0114	=VM17+S1-P1 Pirani Pressure Gauge	see fault number A_0104
A_0116	=VM18+S1-P1 Pirani Pressure Gauge Backup Pump	see fault number A_0104
A_0117	=VM18+S1-P1 MKS Pressure Gauge Backup Pump	see fault number A_0105
A_0120	=VM19+S1-P1 Pirani Pressure Gauge Backup Pump	see fault number A_0104
A_0124	=VM21+S1-P1 Fault at Pressure Gauge	Defective measuring system. • Check measuring tube and cable connection • Check function of measuring tubes • Clean or replace tube if necessary • Check that the plug is firm on the measuring device and measuring head
A_0126	=VM12/=VM14, Pressure not plausible	Plant control system (PLC) carries out an automatic check of trigger point adjustment for vacuum_measuring system. • The fault message indicates an error function in the measuring tube VM12 or VM14 • Check function of measuring tubes • Clean or replace tube if necessary • Check the trigger contact on measuring tube



No	Alarm Message	Alarm Description
A_0127	=VM15/=VM14, Pressure not plausible	Plant control system (PLC) carries out an automatic check of trigger point adjustment for vacuum_measuring system. - The fault message indicates an error function in the measuring tube VM15 or VM14 - Check function of measuring tubes - Clean or replace tube if necessary - Check the trigger contact on measuring tubes
A_0130	Read out of MKS Gauges is switched off	Enable Pressure Reading in Window "Service Vacuum MKS Gauges"
A_0131	Communication Fault RS485 Bus MKS Gauges	- Check Gauges whether two with the same Address are set - Check RS485 Bus  Example of good signals of the two lines of the interface. A (+) = yellow, B (-) = blue
A_0133	=VB20+S1-Y1 Fault at Vacuum Valve V1	see fault number A_0134 Use maintenance compressed air valve to ensure proper function of V1, see maintenance instructions on CD-R
A_0134	=VB10+S1-Y2.1 Fault at Vacuum Valve V2.1	Valve is standing in an undefined or wrong position (OPEN/CLOSE). - Check valve position on plant - Check limit switch function for valve position
A_0135	=VB10+S1-Y2.2 Fault at Vacuum Valve V2.2	see fault number A_0134
A_0136	=VB10+S1-Y2.3 Fault at Vacuum Valve V2.3	see fault number A_0134
A_0137	=VB10+S1-Y2.4 Fault at Vacuum Valve V2.4	see fault number A_0134



No	Alarm Message	Alarm Description
A_0140	=VB30+S1-Y3.1 Fault at Vacuum Valve V3.1	see fault number A_0134
A_0141	=VB30+S1-Y3.2 Fault at Vacuum Valve V3.2	see fault number A_0134
A_0142	=VB30+S1-Y3.3 Fault at Vacuum Valve V3.3	see fault number A_0134
A_0143	=VB30+S1-Y3.4 Fault at Vacuum Valve V3.4	see fault number A_0134
A_0144	=VB30+S1-Y3.5 Fault at Vacuum Valve V3.5	see fault number A_0134
A_0150	=VB11+E1-Q1 Diffusion Pump: Circuit Breaker tripped	see fault number A_0200
A_0151	=VB11+E1-K1 Diffusion Pump: Fault Contactor	see fault number A_0040 Check 350°C Switch –F2
A_0153	=VB11+E1-F1/2 Diffusion Pump: Fault Heater	One of the three heaters carries no current. • Switch off pump state • Check for broken heating resistor • Check for even power supply on each phase
A_0154	=VB11+E1-F3/4 Diffusion Pump: Water Fault	see fault number A_0065
A_0155	=VB11+S1-E1 Diffusion Pump: Fault Temperature Sensor	Fault message occurs when the pre-adjusted cool-down time to <100°C or the heat-up time to >100°C is exceeded. • Check function of temperature sensor on diffusion pump with display in vacuum window. • Check function of diffusion pump heaters.
A_0160	=VB12+E1-Q1 Diffusion Pump: Circuit Breaker tripped	see fault number A_0200
A_0161	=VB12+E1-K1 Diffusion Pump: Fault Contactor	see fault number A_0040 Check 350°C Switch –F2
A_0163	=VB12+E1-F1/2 Diffusion Pump: Fault Heater	see fault number A_0153
A_0164	=VB12+E1-F3/4 Diffusion Pump: Water Fault	see fault number A_0065
A_0165	=VB12+S1-E1 Diffusion Pump: Fault Temperature Sensor	see fault number A_0155
A_0170	=VB13+E1-Q1 Diffusion Pump: Circuit Breaker tripped	see fault number A_0200



No	Alarm Message	Alarm Description
A_0171	=VB13+E1-K1 Diffusion Pump: Fault Contactor	see fault number A_0040 Check 350°C Switch –F2
A_0173	=VB13+E1-F1/2 Diffusion Pump: Fault Heater	see fault number A_0153
A_0174	=VB13+E1-F3/4 Diffusion Pump: Water Fault	see fault number A_0065
A_0175	=VB13+S1-E1 Diffusion Pump: Fault Temperature Sensor	see fault number A_0155
A_0180	=VB14+E1-Q1 Diffusion Pump: Circuit Breaker tripped	see fault number A_0200
A_0181	=VB14+E1-K1 Diffusion Pump: Fault Contactor	see fault number A_0040 Check 350°C Switch –F2
A_0183	=VB14+E1-F1/2 Diffusion Pump: Fault Heater	see fault number A_0153
A_0184	=VB14+E1-F3/4 Diffusion Pump: Water Fault	see fault number A_0065
A_0185	=VB14+S1-E1 Diffusion Pump: Fault Temperature Sensor	see fault number A_0155
A_0190	=VB15+E1-Q1 Diffusion Pump: Circuit Breaker tripped	see fault number A_0200
A_0191	=VB15+E1-K1 Diffusion Pump: Fault Contactor	see fault number A_0040 Check 350°C Switch –F2
A_0193	=VB15+E1-F1/2 Diffusion Pump: Fault Heater	see fault number A_0153
A_0194	=VB15+E1-F3/4 Diffusion Pump: Water Fault	see fault number A_0065
A_0195	=VB15+S1-E1 Diffusion Pump: Fault Temperature Sensor	see fault number A_0155
A_0200	=VC11+N12-Q1 Rotary Vane Pump: Circuit Breaker tripped	Circuit breaker is breakdown. • Check adjustment of Circuit Breaker • If the breakdown recurs, check the corresponding electric circuit for short circuit
A_0201	=VC11+N12-K1 Rotary Vane Pump: Fault Contactor	see fault number A_0040



No	Alarm Message	Alarm Description
A_0202	=VC11+S1-F1 Rotary Vane Pump: Overtemperature	Pump has exceeded operating temperature. • Allow pump to cool down • Check oil status • Check operating pressure and temperature of the coolant supply • Check function and setting of thermostat valve by comparing coolant inlet and outlet line temperature Remove possible fault by referring to the manufacturer's documentation.  Documentation for "Vacuum Pumps" see CD-R, Section "Supplier's Documentation".
A_0203	=VC11+S1-F2 Rotary Vane Pump: Overpressure in Oil case	• Change exhaust filter • Check exhaust line for blockage • Check sensor
A_0204	=VC11+S1-F11/F12 Rotary Vane Pump: No Inertgas Flow	The inert gas air flow on prevacuum pump is monitored by a gas flow controller directly on the gas ballast valve. It prevents for high concentration of Oxygene in the pump exhaust. • Check function of gas ballast valve and filter. Clean if necessary. Use manufacturer documentation • Check gas flow controller (meter) on pump.  Documentation for "Flow Meters" see CD-R, Section "Supplier's Documentation".
A_0210	=VC12+N12-Q1 Rotary Vane Pump: Circuit Breaker tripped	see fault number A_0200
A_0211	=VC12+N12-K1 Rotary Vane Pump: Fault Contactor	see fault number A_0040
A_0212	=VC12+S1-F1 Rotary Vane Pump: Overtemperature	see fault number A_0202
A_0213	=VC12+S1-F2 Rotary Vane Pump: Overpressure in Oil case	see fault number A_0203
A_0214	=VC12+S1-F11/F12 Rotary Vane Pump: No Inertgas Flow	see fault number A_0204
A_0220	=VC13+N12-Q1 Rotary Vane Pump: Circuit Breaker tripped	see fault number A_0200
A_0221	=VC13+N12-K1 Rotary Vane Pump: Fault Contactor	see fault number A_0040

No	Alarm Message	Alarm Description
A_0222	=VC13+S1-F1 Rotary Vane Pump: Overtemperature	see fault number A_0202
A_0223	=VC13+S1-F2 Rotary Vane Pump: Overpressure in Oil case	see fault number A_0203
A_0224	=VC13+S1-F11/F12 Rotary Vane Pump: No Inertgas Flow	see fault number A_0204
A_0230	=VC14+N12-Q1 Rotary Vane Pump: Circuit Breaker tripped	see fault number A_0200
A_0231	=VC14+N12-K1 Rotary Vane Pump: Fault Contactor	see fault number A_0040
A_0232	=VC14+S1-F1 Rotary Vane Pump: Overtemperature	see fault number A_0202
A_0233	=VC14+S1-F2 Rotary Vane Pump: Overpressure in Oil case	see fault number A_0203
A_0234	=VC14+S1-F11/F12 Rotary Vane Pump: No Inertgas Flow	see fault number A_0204
A_0240	=VC15+N12-Q1 Backup Pump: Circuit Breaker tripped	see fault number A_0200
A_0241	=VC15+N12-K1 Backup Pump: Fault Contactor	see fault number A_0040
A_0242	=VC15+S1-F1 Backup Pump: Overtemperature	see fault number A_0202
A_0243	=VC15+S1-F2 Backup Pump: Overpressure in Oil case	see fault number A_0203
A_0244	=VC15+S1-F11/F12 Backup Pump: No Inertgas Flow	see fault number A_0204
A_0250	=VC21+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0251	=VC21+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0252	=VC21+N12-F1 Roots Pump: Overtemperature	• Check motor cooling, free air flow • Check free move of the shaft • Check for leaks (too much gas in pump)
A_0253	=VC21+N12-G1 Inverter Fault	see fault number A_0384
A_0254	=VC22+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200



No	Alarm Message	Alarm Description
A_0255	=VC22+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0256	=VC22+N12-F1 Roots Pump: Overtemperature	see fault number A_0252
A_0257	=VC22+N12-G1 Inverter Fault	see fault number A_0384
A_0260	=VC23+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0261	=VC23+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0264	=VC24+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0265	=VC24+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0270	=VC25+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0271	=VC25+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0280	=VC31+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0281	=VC31+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0282	=VC31+N12-F1 Roots Pump: Overtemperature	see fault number A_0252
A_0283	=VC31+N12-G1 Inverter Fault	see fault number A_0384
A_0284	=VC32+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0285	=VC32+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0286	=VC32+N12-F1 Roots Pump: Overtemperature	see fault number A_0252
A_0287	=VC32+N12-G1 Inverter Fault	see fault number A_0384
A_0290	=VC33+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0291	=VC33+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0292	=VC33+N12-F1 Roots Pump: Overtemperature	see fault number A_0252



No	Alarm Message	Alarm Description
A_0293	=VC33+N12-G1 Inverter Fault	see fault number A_0384
A_0294	=VC34+N12-Q1 Roots Pump: Circuit Breaker tripped	see fault number A_0200
A_0295	=VC34+N12-K1 Roots Pump: Fault Contactor	see fault number A_0040
A_0296	=VC34+N12-F1 Roots Pump: Overtemperature	see fault number A_0252
A_0297	=VC34+N12-G1 Inverter Fault	see fault number A_0384
A_0300	=HA10+G6-Q1 Trolley Drive Motor: Circuit Breaker tripped	see fault number A_0200
A_0301	=HA10+G6-K3 Trolley Drive: Slow Contactor Fault	see fault number A_0040
A_0302	=HA10+G6-K4 Trolley Drive: Fast Contactor Fault	see fault number A_0040
A_0303	=HA10+S2-Bx Fault Limit Switches	Switching of limit switches of trolley drive is not plausible. • Check limit switches • Check for metal dirt in front of or beside sensors
A_0305	=WC20+S2 Spreader Roll Unwinder in restricted Area	See Window "Spreader Rolls" Restricted Area is defined by Supervisor, to avoid collision with guide rollers or tension on hydraulic hose. To Reset move other direction first to leave restricted area.
A_0307	=WE10+S2-M1 Axial Adjustment Timeout	Axial adjustment doesn't reach a limit switch • Check motor function • Check for blockage • Check limit switches
A_0311	=WF10+S2-B1 Nip Roller Function	Check whether nip roller reaches normal position (down), when chamber is open
A_0313	=WC10+G6-Q1 Pivot Arm Motor: Circuit Breaker tripped	see fault number A_0200
A_0314	=WC10+G6-K3 Pivot Arm: Slow Contactor Fault	see fault number A_0040
A_0315	=WC10+G6-K4 Pivot Arm: Fast Contactor Fault	see fault number A_0040
A_0320	=HB11+G6-Q1 Coating Window: Circuit Breaker tripped	see fault number A_0200
A_0321	=HB11+G6-K11 Coating Window Cleaning Position: Fault Contactor	see fault number A_0040



No	Alarm Message	Alarm Description
A_0322	=HB11+G6-K12 Coating Window Operation Position: Fault Contactor	see fault number A_0040
A_0324	=HB11+S2-B1/B2 Coating Window: Undefined Position	Coating window is in undefined position. - Before open/close the chamber make sure that the coating window is completely in working position. - Check limit switches
A_0330	=VF10+G6-Q1 Coldtrap Coating Chamber: Circuit Breaker tripped	see fault number A_0200
A_0331	=VF10+S2-A1 Coldtrap Coating Chamber: Not in Remote Mode	Operating mode LOCAL is selected. Set operating selection switch on Coldtrap panel to position REMOTE
A_0332	=VF10+S2-A1 Coldtrap Coating Chamber: Defrosting Time exceeded	Defrosting time for cold trap has been exceeded. Check functionality of the Coldtrap. For information use manufacturer's documentation.  Documentation for Coldtrap: see CD-R, Section "Supplier's Documentation".
A_0334	=VF11+G6-Q1 Coldtrap #2 Coating Chamber: Circuit Breaker tripped	see fault number A_0200
A_0335	=VF11+S2-A1 Coldtrap #2 Coating Chamber: Not in Remote Mode	see fault number A_0331
A_0336	=VF11+S2-A1 Coldtrap #2 Coating Chamber: Defrosting Time exceeded	see fault number A_0332
A_0340	=VH10+N12-Q1 Coldtrap Winding Chamber: Circuit Breaker tripped	see fault number A_0200
A_0341	=VH10+S1-A1 Coldtrap Winding Chamber: Not in Remote Mode	see fault number A_0331
A_0342	=VH10+S1-A1 Coldtrap Winding Chamber: Defrosting Time exceeded	see fault number A_0332
A_0343	=VH10+S1-A1 Lauda: Common Fault	Type and cause of the fault will be displayed by indicator lamps on the Coldtrap. Read the fault and remove possible fault by referring to the manufacturer's documentation  Documentation for "Lauda, Coldtrap" see CD-R, Section "Supplier's Documentation".
A_0350	=KA01+E1-Q2 Cooling and Heating Unit: Circuit Breaker Heater tripped	see fault number A_0200



No	Alarm Message	Alarm Description
A_0351	=KA01+G6-Q1 Cooling and Heating Unit: Circuit Breaker Compressor tripp	see fault number A_0200
A_0352	=KA01+S2-A1 Fault at Cooling and Heating Unit	Type and cause of the fault will be displayed directly on the display of the cooling and heating unit. Read the fault and remove possible fault by referring to the manufacturer's documentation  Documentation for "Cooling- and Heating unit" see CD-R, Section "Supplier's Documentation".
A_0355	=WG10+G6-K11 Bias Voltage: Contactor Fault	see fault number A_0040
A_0356	=WG10+G6-G1 Bias: Voltage too low	Actual Output Voltage lower than 70% of the Setpoint Voltage. - Check Power Supply - Check Cabling for short circuit
A_0380	=HB20+S2-B1/2 Shutter Limit Position not reached	Shutter is standing in an undefined or wrong position (OPEN/CLOSE). - Check free movement of shutter - Check limit switch function - Check drive cylinder
A_0381	=PH10+N12-Q1 Stroboscope Motor: Circuit Breaker tripped	see fault number A_0200
A_0382	=PD00+G4-A0 Aluminum Feeder: Fault Communication PAC-F	Control unit "PAC-F" indicates defect. Type and cause of the defect will be displayed directly on the component unit by a fault display. - Reset fault by pressing the reset button on PAC-F CPU =PD00+G5-A0.9, located behind a hole above the orange and red LED on the card - Read the fault number and remove possible fault by referring to the manufacturer's documentation.  Documentation for "PAC-F Controller" see CD-R, Section "Supplier's Documentation". IMPORTANT ! After clearing the fault perform an automatic restart. - Press button "FAILURE/RESET" on the operating field again



No	Alarm Message	Alarm Description
A_0383	=PD01..48+G4-G1 Aluminum Feeder: Axis not ramped up	Press button "FAILURE/RESET" on the operating field. - Switch off all feeders in window "Evaporator Selection" and switch them on again If fault message immediately reappears: - Check info on display on PAC-F controller - Shut down plant and turn main switch off - Turn plant on again. PLC will be restarted
A_0384	=PE10+E1-G1 Aluminum Feeder: Inverter Fault	Inverter „SIMOVERT-MICROMASTER“ indicates a defect. The cause of the defect is shown directly on the display of the unit Verify the fault number. Use manufacturer documentation to check the fault and remove it.  Documentation for "SIMOVERT-MICROMASTER" see CD-R, Section "Supplier's Documentation".
A_0386	=PE10+E1-Q1 Aluminum Feeder: Circuit Breaker tripped	see fault number A_0200
A_0387	=PE10+E1-K1 Aluminum Feeder: Fault Contactor	see fault number A_0040
A_0390	=PD00+G5-A0 Aluminum Feeder #2: Fault Communication PAC-F	see fault number A_0382
A_0391	=PD01..48+G5-G1 Aluminum Feeder #2: Axis not ramped up	see fault number A_0383
A_0392	! Formation of Hydrogen Sulfide (H ₂ S) possible !	Attention Hydrogen Sulfide is a unhealthy gas. It results if water reacts with hot Zinc Sulfide (ZnS). It smells like putrid eggs. Before entering the chamber make sure that it is enough vented.
A_0400	=PD01+G4-G1 Aluminum Feeder 01: Power not ready	Check power supply for stepper motor amplifier in cabinet +G4/5 (green LED).
A_0401	=PD02+G4-G1 Aluminum Feeder 02: Power not ready	see fault number A_0400
A_0402	=PD03+G4-G1 Aluminum Feeder 03: Power not ready	see fault number A_0400
A_0403	=PD04+G4-G1 Aluminum Feeder 04: Power not ready	see fault number A_0400
A_0404	=PD05+G4-G1 Aluminum Feeder 05: Power not ready	see fault number A_0400

No	Alarm Message	Alarm Description
A_0405	=PD06+G4-G1 Aluminum Feeder 06: Power not ready	see fault number A_0400
A_0406	=PD07+G4-G1 Aluminum Feeder 07: Power not ready	see fault number A_0400
A_0407	=PD08+G4-G1 Aluminum Feeder 08: Power not ready	see fault number A_0400
A_0410	=PD09+G4-G1 Aluminum Feeder 09: Power not ready	see fault number A_0400
A_0411	=PD10+G4-G1 Aluminum Feeder 10: Power not ready	see fault number A_0400
A_0412	=PD11+G4-G1 Aluminum Feeder 11: Power not ready	see fault number A_0400
A_0413	=PD12+G4-G1 Aluminum Feeder 12: Power not ready	see fault number A_0400
A_0414	=PD13+G4-G1 Aluminum Feeder 13: Power not ready	see fault number A_0400
A_0415	=PD14+G4-G1 Aluminum Feeder 14: Power not ready	see fault number A_0400
A_0416	=PD15+G4-G1 Aluminum Feeder 15: Power not ready	see fault number A_0400
A_0417	=PD16+G4-G1 Aluminum Feeder 16: Power not ready	see fault number A_0400
A_0420	=PD17+G4-G1 Aluminum Feeder 17: Power not ready	see fault number A_0400
A_0421	=PD18+G4-G1 Aluminum Feeder 18: Power not ready	see fault number A_0400
A_0422	=PD19+G4-G1 Aluminum Feeder 19: Power not ready	see fault number A_0400
A_0423	=PD20+G4-G1 Aluminum Feeder 20: Power not ready	see fault number A_0400
A_0424	=PD21+G4-G1 Aluminum Feeder 21: Power not ready	see fault number A_0400
A_0425	=PD22+G4-G1 Aluminum Feeder 22: Power not ready	see fault number A_0400
A_0426	=PD23+G4-G1 Aluminum Feeder 23: Power not ready	see fault number A_0400
A_0427	=PD24+G4-G1 Aluminum Feeder 24: Power not ready	see fault number A_0400



No	Alarm Message	Alarm Description
A_0430	=PD25+G4-G1 Aluminum Feeder 25: Power not ready	see fault number A_0400
A_0431	=PD26+G4-G1 Aluminum Feeder 26: Power not ready	see fault number A_0400
A_0432	=PD27+G4-G1 Aluminum Feeder 27: Power not ready	see fault number A_0400
A_0433	=PD28+G4-G1 Aluminum Feeder 28: Power not ready	see fault number A_0400
A_0434	=PD29+G4-G1 Aluminum Feeder 29: Power not ready	see fault number A_0400
A_0435	=PD30+G4-G1 Aluminum Feeder 30: Power not ready	see fault number A_0400
A_0436	=PD31+G4-G1 Aluminum Feeder 31: Power not ready	see fault number A_0400
A_0437	=PD32+G4-G1 Aluminum Feeder 32: Power not ready	see fault number A_0400
A_0440	=PD33+G5-G1 Aluminum Feeder 33: Power not ready	see fault number A_0400
A_0441	=PD34+G5-G1 Aluminum Feeder 34: Power not ready	see fault number A_0400
A_0442	=PD35+G5-G1 Aluminum Feeder 35: Power not ready	see fault number A_0400
A_0443	=PD36+G5-G1 Aluminum Feeder 36: Power not ready	see fault number A_0400
A_0444	=PD37+G5-G1 Aluminum Feeder 37: Power not ready	see fault number A_0400
A_0445	=PD38+G5-G1 Aluminum Feeder 38: Power not ready	see fault number A_0400
A_0446	=PD39+G5-G1 Aluminum Feeder 39: Power not ready	see fault number A_0400
A_0447	=PD40+G5-G1 Aluminum Feeder 40: Power not ready	see fault number A_0400
A_0450	=PD41+G5-G1 Aluminum Feeder 41: Power not ready	see fault number A_0400
A_0451	=PD42+G5-G1 Aluminum Feeder 42: Power not ready	see fault number A_0400
A_0452	=PD43+G5-G1 Aluminum Feeder 43: Power not ready	see fault number A_0400



No	Alarm Message	Alarm Description
A_0453	=PD44+G5-G1 Aluminum Feeder 44: Power not ready	see fault number A_0400
A_0454	=PD45+G5-G1 Aluminum Feeder 45: Power not ready	see fault number A_0400
A_0455	=PD46+G5-G1 Aluminum Feeder 46: Power not ready	see fault number A_0400
A_0456	=PD47+G5-G1 Aluminum Feeder 47: Power not ready	see fault number A_0400
A_0457	=PD48+G5-G1 Aluminum Feeder 48: Power not ready	see fault number A_0400
A_0460	=PD49+G5-G1 Aluminum Feeder 49: Power not ready	see fault number A_0400
A_0461	=PD50+G5-G1 Aluminum Feeder 50: Power not ready	see fault number A_0400
A_0462	=PD51+G5-G1 Aluminum Feeder 51: Power not ready	see fault number A_0400
A_0463	=PD52+G5-G1 Aluminum Feeder 52: Power not ready	see fault number A_0400
A_0464	=PD53+G5-G1 Aluminum Feeder 53: Power not ready	see fault number A_0400
A_0465	=PD54+G5-G1 Aluminum Feeder 54: Power not ready	see fault number A_0400
A_0466	=PD55+G5-G1 Aluminum Feeder 55: Power not ready	see fault number A_0400
A_0467	=PD56+G5-G1 Aluminum Feeder 56: Power not ready	see fault number A_0400
A_0470	=PC01+G1-G1 Overtemperature: Thyristor Controller 01	Heatsink of the Thyristor controller in cabinet +G1 is hotter than 80°C ! • Check temperature in cabinet • Check the function of the controller fan • Check free air intake of the controller Appears also if 24VDC supply or bus connection is lost
A_0471	=PC02+G1-G1 Overtemperature: Thyristor Controller 02	see fault number A_0470
A_0472	=PC03+G1-G1 Overtemperature: Thyristor Controller 03	see fault number A_0470
A_0473	=PC04+G1-G1 Overtemperature: Thyristor Controller 04	see fault number A_0470



No	Alarm Message	Alarm Description
A_0474	=PC05+G1-G1 Overtemperature: Thyristor Controller 05	see fault number A_0470
A_0475	=PC06+G1-G1 Overtemperature: Thyristor Controller 06	see fault number A_0470
A_0476	=PC07+G1-G1 Overtemperature: Thyristor Controller 07	see fault number A_0470
A_0477	=PC08+G1-G1 Overtemperature: Thyristor Controller 08	see fault number A_0470
A_0480	=PC09+G1-G1 Overtemperature: Thyristor Controller 09	see fault number A_0470
A_0481	=PC10+G1-G1 Overtemperature: Thyristor Controller 10	see fault number A_0470
A_0482	=PC11+G1-G1 Overtemperature: Thyristor Controller 11	see fault number A_0470
A_0483	=PC12+G1-G1 Overtemperature: Thyristor Controller 12	see fault number A_0470
A_0484	=PC13+G1-G1 Overtemperature: Thyristor Controller 13	see fault number A_0470
A_0485	=PC14+G1-G1 Overtemperature: Thyristor Controller 14	see fault number A_0470
A_0486	=PC15+G1-G1 Overtemperature: Thyristor Controller 15	see fault number A_0470
A_0487	=PC16+G1-G1 Overtemperature: Thyristor Controller 16	see fault number A_0470
A_0490	=PC17+G1-G1 Overtemperature: Thyristor Controller 17	see fault number A_0470
A_0491	=PC18+G1-G1 Overtemperature: Thyristor Controller 18	see fault number A_0470
A_0492	=PC19+G1-G1 Overtemperature: Thyristor Controller 19	see fault number A_0470
A_0493	=PC20+G1-G1 Overtemperature: Thyristor Controller 20	see fault number A_0470
A_0494	=PC21+G1-G1 Overtemperature: Thyristor Controller 21	see fault number A_0470
A_0495	=PC22+G1-G1 Overtemperature: Thyristor Controller 22	see fault number A_0470
A_0496	=PC23+G1-G1 Overtemperature: Thyristor Controller 23	see fault number A_0470



No	Alarm Message	Alarm Description
A_0497	=PC24+G1-G1 Overtemperature: Thyristor Controller 24	see fault number A_0470
A_0500	=PC25+G1-G1 Overtemperature: Thyristor Controller 25	see fault number A_0470
A_0501	=PC26+G1-G1 Overtemperature: Thyristor Controller 26	see fault number A_0470
A_0502	=PC27+G1-G1 Overtemperature: Thyristor Controller 27	see fault number A_0470
A_0503	=PC28+G1-G1 Overtemperature: Thyristor Controller 28	see fault number A_0470
A_0504	=PC29+G1-G1 Overtemperature: Thyristor Controller 29	see fault number A_0470
A_0505	=PC30+G1-G1 Overtemperature: Thyristor Controller 30	see fault number A_0470
A_0506	=PC31+G1-G1 Overtemperature: Thyristor Controller 31	see fault number A_0470
A_0507	=PC32+G1-G1 Overtemperature: Thyristor Controller 32	see fault number A_0470
A_0510	=PC33+G1-G1 Overtemperature: Thyristor Controller 33	see fault number A_0470
A_0511	=PC34+G1-G1 Overtemperature: Thyristor Controller 34	see fault number A_0470
A_0512	=PC35+G1-G1 Overtemperature: Thyristor Controller 35	see fault number A_0470
A_0513	=PC36+G1-G1 Overtemperature: Thyristor Controller 36	see fault number A_0470
A_0514	=PC37+G1-G1 Overtemperature: Thyristor Controller 37	see fault number A_0470
A_0515	=PC38+G1-G1 Overtemperature: Thyristor Controller 38	see fault number A_0470
A_0516	=PC39+G1-G1 Overtemperature: Thyristor Controller 39	see fault number A_0470
A_0517	=PC40+G1-G1 Overtemperature: Thyristor Controller 40	see fault number A_0470
A_0520	=PC41+G1-G1 Overtemperature: Thyristor Controller 41	see fault number A_0470
A_0521	=PC42+G1-G1 Overtemperature: Thyristor Controller 42	see fault number A_0470



No	Alarm Message	Alarm Description
A_0522	=PC43+G1-G1 Overtemperature: Thyristor Controller 43	see fault number A_0470
A_0523	=PC44+G1-G1 Overtemperature: Thyristor Controller 44	see fault number A_0470
A_0524	=PC45+G1-G1 Overtemperature: Thyristor Controller 45	see fault number A_0470
A_0525	=PC46+G1-G1 Overtemperature: Thyristor Controller 46	see fault number A_0470
A_0526	=PC47+G1-G1 Overtemperature: Thyristor Controller 47	see fault number A_0470
A_0527	=PC48+G1-G1 Overtemperature: Thyristor Controller 48	see fault number A_0470
A_0530	=PC49+G1-G1 Overtemperature: Thyristor Controller 49	see fault number A_0470
A_0531	=PC50+G1-G1 Overtemperature: Thyristor Controller 50	see fault number A_0470
A_0532	=PC41+G1-G1 Overtemperature: Thyristor Controller 41	see fault number A_0470
A_0533	=PC52+G1-G1 Overtemperature: Thyristor Controller 52	see fault number A_0470
A_0534	=PC53+G1-G1 Overtemperature: Thyristor Controller 53	see fault number A_0470
A_0535	=PC54+G1-G1 Overtemperature: Thyristor Controller 54	see fault number A_0470
A_0536	=PC55+G1-G1 Overtemperature: Thyristor Controller 55	see fault number A_0470
A_0537	=PC56+G1-G1 Overtemperature: Thyristor Controller 56	see fault number A_0470
A_0540	=PC01+G1-G1 Fuse Fault: Thyristor Controller 01	Fuse for thyristor unit is tripped • Check clamping of boat for short circuit by aluminum dirt • Remove boat and check electrical circuit for a short circuit with a ohmmeter • Check electrical connection at the transformers and the insulation of the copper bands • Check fuse in cabinet +G1 • If fuse is not tripped, check 230 VAC controller enable at connectors 5, 6
A_0541	=PC02+G1-G1 Fuse Fault: Thyristor Controller 02	see fault number A_0540



No	Alarm Message	Alarm Description
A_0542	=PC03+G1-G1 Fuse Fault: Thyristor Controller 03	see fault number A_0540
A_0543	=PC04+G1-G1 Fuse Fault: Thyristor Controller 04	see fault number A_0540
A_0544	=PC05+G1-G1 Fuse Fault: Thyristor Controller 05	see fault number A_0540
A_0545	=PC06+G1-G1 Fuse Fault: Thyristor Controller 06	see fault number A_0540
A_0546	=PC07+G1-G1 Fuse Fault: Thyristor Controller 07	see fault number A_0540
A_0547	=PC08+G1-G1 Fuse Fault: Thyristor Controller 08	see fault number A_0540
A_0550	=PC09+G1-G1 Fuse Fault: Thyristor Controller 09	see fault number A_0540
A_0551	=PC10+G1-G1 Fuse Fault: Thyristor Controller 10	see fault number A_0540
A_0552	=PC11+G1-G1 Fuse Fault: Thyristor Controller 11	see fault number A_0540
A_0553	=PC12+G1-G1 Fuse Fault: Thyristor Controller 12	see fault number A_0540
A_0554	=PC13+G1-G1 Fuse Fault: Thyristor Controller 13	see fault number A_0540
A_0555	=PC14+G1-G1 Fuse Fault: Thyristor Controller 14	see fault number A_0540
A_0556	=PC15+G1-G1 Fuse Fault: Thyristor Controller 15	see fault number A_0540
A_0557	=PC16+G1-G1 Fuse Fault: Thyristor Controller 16	see fault number A_0540
A_0560	=PC17+G1-G1 Fuse Fault: Thyristor Controller 17	see fault number A_0540
A_0561	=PC18+G1-G1 Fuse Fault: Thyristor Controller 18	see fault number A_0540
A_0562	=PC19+G1-G1 Fuse Fault: Thyristor Controller 19	see fault number A_0540
A_0563	=PC20+G1-G1 Fuse Fault: Thyristor Controller 20	see fault number A_0540
A_0564	=PC21+G1-G1 Fuse Fault: Thyristor Controller 21	see fault number A_0540



No	Alarm Message	Alarm Description
A_0565	=PC22+G1-G1 Fuse Fault: Thyristor Controller 22	see fault number A_0540
A_0566	=PC23+G1-G1 Fuse Fault: Thyristor Controller 23	see fault number A_0540
A_0567	=PC24+G1-G1 Fuse Fault: Thyristor Controller 24	see fault number A_0540
A_0570	=PC25+G1-G1 Fuse Fault: Thyristor Controller 25	see fault number A_0540
A_0571	=PC26+G1-G1 Fuse Fault: Thyristor Controller 26	see fault number A_0540
A_0572	=PC27+G1-G1 Fuse Fault: Thyristor Controller 27	see fault number A_0540
A_0573	=PC28+G1-G1 Fuse Fault: Thyristor Controller 28	see fault number A_0540
A_0574	=PC29+G1-G1 Fuse Fault: Thyristor Controller 29	see fault number A_0540
A_0575	=PC30+G1-G1 Fuse Fault: Thyristor Controller 30	see fault number A_0540
A_0576	=PC31+G1-G1 Fuse Fault: Thyristor Controller 31	see fault number A_0540
A_0577	=PC32+G1-G1 Fuse Fault: Thyristor Controller 32	see fault number A_0540
A_0580	=PC33+G1-G1 Fuse Fault: Thyristor Controller 33	see fault number A_0540
A_0581	=PC34+G1-G1 Fuse Fault: Thyristor Controller 34	see fault number A_0540
A_0582	=PC35+G1-G1 Fuse Fault: Thyristor Controller 35	see fault number A_0540
A_0583	=PC36+G1-G1 Fuse Fault: Thyristor Controller 36	see fault number A_0540
A_0584	=PC37+G1-G1 Fuse Fault: Thyristor Controller 37	see fault number A_0540
A_0585	=PC38+G1-G1 Fuse Fault: Thyristor Controller 38	see fault number A_0540
A_0586	=PC39+G1-G1 Fuse Fault: Thyristor Controller 39	see fault number A_0540
A_0587	=PC40+G1-G1 Fuse Fault: Thyristor Controller 40	see fault number A_0540



No	Alarm Message	Alarm Description
A_0590	=PC41+G1-G1 Fuse Fault: Thyristor Controller 41	see fault number A_0540
A_0591	=PC42+G1-G1 Fuse Fault: Thyristor Controller 42	see fault number A_0540
A_0592	=PC43+G1-G1 Fuse Fault: Thyristor Controller 43	see fault number A_0540
A_0593	=PC44+G1-G1 Fuse Fault: Thyristor Controller 44	see fault number A_0540
A_0594	=PC45+G1-G1 Fuse Fault: Thyristor Controller 45	see fault number A_0540
A_0595	=PC46+G1-G1 Fuse Fault: Thyristor Controller 46	see fault number A_0540
A_0596	=PC47+G1-G1 Fuse Fault: Thyristor Controller 47	see fault number A_0540
A_0597	=PC48+G1-G1 Fuse Fault: Thyristor Controller 48	see fault number A_0540
A_0600	=PC49+G1-G1 Fuse Fault: Thyristor Controller 49	see fault number A_0540
A_0601	=PC50+G1-G1 Fuse Fault: Thyristor Controller 50	see fault number A_0540
A_0602	=PC51+G1-G1 Fuse Fault: Thyristor Controller 51	see fault number A_0540
A_0603	=PC52+G1-G1 Fuse Fault: Thyristor Controller 52	see fault number A_0540
A_0604	=PC53+G1-G1 Fuse Fault: Thyristor Controller 53	see fault number A_0540
A_0605	=PC54+G1-G1 Fuse Fault: Thyristor Controller 54	see fault number A_0540
A_0606	=PC55+G1-G1 Fuse Fault: Thyristor Controller 55	see fault number A_0540
A_0607	=PC56+G1-G1 Fuse Fault: Thyristor Controller 56	see fault number A_0540
A_0610	=PG15+E1-Q1 Power Supply Pretreatment: Circuit Breaker tripped	see fault number A_0200
A_0611	=PG15+E1-K1 Power Supply Pretreatment: Contactor Fault	see fault number A_0040



No	Alarm Message	Alarm Description
A_0612	=PG15+S2-G1 Power Supply Pretreatment not ready	- Check Interlock bridges at -X3 of Power Supply. - Check the function (Main breaker and LEDs) of ADL-power supply. For trouble shooting refer to the manufacturer handbook.  Documentation for "Power supply – ADL" see CD-R, Section "Supplier's Documentation".
A_0613	=PG15+S2-G1 Pretreatment Power Supply: Communication	No Answer from Pretreatment Power Supply, Profibus connection lost. (together with A_0875) - Check LEDs on Profibus Card on the back of the Power Supply. The left one must blink green with 0.5 Hz, the right one must light red. - Check Configuration: Connect a PC with the Power Supply with the PS2 cable. Start Hyper terminal Program with 9600 8-N-1 and press spacebar for help: <pre> --- ADL Config V2.01 --- EEPROM: z: clear e: init d: download CONFIG: p: mode 1/P u: mode 2/U i: mode 3/I t: conf.-bits f: average-mode SET n: devicename j: DP-address TEST: a: setpoint=0 b: setpoint=1/2 c: setpoint=1 1: START ON 2: START OFF 3: ENABLE ON 4: ENABLE OFF 5: PRGM ON 6: PRGM OFF 7: MODE U ON 8: MODE U OFF 9: MODE I ON 0: MODE I OFF SHOW: w: IRAM -> SCI y: ADC 1-4 x: all Pins v: variables m: last error l: DP In/Out s: mode values g: DP-address SPEZIAL: r: restart h: hardware errors q: quit </pre> <pre> command -> g DP address: 30 command -> s device name: Test-/EE-Bits: 00H/00H modename P : 'P', unit : 'Rw', max. 4095.00, limit : 4095.00 modename U : 'U', unit : 'Rw', max. 4095.00, limit : 4095.00 modename I : 'I', unit : 'Rw', max. 4095.00, limit : 4095.00 command -> _ </pre>
A_0614	=PG15+S2-G1 Pretreatment Power Supply: Overtemperature	- Check function of Power Supply fan - Check free air intake of Power Supply
A_0615	=PG15+S2-G1 Pretreatment Power Supply: Not Remote	Operating mode LOCAL is selected. Set Switch on the Profibus card on the back of the Power Supply to Rem. (Remote).
A_0616	=PG15+S1-F1 Water Fault: Cathode Pretreatment	see fault number A_0065
A_0620	=PG15+S2 Pretreatment: Oxygen empty	- Connect new gas cylinder - Ensure that the manual valves at the gas cylinder, flow controller and manometer are opened

No	Alarm Message	Alarm Description
A_0621	=PG15+S2 Pretreatment: Argon empty	see fault number A_0620
A_0700	=WH11+S2 Unwinder: Web Tear during Standstill	• Check film for possible damage or faulty spots • Check parameter set-up for winder tension, diameter and winding direction
A_0701	=WH12+S2 Rewinder: Web Tear during Standstill	see fault number A_0700
A_0702	=WH11+S2 Unwinder: Web Tear	see fault number A_0700 • Check Film type in Window Parameters • Check Taper tension • Check Aluminum Wire speed in relation to Web speed • Check Vacuum • Check temperature of coating drum
A_0703	=WH12+S2 Rewinder: Web Tear	see fault number A_0700 • Check Film type in Window Parameters • Check Taper tension • Check Aluminum Wire speed in relation to Web speed • Check Vacuum • Check temperature of coating drum
A_0705	=WH10+G6-G1 Power Feeder Module Winding: Common Fault	Rectifier Regenerative unit SIMOVERT-MASTERDRIVE indicates defect. Type and cause of the defect will be displayed directly on the component unit by a fault display. • Verify the fault number and remove possible fault by referring to the manufacturer's documentation. • Check power supply on each phase  Documentation for "SIMOVERT-MASTERDRIVE" see CD-R, Section "Supplier's Documentation".
A_0706	=WH01+G6-Q1 Motor Fans: Circuit Breaker released	see fault number A_0200
A_0707	=WH11+G6-Q1 Chucking drive: Circuit Breaker released	see fault number A_0200
A_0711	=WH60+G6-Q1 Core Adjustment: Circuit Breaker released	see fault number A_0200
A_0712	=WH60+S2-M1 Core Adjustment: Undefined position	• Check free movement of Core Adjustment • Check Position of Limit switches at Motor



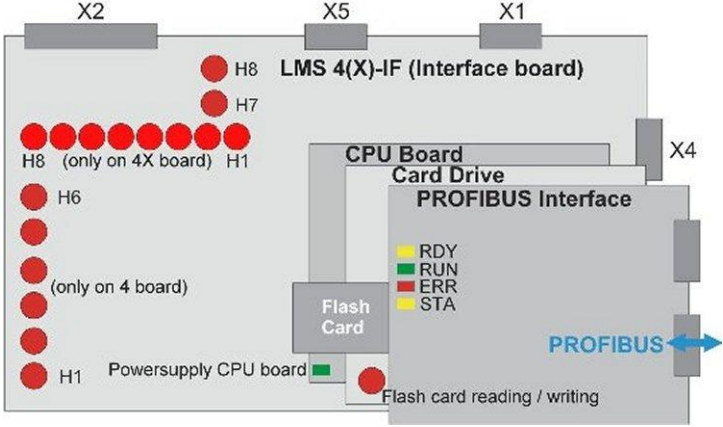
No	Alarm Message	Alarm Description
A_0713	=WH10+G6-K1 Winding System: Fault Main Contactor	see fault number A_0040
A_0714	=WH05+G6-Q1 Threading Device: Circuit Breaker tripped	see fault number A_0200
A_0715	=WH05+G6-K1 Threading Device: Contactor Fault	see fault number A_0040
A_0716	=WH60+GS2-B43/44 Core Adjustment Lock Base Plate: Undefined position	see fault number A_0734
A_0717	=WH60+GS2-B41/42 Core Adjustment Lock Bearing Plate Undefined position	see fault number A_0734
A_0720	=WH11+G6-F4 Unwinder Motor Fan: Bi-Metal tripped	Fan motor is too hot or works incorrect. Circuit breaker in cabinet is tripped (Bi-Metal - function) and the fan motor switches off. • Check setting of Bi-Metals • Check temperature
A_0721	=WH11+G6-G1 Unwinder Inverter: Fault Bus Communication	• Check inverter power supply • Check for wrong parameters in inverter • Check for bus problems, compare A_0840
A_0722	=WH11+S2-M1 Unwinder Motor: Overtemperature	• Allow motor to cool down • Check free air intake • Check ventilation
A_0723	=WH11+G6-G1 Unwinder Inverter: Overtemperature	• Allow Inverter to cool down • Check cabinet temperature • Check ventilation
A_0724	=WH11+G6-G1 Unwinder Inverter: Common Fault	Unit SIMOVERT-MASTERDRIVE indicates a defect. The cause of the defect is displayed directly on the unit through a display. • Verify the fault number and remove possible fault by referring to the manufacturer's documentation. Documentation for "SIMOVERT-MASTERDRIVE" see CD-R, Section "Supplier's Documentation".
A_0725	=WH11+G6-F11 Chuck Motor Unwinder Base Plate: Bi-Metal tripped	see fault number A_0200

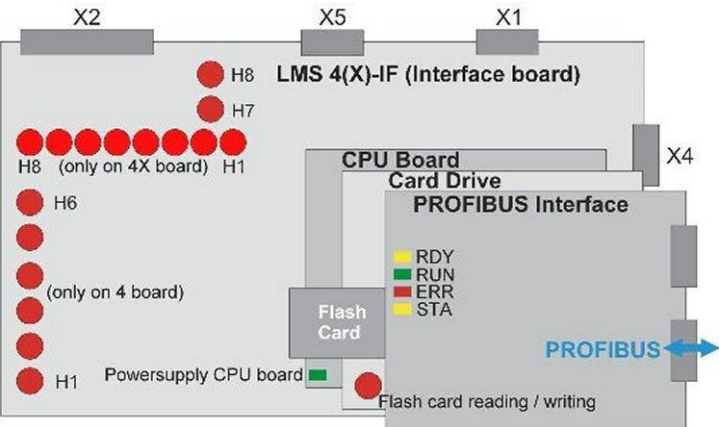
No	Alarm Message	Alarm Description
A_0726	=WH11+G6-F21 Chuck Motor Unwinder Bearing Plate: Bi-Metal tripped	see fault number A_0200
A_0730	=WH11+S2-B50/52 Cradle Unwinder not up	Cradle not up while trying to close the chucks Lift cradles
A_0731	=WH11+S2-B51/53 Cradle Unwinder not down	Cradle not down while trying to switch on the tension Lower cradles
A_0732	=WH11+S2-R11/12 Chuck Unwinder: not closed	Chucks not closed while trying to switch on the tension - Close Chucks or - Check motor function. - Check free movement of Chuck. - Check position and function of the zero position limit switch. - Check function of potentiometer. - Check assignment between potentiometer value and position in window "Core Clamping and Adjustment".
A_0734	=WH11+S2-B3.1 Interlock Unwinder Base Plate: Undefined position	Lock bolt do not reach the limit switch. - Check function of pneumatic piston. - Check free movement of bolt. - Check position and function of limit switches.
A_0735	=WH11+S2-B3.2 Interlock Unwinder Bearing Plate: Undefined position	see fault number A_0734
A_0736	=WH11+S2-B11/21 Core Unwinder: Undefined position	After Chucks are moved in while loading, no Core is detected or While Tension is switched on and Cradle is moved down, Core is detected. - Check position and condition of loaded Core. - Check position and function of limit switches.
A_0740	=WH12+G6-F4 Rewinder Motor Fan: Bi-Metal tripped	see fault number A_0720
A_0741	=WH12+G6-G1 Rewinder Inverter: Fault Bus Communication	see fault number A_0721
A_0742	=WH12+S2-M1 Rewinder Motor: Overtemperature	see fault number A_0722
A_0743	=WH12+G6-G1 Rewinder Inverter: Overtemperature	see fault number A_0723



No	Alarm Message	Alarm Description
A_0744	=WH12+G6-G1 Rewinder Inverter: Common Fault	see fault number A_0724
A_0745	=WH12+G6-F11 Chuck Motor Rewinder Base Plate: Bi-Metal tripped	see fault number A_0200
A_0746	=WH12+G6-F21 Chuck Motor Rewinder Bearing Plate: Bi-Metal tripped	see fault number A_0200
A_0750	=WH12+S2-B50/52 Cradle Rewinder not up	see fault number A_0730
A_0751	=WH12+S2-B51/53 Cradle Rewinder not down	see fault number A_0731
A_0752	=WH12+S2-R11/12 Chuck Rewinder: not closed	see fault number A_0732
A_0754	=WH12+S2-B3.1 Interlock Rewinder Base Plate: Undefined position	see fault number A_0734
A_0755	=WH12+S2-B3.2 Interlock Rewinder Bearing Plate: Undefined position	see fault number A_0734
A_0756	=WH12+S2-B11/21 Core Rewinder: Undefined position	see fault number A_0736
A_0757	=WH12+N29-X22/+N31-X12 Rewinder Limit Switches not connected	Plug in connectors =WH12+N29-X22 or =WH12+N31-X12
A_0760	=WH20+G6-F4 Tension Roll Motor Fan: Bi-Metal tripped	see fault number A_0720
A_0761	=WH20+G6-G1 Tension-Roll: Fault Bus Communication	see fault number A_0721
A_0762	=WH20+S2-M1 Tension-Roll Motor: Overtemperature:	see fault number A_0722
A_0763	=WH20+G6-G1 Tension Roll Inverter: Overtemperature	see fault number A_0723
A_0764	=WH20+G6-G1 Tension-Roll Inverter: Common Fault	see fault number A_0724
A_0770	=WH30+G6-F4 Coating Drum Motor Fan: Bi-Metal tripped	see fault number A_0720
A_0771	=WH30+G6-G1 Coating-Drum: Bus Communication Fault	see fault number A_0721
A_0772	=WH30+S2-M1 Coating-Drum Motor: Overtemperature	see fault number A_0722

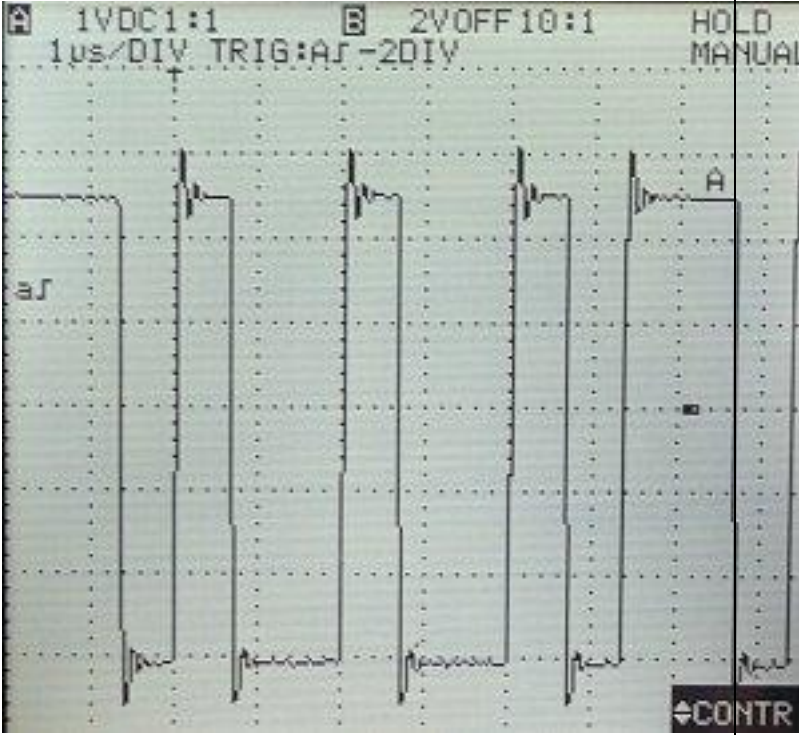


No	Alarm Message	Alarm Description
A_0773	=WH30+G6-G1 Coating-Drum Inverter: Overtemperature	see fault number A_0723
A_0774	=WH30+G6-G1 Coating-Drum Inverter: Common Fault	see fault number A_0724
A_0776	External Handling System: Collective Fault	See alarms displayed on the handling system
A_0800	=PM20+S2-A1 LMS: Device Error	No Data transfer between LMS device and PLC • Check power supply of LMS • Check LEDs on Profibus Interface card • RDY off: No power, hardware defect • RUN off or flashing: No Profibus connected • ERR on: Connection to LMS board lost • STA off: No parameters from PLC  • Compare A_0840 • Check LMS Status by connecting a keyboard and a monitor to the Interface box. If "CMOS checksum bad" boot manually and replace battery on CPU board. More detailed information about Service Mode, see under manufacturer documentation:  Documentation for "Layer Monitoring System" see CD-R, Section "Supplier's Documentation".

No	Alarm Message	Alarm Description
A_0802	=PM20+S2-A1 LMS: Timeout at RS485-Interface	No data transfer between measuring bar inside chamber and control unit =PM20+S2-A1 - Check red LEDs on LMS Interface board - H1...H6 Address-LEDs of the sending and receiving boards, runs not through while operating: Interface board broken - H7 off: System not active, switch to Service Mode in General Parameters Screen to test.  - Check interface cables and feedthrough - Check broken channel in LMS Service Screen and deselect broken channel in LMS Selection Screen
A_0803	=PM20+S2-A1 LMS: Calibration Error Measuring Head	- Select monitor screen „LMS-Service“, light intensity must be gain < 5, respectively gain < 8 for 4X - Cleaning or re-adjust measuring head If cleaning or readjustment is not effectual: - Replace transmitter- and receiver head or deselect broken channel More detailed information of defective head, see under manufacturer documentation:  Documentation for “Layer Monitoring System” see CD-R, Section “Supplier’s Documentation”.
A_0804	=PM20+S2-A1 LMS: Intensity Underflow Transmitter	Transmitter not bright enough. Trim with potentiometer in Transmitter. If not effectual: Replace or deselect channel. Lifetime of transmitter is approximately 10000h
A_0805	=PM20+S2-A1 LMS: Measuring Bar Power Failure	No power supply for measuring bar inside chamber. Check cables and feed through
A_0806	=PM20+S2-A1 LMS: Measuring Value out of Range	Counts or gain invalid format. Internal Fault. Possible reason: Receiver module missing or defect.

No	Alarm Message	Alarm Description
A_0807	=PM20+S2-A1 LMS: Receiver Intensity Underflow	Check the intensity in screen "LMS - Service". The intensity has to be higher than 100 Counts by gain = 9. If not, cleaning and readjustment of measuring heads is necessary. If not effectual: Replace or deselect channel.
A_0820	=PM20+S2-A2 LMS #2: Device Error	see fault number A_0800
A_0822	=PM20+S2-A2 LMS #2: Timeout at RS485-Interface	see fault number A_0802
A_0823	=PM20+S2-A2 LMS #2: Calibration Error Measuring Head	see fault number A_0803
A_0824	=PM20+S2-A2 LMS #2: Intensity Underflow Transmitter	see fault number A_0804
A_0825	=PM20+S2-A2 LMS #2: Measuring Bar Power Failure	see fault number A_0805
A_0826	=PM20+S2-A2 LMS #2: Measuring Value out of Range	see fault number A_0806
A_0827	=PM20+S2-A2 LMS #2: Receiver Intensity Underflow	see fault number A_0807



No	Alarm Message	Alarm Description
A_0840	Bus Address 1:	Bus member lost or defect. • Check the status of the component by its LEDs • Check the status of the OBTs by their LEDs, must be yellow if channel is ok. Check red light of fiber optic cable (in/out) • Check the bus connectors and their resistor switch: Must be on if last device on bus segment (one cable in connector), must be off if in between the bus segment (two cables in connector) • Ensure that devices at the end of an bus segment are power supplied • Check shielding ground of bus cable • Check bus signals with oscilloscope, signal should have rectangle form with straight edges and no disturbances  Example of good Profibus signal, 1.5MBaud, between lines A (-, green) and B (+, red). Attention: scope must be potentialfree, otherwise measure between a line and ground.
A_0845	Bus Address 6: =AS02+G6 Winding Cabinet	see fault number A_0840
A_0846	Bus Address 7: =AS03+G6 Winding Cabinet	see fault number A_0840
A_0847	Bus Address 8: =AS08+R11 Operator Panel	see fault number A_0840
A_0850	Bus Address 9: =AS09+R9 Operator Panel	see fault number A_0840

No	Alarm Message	Alarm Description
A_0851	Bus Address 10: =AS10+R10 Operator Panel	see fault number A_0840
A_0852	Bus Address 11: =AS11+N11 PLC Box Fixed Part	see fault number A_0840
A_0853	Bus Address 12: =AS12+N12 PLC Box Pump Unit	see fault number A_0840
A_0854	Bus Address 13: =AS13+N12 PLC Box Pumping Unit	see fault number A_0840
A_0855	Bus Address 14: =VC21+G6 Pump Inverter	see fault number A_0840
A_0856	Bus Address 15: =VC22+G6 Pump Inverter	see fault number A_0840
A_0857	Bus Address 16: =VC23+G6 Pump Inverter	see fault number A_0840
A_0860	Bus Address 17: =VC31+G6 Pump Inverter	see fault number A_0840
A_0861	Bus Address 18: =VC32+G6 Pump Inverter	see fault number A_0840
A_0875	Bus Address 30: =PG15+S2 Pretreatment Power Supply	see fault number A_0840
A_0882	Bus Address 35: =WC20+S2 Gas Pipe Flow Controller	see fault number A_0840
A_0887	Bus Address 40: =PD00+G5 PAC-F Stepper Wire Feeder 1	see fault number A_0840
A_0890	Bus Address 41: =WH11+G6 Unwinder Inverter	see fault number A_0840
A_0891	Bus Address 42: =WH12+G6 Rewinder Inverter	see fault number A_0840
A_0892	Bus Address 43: =WH20+G6 Tension Roll Inverter	see fault number A_0840
A_0893	Bus Address 44: =WH30+G6 Coating Drum Inverter	see fault number A_0840
A_0897	Bus Address 48: =PE10+N11 Com. Shaft Wire Feeder Inverter	see fault number A_0840
A_0900	Bus Address 49: =KA01+S2 Cooling and Heating Unit Controller	see fault number A_0840
A_0962	Bus Address 99: =PM20+S2-A1 LMS-OD 1	see fault number A_0840



No	Alarm Message	Alarm Description
A_1047	Bus 2 Address 40: =PD00+G4 PAC-F Stepper Wire Feeder 2	see fault number A_0840
A_1050	Bus 2 Address 41: =PC01+G1 Evaporator 01 Thyristor Controller	see fault number A_0840
A_1051	Bus 2 Address 42: =PC02+G1 Evaporator 02 Thyristor Controller	see fault number A_0840
A_1052	Bus 2 Address 43: =PC03+G1 Evaporator 03 Thyristor Controller	see fault number A_0840
A_1053	Bus 2 Address 44: =PC04+G1 Evaporator 04 Thyristor Controller	see fault number A_0840
A_1054	Bus 2 Address 45: =PC05+G1 Evaporator 05 Thyristor Controller	see fault number A_0840
A_1055	Bus 2 Address 46: =PC06+G1 Evaporator 06 Thyristor Controller	see fault number A_0840
A_1056	Bus 2 Address 47: =PC07+G1 Evaporator 07 Thyristor Controller	see fault number A_0840
A_1057	Bus 2 Address 48: =PC08+G1 Evaporator 08 Thyristor Controller	see fault number A_0840
A_1060	Bus 2 Address 49: =PC09+G1 Evaporator 09 Thyristor Controller	see fault number A_0840
A_1061	Bus 2 Address 50: =PC10+G1 Evaporator 10 Thyristor Controller	see fault number A_0840
A_1062	Bus 2 Address 51: =PC11+G1 Evaporator 11 Thyristor Controller	see fault number A_0840
A_1063	Bus 2 Address 52: =PC12+G1 Evaporator 12 Thyristor Controller	see fault number A_0840
A_1064	Bus 2 Address 53: =PC13+G1 Evaporator 13 Thyristor Controller	see fault number A_0840
A_1065	Bus 2 Address 54: =PC14+G1 Evaporator 14 Thyristor Controller	see fault number A_0840
A_1066	Bus 2 Address 55: =PC15+G1 Evaporator 15 Thyristor Controller	see fault number A_0840
A_1067	Bus 2 Address 56: =PC16+G1 Evaporator 16 Thyristor Controller	see fault number A_0840
A_1070	Bus 2 Address 57: =PC17+G1 Evaporator 17 Thyristor Controller	see fault number A_0840
A_1071	Bus 2 Address 58: =PC18+G1 Evaporator 18 Thyristor Controller	see fault number A_0840



No	Alarm Message	Alarm Description
A_1072	Bus 2 Address 59: =PC19+G1 Evaporator 19 Thyristor Controller	see fault number A_0840
A_1073	Bus 2 Address 60: =PC20+G1 Evaporator 20 Thyristor Controller	see fault number A_0840
A_1074	Bus 2 Address 61: =PC21+G1 Evaporator 21 Thyristor Controller	see fault number A_0840
A_1075	Bus 2 Address 62: =PC22+G1 Evaporator 22 Thyristor Controller	see fault number A_0840
A_1076	Bus 2 Address 63: =PC23+G1 Evaporator 23 Thyristor Controller	see fault number A_0840
A_1077	Bus 2 Address 64: =PC24+G1 Evaporator 24 Thyristor Controller	see fault number A_0840
A_1080	Bus 2 Address 65: =PC25+G1 Evaporator 25 Thyristor Controller	see fault number A_0840
A_1081	Bus 2 Address 66: =PC26+G1 Evaporator 26 Thyristor Controller	see fault number A_0840
A_1082	Bus 2 Address 67: =PC27+G1 Evaporator 27 Thyristor Controller	see fault number A_0840
A_1083	Bus 2 Address 68: =PC28+G1 Evaporator 28 Thyristor Controller	see fault number A_0840
A_1084	Bus 2 Address 69: =PC29+G1 Evaporator 29 Thyristor Controller	see fault number A_0840
A_1085	Bus 2 Address 70: =PC30+G1 Evaporator 30 Thyristor Controller	see fault number A_0840
A_1086	Bus 2 Address 71: =PC31+G1 Evaporator 31 Thyristor Controller	see fault number A_0840
A_1087	Bus 2 Address 72: =PC32+G1 Evaporator 32 Thyristor Controller	see fault number A_0840
A_1090	Bus 2 Address 73: =PC33+G1 Evaporator 33 Thyristor Controller	see fault number A_0840
A_1091	Bus 2 Address 74: =PC34+G1 Evaporator 34 Thyristor Controller	see fault number A_0840
A_1092	Bus 2 Address 75: =PC35+G1 Evaporator 35 Thyristor Controller	see fault number A_0840
A_1093	Bus 2 Address 76: =PC36+G1 Evaporator 36 Thyristor Controller	see fault number A_0840
A_1094	Bus 2 Address 77: =PC37+G1 Evaporator 37 Thyristor Controller	see fault number A_0840



No	Alarm Message	Alarm Description
A_1095	Bus 2 Address 78: =PC38+G1 Evaporator 38 Thyristor Controller	see fault number A_0840
A_1096	Bus 2 Address 79: =PC39+G1 Evaporator 39 Thyristor Controller	see fault number A_0840
A_1097	Bus 2 Address 80: =PC40+G1 Evaporator 40 Thyristor Controller	see fault number A_0840
A_1100	Bus 2 Address 81: =PC41+G1 Evaporator 41 Thyristor Controller	see fault number A_0840
A_1101	Bus 2 Address 82: =PC42+G1 Evaporator 42 Thyristor Controller	see fault number A_0840
A_1102	Bus 2 Address 83: =PC43+G1 Evaporator 43 Thyristor Controller	see fault number A_0840
A_1103	Bus 2 Address 84: =PC44+G1 Evaporator 44 Thyristor Controller	see fault number A_0840
A_1104	Bus 2 Address 85: =PC45+G1 Evaporator 45 Thyristor Controller	see fault number A_0840
A_1105	Bus 2 Address 86: =PC46+G1 Evaporator 46 Thyristor Controller	see fault number A_0840
A_1106	Bus 2 Address 87: =PC47+G1 Evaporator 47 Thyristor Controller	see fault number A_0840
A_1107	Bus 2 Address 88: =PC48+G1 Evaporator 48 Thyristor Controller	see fault number A_0840
A_1110	Bus 2 Address 89: =PC49+G1 Evaporator 49 Thyristor Controller	see fault number A_0840
A_1111	Bus 2 Address 90: =PC50+G1 Evaporator 50 Thyristor Controller	see fault number A_0840
A_1112	Bus 2 Address 91: =PC51+G1 Evaporator 51 Thyristor Controller	see fault number A_0840
A_1113	Bus 2 Address 92: =PC52+G1 Evaporator 52 Thyristor Controller	see fault number A_0840
A_1114	Bus 2 Address 93: =PC53+G1 Evaporator 53 Thyristor Controller	see fault number A_0840
A_1115	Bus 2 Address 94: =PC54+G1 Evaporator 54 Thyristor Controller	see fault number A_0840
A_1116	Bus 2 Address 95: =PC55+G1 Evaporator 55 Thyristor Controller	see fault number A_0840
A_1117	Bus 2 Address 96: =PC56+G1 Evaporator 56 Thyristor Controller	see fault number A_0840



No	Alarm Message	Alarm Description
W_2000	Warning: PLC S7-400 Buffer Battery low	- Change buffer battery on S7-400 Power supply module. - Information and safety rules for carry out the maintenance work see on section „Maintenance Instruction“ and on manufacturer documentation.  Documentation for “SIMATIC S7-400” see CD-R, Section “Supplier’s Documentation”.
W_2001	Warning: Service-Mode is selected!	Before further plant operation or after ending the service work, reset service mode: Select monitor window "General Parameters" and set function field Service Mode to OFF.
W_2002	Warning: =AA00+G6-Q1 Maintenance Switch for Winding is off	The following components are switched off: - Winding System - Threading - Pivot Arm - Spreader rolls - Chucking System - Axial Adjustment - Core Adjustment - Trolley Drive - Coating Window
W_2003	Warning: =AA00+N12-Q1 Maintenance Switch for Pumping is off	All mechanical Pumps without the backup unit for the Diffusion Pumps (=VC15/25) are switched off
W_2004	Warning: Coated Width bigger than Film Width!	Check Width in Parameter Window.
W_2005	Warning: PIMS-SERVER Connection lost	You have no access to - Historical Alarms - Event Logging - Recipes - Historical Values for Friction, Leak Rate, Scope etc. - PIMS functions - Check PIMS-SERVER connection with Internet Explorer: http://PIMS-SERVER, Terminal Server Access window must appear. - Check PIMS-SERVER PC is running. - Check cable connection. - Check the “System Management Console“ on PIMS-SERVER



No	Alarm Message	Alarm Description																				
		- Go to "IndustrialSQL Server" ->"IndustrialSQL Server Group"->"PIMS-Server"->"Management Console"->"Status" - Check if all modules are started in the upper right pane <table><tr><th>Module</th><th>Status</th></tr><tr><td> Storage</td><td>Started</td></tr><tr><td> Manual storage</td><td>Started</td></tr><tr><td> Event system</td><td>Started</td></tr><tr><td> Retrieval</td><td>Started</td></tr><tr><td> Indexing</td><td>Started</td></tr><tr><td> OLE-DB provider</td><td>Started</td></tr><tr><td> InSQLIOS</td><td>Started</td></tr><tr><td> System driver</td><td>Started</td></tr><tr><td> Data acquisition on \\PIMS-SE...</td><td>Started</td></tr></table> - If one ore more modules are stopped, check the log in the lower pane for errors - To restart the InSQL Server right click on „Status“ then choose „Stop InSQL...“ - In the following dialog please enter administrator as „Login Name“ and the corresponding Password Stop InSQL Server General Stop InSQL Server Please enter an authorized user ID to Stop InSQL S the following InSQL Server: Server: PIMS-SERVER Domain: Login Name: administrator Password: xxxxxxxx ☐ Do not stop IDAS configured for Store Forward OK Cancel - Wait until are modules are stopped, this can take a few minutes - When all modules stopped, right click on „Status“ then choose „Start InSQL...“ - In the following dialog please enter administrator as „Login Name“ and the corresponding Password - Wait until are modules are started, this can take a few minutes	Module	Status	Storage	Started	Manual storage	Started	Event system	Started	Retrieval	Started	Indexing	Started	OLE-DB provider	Started	InSQLIOS	Started	System driver	Started	Data acquisition on \\PIMS-SE...	Started
Module	Status																					
Storage	Started																					
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OLE-DB provider	Started																					
InSQLIOS	Started																					
System driver	Started																					
Data acquisition on \\PIMS-SE...	Started																					

No	Alarm Message	Alarm Description
		- In the case that some modules would not start please check log, there also a log with more details under "Log Viewer"->"Default Group"->"Local" - If all that does not solve the problem please restart Windows
W_2007	Warning: Data logging: Drive, Path or Filename not valid	Data cannot be saved. - Incorrect Filename - The drive is possibly write-protected - No network connection available
W_2010	Warning: =VM21+S1-P1 High Vacuum Gauge not ignited	High Vacuum Gauge not ignited within 5 minutes after switching on. Clean or replace tube if necessary
W_2011	Warning: Leak-Rate-Test is finished!	Information that the leak rate test is finished. A printout is possible.
W_2012	Warning: =VB11+S1-E1 Diffusion Pump: Check Oil	- Check oil level and color of oil in diffusion pump. - Change oil if the color is brown or the pressure rises while evaporator is switched on
W_2013	Warning: =VB12+S1-E1 Diffusion Pump: Check Oil	see warning number W_2012
W_2014	Warning: =VB13+S1-E1 Diffusion Pump: Check Oil	see warning number W_2012
W_2015	Warning: =VB14+S1-E1 Diffusion Pump: Check Oil	see warning number W_2012
W_2016	Warning: =VB15+S1-E1 Diffusion Pump: Check Oil	see warning number W_2012
W_2020	Warning: =VC11+S1-M1 Rotary Vane Pump: Oil Change is due	Warning message indicates that the maintenance interval has been reached for the pump. Change oil / change oil filters.
W_2021	Warning: =VC12+S1-M1 Rotary Vane Pump: Oil Change is due	see warning number W_2020
W_2022	Warning: =VC13+S1-M1 Rotary Vane Pump: Oil Change is due	see warning number W_2020
W_2023	Warning: =VC14+S1-M1 Rotary Vane Pump: Oil Change is due	see warning number W_2020
W_2024	Warning: =VC15+S1-M1 Rotary Vane Pump: Oil Change is due	see warning number W_2020
W_2025	Warning: =VC21+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020



No	Alarm Message	Alarm Description
W_2026	Warning: =VC22+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2027	Warning: =VC23+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2030	Warning: =VC24+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2031	Warning: =VC25+S1-M1 Rotary Vane Pump: Oil Change is due	see warning number W_2020
W_2032	Warning: =VC31+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2033	Warning: =VC32+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2034	Warning: =VC33+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2035	Warning: =VC34+S1-M1 Roots Pump: Oil Change is due	see warning number W_2020
W_2040	Warning: Trolley not locked with Handling System	Trolley Lock with Handling System manually switched off. Attention: Be sure that Handling System is not in Chamber when close it.
W_2041	Warning: Trolley locked with Handling System	Remove Handling System out of chamber.
W_2044	Warning: Pivot Arm in Outside-Position!	Move pivot arm to rewinder so that vacuum chamber can be closed.
W_2045	Warning: Can't move Pivot Arm: Chamber not open!	Movement only possible when chamber is complete opened.
W_2046	Warning: Pivot Arm is outside: Winding stopped!	Rewinder roll diameter is too large.
W_2047	Warning: Separation Plate not closed!	Close Separation Plate at Coating Drum so that vacuum chamber can be closed.
W_2050	Warning: Can't move Coating Window: Chamber not open!	Open chamber complete.
W_2051	Warning: Coating Window in Cleaning Position!	Move coating window in working position so that vacuum chamber can be closed.



No	Alarm Message	Alarm Description
W_2052	Warning: Coldtraps too cold to open chamber	The cold traps have not yet heated up to the correct setpoint temperature (trigger point). Chamber cannot open! Danger of frostbite! • Check the coldtraps are active in DEFROST mode • Check function of coldtraps • To open the chamber nevertheless, set all coldtraps to local mode or switch it off in Parameter Window of Visualisation. Attention: Danger of frostbite!
W_2060	Warning: Cooling and Heating Unit :Maintenance is due	The working hours of the compressor reaches 5000h. Follow the Cool & Heat Unit documentation to carry out the maintenance.
W_2061	Warning: Cooling and Heating Unit: too cold for Venting!	The coating drum has not yet heated up to the correct temperature (trigger point). Danger of condensation of water -> rust Check whether: • The cooling and heating equipment is active in HEAT mode • The trigger point for the heating up parameter is correctly set
W_2062	Warning: Cooling and Heating Unit is not remote controlled!	Switch operating selection switch on the cooling and heating equipment to REMOTE. Remove possible fault by referring to the manufacturer's documentation.  Documentation for "Cooling- and Heating Unit" see CD-R, Section "Supplier's Documentation".
W_2063	Warning: No Cooling, Evaporation Not Possible	Cooling water of Evaporator or Cooling & Heating Unit not in function. Evaporator will shut off in 5 minutes. • Check coolant supply for evaporation station • Check function of Cooling & Heating Unit
W_2064	Warning: Not enough Aluminum Wire for the following Run	See window "Wire Consumption" from Window "Service Evaporator" for which coil is to short. Here you can disable the warning too.
W_2065	Warning: Boat Conditioning is finished!	Information that the conditioning procedure is finished.
W_2070	Warning: ZnS/SiOx Evaporator Temperature Sensor 1 defective	Check Sensor, Cabling and Feed through.
W_2071	Warning: ZnS/SiOx Evaporator Temperature Sensor 2 defective	see warning number W_2070
W_2072	Warning: ZnS/SiOx Evaporator Temperature Sensor 3 defective	see warning number W_2070



No	Alarm Message	Alarm Description
W_2075	Warning: Pretreatment: No Plasma	The Output Power of the Power supply is smaller than 3% - Check Power Supply Device =PG15 for internal errors - Check Cabling and Connectors - Check Cathode for short circuit or too much Arcing - Check Gases
W_2076	Warning: =PG15+S2 Pretreatment: Low Voltage / Short Circuit	- Check cathode and environment for dust. Clean if necessary. - Measure the resistance between target and ground. The resistance must be greater than 80 kOhm
W_2077	Warning: Unwinder Alarm Diameter is reached!	To reset warning hit the "Acknowledge Alarm" or the "Decrease CD Speed" button on the operator panel.
W_2080	Warning: Stop Length is reached!	The winding system decelerates to reach the preselected stop length of film. Winding will be stopped.
W_2081	Warning: Film Slip in Winding Direction!	see fault number A_0702
W_2082	Warning: Film Slip against Winding Direction!	see warning number W_2081
W_2083	Warning: Web Tear Monitoring is disabled!	Remember to switch on the web tear monitoring after the service work is finished. Important: If web tear monitoring has been switched off (during normal operation mode) serious damages can occur in case of web tear.
W_2084	Warning: Winding System in Friction Recording Mode	Before further operation of the plant or after ending the friction curve entry: Select monitor display "Winding Parameters" and reset Friction recording mode
W_2085	Warning: Friction-Recording only under Vacuum!	Remark: Necessary pressure level for friction curve acceptance has not yet been reached. The release for entry takes place at a pressure of 2×10^{-2} mbar. Corrective Action: Start friction curves only when 2×10^{-2} mbar has been reached.
W_2086	Warning: Stop Diameter will be reached!	The winding system decelerates to reach the preselected stop diameter on the winder. Winding will be stopped.
W_2087	Warning: Winding-System is in Test-Mode!	Before further plant operation or after ending the service work, reset test mode: Select monitor window "Winding Parameters" and set function field test Mode to OFF.



No	Alarm Message	Alarm Description
W_2090	Warning: Winding System is disabled for Lockout!	Now the system is prepared to switch the lockout off
W_2091	Warning: Mechanical Pumps are disabled for Lockout!	see warning number W_2090
W_2093	Warning: Layer Thickness out of Tolerance	The Layer Thickness is out of Tolerance longer than the time set in Window "Parameters". • Check if Layer lengthwise controller and cross controller are switched on • Check Tolerance Value in Window "Parameters" • Check Tolerance time in Window "LMS Service"

